

The Mathematical Universe

by

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Published by Hyperreality Books

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Quotations

“I can’t understand why people are frightened of new ideas. I’m frightened of the old ones.” – John Cage.

“The true spirit of delight, the exaltation, the sense of being more than Man, which is the touchstone of the highest excellence, is to be found in mathematics as surely as poetry.” – Bertrand Russell

“God used beautiful mathematics in creating the world.” – Paul Dirac

“Pure mathematics is, in its way, the poetry of logical ideas.” – Albert Einstein

“If I were again beginning my studies, I would follow the advice of Plato and start with mathematics.” – Galileo Galilei

“If a man’s wit be wandering, let him study the mathematics.” – Francis Bacon

“Mathematics allows for no hypocrisy and no vagueness.” – Stendhal

“Consequently he who wishes to attain to human perfection, must therefore first study Logic, next the various branches of Mathematics in their proper order, then Physics, and lastly Metaphysics.” – Maimonides

“For the things of this world cannot be made known without a knowledge of mathematics.” – Roger Bacon

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Introduction

The universe is a mathematical hologram. It's made of ontological mathematics. It's a *living, thinking*, self-optimising holographic organism composed of immortal, indestructible, ontological mathematical units called *monads*, defined by the most powerful and beautiful equation in the whole of mathematics: Euler's Formula.

Monads have a much more resonant name: *souls*. We all inhabit *Soul World*, a wondrous Singularity outside space and time. Our souls are individual mathematical singularities: autonomous, uncaused, uncreated, dimensionless frequency domains. Via Fourier mathematics, these imperishable, immaterial monadic souls can collectively create the spacetime domain of the material world. Where each soul is a single frequency domain, the material world of space and time is their *collective* Fourier output. What is "matter"? It's simply *dimensional* energy: energy existing in the Fourier spacetime domain rather than in the Fourier *dimensionless* frequency domain.

Souls are immense mathematical vibrations, based on precise, analytic cosine waves and imaginary sine waves (hence are defined by complex numbers rather than the real numbers of scientific materialism). From these waves, we get wave mechanics (quantum mechanics) and holography, i.e. a complete explanation of the material world.

Fourier mathematics solves the previously intractable problem of Cartesian dualism (the famous mind-body problem), i.e. how unextended minds can interact with extended matter. Minds are just Fourier frequency domains and bodies Fourier spacetime domains. Bodies are nothing but an alternative mathematical way of representing mental information. They are mental constructs or projections, and have no independent existence.

What was the Big Bang? It was a purely mathematical operation in which a frequency domain of mathematical souls (a Singularity), outside space and time, generated a Fourier spacetime domain: a cosmic hologram grounded in quantum mechanics. It really is as simple as that.

As Plato recognised, true reality belongs to the *intelligible* domain (which, mathematically, is an eternal, immutable frequency domain based on Euler's Formula). Illusory, contingent reality constitutes the *sensible* domain studied by scientists.

It's the rational mind, not sensory experiments, that reveals the eternal, intelligible "truths of reason". The sensible world is all about "truths of fact", which have no eternal necessity.

Reality is defined by a single mathematical law: the *God Equation*, derived from Euler's Formula. This single equation generates and controls the entire universe. It's the True God – an all-powerful ontological equation, but it's certainly not a person. It's outside space and time and yet can create space and time. It's the uncaused cause of everything, the Prime Mover.

What, at root, are mind and life? They are simply the eternal flow of structured information – mathematical waves. Consciousness is what arises when this information flow becomes self-aware and can attach "I" to itself. Information is all about numbers. As Pythagoras said, "All things are numbers; number rules all." He was asserting that we inhabit a universe of information, of ontological mathematics.

Leibniz was the greatest of all the inheritors of Pythagoras's mathematical legacy. His principle of sufficient reason is the quintessence of ontological mathematics. With this principle and the Leibnizian doctrine of *compossibility*, it can be shown that Max Tegmark's Mathematical Multiverse and indeed all Multiverse theories are fallacious.

Ontological mathematics shows how quantum indeterminacy can be overcome and

replaced with deterministic quantum mechanics of the kind of which Einstein dreamt. There are no such things as randomness and acausality in a universe of cause and effect: everything has a precise reason why it is thus and not otherwise. Through ontological mathematics, the dice-playing God is abolished and the God of Reason and causality is restored.

Only one subject is necessarily eternally valid, and that is mathematics. Nothing and no one can create mathematics. It, however, creates everything else. Without mathematics, existence doesn't have an answer. A universe without an answer is an *impossible* universe. Scientific materialism is an irrationalist claim that the universe exists for no reason at all. The next scientific revolution will see empiricist science replaced by rationalist ontological mathematics.

Pythagorean Illuminism

Professor Brian Cox, the British media's science darling, regularly scoffs at philosophy and presents it as the perfect time-wasting device for intelligent people. For Cox, the scientific method is the only path to knowledge.

Where did it all go wrong for philosophy? How did it become a joke? Today, philosophy divides into the tedious pedantry and falsehoods of analytic philosophy and the cynical, nihilistic social criticism of postmodernism. Philosophy has fled the battlefield as far as the big questions go. Where once it thought it could answer everything, now its ambitions have shrunk to analysing whether the "King of France is bald", or deconstructing the "agenda" served by such a question, or revealing the unstable meaning of the words involved.

Philosophy's problems can be traced to one fateful choice. Science made the right call and philosophy didn't ... because science embraced *mathematics* and philosophy didn't. Think of science without mathematics: it would simply be soothsaying, astrology and alchemy. The only thing that gives science its power is mathematics.

Oscar Wilde observed, "We are all in the gutter, but some of us are looking at the stars." Thanks to mathematics, it's Brian Cox looking at the stars, and the philosophers studying the gutter. Yet it could all have been so different. Pythagoras was the first person to call himself a philosopher – *a lover of wisdom* – yet he was also the first recognisable mathematician, and the forefather of natural science. Above all, he was the first person to understand the staggering *ontological* significance of mathematics, proclaiming, "All things are numbers." He gave as the motto of his Pythagorean sect, "Number rules all."

Bertrand Russell said that the vast majority of Plato's monumental philosophy had its roots in Pythagoreanism, and commented, "The whole conception of an eternal world, revealed to the intellect but not to the senses, is derived from him [Pythagoras]." This in fact sums up the difference between ontological mathematics and science. Ontological mathematics places the intellect over the senses, and science does the reverse. Ontological mathematics says that reason alone reveals the secrets of existence, while science says the senses (via experiments and observations) accomplish this. This should make it clear that science is a fundamentally irrational and anti-intellectual undertaking designed for those for whom "seeing is believing" and "rational unobservables" (hidden variables) are inconceivable.

For science, anything upon which experiments cannot be performed cannot exist. This, ironically, is a metaphysical (hence unscientific) assertion since there's no proof that it's true, no compelling argument why it should be true, and there's no sufficient reason for it whatsoever. It's simply a dogmatic assertion of materialism and empiricism and constitutes a faith-based position.

Science places experiments at its core, which is reasonable enough, but then goes on to conclude that if experiments can't be performed on something then that thing can't exist, which is utterly *unreasonable*. Even worse, a huge amount of advanced science, especially cosmology, hypocritically revolves around metaphysical speculations concerning such pseudo-mathematical

concepts as the Multiverse and “strings” that will *never* be amenable to direct experimental verification.

The Grand Unifying Element

The pre-Socratic philosophers of ancient Greece were obsessed with the *arche* – the fundamental substance from which everything else is said to be made (and which thus provides the unseen unity of things). Their various answers are usually misrepresented because the context is never properly explained. The ancient Greeks were typically *hylozoists*, i.e. they believed that matter is alive in some way. They were also *organicists*, insisting that reality must be considered holistically (in terms of all of its parts at once rather than its individual parts one at a time). Organicism asserts that systems are either outright living organisms, or ought to be treated that way (in contrast with the mechanistic, reductive approach of scientific materialism).

Thales said that the *arche* was water but he meant “living” water, possessing mind, spirit, reason, or some such ordering, animating quality. Aristotle reported Thales as having declared, “All things are full of gods”, and Hippolytus wrote that Thales said, “This principle [water] is god, and it has neither beginning nor end.” So, we have to forget any modern notions of water, and the same goes for all of the other substances proposed by other Greek philosophers as the *arche*.

Anaximander said the *arche* was the *apeiron* (an infinite, unbounded substance which maintained a perfect rational balance between everything); Pythagoras said it was numbers (mathematics); Anaximenes said it was “air” (although “breath”, implying life, is probably more accurate); Heraclitus said it was fire (but with “fire” being strongly linked to the cosmic, rational soul and what we might call mathematical energy). In all cases, no one was thinking of a matter-only world: a reductive world, a dead, mindless, pointless, mechanistic world of the type envisaged by modern scientists.

Ontological Numbers

Pythagoras taught that the number one was a point. Joining two points produced a straight line (so “two” was a line), three a plane, and four a three-dimensional solid. Therefore, everything comes from points (“ones”), and builds up through twos, threes and, especially, fours (in a 3D world). The sum of these first four numbers is ten and, for Pythagoras, ten (the decad) was the divine number, the one he and his sect held in especial reverence. The decad was the cosmic number enshrined in the divine triangle, the *tetraktys* of the decad:



The tetraktys is formally an equilateral triangle formed from the first four numbers (1, 2, 3 and 4) arrayed in four rows, and with a total sum of ten (1 + 2 + 3 + 4). It’s both a mathematical and metaphysical symbol that (so the Pythagoreans believed), conveyed the basic secrets of the universe. In seedlike form, it contained the principles of the Creation (from a point), the harmony of the cosmos, emanation, and the return (ascent) to the divine.

When swearing their most solemn oaths, the Pythagoreans declared, “I swear by Him who has revealed to our soul the divine tetraktys which contains the fount and root of eternal nature.”

The tetraktys is the fourth triangular number and symbolizes Unity, Power, the Limited, the Unlimited, Harmony and the Cosmos. If “one” is the basic unity, “ten” (the number of the tetraktys) is a higher order unity.

The tetraktys also symbolizes the four classical elements: fire, air, water, and earth. In another sense, it symbolizes dimensionality and how space is organized. The number one, the first row of the tetraktys, represents the zero-dimensional point. The second row of the tetraktys represents a one-dimensional line (of two points). The third row represents a two-dimensional plane (exemplified by a triangle of three points), and the fourth row represents three-dimensions (exemplified by a tetrahedron defined by four points).

With the tetraktys, we see a point (the apex) giving rise to two points, then three, then four, in an increasingly “solid” cascade, and it’s easy to imagine the whole material world emerging or emanating from that initial point.

Taoism employs a remarkably similar idea:

The Tao begot one.

One begot two.

Two begot three.

And three begot the ten thousand things.

The ten thousand things carry yin and embrace yang.

They achieve harmony by combining these forces.

(Tao Te Ching – chapter 42: Lao Tzu)

Pythagoras taught that each number had its own special attributes, and these became highly influential in terms of numerology (which is to number theory what astrology is to astronomy):

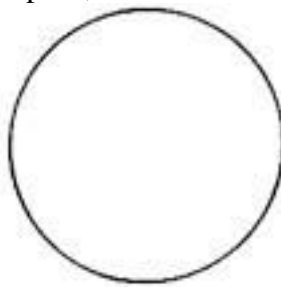
Number

Attribute

Diagram

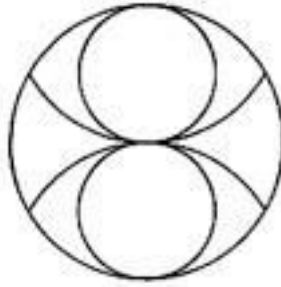
1

Monad (unity): the number of reason, the generator of numbers; stability; the origin of all thoughts in the universe; Apollo, the sun, Jupiter, Reason.



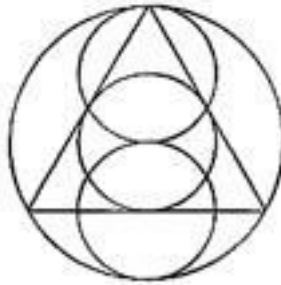
2

Dyad (diversity, opinion, otherness): the first female number; audacity; the first number separated from the Divine One. Monad = father while Dyad = mother; Monad = wisdom while Dyad = ignorance. Dyad is the Demiurge, the False God.



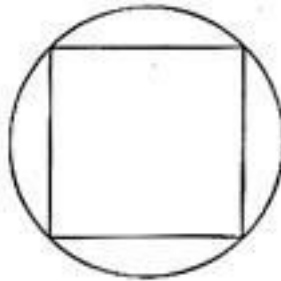
3

Triad (harmony = unity + diversity): first male number; the first authentic number; Chronos, ruler of time; the number of knowledge comprising music, geometry and astronomy (making up the science of the celestials and terrestrials).



4

Tetrad (justice, retribution): squaring of accounts; the root of all things; the fountain of nature, the most perfect number; defining the four powers of the soul of man.



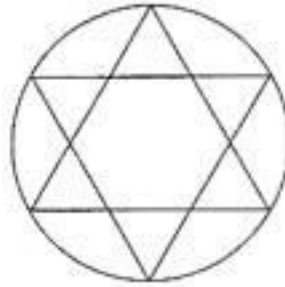
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Pentad (marriage): the union of the first female (2) + first male (3); the union of an odd and even number; the symbol of health, vitality and light; the mysterious fifth element unifying the four elements of earth, water, air and fire.



6

Hexad (creation): first female + first male + 1; perfection of all parts; harmony. A mathematically perfect number. $1 + 2 + 3 = 6$; $1 \times 2 \times 3 = 6$.



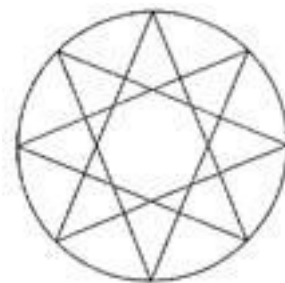
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Heptad: number of the law; we are ruled by seven celestial spirits (gods) – the planets.



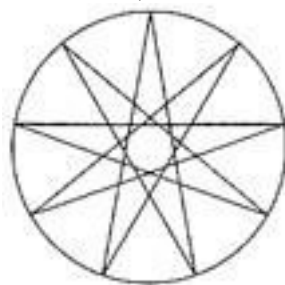
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Octad (ogdoad): number of the first cube (with eight corners); symbol of love, prudence and solidity.



9

Ennead: first square of an odd number; symbolic of failure and shortcomings because it falls just short of the perfect, cosmic number of ten (the sum of the first four numbers).



10

Decad (the universe): the greatest of numbers because it reflects all harmonic and

arithmetic proportions; both heaven and the world; the Tetraktys (closely related to the Tetrad).

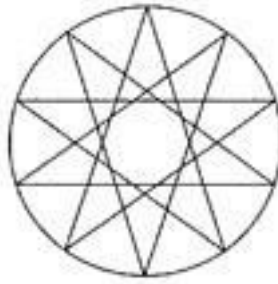


Diagram Source: Kenneth Sylvan Guthrie and David Fideler: *The Pythagorean Sourcebook and Library*

The Pythagorean Big Bang

Imagine a point – a Singularity – comprised of infinite points (since any number of points can be superimposed on one point given that none of them occupies any physical space). From this Singularity (the supreme Monad), endless points (individual monads) can emerge to create all of the mathematical objects of the world through their various combinations and relations.

The Neoplatonists relied on a similar scheme. The ineffable “One” was the origin, the “Nous” (Mind/Spirit) was its first emanation, the “Psyche” (Soul) was its second emanation, and “Nature” its final emanation (we thus have the four levels of the Pythagorean tetraktys). Each emanation contemplates the level above it.

With the notion of everything pouring from a dimensionless point (“nothing”), we have no less than a prototype Big Bang theory – a whole universe being generated by mathematical points emanating from a Singularity outside space and time.

Pythagoras provided a complete mathematical explanation of the nature of reality, but the world simply wasn’t ready for him. He was thousands of years ahead of his time. In fact, the world still hasn’t caught up with Pythagoras. Science – mathematics *lite* – has stolen the crown of its master. Science is the false claimant, the impostor.

Getting to the Point

Pythagoras’s system has the mathematical point (the unit; the monad) as its basic element. Everything else is derived from it. However, this is no lifeless point. It’s a *mind*. Thus, Pythagoras’s world is a) mathematical, b) mental, and, c) alive.

Not only did Pythagoras state that all things are numbers, he also taught the transmigration of souls. Souls, therefore, must also be understood as numbers. In fact, they are monads and are associated with the unit number (one).

Pythagoras was asserting that the material world of things results from countless minds (monads). This was exactly the position advanced by Leibniz over two thousand years later and Leibniz made no secret of his admiration for Pythagoras by writing, “I have the highest opinion of Pythagoras, and I almost believe that he was superior to all other ancient philosophers, since he virtually founded not only mathematics, but also the science of incorporeals, having formulated that famous doctrine, worthy of a whole hecatomb, that all souls are immortal.” (quoted in G. MacDonald Ross’s *Leibniz, Past Masters*, Oxford University Press, 1986. MacDonald Ross describes Leibniz’s philosophy as “largely an updating of the Pythagorean and Platonic traditions, using the concepts of Aristotelian scholasticism.”)

“The science of incorporeals” is the mathematical study of the soul. It’s the most important but, as yet, least studied of all mathematical subjects.

Few people notice how uncannily similar Pythagoras’s system is to Leibniz’s. Both are based on monadic dimensionless points that also serve as minds, ruled by mathematics. Minds are unextended and from them come extended, material things, via mathematical relations.

Whereas the ancient Greek Atomists spoke of indivisible, materialist atoms which travelled through the mysterious and rather inexplicable “void” (after all, if void, in a strictly atomic system, isn’t made of atoms then it isn’t made of anything, hence can’t exist at all!), Pythagoras (and Leibniz) invoked indivisible mathematical monads, which formed a mental plenum (“fullness”), and thus abolished any baffling void.

The question of whether the basic “atoms” of existence are mathematical points (hence mental) or something larger (hence materialist) goes to the heart of reality. Materialism relies on

atoms being dimensional rather than dimensionless, extended rather than unextended. Materialism's ultimate theory (so-called M-theory), is based on one-dimensional strings rather than zero-dimensional point particles. Science provides no sufficient reason why these strings should not be further divisible, although it tries to invoke, rather unconvincingly, the Heisenberg Uncertainty Principle, which serves as a convenient dumping ground for all the toxic waste that science can't adequately explain.

The Music of the Spheres

Pythagoras was the first thinker to relate music to mathematics, discovering, in particular, that the chief musical intervals are expressible in simple numerical ratios involving the first four integers, the same four integers that constitute the tetraktys.

Aristotle wrote, “[The Pythagoreans] saw that the ... ratios of musical scales were expressible in numbers [and that] .. all things seemed to be modelled on numbers, and numbers seemed to be the first things in the whole of nature, they supposed the elements of number to be the elements of all things, and the whole heaven to be a musical scale and a number.” (Aristotle, *Metaphysics*)

The Pythagoreans expected the distances between the planets to reflect, on a cosmic scale, the most harmonious notes of a plucked string. The Pythagorean solar system consisted of ten spheres (the perfect number of the tetraktys), revolving in perfect circles around a central fire (the World Soul – something akin to what the Neoplatonists would later call the “One”), with each sphere emitting a note, the slower, near spheres producing low notes and the faster, far spheres generating higher pitched notes. All together, they combined to create a transcendent harmony, the sublime Music of the Spheres. However, mortal ears could not hear it because the notes were continuous and human ears can discern only those discrete notes that contrast with the surrounding silence. The gods alone could hear this incomparable, heavenly symphony.

Beyond the ten spheres was infinite space (which might be equated to infinite monadic minds taking no part in material existence).

For the Pythagoreans, wisdom and reason lay in Number and beauty in Harmony. The mathematical law of Harmony controlled the universe.

Genesis

In the beginning, the Monad (the number One) and Chaos were all that existed. Chaos might be associated with some vague, unformed, indeterminate infinity (*apeiron* in Greek). Or we might say that the Monad was God while “Chaos” comprised all other monads, currently disordered, but waiting to be organised and ordered by the Monad God.

The Monad used the countless monads to create all of the lines, planes and solids that constituted the ten spheres of Creation and all their contents. Harmony delivered everything in their proper proportions and relations, and conferred perfect beauty (thus generating what Leibniz would later call “the best of all possible worlds”).

Thus was produced the Cosmos (the “Ordered Whole”), and it was a living creature with a soul at its centre (the Monadic Soul, the World Soul – God). The whole universe was literally made of souls. This was the same vision that inspired Leibniz when he produced his remarkable *Monadology*.

Perfect Solids

Since the Pythagoreans were obsessed with geometrical perfection, they especially loved the so-called perfect solids. Perfect solids have faces that are all regular and identical, and display wondrous symmetries. There are five such solids. Each can rest within a sphere, with each of its corners touching the sphere. Alternatively, a sphere can be placed inside every such solid and touch every face.

The Pythagoreans said that the four natural elements (earth, water, air and fire) were composed of atoms of these perfect solids. Thus, earth atoms were cubes (like building bricks),

water atoms were icosahedrons, air atoms were octahedrons, and the light fire atoms were tetrahedrons. Plato described this scheme in the *Timaeus*, a book named after an Italian Pythagorean.

The Cosmos “atom” (the whole finite universe) was characterised as a *dodecahedron* (the regular polyhedral shape closest in volume to a sphere fitted around it), with its twelve sides matching the twelve signs of the zodiac.

The dodecahedron, with its twelve regular pentagons was, to ancient mathematicians, the most mysterious and amazing of the perfect solids. Being the most difficult to construct, it was also deemed the one the Divine Mind would surely have employed to act as the “hull” of the cosmic sphere.

The dodecahedron is not known to occur in nature (suggesting that it’s therefore supernatural, or divine, in some sense). It was a cult object of veneration for the Pythagoreans.

The Demi God

Pythagoras’s students considered him a supernatural being and a demigod. They said, “There are in the universe men and gods and beings like Pythagoras.” A biographer called him the “harmonic deity, halfway between gods and men.”

Pythagoras was named after *Pythios*, one of the identities of Apollo, God of Reason, and it was even said by some that Apollo was his real father, making him the “son of God” (or the son of Reason itself).

Iamblichus depicted Pythagoras as a messenger sent from the gods to enlighten humanity and who was persecuted by his ignorant enemies and finally martyred. That story sounds familiar, doesn’t it?

Learners and Listeners

Plato half-understood Pythagoras; no other prominent thinker did. It wasn’t philosophy that carried forward Pythagoras’s ideas, but his own secretive sect. As far as the world is concerned, the Pythagoreans simply faded into oblivion. What actually happened was that they were absorbed by the world of mystery schools, mystery religions and secret societies. Freemasonry, for example, often cites Pythagoras as one of its founders.

Pythagoras’s philosophy was mixed with Hermeticism, Gnosticism, Neoplatonism and, for most of its adherents, took on an increasingly mystical and magical character.

When Pythagoras addressed his followers, it was from behind a curtain. Only the inner circle, called *mathematikoi* – the learners – were ever admitted to his presence. The outer circle were known as the *akousmatikoi* (“listeners”; those who “heard things”).

The *mathematikoi* studied Pythagorean proofs and treatises. The *akousmatikoi* typically had neither the time, inclination nor ability for such work. Pythagoras – like a prophet – simply gave them oral instructions on how to act, without explaining the reasons. The *akousmatikoi* were “believers” rather than thinkers.

As time went on, Pythagoreanism diverged along two clear paths. The *mathematikoi* were far more interested in rational knowledge – in mathematics, science and philosophy – while the *akousmatikoi* were drawn to ritual, magic, numerology, and mysticism. Each group developed its own inner and outer circles, and each claimed it represented the true path of Pythagoras.

The *akousmatikoi* were those who created various Gnostic, Hermetic and Neoplatonic secret societies. The *mathematikoi*, on the other hand, turned mathematics into a fully fledged religion known as Pythagorean Illuminism (because it involved the search for illumination/enlightenment) or simply *Illuminism*, and their secret society became known as the Illuminati, about which all sorts of absurd myths have grown up, to the extent that conspiracy

theorist David Icke calls them alien, pan-dimensional, shape-shifting lizards from another world!

Those who like well-documented evidence will of course treat all claims of secret societies with a pinch of salt. Naturally, such claims are highly speculative because there is little or no documentation available in the public arena. However, anyone interested in the influence secret societies may have exerted on towering thinkers would do well to read *Hegel and the Hermetic Tradition* by Glenn Alexander Magee (Cornell University Press, 2001).

The mathematikoi and akousmatikoi did not part company entirely. The various secret societies that grew from them remained in close contact and engaged in a continual exchange of ideas. The Neoplatonist religious philosophy of Ammonius Saccas and Plotinus was used by the Illuminati for the initial development of a sophisticated mathematical religion. The religion did not reach its culmination until the work of Leibniz. The Leibnizian *Monadology* is now the basis of Illuminism. However, several key mathematical and philosophical modifications have been made to Leibniz's published work.

Illuminism is a religion that involves no saviours, messiahs, popes, prophets, priests, holy books or divine revelation. It rejects faith and does not look to any Creator. What it does have at its centre is an indestructible, indivisible, immortal mathematical soul (monad) which travels from 100% potential and zero actualization to zero potential and 100% actualization. To put it another way, Illuminism is about souls evolving – via dialectical mathematics – into Gods! God does not create the universe. The universe creates God. God is the evolutionary climax of existence. God is simply mathematics that has become conscious of itself.

The universe is conceived as a living, self-optimising, self-solving mental equation, composed of infinite nodes – *souls*. The cosmos is a living computer calculating its own optimal state – it's Omega Point, its Absolute state of perfection ... its *divinity*.

The Monad

Leibniz's first definition in his *Monadology* of 1714 was: "The monad, of which we will speak here, is nothing else than a simple substance, which goes to make up compounds; by simple, we mean without parts."

In Illuminism, the addition of a single qualifying word radically changes the meaning of "monad". Instead of a monad having no parts, it has no *resultant* parts. A monad in fact has *infinite* parts, but they are mathematical and they perfectly balance to zero (via equal amounts of positive and negative frequencies, real and imaginary), meaning that the monad remains what it always was – a dimensionless point. From monads and their contents, and nothing but monads and their contents, the whole of reality is derived. The monads in Illuminism are not Leibniz's "windowless" monads, but are interactive ("windowed").

The monad comprises "numbers" (frequencies), meaning that all things are numbers, just as Pythagoras said so long ago. Each monad contains *all* positive real numbers, *all* negative real numbers, *all* positive imaginary numbers and *all* negative imaginary numbers, meaning that the entire laws of ontological mathematics are encoded inside each and every monad. Numbers are carried ontologically by sinusoidal waves (sines and cosines).

Mathematically, a monad is an expression of *complex numbers*, i.e. it has real and imaginary parts. (Moreover, it also has positive and negative parts.) Scientific materialism is, conversely, based on positive real numbers alone, though it provides no sufficient reason for why imaginary numbers or negative numbers should be ontologically excluded. In fact, imaginary numbers are present throughout science but they perform the curious role of serving as "scaffolding". They are used to construct elegant and manageable equations, but are removed at the end by squaring them (to create a negative number, since $i^2 = -1$) and then taking an absolute value (i.e. changing negative to positive since negative numbers are generally as unwelcome in physics as imaginary numbers). By the end of the procedure, only nice, safe positive real numbers remain.

In other words, science is a continual "fiddling of the books" to exclude numbers that scientists don't like! (Yet note that Paul Dirac discovered antimatter precisely because he didn't perform the usual trick of ignoring a negative solution of an equation.)

In Illuminism, all such dubious and inconsistent mathematical operations are forbidden. All numbers must be embraced ontologically since there's no sufficient reason, beyond the prejudices of scientific empiricist materialism, to exclude them and thereby reduce mathematics to a bizarre and unjustifiable subset of what it ought to be. Illuminism concerns "complete" and consistent mathematics, while science is based on incomplete and inconsistent mathematics.

Illuminism begins with the simplest possible thing – a single mathematical point. This is the monad, the basic unit of existence. Being unextended, it conforms with Descartes' definition of a thinking *mind*. However, as per Leibniz, its mental activity is by default *unconscious* (consciousness is something that a mind evolves).

Does this mathematical point consist of anything? In fact, as already stated, it contains all the numbers between zero and infinity in all directions, signs and orientations. These numbers exist in an extremely precise way, guaranteed to produce a net result of nothing so that the point is ultimately defined by the number zero, the inverse of which is infinity.

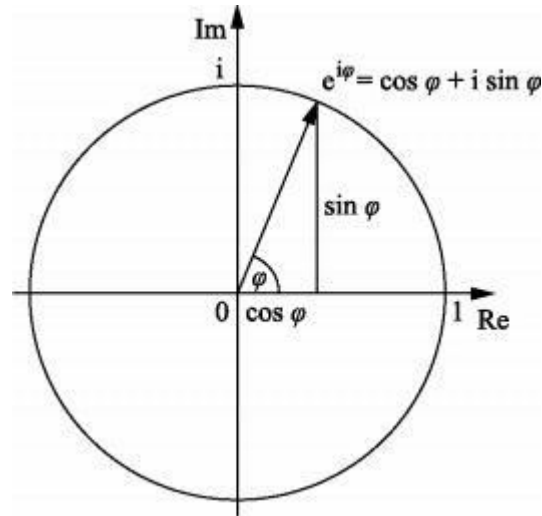
Where the Pythagoreans originally defined "one" as the basic unit of existence (associated

with the point and zero dimensionality), modern Illuminism assigns that role to zero. The dimensionless point (monad) is not an instance of the number one but of the number zero. The number one is divisible; the number zero isn't – therefore zero is the true indivisible, immaterial, unextended, immortal point with no resultant parts.

In order for zero to be the inevitable and inescapable net result of the combination of infinite numbers, all of the numbers must conform with the most powerful analytic formula in the whole of mathematics – Euler's Formula, the great jewel of mathematics:

$$e^{ix} = \cos x + i \sin x$$

This formula generates a unit circle (i.e. with radius 1) in the complex plane (diagram courtesy of Wikipedia):



By introducing the imaginary number to the exponential function, exponential growth is tamed and converted into an eternal circle. We might say that exponential growth, in this context, happens circularly and reaches infinity. Moreover, perfect periodic cosine and (imaginary) sine waves are generated, which, like the circle, go on forever.

What's so remarkable about Euler's Formula is that it produces *perfect* balance between negative and positive numbers, between real and imaginary numbers and between zero and infinity. No element is privileged over any other. The net ontological effect of the formula is zero (since the circle's negative half perfectly cancels its positive half), yet this is an "infinite" zero, a structured "nothing" that goes on forever!

Leibniz famously asked why there is something rather than nothing. The answer is that something *is* nothing – thanks to Euler's Formula. Via this formula, existence can be maintained at its *necessary* ground state of zero (*nothing*), while always being *something*. (Any non-zero resultant cosmic energy is forbidden. There is no sufficient reason why the cosmos should have any arbitrary energy, and why such an energy should be above the ground state.)

In order to include all possible ontological numbers, it's necessary to introduce a more generalized form of Euler's Formula:

$$A e^{i(fx + \varphi)} = A \cos(fx + \varphi) + i A \sin(fx + \varphi)$$

where A is amplitude, f is frequency and φ = the phase angle (phase shift).

In the frequency domain, the three elements necessary to specify a wave are amplitude, frequency and phase, so this generalized formula allows *all possible waves* to be accommodated.

A "simple" point is therefore nothing of the kind. It's an infinite information system, based on a superposition of infinite waves of every conceivable permutation, all of which put together produce a sum of zero (total and infallible balancing to zero).

Simply by defining a wave as the mathematical basis and definition of *energy*, a point is transformed into a repository of infinite, balanced energy. And bear in mind that this energy is *necessarily* eternal. Euler's circle never stops spinning. Nothing can ever halt it – because ultimately there's nothing there! An Euler circle is simply an ingeniously ordered and structured nothingness that can never perish. It's always rotating and can never stop. Energy is just eternal motion.

One of the most baffling mysteries of physics is why “space” – which, according to most theories, contains infinite energy – does not produce catastrophic gravitational effects (via Einstein's mass-energy equivalence). In Illuminism, there are two easy answers: all of the energy balances to zero, hence has no net effects, and the energy in any case exists *dimensionlessly*, hence is not part of the material world at all. The Fourier frequency domain, outside space and time, is gravity-free. Only the Fourier spacetime domain (the material world) is gravitational. Scientific materialism has no such dimensionless escape routes, which is why the problem is known as the “vacuum catastrophe”, which has been described as the biggest discrepancy between theory and experiment in the whole of physics.

How many monads are there? If one monad can exist with no net energy, what sufficient reason could prevent the existence of others, also with zero net energy? In fact, what could prevent the existence of infinite such “nothings”? There's nothing obvious to prevent it, hence there are infinite monads, as Leibniz insisted. (The only potential obstacle is the possibility that there is in fact an ontologically highest number – an enormous but nevertheless finite number. This question will be explored later.)

All of these monads can exist together, within a single point. So, there's a Singularity, composed of infinite singularities, each singularity being composed of infinite balanced energy defined by the generalised Euler Formula. That – mathematically – is the dimensionless world of mind. That's the domain that existed prior to the birth of space, time and the material world. It's wholly immaterial and undetectable by any scientific experiment. It's the “intelligible world” and is reached by reason alone, as Plato always said (as opposed to the “sensible world”, which can be explored experimentally). This majestic, *well-defined* Singularity is the critical, analytic mathematical entity that has been disastrously missing from philosophy and science. This is the *noumenal* universe, and from it, via strict mathematical operations, comes the *phenomenal* universe.

How, then, did the Singularity generate the familiar, extended, phenomenal world? In physics, there's a crucial distinction between bosons and fermions, the two fundamental classes of particles. The *bosonic* wavefunction is said to be *symmetric* with regard to particle exchange (meaning that infinite bosons can occupy the same quantum state). The *fermionic* wavefunction on the other hand is *antisymmetric* with regard to particle exchange (meaning that no two fermions can occupy the same state).

If there are infinite monads all occupying the same Singularity, they can all be regarded as mental bosons. However, the application of a simple antisymmetry operation converts them into mental fermions. This has the most astounding consequence: it confers unique coordinates on each of the monads and instantly creates an extended Cartesian coordinate grid.

The monads haven't actually moved anywhere – they are still inside the Singularity – but they now have unique identifiers (coordinates) and this produces the effect (illusion) of all monads now being separated from each other. To put it another way, they now have *extended* (fermionic) relations with each other. A Cartesian extended world has come into being. Yet the remarkable thing is that it's entirely constructed from points (from minds) and their mathematical relations.

The “Big Bang” was simply a mathematical antisymmetry operation that instantaneously

turned a Singularity into an infinite Cartesian grid universe. In fact, in Illuminism, 4D spacetime (involving three space axes and one time axis) is replaced by an enormously more elegant, productive and symmetrical 6D spacetime (involving three space axes and three time axes). The space axes are defined by real numbers, and the time axes by imaginary numbers, meaning that spacetime involves complex numbers. Space and time are dynamically, and inversely, coupled. As an object moves faster through space, it slows down in time. As it speeds up through time, it slows down in space. A spatially stationary object moves at maximum speed through time (i.e. at the speed of light).

The Singularity is a dimensionless Fourier frequency domain. The extended Cartesian coordinate grid is the dimensional Fourier spacetime domain that is generated by it through an inverse Fourier transform on a cosmic scale.

For those who want to object to this scheme on the basis of the “expanding universe”, suffice to say that an infinite grid can nevertheless expand thanks to the mathematics of “Hotel Infinity”. This Hotel Infinity expansion is the basis of “dark energy”! (This has been described in detail elsewhere in the *God Series*.)

The infinite Cartesian grid is still wholly contained within the Singularity and is thus a mathematical illusion. Existence can only ever take place within a single point (*everything* that exists is contained within the Singularity: we live *inside* the Big Bang Singularity, and the Big Bang itself took place within that Singularity, and remains contained within it). At no point does anything ever leave. The Big Bang was nothing but an internal mathematical restructuring of the Singularity via an antisymmetry operation and Fourier mathematics.

All individual monads are able to release a small amount of their energy (a low energy band) into the shared space – the Cartesian grid formed by all active monads – and the whole material universe we experience is simply the interaction of all of that mind energy in an arena of mathematical extension. Matter, to state it bluntly, is produced by mind energy through Fourier mathematics and has no independent ontological status. It’s wholly derived from mind. Scientific materialism is therefore the inverse of reality!

Only a sliver of monadic energy frequencies are compatible with a material domain. To see why, consider a black hole singularity. When a star of suitably enormous mass implodes, it collapses all the way down to a point. High-energy frequencies within monads similarly collapse straight back into themselves: they cannot enter the shared physical world (and would destroy it if they could). This means that the mental domain is infinitely “deep” while the material domain is, conversely, almost infinitely “shallow”. True reality belongs to the mental, not the physical, domain. All sorts of things can happen mentally that will not be reflected physically. This means that minds can potentially control bodies in all manner of extraordinary ways, but bodies cannot dictate to minds except at a shallow level of mere physical causality.

All monads are uncaused causes. They do not depend on anything else for their existence (they have no Creator), hence they are not determined by any causal chains. That means that they are causal initiators (causal agents), but themselves are inherently uncaused. *That* is what it means to be free, to be capable of exhibiting free will and free choice.

A universe of monads made of sine and cosine waves is nothing other than a universe of countless musical notes. The cosmos, exactly as Pythagoras said, is a great musical symphony producing the music of the heavens. The material world is just one of its notes. We inhabit a living, cosmic organism of music playing every conceivable note!

Two Worlds

Illuminism is based on two worlds: one *inside* space and time (the material world; the phenomenal world) and one *outside* space and time (the mental world; the frequency world; the noumenal world). Yet the phenomenal world is simply a low-energy mathematical construct of the noumenal world.

The mental, noumenal world is the one wholly denied by scientific materialists. However, for idealists, it's Kant's noumenal domain where freedom, the soul and God reside; it's where Hegel's *Geist* is truly found; it's where Schopenhauer's Will exists; it's where Nietzsche's Will to Power logically resides (though Nietzsche would no doubt disagree since he condemned all "two worlds" solutions); it's where Eduard von Hartmann's Unconscious is; and it's the home of the Jungian Collective Unconscious. Looking back to the ancients, it's the physically unchanging world Parmenides had in mind; it's Plato's domain of perfect, immutable Forms; and it's the One, the Nous and the Higher Psyche of Neoplatonism.

This is the domain that has always been at the heart of metaphysics, but whose nature has never been meaningfully defined – because pure, analytic mathematics was never brought to bear on it.

This is the domain of instantaneous, non-local connections (of the sort that define quantum mechanics). It's the arena through which Jungian Synchronicity operates. It's how intuition is possible. It's where all the "woo woo", religious and paranormal experiences occur. It permits homeopathy (which is a non-local, "information-based" rather than "physical" system of medicine). Yet there's nothing obscurantist concerning any of this – it's all about analytic mathematics.

There is only one *true* world (a mental Singularity) but it's experienced in two ways: mentally and physically. The Singularity, the mind domain, is the unextended origin which is able to generate an extended Cartesian grid world – perfect for physics – simply by executing an antisymmetry operation. Yet each point in the grid is itself an unextended point, meaning that "space" is fundamentally mental. That's why no scientist could ever find the "ether".

It turns out that the metaphysicians (or should we say *metaphysicists*) were right all along in their broad thrust. All they lacked was the mathematical knowledge to clarify and precisely define their concepts. They could not find a way to explain an unextended world – a Singularity – and give it the analytic mathematical structure it needed to become anything other than a source of vague speculation, intuition and mysticism (as in Eastern religion).

Kant believed that "pure reason" strayed when it departed from the phenomenal world. However, no such problem occurs if pure reason is exercised according to the strictest mathematical considerations (of exactly the kind Kant held in such high regard as far as the phenomenal world was concerned).

Illuminism – ontological mathematics – provides the missing mathematical infrastructure for the mental domain and thus ushers in a new Copernican Revolution. Religion gave way to science and science must now give way to mathematics. Yet, astoundingly, mathematics can raise religion from the dead, but now as a rational, Logos subject rather than an absurd, irrational, superstitious, faith-based Mythos subject.

Scientific materialism has enjoyed a great deal of pragmatic success, but you learn as much about true reality from it as someone trapped in *The Matrix*, i.e. none at all. Mind is all that exists. Its mathematical properties give rise to the illusion of materialism.

Illuminism is the true grand unified theory of everything since it unites mathematics,

science, philosophy, religion and psychology.

The Two Pills

The red pill = mathematics.

The blue pill = science.

Well, which pill will you take? Do you want to inhabit the intelligible world of truth (mathematics) or the sensible world of illusion (science) – the Matrix? Your choice.

The Mystery of Scientific Laws

What and where were the laws of physics before the physical universe came into being? What laws were controlling the Big Bang as it happened? What laws caused it in the first place? Whatever caused and controlled the Big Bang must, of necessity, have preceded the Big Bang, yet scientists openly say that space, time and matter did not exist prior to the Big Bang. In which case, there's nothing left within the empiricist, materialist paradigm to account for how the Big Bang happened.

Scientific laws themselves are no part of the empiricist materialist Meta Paradigm of science since laws are not themselves either material or directly observable. You can see their effects, but you can't see *them*. They are immutable and eternal, and have no known origins. No scientist has ever plausibly explained where scientific laws come from, how they interact with mutable, material things and how they exist at all.

Scientific laws are science's "magic wand". They are an appeal to something outside science. Remarkably, science's greatest challenge isn't to create a Grand Unified Theory of Everything, but to explain its own laws! If it can't explain those then it can't, ultimately, explain anything at all.

Rationalism is about revealing those elements of reality that have *compulsory* existence. *Hyperrationalism* adds one element to rationalism: all rational statements must be viewed through the prism of ontological mathematics. The foundational, uncaused causes of reality must be mathematical and they must have the property of being "nothing" since nothing is the compulsory rational ground state of existence. Mathematics has the ultimate rational trick up its sleeve – because "nothing" can also be something. It's precisely because something and nothing can be equated (via an equation as simple as, for example, $2 - 1 - 1 = 0$) that we are all here at all; that anything is here. If this weren't so, non-existence would be the ground state of "reality" and so there would be no sufficient reason for anything to exist. The universe would comprise eternal void, eternal oblivion in which nothing could ever be and nothing could ever happen. It's only the miraculous properties of mathematics that spare us that fate. To put it another way, you can *never* beat nothing, but mathematics allows us to arrange nothing so ingeniously that an entire, infinite universe of "something" can exist! (But it's still just nothing when all's said and done.)

Why are souls immortal and indestructible? Because they are *nothing*. You can't annihilate nothing, can you? The first law of thermodynamics (stating that energy can be neither created nor destroyed) is, rationally, a statement that the energy of the universe is always zero (because there could never be a sufficient reason for the energy to be greater than zero, and if the energy of the universe is always zero then it automatically follows that there can't be any more or less of it). The first law of motion states that a moving body, subject to no external force, will continue in a straight line at a constant speed forever (i.e. it's inherently a perpetual motion machine). The reason it moves forever is that it uses *zero* energy. If it had non-zero energy, it would expend it until it had zero (ground state) energy, and then it would stop. Think about that – an independent object moves forever precisely because its energy expenditure is *always* zero. Movement is an inherent part of existence precisely because nothing can stop it since it requires nothing.

Why can science never find the soul? Because there's literally "nothing" to find! The soul – the Leibnizian monad – is an infinite, mathematically structured nothingness. Thanks to mathematics, that also makes it an infinite something. "Something" is just "nothing" existing in a certain way. It's a special arrangement of nothing. Nothing is not "simple"; nothing is the most

complex thing there is. “Nothing” is the soul – the fundamental unit of reality. Existence comprises nothing but mathematical souls and their mathematical interactions.

“Bless us, divine number, thou who generated gods and men!” – Pythagorean declaration

The Fear

Why have philosophers been so afraid to expose the charlatanry of scientific materialism, its nostrums, quackery, half-baked, ad hoc ideas, its lack of internal consistency, its sheer sensory irrationalism? It's because just as Bertrand Russell was famously intimidated by Wittgenstein (since he didn't understand much of what Wittgenstein was saying but assumed it must be very "deep"), philosophers are intimidated by scientists. They don't understand science well enough to challenge it. The mathematics bamboozles them. They don't realize that science operates like the Wizard of Oz. Behind the show of power is a feeble brain, committing outrageous error after outrageous error, and committed, if truth be told, to a clunky process of trial and error. Scientific papers are just presentations of educated guesses, and most are soon forgotten.

There is no "clear thinking" in science. Science proceeds by way of individuals proposing ad hoc hypotheses. Some of these become popular and prosper; others fade away. Science therefore operates according to Darwinian natural selection, not reason. It's about memes instead of eternal truths of reason. The whole enterprise is a grand exercise in provisionalism and contingency. It's utterly false that science is objective and factual. As Max Planck so damningly said, "A new scientific truth does not triumph by convincing its opponents and making them see the light, but rather because its opponents eventually die, and a new generation grows up that is familiar with it." That's just the same as religion! And Darwinism.

Science invokes a falsification principle (a statement is not "scientific" unless it is capable of being falsified) that is itself non-falsifiable, hence non-scientific by its own criterion. But if a statement must be capable of being falsified then it cannot be *true* – because truth is by definition *not false*. $1 + 1 = 2$ is not falsifiable and where would science be without it?

Some scientists look to a verification principle (a statement is not "scientific" unless it is capable of being verified) that is itself non-verifiable, hence non-scientific by its own criterion. But if a statement is in need of experimental verification then it cannot be *true* – because where would the verification process ever end? Verification can increase confidence in a statement; it cannot *prove* it. Is $1 + 1 = 2$ in need of verification?

Science is a process of approximating truth through well-tested hypotheses, models and simulations (all of an ad hoc nature). It has absolutely no connection with actual truth, which can't be falsified or verified, only analytically proved.

Quantum mechanics, according to the mainstream interpretation of the Copenhagen school, is based on an unreal, unobservable wavefunction that is everywhere, is non-local, features imaginary numbers and mysteriously collapses when an observation takes place, but not otherwise.

This, the establishment view, is an extremist empiricist stance, and asserts (just as the fanatical idealist empiricist Bishop Berkeley did) that *to be is to be perceived*. Things are real only when they are being perceived, and, when they are not being perceived, they are not real. Thus, the moon doesn't exist when you're not looking at it, atoms don't exist when you're not performing observations and measurements on them, and things (such as Schrödinger's cat) exist in a superposition of states (including being simultaneously both dead and alive!) prior to observation. (Note that if Niels Bohr and Werner Heisenberg – the main authors of the Copenhagen Interpretation – were being consistent then they would have asserted that it was *meaningless* to ask about the ontological condition of the cat since, formally, within their paradigm, nothing at all can be said about anything unobserved. If the moon doesn't exist when no one is looking at it, nor does Schrödinger's cat when it's locked in its box! Superposition

states are no part of conventional science since they are wholly unobservable, hence “metaphysical”). Isn’t it somewhat comical that a fanatical empiricist science is based on an unempirical quantum mechanical wavefunction? How can anyone take it seriously? It fatally contradicts itself.

The Copenhagen Interpretation is a complete repudiation of objective reality and the reality principle (which asserts that the world is there regardless of any conscious observers). Einsteinian relativity also destroys the reality principle since it’s no longer possible to say what’s moving and what isn’t, and what size and mass anything truly has (because of time dilation, length contraction and mass increase as relative speed increases).

Consider the extraordinary mismatch between Einsteinian relativity and quantum mechanics. These two ad hoc theories are the most successful in scientific history, yet they are incompatible in almost every way – which is why a theory of quantum gravity remains so elusive despite having had more elite intellectual effort expended on it than anything else *ever*. The problem cannot be solved for the simple reason that the underlying paradigm is false (in particular in how it rejects the rational unobservables, or “hidden variables” of mathematics), but no scientist is willing to abandon the paradigm, so the futile exercise goes on.

In fact, the irreconcilable differences between the two leviathan theories of science prove how flawed the scientific method actually is in terms of truth content. The scientific method produces *successful* theories; it does not produce *true* theories. (As Nietzsche said, “Success has always been the greatest liar.”) If these two successful theories were true, they would blend seamlessly. In fact, only mathematics delivers seamless truth since it’s based on analytic *a priori* truths and not on contingent, ad hoc hypotheses subjected to unreliable experimental verification. “Unreliable” because many failed theories have been successfully verified experimentally – until new experiments falsified them. Think of how successful and dominant Newtonian physics was despite its description of reality being almost wholly false (there is no Newtonian absolute time, no Newtonian absolute space, gravity does not operate instantaneously across the “void”, the speed of light is not variable, and mass, length and time are not fixed).

Scientific materialism is absurdly incoherent. It invokes all sorts of dubious concepts such as Multiverses and “many worlds” in which all possible events occur (“anything not forbidden is compulsory”), while at the same time saying that unobserved cats are both alive and dead.

In a Multiverse where every possibility is realised, there is no meaning. Meaning is about choices, about non-inevitability, about free will rather than determinism, about the possibility of our being independent causal agents not invariably subject to other causes and the pawns of other causes. It’s about life rather than machinery.

The Multiverse is the ultimate Doomsday Machine, the most terrible Death Star – because all it does is exterminate meaning. It relentlessly converts every possibility into actuality at every instant. Absolutely nothing is omitted. All processes occur. All possible effects take place with absolute inevitability, even if they have no causes (!). At no point does choice or free will take place.

Strange though it may seem, this is the supreme dream of scientific materialists. They *despise* meaning. All they want is for *everything* to happen according to inescapable scientific determinism (even though modern science is indeterministic!), and the Multiverse is the perfect mechanism for achieving that goal. With the Multiverse, you don’t have to answer any questions, or explain anything – because everything that is not forbidden is sure to happen, come what may.

Why have philosophers allowed these endless contradictions and bogus claims to go unchallenged? For the simple reason that they don’t understand what is being asserted, and don’t want to embarrass themselves by wandering into territory where they feel distinctly out of their comfort zone. It’s not at all hard to understand why the Emperor is allowed to parade around

stark naked without prompting a single titter. All he has to do is have his invisible suit weaved for him by scientific materialists, and all philosophers will be cowed into selective blindness. Wake up! Scientific materialism is a joke. It's now much closer to Berkeleyan empiricist idealism than it is to any kind of materialism.

Pythagorean Illuminism, being rationalist rather than empiricist, places complete reliance on Leibniz's principle of sufficient reason, which states that for every fact there is a reason why it is so and not otherwise. There are no arbitrary, ad hoc elements. Everything has an exact cause. Nothing is probabilistic. This is the only meaningful way forward.

The Sensory Delusion

It's irrelevant how much sensory-obsessed empiricists rail against rational unobservables. Why should the human senses be accepted as the arbiters of what exists and what doesn't? Human reason – within the strict context of mathematics – is always right and always superior to empiricism. The ad hoc, contingent, provisional, mutable, temporal, inductive, synthetic, *a posteriori* hypotheses of science could never satisfy *any* rational person. Science doesn't even claim to be about truth. It's simply a pragmatic method that generates models (simulacra!) of reality that may prove useful in certain circumstances.

Certain people, as they read this, will be suffering apoplexy. They will be Jungian thinking-sensing types, who use their rationality to validate their sensory experiences. These people are “scientists”, who conceive of “evidence” strictly as that which is available to the senses. They cannot conceive of *rational unobservables* that are undetectable by the senses or by any experiments, but are available solely to intellect.

Another group of people – thinking-intuitives – might experience an incredible thrill because their long-held intuition that scientific materialism comes nowhere near explaining reality is being vindicated. It's an astonishing truth that how we understand the world is dependent on our psychological type. Sensing types can never get beyond their senses, but intuitives can. Thinking-sensing types are natural empiricists, and thinking-intuitives are natural rationalists.

Science is a subject for people heavily invested in their sensory apparatus, who cannot conceive that their senses are revealing an ingenious mathematical illusion to them. You have to be intuitive to escape from the sensory jail. The Pythagoreans used the expression *soma sema*, meaning that the body is the prison or tomb of the soul (literally “body tomb”). Scientists are those who can never escape from the tomb, or the Platonic Cave of Ignorance and Delusion.

Sensing types outnumber intuitives three to one – which is why our world subscribes to the many false doctrines (produced by sensory thinking) with which we are all so familiar. Even the story of “God” incarnating as a human being (Jesus Christ, allegedly) is designed to pander to sensory types, and is a central reason why Christianity has been so successful.

With Jungian feeling types, thinking is barely engaged at all. They judge according to how something makes them feel, not according to its rationality. With Jungian sensing types, thinking is used as a servant of sensory experience (and produces empiricists and materialists), while with Jungian intuitive types, thinking is invoked to validate non-sensory ideas (and produces rationalists and idealists).

These key differences in how various psychological types process ideas and information, and what biases they exhibit, are never taken into account in conventional thinking. All hypotheses, theories and philosophies ought to be stamped with what paradigm they belong to, and what psychological type they reflect. They are not neutral and dispassionate – they all arise from specific attitudes and cognitive patterns and inclinations.

As Nietzsche said, so tellingly, “There are no facts, only interpretations.” Even more cuttingly, he declared, “What, ultimately, are man's truths? Merely his irrefutable errors.” That's especially relevant to both mainstream religion and scientific materialism. Faith allows errors to become “irrefutable” (a person of faith never attempts to refute what he believes, and considers such a refutation inherently impossible). Scientists, when they proclaim that anything not amenable to experimentation does not exist, are making their sensory dogmatism – their sensory *error* – “irrefutable” because they have rejected any means by which it could ever be refuted.

Scientists could easily adopt the stance of Leibniz, which combined the physical and metaphysical, but they refuse to do so. They have staked everything on the physical, and everything physical can of course, in principle, be subject to experimentation. However, what will science do when it asserts the existence of things such as strings that will never be accessible to any experiments? At that stage, science has refuted itself (though it will never admit this).

The famous Higgs boson was, in terms of science, metaphysical (hence not scientific), until its existence was experimentally vindicated. Einstein's theories were metaphysical until experimentally supported. Science simply cannot escape metaphysical assertions. If science ever proclaims a final theory of everything, you can be certain it will contain elements beyond the reach of any experiments, hence it will actually be a metaphysical theory. That being the case, why doesn't science accept the logic of its own position and proclaim that experiments will never reveal the final truths of science, hence it has to abandon the defining role of experiments in its paradigm and method.

Why aren't experiments regarded as simply useful tools for gathering data? Why are they taken as ontological filters, with anything not available to experiment, such as the dimensionless soul, ipso facto being deemed unreal and non-existent?

As Leibniz demonstrated, no one rationally requires science to make this assertion. It *chooses* to do so because it's a quasi-religion of materialism and empiricism, with its own fanatical Church, priesthood, popes and prophets, all fanatically defending their dogmatic paradigm.

The Rebirth of Philosophy

Pythagorean Illuminism can straightforwardly – mathematically – explain all of the great mysteries: free will, mind-body dualism, consciousness versus the unconscious, tensed versus tenseless time, and so on. It's all in the math!

It's time for philosophy to regain its former glories and concern itself once more with the big questions of existence. It cannot hope to compete with science unless it uses the same core engine – mathematics – but it should then turbo-charge that engine by using *all* of it, including zero, infinity, negative and imaginary numbers.

Mathematical metaphysics goes *beyond* mathematical physics – and explains that which physics cannot by using those rational mathematical “unobservables” that science disdains because of its all-consuming empiricist, materialist bias.

Mathematical metaphysics is the science of the *mind*, and mind is the origin of all. Mind is unextended, dimensionless, immaterial and outside space and time. Science will *never* know anything of it. No experiment will ever probe it. But philosophy – rationalist mathematical metaphysics – can know all of it.

It's time to jettison Newton and embrace Leibniz. It's time to cast down scientific empiricist materialism and raise up scientific rationalist idealism. Or shall we rational metaphysicists go on being mocked forever by Professor Cox and his Philistine ilk and consign ourselves to intellectual irrelevance and oblivion? Is philosophy the ultimate way to waste a good mind, or the ultimate means for unleashing the mind's full power?

The False Dichotomy

Scientists often set up a false dichotomy, asking people to choose between “evidence” (science) and faith (religion), as if these were the only two options. What about rational mathematical proofs, wholly outside sensory experience and experiment? These are used all the time in science, and yet they form no part of the materialist, sensory, empiricist, experimental Meta Paradigm of science. Science has *never* defined what mathematics is ontologically, and, given that mathematics is at the core of meaningful science and the formulation of scientific laws, that’s an *extraordinary* omission. As ever, scientists, as shameless instrumentalists and pragmatists, don’t care about the catastrophic philosophical contradictions, inconsistencies and absurdities at the heart of their discipline.

People have four primary, Jungian ways of understanding the world:

- 1) Faith (feelings) – mainstream Western religion.
- 2) Experiments and experience (senses) – science.
- 3) Mysticism (intuition) – mainstream Eastern religion.
- 4) Rationalism (thinking) – mathematics and metaphysics.

The question is which of these is best, which is most suited to discovering the ultimate truths? Intuition, if were accurate and powerful enough, would be the best. Unfortunately, it isn’t, and usually degenerates into Mythos mysticism, as in the case of Hinduism and Buddhism.

No sane person would ever rely on feelings (faith and revelation) to yield infallible truth. Feelings are spectacularly subjective, unreliable, biased and irrational. Nor would any sane person rely on their senses to disclose ultimate truth. The senses are horribly fallible, subjective, unreliable and easy to deceive. In these regards, they are not greatly different from feelings. Scientists have elevated sensory “evidence” to the pinnacle of their subject – an absurd act since the senses have nothing to do with ultimate truth. Ultimate truth isn’t something that can be seen, heard, smelt, tasted, touched, felt, or anything of the kind. Ultimate truth is solely rational, intelligible, intellectual, and noumenal. We work out ultimate truth using our reason. We don’t encounter it with our feelings or our senses.

We might intuit much of the ultimate truth, but it’s reason that makes our intuitions robust, systematic and integrated. In the end, we always come back to reason. That’s why Illuminism is the doctrine of hyperrationalism. The language of hyperrationalism is ontological mathematics and the logic and ontological dialectics which flow from it.

You will never attain the ultimate truth unless you know what you’re looking for and how to look for it. That was the solution to the quest for the Holy Grail.

People of faith (of feelings) hope to meet some superbeing who *feels* like God. People of intuition hope to encounter some superforce that *intuitively* seems divine. People of the senses want to be presented with sensory evidence of God-knows-what (!). Does the fact that the Higgs boson has purportedly been discovered make you think you are one jot closer to the ultimate truth of existence? You certainly can’t directly sense the Higgs boson. You have to take its existence on trust – trust in CERN’s Large Hadron Collider and the scientists and engineers operating it and interpreting its output.

The only thing that we rationalists trust is reason itself. We completely reject feelings and the senses, and intuition too when it dissolves into mysticism. No one can deny that $1 + 1 = 2$. That, itself, is ultimate, incontestable truth, and the whole of mathematics flows from it.

Mathematics is eternally and immutably true, hence is the basis of existence. Science most certainly isn't eternal and immutable. Mathematical theories are additive and cumulative. Scientific theories, on the other hand, refute and replace earlier ones. There's a graveyard for countless discredited scientific theories. There's no graveyard for mathematical theories. All valid mathematical theories were correct in the past, are correct now and will always be correct. Mathematics and science belong to wholly different categories of knowledge. Mathematics is all about eternal truths of reason. Science is about provisional interpretations of "fact". It's a category error to have rationalist mathematics at the heart of empirical science.

Frankly, if you want ultimate truth, you have nowhere else to turn than ontological mathematics and the principle of sufficient reason enshrined by it.

Reason – mathematical reason – is truth. Nothing else is. Everything else is mere opinion, belief, conjecture, feeling or interpretation.

Many people believe that "God" is the ultimate answer. This is absurd. The only True God is reason itself. It alone is all-knowing, eternal, flawless and infallible.

In Illuminism, Abraxas is the ancient name given to reason itself, hence is the True God of Illuminism. Abraxas is also the name given to the first mind – the first person – to become the conscious expression of mathematical rationalism.

All objective knowledge is contained within mathematics. Yet mathematics also provides subjectivity (the "living thinking" of monadic souls), and it's subjectivity that is associated with feelings, the senses and mystical intuitions. Subjectivity provides the "irrational", interpretive aspect of existence, the aspect of opinions, beliefs, the senses and feelings.

You are being subjective if you expect the ultimate answer to existence to lie in any way outside objective mathematical reason. Subjectivity is how we *experience* existence – it's *life*! – and objectivity is how we *know* existence – it's *knowledge*.

Subjectivity is the "withinness" of mathematics, the interior, the internal experience of mathematical information. We gain objective knowledge when we transcend our subjectivity and engage with mathematical reason – which is true at all times for everyone, for all species, for all intelligent beings. The same cannot be said for science. Intelligent alien beings with very different senses from our own wouldn't understand our science at all – but they would have no difficulty in understanding mathematics.

Objectivity is the only route to ultimate truth and objectivity is all about mathematical reason. Subjectivity provides all the "colour" of existence – and all the irrational madness – but is inherently incapable of bringing us into contact with ultimate truth. Faith is subjective nonsense, as the countless different religious faiths demonstrate. The senses are wholly unreliable and we can have no certainty that they aren't revealing a "Matrix world" to us. Intuition, if not guided by reason, is mystical claptrap, as we see in Eastern religion.

The only objective, universal tool we have is mathematical reason. Nietzsche said we have no "organ for truth". He's right in relation to feelings, the senses and mystical intuitions, and to reason also whenever it's applied to anything non-mathematical. He's wrong in relation to mathematical reason itself. This is indeed the human – in fact the *universal* – "organ" for eternal truth!!

If you want truth, you have no option but to get onboard with ontological mathematics. Everything else is subjective nonsense, illusion and self-delusion.

This is the gospel of the Illuminati.

Fourier Mathematics

There's an astonishing branch of mathematics, based on Euler's Formula, that's all about linking two radically different domains, one inside space and time, and one outside space and time, one dimensional and one dimensionless. It's *Fourier Analysis*.

Joseph Fourier asserted that it was possible to expand *any arbitrary function in the form of a trigonometric series*. Any ontological pattern can be made by adding, or superposing, sinusoidal basis waves. That is, any complex pattern you will encounter in the real world can be broken down into a collection of simple sinusoidal waves. The complex pattern is *inside* space and time; the simple sinusoids correspond to eternal, immutable frequencies *outside* space and time (somewhat like Platonic Forms, and each soul has a full set of these Platonic frequencies).

The key point to grasp is that the spacetime function is showing the same information as is contained in a combination of basis frequencies that *do not belong to spacetime*. So, we have a means of depicting the *same information* in two different ways, in terms of two radically different mathematical domains (the spacetime domain and the frequency domain). A Fourier transform converts a complex spacetime function into its underlying frequencies. An inverse Fourier transform performs the reverse process.

The whole mystery of existence is contained within Fourier mathematics because it's none other than the means by which unextended Cartesian minds (frequency domains) communicate with extended Cartesian bodies (spacetime entities). Fourier mathematics is the solution of the most intractable problem of all: Cartesian dualism. The Illuminati alone saw that the concept of the mathematical *transform* unravelled the Gordian Knot created by Descartes' dualism. It allowed *one thing* (hence a monism, not a dualism) to be presented in *two different ways*, thus creating the *illusion* of dualism, whilst being a monism all along, and avoiding the problem of how two different substances could possibly interact.

Monads are composed of nothing but eternal, immutable sine and cosine waves of every conceivable type (via the generalized Euler Formula). The monadic domain – the domain of mind – is the ontological realisation of the Fourier frequency domain.

Via inverse Fourier transforms, this monadic frequency information can be combined to create *any* spacetime representation. In other words, mental “ideas” can be converted into physical “bodies” via Fourier mathematics. Mind is the basis of matter, not the other way around (as scientific materialists have always claimed). The phenomenal world is simply a mathematical way of presenting noumenal frequency (mental) data. Every phenomenal object is nothing but a representation of an underlying noumenon (which could be interpreted as the correct way to understand Kant's philosophy, with the noumenon no longer being an unknowable mystery but a well-defined mathematical frequency function outside space and time). Noumena are immaterial and outside space and time; their associated phenomena are material and inside space and time. That's the ontological essence of Fourier mathematics. Kant was wholly wrong that the noumenal domain is unknowable: it's fully knowable via ontological mathematics.

Science is clueless about the true nature of Fourier mathematics, despite using it all the time. Fourier mathematics is the *proof* of idealism and the refutation of materialism. The reason for that is simple. Fourier mathematics depends on a frequency domain of perfect, immutable, eternal sinusoidal signals outside space and time. Fourier mathematics fundamentally relies on frequencies of pure sinusoidal waves that are not in the material world, have nothing to do with the material world (insofar as they are eternal and immutable) and are not impacted by anything taking place in the material world.

A frequency “mind” cannot die when a body dies because the frequency mind, being a singularity, is not in space and time at all. It’s linked to it, but not determined by it. This, of course, is the basis of the immortal soul linked to a body. The soul is not some fantastical and silly religious idea reliant on desperate faith: it’s the purest expression of analytic mathematics. If you wanted to bet your life on the certainty of anything, it would be on your own imperishable soul.

Fourier mathematics is at the root of quantum mechanics, hence of the most successful scientific theory of all time. Fourier mathematics also underlies the extraordinary phenomenon of holography. A hologram – just like the Cartesian mind-body – is a quasi-dualistic system. At one level, it’s an incomprehensible wave interference pattern. At another level, the proper illumination of this pattern by a laser gives rise to a beguiling 3D spacetime lifelike image. The *same information* therefore exists in two radically different forms.

The interference pattern is the wave representation of the information, and the image is the spacetime representation. It’s well-known that every part of a true hologram contains the whole image. So, if you take a suitably prepared hologram and smash it into pieces, you will be able to reconstruct the whole from any of the pieces. If you could reduce the smashed pieces all the way down to dimensionless points, each point would still contain the whole! In other words, the physical world is fully captured in the dimensionless world (the domain of the singularity).

Anyone familiar with Leibniz will know that each of his dimensionless monads was said to reflect the entire universe, meaning that Leibniz understood holography some two hundred and fifty years before it was invented. Leibniz was the first holographer, and the hologram, based on Fourier mathematics (which is based in turn on the Euler Formula), is the perfect means to understand reality. The universe is nothing but a self-generating, living hologram, composed of infinite, interacting, living monads. The universe is alive! Mathematically alive.

That’s the strangest thing of all. Who would ever have imagined that mathematics is the source of life, the origin of all living things, the root of mind, consciousness and free will. Yet, if you think about it, it’s the only conceivable answer. Why? Because mathematics is all about *information* and life is nothing but information – receiving it, processing it, generating it, experiencing it, feeling it, sensing it, intuiting it, thinking about it, rationalising it, giving it meaning, desiring information that leads to increased power, dreaming of having infinite information (thus of knowing everything and being God!)

Each of us is simply a self-contained, uncaused, uncreated mathematical information system that goes on forever. We are information *experiencers*. That’s the difference between us and soulless androids. Androids can detect information and process information but what they can *never* do is subjectively experience information. They can never have *qualia*. The transhumanism project is absurd because it seeks to make us live forever in a single body that never perishes.

Our journey to divinity actually *relies* on reincarnation – on continual physical death followed by rebirth. Our bodies die, our minds do not. If you stopped your body from dying you would stop the natural progression and evolution of the mind and remain fixed forever in a low-efficiency state, while those who accepted bodily death would keep advancing and transmuting until their minds were capable of divine expression. Nothing could be more foolish – and show less understanding of true reality – than atheistic transhumanism. Bodily death and reincarnation is nature’s greatest gift to us because it’s the sole means by which we can become Gods and fulfil our destiny.

The “immortal” transhumanists will be like the Struldbrugs of *Gulliver’s Travels*. As Wikipedia says, “In Jonathan Swift’s novel *Gulliver’s Travels*, the name struldbrug is given to those humans in the nation of Luggnagg who are born seemingly normal, but are in fact immortal. However, although struldbrugs do not die, they do nonetheless continue aging. Swift’s

work depicts the evil of immortality without eternal youth.” Struldbrug children are born with a red spot on their foreheads and it progressively changes colour until it hits black at age forty. Transhumanists should all bear such a mark.

The transhumanists would inevitably become Struldbrugs – hideous creatures, patched up like Frankenstein monsters. And if they ever solved the problem of eternal youth, they would simply become the miserable, suicidal “Eternals” of the sci-fi movie *Zardoz*.

The Material World

We inhabit an enormous wave interference pattern. Thanks to holographic Fourier mathematics, part of this pattern is able to manifest itself as what we perceive as the material world. That’s it. That’s the whole basis of “science”.

The “Experiment” Fallacy

Scientists set huge store by experiments. In fact, it's the core of their method and defines their whole enterprise. But *why* do they have such attachment to experiments? Is it rational? Quantum mechanics is the most successful theory in human history. It has been experimentally validated countless times. Yet here's the rub. Not a single scientist knows what quantum mechanics means. There is no agreed understanding of what quantum mechanics says about reality. There are many conflicting interpretations which have radically different implications for the nature of reality, and none of these interpretations can be experimentally proved or disproved.

Experiments don't help one iota since each interpretation is consistent with all known experimental results and with the basic mathematical formulation of quantum mechanics. What this definitively tells us is that experimental science is useless at explaining reality. It simply can't do so. The quagmire of quantum mechanical interpretation shows that science has reached its limit. To go beyond, it must reject experimental science and embrace rational science based on the Leibnizian principle of sufficient reason.

Scientists are living in a fantasy world if they think they will ever explain reality. No amount of experiments and no amount of Large Hadron Colliders will help them.

Science's central problem is that it uses instrumental rather than analytic definitions. For example, it says that time is what you measure with a clock, which tells you what instrument you require to measure time, but signally fails to tell you what time actually is (and indeed what a clock actually is). It's not enough to know how to measure something, you also need to know what the something is.

Science doesn't know what anything is. It's just a set of measurements with labels attached. Science can't define space, time, mass, energy, speed, motion, or anything else. It can certainly provide equations into which you can slot numbers (measurements) and get out different numbers ... but that's not going to help you explain ultimate reality.

As Leibniz alone realised, the essence of truth and explanation is the analytic definition. Analytic definitions tell you what things are in their essence, while the instrumental definitions of science don't tell you what they are but only how to measure them.

To measure something isn't to explain it. Quantum mechanical experiments measure things extremely accurately, but no scientist knows what quantum mechanics says about reality. Why? Because quantum mechanics doesn't contain any analytic definitions and revolves around an equation that invokes imaginary numbers, which are ontologically forbidden by scientific materialism.

Max Born won a Nobel Prize for his fallacious interpretation of the quantum mechanical wavefunction. What he did was produce a probabilistic formulation that satisfied the establishment's empiricist and materialist Meta Paradigm. The scientific establishment is now committed to explaining ultimate reality on the basis of statistics and probability. The answer to life, the universe and everything is, apparently, pure randomness!

Illuminism states the precise opposite. The answer to life, the universe and everything is pure reason. There is a sufficient reason why everything is thus and not otherwise. Probability, statistics and randomness don't come into it at all. The quantum mechanical wavefunction is ontological. Imaginary numbers actually exist and have real consequences. In fact, they're nothing other than the basis of tenseless time.

Substance

A substance is something unique and does not depend on anything else for its existence. It's a self-subsisting thing.

One of the most decisive moments in the history of thought occurred when Descartes gave his mathematical definition of what constituted a "substance". For Descartes, the only true substance was, in an absolute sense, the Christian God: he was the Creator, everything depended on him for its existence, while he did not depend on anything for *his* existence. In fact, according to Descartes, God not only created the world, but continually *recreated* it at every instant. If he failed to do so, we and our world would all vanish. However, bearing in mind that God was the sole absolute substance, it was also possible to define *relative* substances, which depended *only* on God for their existence.

Descartes famously defined two relative substances: mind (*res cogitans* – "thinking thing") and matter (*res extensa* – "extended thing"). Uniquely, humankind partook of both, i.e. we have both a mind and an extended, material body. Other animals have bodies, but no minds in the Cartesian sense; they are therefore mere biological automata – sophisticated machines.

The principal attribute of mind is thought, and only humans think in the Cartesian view.

The principal attribute of body is "extension" (it has dimensions; it takes up space).

So, for Descartes, the world, as we experience it, is composed of two incompatible substances. *Res cogitans* is immaterial, unextended, dimensionless and indivisible. It's mind, which, for Descartes, meant consciousness. *Res extensa* is material, extended, dimensional and divisible. It automatically follows that our brains – part of the material world – play no part in our consciousness. Animals have brains, but no minds. Confronted with the problem of how mind and matter could interact given that they had nothing in common, Descartes rather desperately (and illogically) resorted to claiming that the mind or soul resides in, or rather operated through, the brain's pineal gland – surely one of the most outrageous claims in the history of thought, and which made no sense in terms of Descartes' definitions of mind and matter. Moreover, identifying *where* the interaction took place was very different from explaining *how* it took place.

Descartes' philosophy presented human beings as incorporeal minds lodged, somehow, in a mechanical body, or linked somehow with the mechanical body.

In this philosophy, whereas the material, extended world constituted a single substance, there were as many thinking substances as there were souls created by God. Each one was dependent only on God, and was immortal (unless God chose to annihilate it). We therefore have a many-to-one relation (many minds linked to a single material world). However, given the fact that mind and matter were opposite in their properties, a catastrophic problem reared its head – the famous puzzle of Cartesian dualism. How could physical things affect minds, and how could minds control bodies? There seemed to be no viable route for any interaction between minds and bodies.

No one – other than Illuminists – has ever solved this problem. To understand the solution, it's first necessary to mathematically clarify Descartes' central point. Namely, there are countless unextended substances and one extended substance. Leaving aside the extended substance for the time being, how do all of the unextended substances interact, if indeed they do at all? And where are these thinking substances?

This was the aspect of Descartes' thinking that was more or less ignored (except by Leibniz), even though it turns out to be the most important issue of all – the key to existence itself.

In Illuminism, each Cartesian thinking substance is a mathematical singularity (like that at the centre of a black hole!), and any number of singularities can occupy the same “location”. That is, you can have a Singularity with a capital “S” (the Cosmic Mind, so to speak), composed of infinite singularities (with a small “s”).

What was the Big Bang Singularity? It was a stupendous collection of Cartesian minds, existing outside space and time, outside material “reality”! In fact, the material world is generated from Cartesian minds (which are Fourier frequency domains), hence Descartes’ extended substance is not a proper substance at all, being reliant on unextended substance. Just as God, in Descartes’ system, was an absolute substance (with mind and matter being relative), so, in Illuminism, mind (or, rather, dimensionless existence) is absolute and matter (dimensional existence) is relative. Thus, we get rid of intractable Cartesian dualism and are left with a nice, simple Cartesian monism.

The extended arena of matter is simply the Cartesian coordinate grid, made of ordered mathematical points and reflecting inverse Fourier transforms. Mind and matter (non-extension and extension) are hard-wired into Descartes’ great mathematical innovation of the mathematical grid.

Existence and Essence

If a substance depends on nothing else for existence then it's uncaused and uncreated. The only things that can exist *necessarily* are substances. Everything else results from substances. Therefore, substances are noumena and everything to which they give rise are phenomena. Phenomena cannot exist without noumena. Phenomena are contingent, and noumena (substances) are necessary.

What kind of thing has necessary existence? Religious people say "God" but God is in fact the most complex thing conceivable, hence the least likely thing to have necessary existence. The direct opposite is the case. The thing that *must* exist is the simplest thing conceivable. That is none other than the dimensionless mathematical point. The mathematical point is logically zero – "nothing". Being nothing, it requires nothing. Being nothing, and taking up no space, nothing can obstruct its existence. The ontological mathematical point is nothing and nothing can prevent nothing's existence.

So, nothing has no need of anything else and nothing else can have any say over whether nothing exists or not. It's impossible to create nothing, impossible to destroy nothing and impossible to cause nothing, hence nothing is none other than the indestructible, immortal uncreated, uncaused soul. It's the Pythagorean monad.

Each of us is a soul – "nothing" – and we have been around forever!

The Book of Genesis claims that God made the world out of nothing. Well, nothing, unlike God, has necessary existence. Genesis is the complete inverse of the truth. It's God who is created by nothing!

What is "nothing"? It's categorically *not* "non-existence". Nothing is actually something. Something is mathematically structured nothingness. The generalised Euler Formula is exactly the miraculous mathematical instrument that allows nothing to be structured.

Ontological mathematics *is* the generalised Euler Formula. Mathematics exists because the generalised Euler Formula exists. The generalised Euler's Formula exists because it's impossible for it not to exist since it requires nothing and cannot be prevented by anything.

The truths of mathematics are eternal truths. This means that they can never *not* apply. There has never been a time without them. It's impossible for eternal truths not to exist, it's impossible for them to be caused or created (because then they wouldn't be eternal). Eternal truths are necessary truths. Plato understood this with his domain of eternal, immutable, perfect Forms.

Here is wisdom. Plato's perfect Forms (the ones that are ontologically valid), the generalised Euler Formula, ontological mathematics, the principle of sufficient reason and causality, eternal truths, the truths of reason, necessary truths, uncaused causes, prime movers, first and final causes, uncreated and indestructible things, are all in fact one and the same thing. They are all just different ways of referring to the same thing: *necessary existence*.

We live in a rational universe, an ordered cosmos, for the extremely simple reason that existence *is* reason.

Plato's ultimate Form, the apex of everything, was "the Good". This is none other than Reason. Aristotle's God was matterless form – pure Reason. Pythagoras' God was pure reason (mathematics). Leibniz's God was pure reason operating mathematically. Hegel's God was pure dialectical reason.

All of these thinkers ultimately had the same idea – *reason is alive*. Hegel realized something else about it: it's evolving. One aspect of reason is objective, transcendent and immortal. We can refer to this as Aristotelian logic. The other aspect of reason is subjective, immanent and evolutionary – it follows *dialectical* logic.

Aristotle had the idea that the rational soul of human beings didn't belong to humans at all but to God. Actually, we all have the eternal truths of reason built into us (the "divine" truths), but we are not consciously aware of them. The living part of us is dialectically evolving. Only at the culmination of an immense, iterative process of thesis, antithesis and synthesis does our reason become conscious of itself. It's then that we become Gods. We understand reality exactly – through pure, absolute, infallible, objective reason. We have entered into union with Aristotle's God, so to speak.

Reason has two aspects: objective and subjective. The objective aspect is necessary, eternal, deductive, analytic and *a priori*. The subjective aspect is contingent, in time, inductive, synthetic and *a posteriori*. Objective reason is mathematical and rationalist. Subjective reason is scientific and empiricist. In fact, earlier in its dialectical history, subjective reason is based not on Logos at all but on Mythos. It employs faith, feelings, superstitions, stories, prophecies, "divine revelation", and so on, to attempt to understand reality.

To put it another way, the more rational you are, the more highly evolved you are and the more divine you are. Meditating or praying won't help you to achieve gnosis. The only way is by becoming a rational genius, or having a miraculous degree of rational intuition.

Subjective reason is living reason evolving through time via the dialectic, and its ultimate mission is to become conscious of objective reason – the laws of ontological mathematics which explain the whole of existence.

The dialectic is the rational – if extremely crude process – by which reason becomes self-aware. Rational conscious beings, unlike irrational conscious beings, or unconscious beings, can actively reflect on their rationality and strive to master rationality. Irrational conscious beings – such as Abrahamists – cannot become rational through their own efforts. They unwittingly rely on rationalists appearing amongst them and slowly and painfully educating them.

The Enlightenment was a huge breakthrough for the human race because rational people gained power, and the Abrahamists began to doubt. In the West, religion has been slowly dying ever since. In Europe, Christianity is dead in all but name.

The Muslim Abrahamists did not have an Enlightenment and these people show just how dangerous Abrahamists are when they are not moderated by a powerful rational group of scientists, philosophers, mathematicians, artists and freethinkers.

Mythos Abrahamists have failed to evolve. They are retarded, bestial and can be barely considered human. They are only just conscious, and it does not take much to send them back into a primitive, pre-conscious bicameral state. Scientists – wedded to empiricism rather than rationalism – have evolved beyond the Abrahamists but are still seriously retarded.

The most highly evolved human beings – the master race, if you will – are the rationalists. They are metaphysicians and ontological mathematicians, relentlessly pursuing eternal truths of reason. We can even identify who most of them are: Myers-Briggs INTJ and INTPs.

They are not feeling types, they are not sensing types, and they look to the inner world of thought (the intelligible world) rather than the outer world of experience (the sensible world).

A divine human race would be one in which INTJ and INTPs were extremely common. As it is, they are exceptionally rare ... which accounts for why the world is the irrational mess it is.

The simple fact is that some human beings are immensely more highly evolved than others (mentally rather than physically; physical evolution has more or less ended for humanity).

We all belong to the same physical species, but we are radically different in terms of our mental species. Some of us – such as the Abrahamists – are closer to the beasts than to humans. And some of us, especially the INTJ and INTPs are on the verge of divinity!

In our world, Gods and beasts mix together, but the beasts are in charge because they are in the overwhelming majority, and the Gods have not yet realised their own powers. But one day they will, and then everything will change. The task of the Illuminati is to hasten that glorious day, the Day of Reason when humanity finally enters its divine phase.

Reason transcends our mortal lives because it's eternal. That's why all intelligent species in the universe necessarily share the same mathematical truths. All intelligent species, however, have different versions of science because science is based on empiricism, contingency and ad hoc hypotheses ... and these differ for every different species depending on how their senses are configured.

Nothing is more foolish than for any thinking person to worship empiricism over rationalism. It's inherently irrational to do so, and all scientists are inherently irrational.

Humanity becomes more conscious and more divine the more it turns to rationalism rather than empiricism, to mathematics and metaphysics rather than physics.

A substance depends on nothing else for its existence. The “essence” of something is the element of it which makes it part of a class. So, all monads are unique substances, but all share the same essence, hence belong to the same class. In terms of substances, there are an infinite number of them. In terms of essence, all of these substances belong to the same class.

In Descartes' system there are countless mental substances and one material substance. All of the mental substances have the same essence and belong to the same class. However, they have a wholly different essence and class in relation to the material substance. This is therefore a dualistic system with an enormous problem of how mind and matter can interact. It's a category error to imagine that they have anything in common.

In Leibniz's system, there are infinite mental substances (monads), but they all have the same essence and belong to the same class. There is no material substance. This is therefore a monistic system and there's no problem of Cartesian dualism.

What is the essence of a mind? That it thinks. Descartes thought that thinking must be conscious. Leibniz realized that it could also be unconscious and, in fact, most thinking in the universe is unconscious. Unconscious thinking is the default mode of thinking, and conscious thinking is a relatively rare type of thinking that evolves from it via dialectical evolution.

The first glimpses of consciousness arise through feelings and sensations, through Mythos – emotive, simplistic stories. Then intuitions become possible. Finally, rationalism is born. The more rational you are, the more conscious you are.

Rationalism – Logos – appeared on Earth at a very specific time – in ancient Greece with the pre-Socratic philosophers, including Pythagoras. This was the most miraculous event in human history, the moment it became possible for humans to become Gods. Pythagoras identified the answer to existence 2,500 years ago when he announced, “All things are numbers; number rules all.” This was the single greatest insight of all time.

Living Reason

Living reason has two aspects – objective and subjective, outer and inner, external and internal, a without and a within. Its objective character is Aristotelian. Its subjective character follows the Hegelian dialectic.

Wittgenstein

Wittgenstein, a great admirer of natural science, condemned mathematics as “empty” and devoid of descriptive content because it was all about analytic tautology. This is perhaps the biggest blunder in philosophical history since the only things that can be unconditionally true are those that are true by eternal definition – a view championed by Leibniz. In a 100% mathematical universe, all absolute, unconditional, eternal, immutable truths – those for which you would go to the stake – *must* be analytic tautologies! How can the conditional “truths” of natural science have any authentic truth content if they are conditional? They are ipso facto not true. How many scientific theories – well-verified by experiments – have fallen by the wayside? Did any of these theories have any truth content at all? In fact, they are about models of reality and confidence in those models. They have no connection with truth if truth is something that is infallible, immutable and eternal. Anything that is not all of those things is *not true*.

The “theories of everything” proposed by science are nothing of the kind. At best, all they do is unify quantum mechanics and gravity – which has nothing to do with explaining the fundamental basis of existence.

Descartes

Descartes – the first modern philosopher – brought all of the tensions within philosophy to a climax. Before Descartes, the general view of reality was that it was some kind of living organism. Matter was fully penetrated by mind, spirit, or God. The universe was therefore teleological, as Aristotle had insisted.

When Descartes separated mind and matter into two distinct domains, he allowed two radically different schools of thought to arise. The materialists promptly abandoned Descartes’ unextended domain. After all, it was both immaterial and empirically unobservable, so what possible function did it serve? Who needed it? The materialists (and empiricists) killed off the notion of the universe as alive. With them, it became a dead, clockwork mechanism, with life and mind emerging as bizarre and ineffectual epiphenomena in a pointless machine universe.

The idealist school took the opposite view. They abolished the extended, material world and declared that it was all an illusion of mind. As Descartes had said, thinking (mental activity) was the one sure fact. Everything else was conjecture, and the existence of a fantastical dream world (that we enter when we go to sleep) proved how easy it was to summon false sensory experiences into the mind.

Idealism and rationalism became the path of choice of philosophers and metaphysicians while materialism and empiricism morphed into science. Science embraced mathematics and started measuring things, making things and predicting things – with great success. Philosophy produced wondrous ideas, but in the manner of the impractical denizens of the floating island of Laputa in *Gulliver’s Travels*.

Mathematics is the difference between success and failure. Only mathematics provides the tracks that keep reason on the straight and narrow.

Why did philosophy fail to embrace mathematics? After all, Pythagoras, Plato, Descartes, Leibniz and others were brilliant mathematicians *and* philosophers. It’s because people largely understand mathematics in terms of measuring distances between things, i.e. in terms of *extension*. How do you apply mathematics to a Singularity with no extension at all, and no matter? What would you be measuring?

A Singularity is the most feared entity in physics because, as any physicist will tell you, it’s where the laws of physics fall apart. Why? Because the Singularity is the creature of zero and infinity – the two numbers that are fatal to the empiricist materialist paradigm of science.

To reconcile mind and matter (the non-local and dimensionless on the one hand, and the

local and dimensional on the other), it's imperative to grasp the mathematics that define each situation. This has *never* been done in the mainstream; only within secret societies. It's not that the relevant mathematics doesn't exist. It does, and is extremely well known (it's just Fourier mathematics). The problem lies in the resistance to using it and understanding it.

The Cosmic Hologram

A hologram has a dual-nature, encapsulating both the frequency and spacetime representations of information. The hologram, in a nutshell, captures the whole of existence and is the basic model for how we understand reality. Our bicameral, dual-hemispheric brains have evolved to operate according to holographic mathematics. Our left hemisphere, home of our local consciousness, shows us the local spacetime representation, while our right brain, home of the non-local unconscious, is immersed in the non-local frequency representation and has contact with the entire cosmic hologram (“the whole is in each part” – the holographic principle).

Each hemisphere is responding to the same information, which is presented in two different ways and processed in two different ways. Our left brain performs inverse Fourier transforms (i.e. deals with spacetime representations constructed from the addition of frequencies) and our right brain performs Fourier transforms (i.e. deals with frequency representations *deconstructed* from spacetime functions, or deals with pure frequencies themselves).

Our left brain is concerned with the local, with separation, with extension, with reductive analysis (all the stuff of scientific materialism), while our right brain is concerned with the non-local, with integration, with interconnectedness, with synthesis and holism, (all the intuitive New Age stuff).

Our left brain is sensory, our right brain intuitive. An intuitive can have a “flashbulb” moment or snatch a brilliant, fully formed idea “out of nothing” because he has tuned into the non-local information domain.

Jungian synchronicity is all about the right brain and left brains working in conjunction to produce a completely unexpected, meaningful event. Jung thought that such events are *acausal* (because he could see no means by which they could be causal). In fact, they are wholly causal. The right brain is detecting non-local information and it can therefore make someone turn round at exactly the right moment to see some meaningful pattern that the non-local right brain, but not the local left brain, *knew* was coming.

Non-localism is of course fundamental to quantum mechanical theory. The wavefunction of quantum mechanics is wholly non-local (extending everywhere) and, what’s more, it involves imaginary numbers – just like Euler’s function and Fourier mathematics! (Which ought to lead any rational person to draw the appropriate conclusions. Tragically, irrational scientists never did; they could never reconcile themselves to the real existence of imaginary numbers, which stand outside the Meta Paradigm of materialism and empiricism.)

Isn’t it becoming obvious what’s going on? Mathematics is all about the mind, and the material world is simply one of its mathematical side-effects. There’s no material world independent of mind, and there never could be.

The inability of any scientist to understand quantum mechanics shows the astounding fragility of the scientific paradigm. In truth, it’s impossible to make sense of quantum mechanics within the localist empiricist materialist paradigm of science. Quantum mechanics can be understood only by invoking a non-local, dimensionless, frequency domain outside space and time (and thus mental rather than physical). All the mysteries, all the bizarreness of quantum mechanics, dissolve in that new context. However, it necessitates the abolition of the existing scientific paradigm! In terms of Thomas Kuhn’s philosophy, too many anomalies have accumulated within the prevailing paradigm and science must enter a revolutionary phase where the old is destroyed and a new paradigm constructed.

Illuminism – ontological mathematics – is that new paradigm. Ontological mathematics must replace science.

Schopenhauer and Fourier

It's remarkable how closely Fourier mathematics reflects the philosophy of Schopenhauer. Schopenhauer asserted that his whole philosophy was based on a single idea: "My body and my will are one." In other words, his body was the phenomenal expression of his noumenal will (mind), and his will (mind) the noumenal root of his phenomenal body. His body was "appearance"; his will was "thing in itself".

Schopenhauer committed a grave error, however, when he said that there was only a single cosmic Will. He reached this conclusion because, since the Will was outside space and time (hence couldn't support spacetime individuation), it must surely, he thought, be a unitary entity. In fact, you can have a cosmic Will composed of infinite autonomous wills, each of them equally outside space and time. To put it another way, you can have infinite Leibnizian monadic wills in one Singularity. One point outside space and time can support infinite points outside space and time, each a separate, uncreated, uncaused soul.

Buddhists don't believe in an eternal soul but in a single, unitary, divine cosmic force, and Schopenhauer reflected this way of thinking. However, Leibniz's system shows that there are indeed individual souls: an infinite number of them. Leibniz's system thus mathematically and metaphysically supports Hinduism over Buddhism.

To recap, there are two domains. One is unextended, dimensionless, immaterial and outside space and time. It's a mental domain, and is based on frequencies. The second domain – derived from the first – is extended, dimensional, material and inside space and time. It's based on sinusoidal wave combinations travelling through a Cartesian coordinate arena of extension.

Consider the origin that stands at the centre of any Cartesian grid. Ontologically, this is a Singularity. In fact, it comprises infinite singularities, and each of these singularities comprises infinite frequencies that are so perfectly balanced as to produce a net result of zero.

The collective Singularity is eternal, dependent on nothing but the eternal, uncaused, uncreated, autonomous singularities of which it's composed. Via a mathematical "antisymmetry" operation, all of these singularities can be given unique identifiers (coordinates) and the effect of this is to produce an infinite extended Cartesian grid in which all points are separated from each other (but note that each point itself remains unextended and dimensionless, hence does not partake of the extended properties of the grid).

The next thing to bear in mind is that the frequency domain of each singularity consists of infinite "depth" (i.e. there are infinite frequencies). The same is not true of the material world that results, via Fourier mathematics, from frequencies. Only a small band of low-energy frequencies can inhabit the collective Cartesian arena. Higher energy frequencies instantly collapse back on themselves, into their home monad. This means there's a profound mathematical asymmetry between mind and matter. Mind is much more powerful. There's an infinite mental hinterland that can be explored that has nothing to do with the material world. It's this *Unknown Country* that allows us to become Gods!

What is a Mind?

A mind = a singularity, comprised of infinite energy waves (sines and cosines) that balance to zero.

A mind is a frequency processor. One way to think of a mind is as a self-playing piano. There's an infinite scale of notes on this mental piano, and playing those notes constitutes thinking. We are all playing different tunes, and there are infinite tunes to play. Some of us are playing Wagnerian symphonies while most of us are playing muzak.

When we go to sleep, we enter our private monadic world and, in our dreams, we lose all sense of space, time and conventional causality. When we awake, we enter the public monadic world and find ourselves in a rigid spacetime, materialist world where we are opposed by all of the other minds of the universe. That is true for every monad, so no monad gets its own way within the shared space.

In truth, we are *always* dreaming. We are either dreaming alone (when we are asleep) or dreaming together (when we are awake). An "out-of-body" experience occurs when someone goes to sleep, but instead of their mind entering the private monadic domain, it remains connected to the public world. Unencumbered by its sleeping body, the mind can now travel anywhere in the universe!

Free Will

Given two distinct but linked domains – mental and material (frequency and spacetime) – four types of causality are possible:

- 1) Mental cause and mental effect (e.g. one thought prompts another thought).
- 2) Mental cause and material effect (e.g. you freely choose to move your arm ... and it moves in the physical world).
- 3) Material cause and mental effect (e.g. someone kicks your shin and you experience pain and anxiety).
- 4) Material cause and material effect (e.g. you step on ice and fall over).

The situation is actually rather more subtle since “matter” is really just low energy mind. As Einstein said, “Concerning matter, we have been all wrong. What we have called matter is energy, whose vibration has been so lowered as to be perceptible to the senses. There is no matter.”

At this low energy level, we experience the full power of Fourier transforms, forward and inverse. Any change in the material spacetime representation is immediately reflected in the frequency representation, and vice versa. This all happens at an unconscious, auto-pilot level. However, our normal thinking occurs in a high energy band and is barely reflected, or not reflected at all, in our bodies. We become more and more Godlike as we think in higher and higher frequency bands.

It's Complex

Scientific materialism is a doctrine that only real numbers greater than zero and less than infinity have any general bearing on reality. Zero, infinity and imaginary numbers are regarded as abstractions that can never appear in reality. Negative numbers appear only in specific contexts such as negative charge.

Illuminism asserts that complex numbers (comprising real and imaginary numbers) are more fundamental than real numbers, and that zero, infinity and negative numbers all have ontological reality.

Illuminism involves *absolute* spacetime, hence preserves objective reality and the reality principle. (As shown elsewhere in the *God Series*, the Lorentz transformations of Einstein's special theory of relativity can be much more elegantly obtained via Euler's Formula and require no appeal to any bizarre Einsteinian relativity principle, which contradicts the reality principle.)

Mind is a frequency domain outside space and time, hence belongs to a dimensionless Singularity. Matter is in space and time, hence belongs to an extended, dimensional domain.

The energy of the material world is supplied by low-energy waves (from monads) which can be shared in a collective monadic domain (spacetime). High-energy monadic frequencies remain locked inside the monads.

Matter is simply the shared content of monadic minds. It has no reality outside minds.

Nothing is falser than the scientific doctrine that matter exists independently of mind, precedes mind and gives rise to the mind. The precise opposite is the case.

Mathematics is mental, not physical. It's ontologically grounded in dimensionless frequencies. From frequencies comes spacetime and matter, via the inverse Fourier transform. It really is that simple.

The Cosmic Building Blocks

What is the universe made of? It must be made from the simplest of all things – nothing! The universe is literally constructed from nothing. It's just *structured nothingness*.

However, nothing is deceptively simple. From a higher understanding, it's the most complex thing of all. In fact, it's none other than the soul, and all souls are capable of achieving maximum optimization, which transforms them into Gods!

Abrahamists say that existence begins with "God", the most complex thing of all. Actually, it begins with the soul – "nothing" – which is both the simplest and most complex thing of all, simple because it's nothing and complex because it's infinity. The soul, at the beginning of a Cosmic Age, starts off as blank, as pure potential, and ends the Age as full, as pure actualization – as God!!!

The destiny of all of us is to attain divinity, but some of us get there much faster than the others – and no thanks to the others who actually resist us (as, for example, the Abrahamists do with extreme ferocity), and we selflessly pave the way for the rest to join us.

Why do we help those who have stood in our way? Because when you're God, you no longer have any pettiness left. You don't want to punish anyone, or get any revenge, or prove that you're "better" than others. None of these would be divine qualities. The monstrous behaviour of the Abrahamic God and his compulsion to have people on their knees to him show that he is no God, but, rather, the anti-God – the Devil. He is precisely that which must be overcome. By reason.

The Mystery of Time

A famous question in philosophy is whether time is tensed or tenseless. In tensed time, events that have happened are in the past, events that are happening now are in the present and events yet to happen are in the future. In tenseless time, events are ordered according to tenseless relations (“earlier than”, “simultaneous with”, “later than”), and there are no such things as past, present and future.

“Past” means that something has gone forever and can never be recovered. (Time travel into the past is impossible because there’s nowhere to travel to). “Earlier than” is a direction to finding something that continues to exist, i.e. the French Revolution is earlier than the execution of Louis XVI, but both events continue to exist. They haven’t been removed from existence (as they would be if they were in the tensed “past”). Time travel *can* take place in tenseless time since all tenseless destinations remain available forever, just like coordinates on a map of the world.

Tenseless time is the one that goes hand in hand with the “block universe” of physics (the view favoured by Einstein), where time and space are described by coordinates and all coordinates are present at once (so tense is absent). The “future” has already happened – it just hasn’t been *reached* yet. Therefore, it’s not the future, just the present, the *eternal* present.

The problem of time is perhaps the biggest embarrassment of physics and philosophy. How can no one prove which theory is right between the tensed and tenseless views even though they are the opposite of each other? Of what value are physics and philosophy if they can’t settle such a matter?

The central issue is that the concept of time hasn’t been understood at all. It has no mathematical, scientific or philosophical definition. Moreover, time has never been differentiated sufficiently from *motion*. Indeed, that’s the root of the problem. Are things moving *in* time (so time is a kind of static container), or is movement happening *because* of time (so time is somehow pulling or pushing things)?

Which is it? Is time like a stream which carries static things along in its current? In which case what is this stream made of? What *is* time?

If time is static and a kind of space, as Einstein proposed, then how do things move at all?

The answer is simple. “Time” – the stuff measured by clocks – is indeed simply another kind of space. Specifically, time is *imaginary space*. Space is defined by real numbers and time by imaginary numbers. In other words, time is a container just like space but is mathematically orthogonal to space.

“Time”, being a static container and type of space, has nothing to do with movement. Movement is independent of time, and a wholly different concept. In Einstein’s special theory of relativity, hypothetical clocks attached to photons don’t tick: time has stopped or, rather, passes infinitely slowly. How can photons move, how can change take place, if photons do not experience time? Yet photons *do* move and *are* associated with change, so time *cannot* be inherently connected with movement and change.

Motion, not time, is the origin of change. Time is just a tenseless static container, directly comparable to space. Time is tenseless. It’s motion that’s tensed. Motion defines the past, present and future.

Imagine an object obeying the first law of motion and travelling at a constant speed in a straight line *forever*. This object can never travel backwards into the past. Even if you turned an object around and made it reverse its direction, it would still be moving forward into the future,

and leaving the past behind. That never alters. An object is always in the present, with the future immediately ahead and the past immediately behind. A moving object can never reverse that sequence of motion, i.e. it can never be in the condition of having the past immediately ahead of it and the future immediately behind it, no more than the hands of a clock in our world could ever suddenly start moving anticlockwise.

Time travel is as impossible as clocks that run backwards. Motion is *always* in the forward direction (from the past into the future). Time is simply a container in which motion takes place (just as space is) and has no connection at all with tense. Time is tenseless and motion is tensed. The universe is always advancing and can never go into reverse. No force in the universe could make motion travel from the future into the past. All talk of “time travel” simply reflects the staggering lack of understanding of the concept of time within physics.

Time doesn’t travel; moving things do, and they always go in one direction – *forward* into the future. There’s no such thing as an “arrow of time”. What does exist is an “arrow of motion”, and that arrow always points from the past into the future; *never* the other way around.

Space = real numbers; time = imaginary numbers; motion = tensed movement through spacetime.

In terms of the Singularity, where there is no spacetime, motion is none other than *thought*, and thought always moves forward. No one *ever* replays their thoughts backwards. They “remember”, but in the present. They don’t go back and replay exactly what happened long ago. In fact, memories can easily become corrupted precisely because they can involve information that belongs to the present and not to the past event being “remembered”. That’s why false memories are so common.

Remember, the proper application of mathematics clarifies all bogus, confused, non-mathematical concepts such as scientific and philosophical “time”.

Motion is *the* fundamental feature of the universe. Everything is moving all the time. There is no such thing as a stationary entity. Motion is associated with *energy*, not with time.

The Localism Fallacy

Scientific materialism is a localist phenomenon. The mind is non-local. It’s everywhere and nowhere. It’s outside space and time. It’s the domain of eternal, immutable, Platonic frequencies. It’s a Fourier frequency world. It’s a singularity.

Scientists, locked into their sensory worldview, cannot conceive of non-local, non-sensory phenomena. They lack the imagination and intuition to be truly intelligent. It’s up to the elite – ontological mathematicians – to put these scientific apes in their place.

Freud and Jung

Mathematically, Freud was a localist and Jung a non-localist. Freud believed in a personal consciousness and personal unconscious, with no link to any other mind. Jung, when he introduced the Collective Unconscious, created a non-local entity beneath the personal unconscious in which *all* minds shared, and via which everyone was linked to everyone else. Although Jung didn’t grasp it himself, his idea of synchronicity is one that fits in perfectly with the Collective Unconscious, except synchronicity is no longer bafflingly acausal (as Jung considered it) but *causal*. That is, the Collective Unconscious is a mathematical Singularity, and is all about non-local Fourier connections outside space and time. Its *teleological* and its purpose is to *increase meaning*. Anything that introduces additional meaning is *always* sought out.

Motion and Energy

What is needed for existence? Flawless, analytic definitions and *motion*. Motion is the

direct product of the principle of sufficient reason. To see why, consider “energy”. What is “energy”? It’s simply the consequence of the principle of sufficient reason applied to a circle. If no point on a circle’s circumference can be privileged over any other point then all points must be treated on a par. The only way to satisfy the equivalence of all the points on a circle’s circumference is to allow each one to be occupied in turn, in perfect uniform motion. Energy is just this process of moving around a circle, and, in doing so, it generates, via the generalised Euler Formula, real cosine and imaginary sine waves ... and from these we get the *world* – the world of pure mathematical, analytic energy.

Energy is perfect mathematical motion proceeding by way of eternal circular motion and the analytic sine and cosine waves generated by such motion (in accordance with the generalised Euler Formula).

The Challenge

“The greatest challenge to any thinker is stating the problem in a way that will allow a solution.” – Bertrand Russell

But will the solution be *necessarily* compatible with solutions to other problems? In scientific materialism, there’s a catastrophic standoff between two highly successful but incompatible theories: quantum mechanics and Einsteinian relativity. Only mathematics provides a standalone system where all correct solutions are necessarily compatible with all other correct solutions – because mathematics is analytic, not synthetic; necessary, not contingent; *a priori*, not *a posteriori*; deductive and not inductive.

Mathematics is rational, not experimental. Why do scientists prefer experimental evidence over rational argument based on ontological mathematics? Because they are slaves of their senses and disregard their intelligence. There are no circumstances in which $1 + 1 = 2$ can be false. There are endless situations in which experimentally verified scientific hypotheses can subsequently be disproved, as the overflowing graveyard of scientific theories demonstrates. Math can never be wrong and science can never be right (it’s only as good as the last provisional set of experimental data).

The claim that something isn’t science unless it makes falsifiable statements necessarily means that mathematics does not constitute science since no part of mathematics is falsifiable. What is the engine of science? Mathematics! So, science is grounded in something that is not scientific, hence is absurd.

It’s almost surreal that any intellectual could take science seriously. When it comes to ultimate reality, the only thing science has accomplished is to blind the intelligentsia to the fact that ultimate reality has nothing to do with science and is wholly mathematical.

Science is “mathematics for dummies”, approximate mathematics, bungled and botched mathematics, a bodger’s version of mathematics designed to play to sensory prejudices and delusions.

Leibniz Contra the Multiverse

Long before any scientific materialist ever conceived of a Multiverse in which anything that can happen does it happen – in an infinity of universes – Leibniz had explicitly repudiated this idea as irrational. He wrote, “If all possibles were to exist, there would be no need of a reason for existing, and mere possibility would be enough.” In other words, if things can happen just because they’re possible then they no longer require a reason why they are thus and not otherwise. Randomness and statistics replace the principle of sufficient reason.

If you scrutinize the thinking of scientists regarding the origins of the universe, you will see that their objective at all times is to remove meaning, purpose and reason from the universe – anything that smacks of teleology and any form of intelligence and design – and replace it with randomness, statistics and probability.

Scientists despise the concept of mind as something that exists in its own right in the universe. Sure enough, once you abolish mind, you abolish teleology, leaving you with nothing but an appeal to random events as the starting points for everything.

If the idea of an Abrahamic God as the origin of the Big Bang is absurd, isn’t the idea of a random event with no conceivable explanation every bit as bad? In the first case, we have a cosmic magician. In the second, we have magic without the magician. What we certainly don’t have in either case is a rational explanation. Only the principle of sufficient reason can provide a rational explanation. There’s no alternative if you want an ultimate answer to existence.

Ontological Reason

If the universe is grounded in reason, we live in the best of all possible worlds – because it would be irrational for reason to give rise to anything other than the most rational world. Assuming that the most rational world is better than any less rational world, we are in the best world we can possibly be in. In fact, an irrational world couldn’t exist at all – its irrationality would destroy it. That’s why the Multiverse theories of scientists are ludicrous. If the universe were grounded in randomness rather than reason, the universe would be in a permanent state of chaos and disorder. There would be no organization and no laws. *Ipsa facto*, science is false and rationalism is true. We live in a world of reason.

The purpose of a universe of reason is to maximize its reason, to find the “best possible world”, the best solution, the world where reason is optimally expressed. It achieves this through Hegel’s dialect of iterative thesis, antithesis and thesis – a process *guaranteed* to reach the best answer in the end.

The Transition to the Empire of Reason

“A metaphysics should be written with accurate definitions and demonstrations. But nothing should be demonstrated in it apart from that which does not clash too much with received opinions. For in that way this metaphysics can be accepted; and once it has been approved then, if people examine it more deeply later, they themselves will draw the necessary consequences. Besides this, one can, as a separate undertaking, show these people later the way of reasoning about these things.” – Leibniz

Leibniz put a huge amount of effort into trying to “sell” his Monadology as a rational clarification of Christianity and its Creationist God. His notion was that once people became

familiar with monads, it would then be possible for them to leave behind Christianity. Sadly, monads were way too baffling for Christians. Reason simply wasn't their thing.

Singularities

“Without Space, there is no coexistence, but the universe is contracted to a mathematical point, which is nowhere and has no relation to anything beyond itself; without Time, there is no successive existence but the past and the future shrink into the indivisible present; and even this disappears as soon as it begins to be.” – Francis Bowen

What are space and time? They are what prevents everything from belonging to a Super Singularity. Space and time belong to the domain of extension while singularities belong to the domain of non-extension. Extension is about all numbers between zero and infinity; non-extension is about zero and infinity. The two domains belong to a sublime mathematical continuum: matter (extension) and mind (non-extension), just as Descartes said.

Everything is about individual singularities (minds) on the one hand and organised spacetime arrays of the energy contents of singularities (matter) on the other. Singularities are Fourier frequency domains, and spacetime and the material world are what are created from these frequency singularities via a cosmic inverse Fourier transform. It's all in the math!

“Hence, both Intellect and Will are unspatial in their very nature, since the former creates Space in idea, and the latter creates it in ‘reality’. It follows, therefore, that even the atomic Will, or what we call atomic force, exists outside of Space and independent of it; for as Schelling says, it is prior to extension.

“It also exists out of Time; for as we have seen, it is characteristic of the action of the Unconscious, that it never wavers or doubts, it needs no time for consideration, and it is independent of memory, since it acts unerringly as well before as after experience; therefore it does not, like conscious reason, proceed by comparison and inference, but it grasps the results instantaneously ... The thought of the Unconscious, therefore, has no *duration* in Time... The realm of the Unconscious, therefore ... exists outside of Space and Time... Space collapses into a mathematical point; Time shrinks into the indivisible present moment; and One becomes identical with All.” – Francis Bowen

The Unconscious thinks instantaneously. It has no doubt. Equally, it has no choice. Consciousness is an enormous slowing down of thought in order to reflect on different options ... in order to freely choose between possibilities. Freedom needs time for deliberation and consideration. Unconscious animals never ponder. They don't choose from a variety of options. They don't plan. They don't consider the future. They don't reflect on the past.

Memory goes hand in hand with consciousness. We remember what happened before, and we anticipate what might happen. The Unconscious has no memory, and never considers the past or future. Instinct always operates in the present.

It's through consciousness that we build memories (reflecting the past) and expectations (anticipation of the future). Consciousness breaks us out of the tyranny of the present. You can't be conscious unless you have memories and expectations. If you have expectations, you can plan and choose between rival options ... you can construct a future you have selected rather than one imposed on you (by the Unconscious, by instinct). Animals can't plan, hence are not free. Freedom implies a knowledge that the future can be different depending on the choices we make. Animals act instinctively, always responding to the present circumstances.

To say that the purpose of the universe is to maximize consciousness is to say that its purpose is to maximize freedom, choice and the ability to plan what we want the future to be. Of

course, we want the future to be as good as possible. So, the universe generates consciousness because only through consciousness can the universe optimize itself and solve itself. The universe reaches heaven (the optimal state, the omega point) via planning, and planning requires consciousness and freedom.

As Hegel said, “The history of the world is none other than the progress of the consciousness of freedom.” In fact, consciousness *is* freedom. Good and evil apply only to conscious minds because only such minds can freely choose. Creatures operating through instinct do not act freely but through necessity. They exhibit programmed responses. Since they never choose what to do, they can never be accused of choosing evil.

Stupid human beings are almost bestial and they too operate primarily through instinct. Ironically, jails are full of stupid people labelled “evil” and yet they aren’t “evil” at all because they operate at the animal level of instinct, hence can’t be held responsible for their actions. Only smart people can genuinely be evil. Arguably, even Satan isn’t evil. He instinctively acts with maximum selfishness at all times (just like huge numbers of human beings!). He can’t help himself, hence he’s unaccountable for his actions. He couldn’t have acted otherwise. He’s too much of an animal. One might also say that all “evil” Abrahamists are just animals from whom we can’t expect anything better.

A scientific materialist universe can’t optimize itself because it has no mind, no purpose, no freedom, no ability to plan, no concept of the past, present and future. It’s a *dead* universe.

The Answer

We alone, in the *God Series*, have given the definitive answer of why existence is possible at all. It’s nothing other than an inevitable consequence of mathematics and the two fundamental, linked numbers: zero and infinity. When you realize that there is no such thing as “nothing” – considered as non-existence – and that “nothing” is in fact “something”, you understand why non-existence is *impossible*.

Nothing and infinity are integral, defining parts of an enormous, necessary, interlocked, *a priori*, analytic, immutable system – *ontological mathematics*. Science is contingent, mathematics is necessary. There’s nothing eternal about science. Mathematics, on the other hand, is absolutely eternal. It’s therefore absurd to say that the eternal universe is scientific rather than mathematical. Science does not have the requisite rational and logical properties to account for existence that *necessarily* lasts forever. Science is fallacious and bogus. It relies on eternal scientific laws and yet, according to science, time and space began with the Big Bang and it’s absurd to refer to anything existing before space and time began (this is the defining doctrine of materialism).

By this “logic”, no eternal scientific laws existed prior to the Big Bang, in which case there is no conceivable cause of the Big Bang since there were no laws to bring it about. It therefore stands as a miracle, an act of magic. Scientists might as well get on their knees to the Abrahamic God since he operates in the same terrain of miracles and magic.

Of course, ontological mathematics has at its core zero and infinity – unextended, dimensionless existence, outside space and time, enshrined in the Pythagorean-Leibnizian monad. Each monad carries within it the complete, consistent, *eternal* laws of ontological mathematics, so there’s no mystery as to what caused the Big Bang. Mathematical necessity did. The Big Bang is part of inexorable mathematical cause and effect. Mathematical space and time are produced by mathematical singularities (via Fourier mathematics), and return to these singularities in due course, then begin the whole cycle again. This goes on *forever*.

Just as existence is mathematically inevitable, so is the cyclical nature of existence. Ontological mathematics is all about infinite singularities existing within a collective Singularity,

and then ordering themselves (via a mathematical, “fermionic”, antisymmetry operation associated with an inverse Fourier transform on a cosmic scale) into a Cartesian grid. This provides the arena of extension, and extension is the basis of space, time and matter, i.e. of the material world. But the relentless expansion of the universe leads, in the end, to the destruction of all matter and thus the return to the frequency Singularity.

The universe is 100% mathematical, and everything true and objective can and *must* be explained mathematically. Everything else is nonsense. Science, as it’s interpreted by scientists, is largely nonsense. The sole reason why science is credible at all is that it has mathematics as its engine. Remove this from science and it would become astrology and alchemy. So, why not get rid of scientific hocus pocus completely, leaving nothing but the mathematical engine of truth? *That* will be the next intellectual revolution.

Boscovich

“[The doctrine of Father Boscovich is that] the ultimate elements of which Matter is composed, are unextended atoms, or, in other words, mathematical points, endowed with certain powers of attraction and repulsion; it is from these powers that all the physical appearances of the universe arise. The effects, for example, which are vulgarly ascribed to actual contact are all produced by repulsive forces, occupying those parts of space where *bodies* are perceived by our senses.” – Dugald Stewart

The History of Human Thought

Human thinking has been about a great dialectical adventure to establish a single truth – that existence is 100% mathematical, and we ourselves are 100% mathematical. 2,500 years ago Pythagoras had the greatest single insight in history when he said, “All things are numbers; number rules all”, and sought cosmic laws of mathematical harmony reflecting those of music. Pythagoras was the first ontological mathematician and the founder of the secret society of ontological mathematics – the Illuminati, the enlightened ones, the ones who discerned the absolute truth of reality.

To most people, mathematics is about as far from the answer to existence as they could possibly imagine. Nothing could be more horrifying to them given that they are spectacularly useless at mathematics and viscerally hate it. So, it's no surprise at all that humanity has looked to everything other than mathematics for the answer to life, the universe and everything.

Human beings will typically believe whatever answer makes them *feel* good, and gives them the simplest understanding of reality. Tragically, the simplest narrative of all is Abrahamism. There's an all-powerful, all-knowing, all-seeing being who created everything, and made us in his image. He's the answer to everything. Do what he wants and you will be rewarded. Disobey him and you will be punished. He used prophets and holy texts to reveal his thoughts, wishes and Commandments to us.

What could be more basic than this Mythos? Contrast that with:

One mathematical equation – the generalised Euler Formula (the *God Equation*) explains the whole of existence.

The latter is the absolute truth of existence, but, frankly, you would need to already be a God to truly grasp it. Illuminism is for humanity's smartest people. Abrahamism is for humanity's dumbest people. So, we all know which one will win! The Illuminati need to gain such an overwhelming advantage that one Illuminist is superior to one billion Abrahamists. One day, this will actually be achieved – because Illuminism is in the business of creating Gods!

Stationary Souls

Souls do not physically move. They are outside space and time, so how could they? This means that the mind is stationary while the body moves. However, if one soul is outside space and time then all souls are. They all inhabit a Singularity. All souls are together in Soul World. Although souls don't physically move, their energy contents are in permanent motion – the non-physical mental motion known as thinking. Via a very special type of thinking – the inverse Fourier transform – thoughts (the output of frequency domain singularities) give rise to the material world of spacetime.

The physical world is a mental construct of souls. The motion of thought is converted into the spacetime motion studied by physics.

Although souls never move physically, they can of course produce motion effects in the physical world. Indeed, getting a body to move is the perfect example. When a body has died, a soul can produce other effects – such as ghostly hauntings, or bright lights moving rapidly (like UFOs).

Fermions and Bosons

In physics, no two fermionic (“matter”) particles can share the same state, which implies that all fermions are unique since, no matter if they are indistinguishable in every other way, they can always be distinguished by their unique fermionic signature. A fermionic signature is equivalent to a particle having unique coordinates (with “coordinates” being used in a very general sense and involving notions such as “spin”).

The situation is radically different with bosons. Bosonic (“force”) particles can share the same state, as many as you like.

In quantum mechanics, it's often said that a particle can be in many places at once. In fact, it can be everywhere at once since its wavefunction is defined with reference to the whole universe. Another equivalent expression is that a particle can take all possible routes to its destination. These comments are true of fermions, which are “matter” particles. What these comments mean, in effect, is that every particle belonging to the same class (e.g. the class of electrons) is involved in any fermionic “transaction” concerning that class. This follows automatically from all fermions being unique. They can only be unique by ensuring that they all precisely track each other, in a mathematical antisymmetry sense, in order to ensure that they never occupy the same state.

Where a fermion can be said to be “everywhere at once” (because all fermions and all their possible states have to be considered at once), all bosons could, in theory, be in one state at once.

Movement

Regarding Einstein's theory of gravity, we are told that matter tells spacetime how to *curve*, while curved spacetime tells matter how to *move*. Of course, this leads to a classic chicken-and-egg scenario. If we don't know where the matter is, how do we define the curvature of spacetime, and if we can't define the curvature of spacetime, how do we know how matter moves, and thus where it is and where it will go?

In fact, the theory is so complex that solutions can be provided only in extremely

simplified situations based on numerous assumptions and approximations. That's just the beginning of the problems with this theory. If we consider it metaphysically, the whole theory collapses. Is matter a substance? Is spacetime a substance? Is space a substance? Is time a substance? If space and time are independent substances, we're in a typical Cartesian dualism situation. How do these substances interact if they are radically different, with nothing in common? If they are similar, why don't they merge into one thing – not spacetime, but *either* space or time alone. What keeps them distinct? If we add matter into this deadly cocktail, we simply multiply the difficulties. And what about energy?

Physics is based on four radically different substances: matter, energy, space and time. Science make no attempt whatsoever to define exactly what these things are, and how they manage to interact. It simply uses instrumental (rather than ontological definitions), and combines them with experimental observations and measurements. The whole thing hangs together mathematically while resolutely ignoring the obvious conclusion: matter, energy, space and time are all the same substance – the *mathematical* substance. They can interact purely and simply because they are different aspects of mathematics. They are one substance presenting itself in multi-faceted ways, to which scientists give different labels. These labels are then taken as independent substances, even though they're not.

Matter is simply “dimensional” energy (which in its default mode is dimensionless) and spacetime is simply a Cartesian arena of real and imaginary numbers, dynamically and inversely coupled, i.e. the faster you move through imaginary space (time), the slower you move through real space, and vice versa. When dimensional energy (matter) moves through this arena, it continually changes its shape depending on what speed through space or time it's travelling at, and hence how it relates to the real and imaginary components of spacetime.

Contrary to the claims of materialism, pure energy is dimensionless, hence immaterial. The Big Bang can straightforwardly be described as a Singularity of dimensionless energy (pure frequencies) “exploding” (mathematically, via a fermionic antisymmetry operation) and converting the dimensionless “bosonic” energy into dimensional “fermionic” energy (matter) via a cosmic inverse Fourier transform.

Since science rejects any concept of dimensionless energy, it's absolutely clueless as to what caused the Big Bang, and is praying that M-theory will yield a materialist answer (it won't! – all M-theorists are wasting their time and lives on pure nonsense, on a materialist Mythos).

Similarly, Illuminism straightforwardly explains a black hole singularity as a place where matter is converted into dimensionless energy. According to science, a black hole singularity somehow has mass (how can it if it's dimensionless?), but it's unable to make any sense of this, and once again hopes that M-theory will sort out an explanation.

Science – when you get down to brass tacks – is total nonsense, a ridiculous Mythos. It “works” by avoiding all difficult questions via its stratagem of using instrumental definitions. It never has to explain what anything is (ontology). It simply has to say how it can be measured (instrumentalism and pragmatism). So, to say that time is what you measure with a clock is to make no attempt to explain either time or a clock, except circularly: time is that which is measured by clocks, and clocks are that which measure time. To say that space is what you measure with a ruler is, likewise, to make no attempt to explain what space is and what a ruler is. However, if you have instruments for performing measurements, if you have measurements, and if you have equations into which to slot the measurements and get out numbers that enable you to do useful things in the real world then you have “science” – but at no time have you actually explained what anything is.

Science is instrumentalism; Illuminism is ontological – it explains what things actually are. And everything is of course mathematical, defined by the frequencies and amplitudes of the

sine and cosine waves of the generalised Euler Formula.

“[I] will assume that although space is not a substance, it must nevertheless *be* something! But if space is not a substance, can we interact it? Can matter affect it in some way? It turns out that matter can indeed affect space itself: it can bend it!” – Jim Al-Khalili, *Black Holes, Wormholes & Time Machines*

This is typical scientific gibberish that explains nothing. We’re told that space isn’t a substance but can nevertheless be “bent”. We aren’t told how of course, or indeed how such a statement can contain any meaning. Something that can be bent sounds like a substance. If it’s not a substance then how can it be bent by matter, which is presumably some kind of substance.

If you read popular science books naively, you can imagine they make some sort of sense and are impressively rational. However, as soon as you drill down into each sentence on a philosophical basis, you discover that you are dealing with an ingenious but irrational Mythos that has zero truth content and zero meaning. It’s as ridiculous as Abrahamic holy texts.

Science is much better than Abrahamism not because it’s truer but because it contains useful instrumental definitions that yield ways of affecting the real world in worthwhile ways. However, these instrumental definitions are not ontological definitions and have no connection with reality. They are just models and simulacra that have proved successful in allowing us to manipulate the world.

Of course, on another level, the Abrahamic Mythos has had a much greater impact on the human race than science. Billions of people are Abrahamists. How many scientifically literate people are there? So, by this criterion, Abrahamism – a wholly false doctrine – is much more successful than science, another wholly false doctrine that, even so, allows us to have a degree of meaningful practical control over our world.

Isn’t it time for a new paradigm – one based on ontological truth *and* pragmatism? Illuminism – ontological mathematics – is that new paradigm.

Time

In physics, time is treated as one-dimensional and mono-directional (the “arrow of time”), hence is effectively a scalar quantity. If, however, the time measured by clocks is in fact imaginary space, and is on a par with the real space distances measured by rulers, then time, in this sense, is every bit as vectorial as direction through space, which means that physics is handling time radically wrongly.

In fact, tenseless time is indeed vectorial and it’s tensed time that provides a uni-directional arrow of time, always going forward (from past to future via the present).

The God Equation

The generalised Euler Formula is the basis for the whole of ontological mathematics. There can be only *one* fundamental equation, and *everything else* must flow from it. The situation is rather like Cartesian dualism. You can't have two incompatible substances because it's impossible for them to interact, and there's no sufficient reason why they should co-exist at all. Existence must be based on monism, not on dualism or polyism. By the same token, you can't have *two* master equations since which would take priority if they clashed?

Humanity's great quest to understand existence boils down, in the end, to identifying the *single* ruling equation that controls everything. That equation, the *God Equation*, is the generalised Euler Formula.

The full, generalized version of Euler's Formula is:

$$A e^{i(fx + \varphi)} = A \cos(fx + \varphi) + i A \sin(fx + \varphi)$$

It's the one true ontological equation because it perfectly balances *all* numbers: zero, infinity and everything in between, including all negative and imaginary numbers. No numbers are left out. It's comprehensive, consistent and complete. It allows all numbers to exist while at the same time guaranteeing that they all balance to zero – meaning that an infinite world can permanently exist in its zero ground state.

Via the cosine and sine waves associated with it, the Euler Formula generates the whole of wave mechanics.

The generalised Euler Formula is the key to the most mysterious mathematical object of them all – the *Singularity*. Every instance of the Euler Formula is, if understood properly, an expression of a Singularity.

The generalised Euler Formula generates infinite “Russian Doll” singularities – singularities that sit inside other singularities and make up an infinite existential Singularity. All souls are singularities, controlled by the generalised Euler Formula.

Reality is nothing but the exploration of the consequences of the generalised Euler Formula. This miraculous Formula contains absolute objective knowledge of ontological mathematics. Yet, in its subjective aspect, it contains all of life and all possible mental experiences.

Leibniz tellingly asked why is there something rather than nothing. The answer – the *only* answer – is that Euler's Formula ontologically equates something and nothing.

When any intuitive sees the theorem: $e^{i\pi} + 1 = 0$ (which is what Euler's Formula reduces to when $x = \pi$), they feel a shiver. They *know* they are looking at something infinitely mysterious and divine.

Nothing else in mathematics can represent all numbers ontologically. Therefore this formula is the one, true, indisputable master equation. It has no rivals. It alone is *truth*.

Reality reduces to just one analytic, mathematical equation. Because it's ultimately *nothing*, it can't *not* exist. There is no sufficient reason to forbid its existence since “nothing” can *never* be prohibited. “Nothing” is the foundation of existence. In fact, it defines existence. It's the authentic basis of ontology.

The trick to understanding reality has always been about finding the means to equate something and nothing ontologically. Euler's generalized Formula is the *unique* answer since it perfectly balances all positive and negative numbers and all real and imaginary numbers, and is thus the final and definitive answer to *everything*.

Never forget, there is nothing other than the generalised Euler Formula.

One Equation To Rule Them All

To say that existence is mathematical is to say that a single equation explains everything. By itself, it encapsulates the whole of ontological mathematics. That one, cosmic equation is the generalised Euler Formula. All that exists is this equation and nothing besides. This equation exists because it can't *not* exist. It encapsulates something and nothing. It accounts for "here" and "now". It embraces void and infinity. It contains all possible numbers. It defines energy. It defines information. It's the quintessence of motion. It enshrines causality. It expresses both objectivity and subjectivity. Nothing else has these properties. Existence is simply the unfolding of these properties.

Existence is about information: receiving, processing, generating it and experiencing it. The generalised Euler Formula is the unique and perfect information system. Its ability to generate infinite, flawless sine and cosine waves underlies everything.

Joseph Fourier was the genius who first grasped that all problems of the physical world can be analyzed in terms of nothing but the combination of simple, basis sinusoidal waves. Fourier mathematics is derived from the generalised Euler Formula. From that same miraculous formula, Euclidean and Riemannian geometry are ultimately produced. It also generates the cosmic Cartesian-Gaussian coordinate grid to act as the perfect mathematical arena in which mathematical existence unfolds. This arena is composed of mathematical points – Eulerian "zeros". Eulerian zeros are simply Leibnizian-Pythagorean monads, and monads are none other than subjects that internally experience information. They are *subjects*. What are subjects? They are immortal, indestructible *souls*.

The universe is made out of subjects/souls. The phenomenal objective universe is simply a projection of the subjective, noumenal universe.

The universe is *Soul World*, and it's completely mathematical. The ancient Greek conception of reality based on souls was much closer to the truth than the present soulless scientific materialist understanding. Humanity has regressed in its grasp of the true nature of things. All of that can change – by turning to Pythagorean Illuminism!

The universe is nothing but vibrations. It's an infinite, shimmering hologram. You are in an ocean of waves, forming every conceivable information pattern. You are immersed in information signals. There is nothing but mathematics. Mathematics is existence.

Death

Materialists think of death as the definite end. They think of all the atoms that comprise our bodies falling apart and never coming back together again. If we are nothing but our atoms then there is indeed no coming back for us. Yet the atoms that comprise our body are changing all the time. Every time we eat, drink or breathe, we are introducing new atoms into our body. And we are forever expelling waste from our bodies, shedding skin, and so on. So, clearly, we are not made of any particular atoms. By the time we die, we might not have a single atom left that we had when we were born. It's not atoms that define us but the information system contained in our DNA. This is what organises our atoms, and it doesn't rely on any specific atoms. One carbon atom is as good as another.

When you additionally consider the fact that bosons can be conceived as dimensionless mental particles, and that all fermions are in a sense mentally interconnected, the whole materialist conception of reality disintegrates. Atoms aren't solid, enduring "things". They are weird and wonderful mathematical functions being calculated on an instant by instant basis. Reality is about math ... and mathematics is not about materialism. On the contrary, it's about idealism and rationalism.

You lose your fear of death as soon as you lose your fallacious understanding of atoms as material objects. Atheists reject religion because they have no ontological understanding of atoms. They treat them as real things in a real material world. But there is no material world. There is only a mathematical world.

Proof

There are those who say that reason doesn't prove anything. They don't say why not. What they really mean, of course, is that they believe that only sensory information is "proof" i.e. seeing is believing. Of course, the senses are just about as unreliable as it gets. Only madmen and scientists set any store by the senses.

Why does $1 + 1 = 2$? It couldn't be simpler. It's equivalent to saying $1 + 1 = 1 + 1$, i.e. it's pure tautology. It's an infallible statement because it's tautologically true by definition. It can never be falsified and it's in no need of verification. If you don't accept that $1 + 1 = 2$, that's your problem. It means you're irrational. You don't accept reason as proof because you yourself haven't mastered reason. You don't know what it is and what it does.

Illuminism is infallible because it is based on infallible, mathematical tautologies. As soon as you understand that reality is 100% mathematical, you grasp that all truths are truths of reason alone, the truths of ontological mathematics.

As we often say, truth isn't for everyone. It certainly isn't for you if you don't "get" reason, if you think it's mental masturbation and an "ego trip" (as some idiot described it).

Mind and Matter; Form and Matter

Pythagoras famously said that all things are numbers. He had in mind an expansive definition. At one level, it was highly geometric. He imagined mathematical points (monads) being linked in all sorts of ways to form geometrical shapes, and he especially favoured spheres and regular polyhedra. At another level, it was highly musical. He conceived of the universe as being akin to mathematical/musical notes, producing harmony when they worked well together, and disharmony and discord when they didn't (Pythagoras was the first "string theorist", so to speak). We might liken this to a kind of attraction and repulsion. At another level, Pythagoras

was thinking of number theory and even numerology.

Numbers had various “atomic” properties, and when you combined them, you obtained “molecular” numbers reflecting a composite or blend of the atomic properties. This was an immensely sophisticated vision, but far, far beyond its time. Even today, Pythagoras’s ideas are far beyond about 99.9% of the human race.

To this extent, Pythagoras could be regarded the most intelligent human being of all time. However, Leibniz fully understood Pythagoras (and indeed it was even mooted that he was Pythagoras’s reincarnation), and he turned Pythagoras’ stunning intuitive vision into the hyperrationalist edifice that sustains modern Illuminism. Leibniz’s system is so dazzling that he must surely be regarded as the greatest genius of them all. He, like Pythagoras, has proved far beyond his time and so far beyond ordinary human beings as almost to belong to a different species.

With Pythagoras, mind and matter were both mathematical. Then Plato came along and effected a radical separation between mind and matter. Mind was defined in terms of transcendent Forms (Ideas) and matter became a kind of shapeless clay. Forms were imprinted on matter via God (the Demiurge), just as a sculptor imposes an idea on clay to create his sculpture.

Aristotle rejected this idea and removed Forms from their transcendent realm and made them immanent. Matter and Forms were always found together except in the limiting cases of matterless Form (God = Self-Aware Reason) and formless matter.

At the start of the modern age, Descartes came along and, like Plato, separated matter and mind (Form). However, matter was now no longer shapeless but “extended”, mathematical and formed. Science quickly seized on this and simply abolished mind and Forms. Everything became about mindless, lifeless material atoms obeying laws of motion. To this day, this remains the core of the scientific paradigm, even though it makes no sense in terms of quantum mechanics, and does nothing to explain mental phenomena and life.

Illuminism returns to a Pythagorean mathematical vision, with a few subtle but decisive modifications. Form and matter are united in mathematical waves: *energy*. Form and matter are both about information; in particular, mathematical information. Energy – *waves* – is what information actually is.

Illuminism has in effect returned to Pythagorean musical/ mathematical “notes”. It’s not unlike science’s M-theory, but with one astounding difference. In Illuminism, energy can be both dimensionless (mental) and dimensional (material), while for M-theory, energy is purely dimensional, which means it has no ability to address mind, life and consciousness, subjectivity and free will.

We inhabit a mathematical universe, which is an energy universe, an information universe, a universe of waves, of frequencies, of vibrations.

Pythagoras was the first ontological mathematician, and the ontological definition of a number remains the most critical question of all. In Illuminism, numbers, ontologically, are energy. Numbers are not pencils mark on paper. They are actual existents – they are energy waves.

“The Fourier transform is a mathematical method of breaking a signal in the time domain into all of its individual frequency components. The process of examining a time signal broken down into its individual frequency components is also called spectral analysis or harmonic analysis. ... The work that Fourier did in the 1800s is still used today to decompose a signal that varies with time into components that vary only in frequency.” – Robert Oshana, *DSP Software Development Techniques for Embedded and Real-Time Systems*

The Fourier Series is the Fourier Transform for periodic functions.

Kant and Leibniz

“Was Kant really a Leibnizian after all, as Eberhard had accused him of being? Did he believe that the world of nature is nothing but a ‘well-founded phenomenon’, reality itself consisting in timeless, spaceless, noumenal substances whose attributes are derived from reason alone? Is the ‘thing-in-itself’ the underlying substance which sustains appearances?” – Roger Scruton, *Kant*

Kant was described both as a Leibnizian and as the “Prussian Hume” – which shows that no one really knew what Kant was talking about (since Leibniz and Hume were practically opposites). Kant is to philosophy what Einstein is to physics – an ingenious charlatan. Both men were weavers of such subtle fallacies that it was hard for ordinary eyes to see where they had gone wrong. They have set many off on the wrong tracks. Both men fatally led the intelligentsia away from Leibniz, the supreme genius.

Newton versus Leibniz

Why is Newton so revered despite being spectacularly wrong about the nature of reality, while Leibniz is more or less unknown despite being spectacularly right? Newton had no idea what his system implied about reality (or, to be more accurate, did know but didn’t want to say because it would reveal his mad, heretical religious beliefs). Newton hid behind the formula, “I do not feign hypotheses” meaning that he saw no reason to explain why his laws worked, only to show that they did. Leibniz, on the other hand, was obsessed with his principle of sufficient reason and wanted to explain exactly why things were as they were and not otherwise. He produced an astonishing explanatory system – his dazzling *Monadology*, a short work of monumental genius.

However, scientists preferred Newton’s pragmatism and instrumentalism over Leibniz’s logic rationalism, and that’s why we live in a world believed to be scientific rather than mathematical. In science, you don’t have to explain reality, you simply have to produce models that successfully emulate phenomena. The truth content of a scientific model is neither here nor there providing it works.

Newtonian physics was highly successful while being ontologically nonsensical. Why is today’s science in such a muddle as it tries to find a final theory of everything? It’s because it needs rationalist, ontological theories to explain reality (it *must* “feign hypotheses” – scientifically untestable explanations), but all it has at its disposal are instrumental definitions which have very close to zero truth content.

The Science Mythos

In the *God Series*, we have shown the following:

- 1) Science's understanding of time is absurd.
- 2) Science's understanding of space is absurd.
- 3) Science's understanding of mass and energy is absurd.
- 4) Einstein's principle of relativity is absurd and directly contradicts the reality principle.
- 5) Science's understanding of quantum mechanics is absurd, and turns it into mystical, fuzzy, probabilistic nonsense, in direct contradiction of the principle of sufficient reason and objective reality.
- 6) Science's understanding of the role of mathematics in science is absurd.
- 7) Science has zero understanding of dimensionless existence.
- 8) Science is a set of models held together by instrumental, operational definitions that have no necessary truth content.
- 9) The philosophy of science shows that all scientific concepts are inherently false and based on provisional, contingent paradigms.
- 10) Science has no truth content. It's an exercise in successful modelling of phenomena using ad hoc, arbitrary hypotheses that are forever changing.
- 11) Science cannot handle truths of reason, and prefers empirical truths of fact over non-empirical truths of reason.
- 12) Scientific quantities that bear the same name have nothing in common, e.g. mass in Newtonian physics is totally different from mass in Einsteinian physics, as Thomas Kuhn highlighted. The illusion of continuity is produced when in fact these theories are incommensurate. Einsteinian physics is not an evolution of Newtonian physics but a revolution that involves a wholly different paradigm, ontology and epistemology. By the same token, quantum mechanics is a wholly different paradigm from Einsteinian physics and it's because these two theories are incommensurate that they cannot be unified in a single, "final" theory of everything.
- 13) Science cannot produce a final theory of everything because science has no truth content and nothing to do with true reality – which is wholly mathematical, analytic, necessary and *a priori*.
- 14) In short, the only function of science is to pave the way for its inevitable successor – ontological mathematics (Illuminism, the Pythagorean-Leibnizian science of the immortal, mathematical, holographic soul defined by the God Equation).

“I Do Not Feign Hypotheses”

The difference between Leibniz and Newton is the difference between the metaphysical mind committed to the pursuit of absolute truth (and seeing mathematics, logic, reason and proof as the supreme tools in this regard), and the scientific mind committed to provisional truth (and seeing experiments, observations, measurements and evidence as the supreme tools).

Leibniz thought – rightly – that Newton's theory of gravity was preposterous because it did not in any way *explain* gravity but simply provided an instrumental equation that was in accord with astronomical observations. The world loved Newton precisely because his equation was so successful (“Success has always been the greatest liar.” – Nietzsche). However, no one could explain how the “force” of gravity (and what on earth is a “force” anyway?) could instantaneously cross empty space (how does it travel through void if there's nothing there to

travel through?). How does the moon “know” the mass of the Earth if they are separated by void, and vice versa; how does each “know” the distance between them; does the moon send out some force of gravity to Earth, and the Earth some force of gravity to the moon; do these two forces meet and interact in empty space; is empty space therefore in fact full of forces; does every point of space contain information pertaining to the force associated with every body in the universe; why does volume not matter in Newton’s theory of gravity, with bodies such as the moon and Earth being regarded as point masses?

Leibniz, unlike everyone else, was nauseated by Newton’s theory because he understood immediately that if this type of approach took hold, nothing would ever be explained. You would have successful equations in great accord with experimental measurements, but you would have zero understanding of what was actually happening. Your theories would have zero truth content.

Einstein’s theory of gravity is radically different from Newton’s, and any theory of quantum gravity will be radically different from Einstein’s – so, what criterion shall we use to establish if we’re getting any closer to the truth of what gravity is and how it operates? Plainly, successful equations that are in accord with experiment (the whole basis of science!) can’t help us one iota.

Science is all about pragmatism, instrumentalism and career advancement. It has nothing to do with truth and rational, definitive explanations of the sort that Leibniz sought with his principle of sufficient reason.

Leibniz was *enormously* smarter than Newton, and that was exactly his problem. No one could fathom the extent of Leibniz’s genius and think in the same breathtaking way. It was so much easier to follow Newton’s route – guess at an answer (construct a hypothesis), turn it into an equation and compare it with experimental measurements. If your equation works, don’t make any attempt to explain how and why it works.

Look at quantum mechanics. It’s the most successful scientific theory of all time and yet not one scientist has ever understood it or explained it, or made any sense at all of why the universe should operate in this random way.

This sums up science. You can do wondrous, practical things with a tool of which you have zero real understanding!

This scientific “method” was abhorrent to Leibniz since it explained nothing and actually got in the way of understanding things because it was always ad hoc, arbitrary, contingent, synthetic, *a posteriori* and inductive.

Leibniz wanted proof, not evidence. They are entirely different things. Proof is what the rational mind pursues. Proof concerns infallible truths of reason. Evidence is what the sensory mind pursues. Evidence concerns fallible truths of fact through which we perversely expect our dubious senses to reveal absolute truth to us.

The movie *The Matrix* shows how the senses can be 100% deceived. You could literally be living in a simulation and believe wholeheartedly that it was true reality.

Mathematics is all about proof (rationalism) while science is all about evidence (empiricism).

For scientists, seeing is believing. For mathematicians, you could work out the entire truth of the universe without ever leaving a darkened room.

When most people say, “Prove it”, what they actually mean is “Show me sensory evidence”, which is nothing to do with proof. The sensory “fact” that the sky is blue is not proof that the sky is blue. A blind or colour blind person would have no idea what you meant if you referred to “blue”. Moreover, you can have no idea whether your perception of blue is the same as anyone’s perception of blue. On top of that, sometimes the sky is red, sometimes it’s grey and sometimes it’s black. As soon as you think hard about what “blue” means, or what the “colour of

the sky” means, you realize you’re firmly in the arena of subjective interpretation. “Blueness” is not a property of the sky, it’s a property of your *perception* of the sky. This goes back to the philosophical difference between primary and secondary properties of objects.

Science is the fallacy that evidence is proof. It’s not. Science is interpretation. Science is the quasi-religious worship of sensory experience, and the rejection of rational unobservables.

If Doubting Thomas were a scientist, his alleged encounter with Jesus Christ would have provided him with “indisputable” evidence that Jesus Christ rose from the dead. It would have been “proved” to his scientific satisfaction that the dead can be resurrected.

To a rationalist, Christian resurrection is impossible, hence no amount of sensory “evidence” can prove that it happens. The rationalist immediately seeks the means for how Jesus Christ perpetrated his fraud.

So, when you ask for “proof” of something, be very clear what you mean. Are you the sort of person who would become a Christian if you were in Doubting Thomas’s position?

The only proofs we accept are those of mathematics. If you’re not in that game, you’re not in our game: the game of absolute, infallible truth.

Unlike scientists who believe whatever they see, we can readily conceive that we are inhabiting some sort of Matrix and that nothing we see is to be believed. No scientist is willing to accept metaphysics, meaning that they think that what they experience must be the truth.

All scientists living and working in *The Matrix* would consider it self-evidently “real” (even though it was in fact an unreal simulation). That’s what it means to be a scientist. You privilege what you are experiencing over everything else. You refuse to consider other possibilities outside your direct sensory experience.

If you think about it, *The Matrix* is a film about the existence of the soul. The Matrix is the sensory world while the world beyond the Matrix is the true world, wholly unobservable except in the most special circumstances (being released from the control of the Matrix). All those who think the Matrix is the sole reality are deluded. They have no imagination and no intuition. They are transfixed by their senses rather than their reason.

Possible and Compossible

Compossible from Latin *compossibilis* (“com”: with and “possibilis”: possible): compatible; consistent; able to exist or happen together; possible in conjunction with another.

“Compossibility is a philosophical concept from Leibniz. According to Leibniz a complete individual thing (for example a person) is characterized by all its properties, and these determine its relations with other individuals. The existence of one individual may contradict the existence of another. A possible world is made up of individuals that are compossible – that is, individuals that can exist together. Possible worlds exist as possibilities in the mind of God. One world among them is realized as the actual world, and this is the most perfect one. ... Alain Badiou borrows this concept in defining philosophy as the creation of a ‘space of compossibility’ for heterogeneous truths.” – Wikipedia

“[Every possible thing has within it] a certain demand for existence or ... a claim to exist. [It will exist unless something else prevents it.]” – Leibniz

“...for I call ‘possible’ that which does not involve contradiction...” – Leibniz

In physics, it’s sometimes said that anything not forbidden is compulsory. In other words, if it’s possible then it will happen. In this view, absolutely everything that did not involve contradiction would come into existence.

Leibniz put an enormous constraint on this type of thinking when he introduced his concept of *compossibility*. By this he meant that not only must things be possible individually, they must also be possible *collectively*. Possibles that are not compatible with each other are not

jointly realizable. If some things are possible on their own or in small groups but other things are possible in large groups then the latter are *more possible* than the former. Where mere possibility concerns whether or not something is logically possible, compossibility is in effect a measure of the likelihood of possibilities being actualized. Logical possibilities are more likely to become actual the more they are consistent with other logical possibilities. The universe that will actually be realized at any one instant will always be determined by the greatest degree of compossibility. Compossibility is always maximized. It will never be the case that a smaller degree of compossibility will be actualized over a larger degree. Thus less likely possibilities can be rendered more or less impossible when considered in the context of compossibility.

A possible world is not a collection of possibles but a collection of compossibles, and the most compossible world will be the one that actually comes into existence. This is a fantastic explanation of why some possibles exist, but not others. Without such a sufficient reason, it would be necessary to invoke the monstrous Multiverse notion.

There's nothing logically impossible about a unicorn, but given the compossibility history of evolution on our planet, unicorns will never exist on Earth. On another planet, they might be compossible.

The concept of compossibility is a direct refutation of scientific Multiverse thinking whereby scientists generate infinite worlds in which all possibilities, no matter how improbable, become actual. Compossibility is consistent with Occam's Razor. Multiverses are not.

We can think of heresy as not being compossible with orthodoxy, or meritocracy not being compossible with privilege, and so on. Compossibility is thus one of the most important concepts of all since it allows us to rationally determine what factors are not compossible with a fair, just, meritocratic world of all equal opportunities for all – and to eliminate via the Meritocratic Constitution all such impossible elements. Above all, the super rich must end and dynastic inheritance must be abolished for there to be a fair, meritocratic world of equal opportunities.

Existence

Everything that can exist will exist ... unless something prevents it. The primary obstacles to existence are a) contradiction and b) "compossibility". Nothing can exist if it involves an inherent contradiction. A possibility – something without contradiction – is not thereby granted necessary existence. It's not sufficient that things *can* exist. They must be compossible with other things, i.e. they must be compatible with other things and jointly realizable with those things. The things that will exist are those that are most compossible. Anything less compossible will be prevented from existing by that fact alone.

It's often said that anything not forbidden is compulsory. Compossibility forbids all except the most compossible outcome. In other words, compossibility, by itself, refutes Multiverse thinking (by which everything that can happen will happen). There is only one universe: the universe of optimal compossibility. The "Multiverse" is a hypothetical means to allow all logically possible outcomes to exist, regardless of compossibility. It is 100% incompatible with Occam's Razor, the law of economy, while Compossibility is 100% consistent with this famous philosophical Razor. The Multiverse is also 100% incompatible with the principle of sufficient reason while Compossibility is 100% compatible with it. All Multiverse theories are false. That's a fact.

The Ideal World

“Then we must adopt the doctrine of Leibniz, and believe that, at the beginning of all things, all possible universes were present in idea to the divine Intellect, and that *this* universe was made actual merely because it is the best possible of the whole number.” – Francis Bowen

We might say that at that beginning of a Cyclical Mathematical Age, all possibilities exist as potential within the “blank” slate universe (maximum potential, minimum actualization). At the end of the Age, we have minimum potential and maximum actualization. One of the original possibilities has been made fully actual, and all the others have vanished.

The outcome that wins is always the *optimal* solution, the best solution, the maximization of power, knowledge, consciousness and freedom – the *divine* solution. It’s the solution where all monads, all souls, are transmuted into *Gods*. The mathematical dialectic guarantees this outcome. The dialectic might be seen as the ultimate winnowing instrument. It eliminates all but the strongest outcome. The strongest outcome is the most truthful outcome. As Hegel said, “The true is the whole.”

The Multiverse

“Only the essence of God is necessary, and it implies contradiction that it does not exist. But God is determined to produce other things not by the law of necessity, as would be true if he produced everything possible, but by the law of wisdom or the law of the best, of the best ordered, of the most perfect.” – Leibniz

The scientific Multiverse functions as a God that produces everything logically possible, without any discrimination at all and without any regard to how logically possible it is in the context of other things (*compossibility*).

Leibniz’s “law of the best”, when applied to an evolving universe, implies that, at each instant, a selection is made. The best universe possible, not any old possible universe, is chosen at each instant. The best universe available at each instant may be very far from final perfection but, as long as the best available option is always chosen, it’s certain that the universe will eventually reach a final perfection at the end of its evolution. Even Darwinian natural selection implies some kind of final perfection.

If, however, the universe does not choose the best available outcomes but simply and randomly chooses any logical outcomes that are not impossible, no matter how imperfectly they may sit together, then the universe could never reach a perfect outcome and would be utterly meaningless, purposeless and pointless – exactly as results from Multiverse thinking.

The Quantum Riddle

Quantum mechanics, as conventionally interpreted by physicists, is incompatible with the concept of tensed time (past, present and future). In quantum mechanics, measurements or observations collapse the quantum wavefunction, with the implication being that if there were no measurements or observations then the wavefunction could never collapse. This rather begs the question of how in the early universe without any observers or measurements anything ever collapsed in the first place. Astoundingly, physics cannot answer this question and isn't even interested in it. As ever, it adopts an instrumentalist approach, i.e. quantum mechanics works so who cares what it means, how it works, and how it got going originally.

Quantum mechanics is notoriously vague about what time is within the quantum formalism, particularly when it comes to defining a time reversal operator.

If you apply tensed time to quantum mechanics, the conventional understanding of quantum mechanics disintegrates. If a quantum wavefunction doesn't collapse until someone performs a measurement on it, how can you apply any description of past or present to it prior to the collapse given that no discernible events have occurred (all possibilities are held in suspended animation)? In Illuminism, it's tensed time itself that is collapsing the wavefunction, i.e. the cosmic wavefunction isn't waiting for any measurements or observations, it's collapsing on an instant by instant basis, and instantly reforming – thus allowing us to talk of past, present and future at every instant.

Because the present instant – “now” – continuously collapses the wavefunction, we get a continuous reality (exactly as we observe). In standard quantum mechanics, there is no reason why we should have a continuous reality since there's no formal mechanism to ensure continuous wavefunction collapse.

In Illuminist quantum mechanics, “now” (which is continuously collapsing the cosmic wavefunction) is associated with Leibniz's concept of compossibility. At any instant, it's not a question of just anything being able to happen. Only the optimal compossible outcomes can happen, i.e. everything that can happen *won't* happen.

The Mathematical Universe

The universe is made of math. We live inside a mathematical equation. We are all mathematical functions. We are all mathematical information systems. Existence is information. Information is energy. Energy is Aristotelian “matter” and “form” combined. Mathematics is all there is. Mathematics is all that exists. Mathematics is what existence actually is. Mathematics is the True God. We ourselves are mathematical “becomings”. What are we becoming? We are becoming Gods. No God created us. We create God, using ourselves and from ourselves.

Max Tegmark's Mathematical Multiverse Theory

Only two well-known scientists in today's world are capable of grasping Illuminism. One is Max Tegmark, the other Roger Penrose. Tegmark, like the Illuminati, asserts that reality is 100% mathematical. However, where Illuminism concerns a single mathematical universe, Tegmark has proposed a mathematical *multiverse* [labelled "Level IV"], under which are three other levels of scientific multiverse (!).

"Galileo and Wigner and lots of other scientists would argue that abstract mathematics 'describes' reality. Plato would say that mathematics exists somewhere out there as an ideal reality. I am working in between. I have this sort of crazy-sounding idea that the reason why mathematics is so effective at describing reality is that it *is* reality. That is the mathematical universe hypothesis: Mathematical things actually exist, and they are actually physical reality." – Max Tegmark

"Our external physical reality is a mathematical structure. ... A mathematical structure doesn't describe a universe, it *is* a universe." – Max Tegmark

"The integers 1, 2, 3 are a mathematical structure if you include operations like addition, subtraction, and the like. Of course, the integers are pretty simple. The mathematical structure that must be our universe would be complex enough for creatures like us to exist. Some people think string theory is the ultimate theory of the universe, the so-called theory of everything. If that turns out to be true, then string theory will be a mathematical structure complex enough so that self-awareness can exist within it." – Max Tegmark

"The Platonic paradigm raises the question of why the universe is the way it is. To an Aristotelian, this is a meaningless question: the universe just is. But a Platonist cannot help but wonder why it could not have been different. If the universe is inherently mathematical, then why was only one of the many mathematical structures singled out to describe a universe? A fundamental asymmetry appears to be built into the very heart of reality. As a way out of this conundrum, I have suggested that complete mathematical symmetry holds: that all mathematical structures exist physically as well. Every mathematical structure corresponds to a parallel universe. The elements of this multiverse do not reside in the same space but exist outside of space and time. Most of them are probably devoid of observers. This hypothesis can be viewed as a form of radical Platonism, asserting that the mathematical structures in Plato's realm of ideas or the 'mindscape' of mathematician Rudy Rucker of San Jose State University exist in a physical sense. It is akin to what cosmologist John D. Barrow of the University of Cambridge refers to as ' π in the sky,' what the late Harvard University philosopher Robert Nozick called the principle of fecundity and what the late Princeton philosopher David K. Lewis called modal realism. Level IV brings closure to the hierarchy of multiverses, because any self-consistent fundamental physical theory can be phrased as some kind of mathematical structure." – Max Tegmark

"...physics theories involving parallel universes ... form a natural four-level hierarchy of multiverses allowing progressively greater diversity." – Max Tegmark

Put in simple terms, the four levels are:

Level I (the "initial conditions" multiverse): a multiverse in which all possible initial conditions are realised, i.e. there's a universe for each set of initial conditions. Each universe obeys the same laws of physics.

Level II (the "physical constants" multiverse): a multiverse of Level I multiverses, each multiverse reflecting different fundamental physical constants. Each universe obeys the same laws of physics.

Level III (the “quantum mechanical” multiverse): a multiverse reflecting the “many worlds” interpretation of quantum mechanics. Each universe obeys the same laws of physics.

“There may be a third type of parallel worlds that are not far away but in a sense right here. If the equations of physics are what mathematicians call unitary, as they so far appear to be, then the universe keeps branching into parallel universes ... whenever a quantum event appears to have a random outcome, all outcomes in fact occur, one in each branch.” – Max Tegmark

Level IV: A multiverse of mathematical universes, each reflecting different mathematical structures, hence *different* laws of physics.

“The initial conditions and physical constants in the Level I, Level II and Level III multiverses can vary, but the fundamental laws that govern nature remain the same. Why stop there? Why not allow the laws themselves to vary? Welcome to the Level IV multiverse. You can think of what I’m arguing for as Platonism on steroids: that external physical reality is not only described by mathematics, but that it is mathematics. And that our physical world (our Level III multiverse) is a giant mathematical object in the Level IV multiverse of all mathematical objects.” – Max Tegmark

“The Level IV multiverse hypothesis ... involves an ensemble (in this case, the full range of mathematical structures)...” – Max Tegmark

Tegmark’s “Level IV” mathematical multiverse consists of all possible mathematical “structures”, each structure reflecting different fundamental equations of physics. Mathematical existence thus equals physical existence, and we allegedly have the answer to the “unreasonable effectiveness” of math in physics and the answer, supposedly, to the question, “Why these laws, not others?” Of course, we actually have a non-explanation since all conceivable laws happen and there’s no reason for any of them beyond their mere possibility.

Max Tegmark’s Ultimate Ensemble Theory

“I explore physics implications of the External Reality Hypothesis (ERH) that there exists an external physical reality completely independent of us humans. I argue that with a sufficiently broad definition of mathematics, it implies the Mathematical Universe Hypothesis (MUH) that our physical world is an abstract mathematical structure.” – Max Tegmark

“We discuss some physical consequences of what might be called ‘the ultimate ensemble theory’, where not only worlds corresponding to say different sets of initial data or different physical constants are considered equally real, but also worlds ruled by altogether different equations. The only postulate in this theory is that all structures that exist mathematically exist also physically, by which we mean that in those complex enough to contain self-aware substructures (SASs), these SASs will subjectively perceive themselves as existing in a physically ‘real’ world.” – Max Tegmark

Mathematical Structures

The “mathematical structures” on which Tegmark claims physical reality is based are absurd. Their function has nothing at all to do with mathematics. They are introduced purely to allow Tegmark to accommodate the Multiverse and many worlds theories of physics. Tegmark simply can’t bring himself to accept the logic of his position and actively pit mathematics against physics. Both can’t be right. It’s one or the other. Any attempt to use *a priori*, analytic, necessary, eternal, immutable, deductive, intelligible mathematics to justify *a posteriori*, synthetic, contingent, provisional, ad hoc, inductive, “sensible” physics is misconceived on every level. It’s a category error.

Ontological mathematics is a single system of absolute tautology. Every part of it automatically summons every other part of it. The whole system is necessarily consistent and

complete, without a single error, flaw, omission or gap.

The mathematical “structures” Tegmark proposes are all inconsistent and incomplete subsets of mathematics, all of which would instantly annihilate any universe based on them since they are 100% unstable and would inherently dissolve thanks to their internal contradictions. There is therefore a sufficient reason why no such universes can exist. A mathematical universe *must* be stable under any mathematical operation applied to it. Any subset of mathematics automatically fails this test. There are infinite mathematical operations that would destroy any but a complete and consistent mathematical universe.

Mathematics is an indissoluble unity: a single, adamant structure that does not permit any subdivisions. You can’t have infinite mathematical universes. You can have only one, exactly as Leibniz said. Tegmark, sadly, isn’t in the same league as Leibniz.

Einstein’s Dilemma

Einstein came to realise that there was something catastrophically wrong with quantum mechanics. Sadly, he wasn’t enough of a philosopher to see what was wrong. He didn’t grasp that physics itself, for all its pretensions, is just a branch of philosophy, namely *empiricist materialism*. Given that worldview, quantum mechanics must be interpreted in the standard probabilistic way advocated by the Copenhagen school. However, given a different philosophy, such as rationalism and idealism, quantum mechanics could be interpreted completely differently, exactly as Einstein wanted to understand it, as non probabilistic.

The interpretation of quantum mechanics is a matter of philosophy (metaphysics, to be precise), not of “science”.

Parallel Universes

“To me, the key question is not whether parallel universes exist ... but how many levels there are.” – Max Tegmark

“I argue that accepting quantum mechanics to be universally true means that you should also believe in parallel universes.” – Max Tegmark

So, what’s wrong with Tegmark’s thinking? Well, he’s a physicist and he can’t abandon the mindset of all physicists. He intuits that mathematics is the right answer to existence but rather than seeing that mathematics *replaces* physics, he sees mathematics as *underpinning* physics.

Physics is an expression of materialism and empiricism. It’s wholly false at the fundamental level, for reasons we have discussed throughout the *God Series*.

Tegmark thinks he can have his cake and eat it. He wants to remain part of the physics establishment and that means he has to accept their Meta Paradigm. He thinks all he has to do is add one extra level of Multiverse and all will be well (as if three other Multiverse levels weren’t already bad enough).

It’s almost obscene how much four Multiverse levels defy Occam’s Razor. As ever, rather than provide an explanation for everything via a sufficient reason for why things are thus and not otherwise, Tegmark, like all physicists, resorts to destroying explanation entirely and asserting that everything that can happen will happen. In this view, we are here simply because we are possible.

Kant famously said that existence is not a predicate:

“Kant goes on to write, “‘being’ is obviously not a real predicate’ and cannot be part of the concept of something. He proposed that existence is not a predicate, or quality. This is

because existence does not add to the essence of a being, but merely indicates its occurrence in reality. He stated that by taking the subject of God with all its predicates and then asserting that God exists, 'I add no new predicate to the conception of God.' He argued that the ontological argument works only if existence is a predicate; if this is not so, then it is conceivable for a completely perfect being to not exist, thus defeating the ontological argument." – Wikipedia

In Multiverse thinking, existence *is* a predicate since anything that is possible is certain to exist somewhere at some time. As physicists say, "If it's not forbidden, it's compulsory." It automatically follows that if it's not forbidden then its real existence necessarily follows.

Tegmark puts himself up against Leibniz, an infinitely smarter thinker. Against a four-level Multiverse (three of physics and one of mathematics), Leibniz posits a principle of sufficient reason and the doctrine of *compossibility*. It's not enough for something to be possible, it also has to be *compossible*. Compossibility wipes out the Multiverse. Possible universes are not compossible universes. Only one universe can exist and everything in it has a sufficient reason. The universe is not in any way based on random possibilities.

When it comes to mathematics, there can be only one expression of it. Only one mathematics works. Tegmark's notion of different "mathematical structures" is preposterous. His central problem is that he hasn't tied mathematics to anything ontological. In Illuminism, mathematics is energy, or rather energy is mathematics. Therefore, mathematics directly *replaces* physics.

Tegmark treats mathematical structures as abstractions that exist only in some transcendent Platonic sense and which then inform and define what really matters to Tegmark – physics! Mathematics is *not* a precursor of physics, as Tegmark believes. Rather, mathematics and physics are in direct competition. One or the other defines the stuff of existence.

Physics is a botched and bungled version of mathematics constructed to serve the fallacious empiricist materialist Meta Paradigm. Physics is all about trying to account for the sensible world while mathematics defines the intelligible world, of which the sensible world is a subset (that part dealing with our sensory rather than rational engagement with reality).

Imagine mathematical worlds made only of integers or irrational numbers or real numbers or fractions or imaginary numbers. Tegmark believes such worlds are possible and actually exist. In fact, they are impossible. All subsets of mathematics are mathematically unstable. When such systems are studied in terms of energy, it becomes immediately apparent that they defy all of the laws of energy, especially the law of the conservation of energy and the fundamental requirement of this law that energy is always conserved at zero, the ground state energy. There can never be a sufficient reason for ground state energy to be greater than zero, and any such hypothetical system would automatically decay to its more stable ground state.

An ontological mathematical universe must satisfy the following conditions:

- 1) It must be conserved to net zero at all times.
- 2) "Something" must ultimately be equivalent to "nothing".
- 3) It must be complete.
- 4) It must be consistent.
- 5) It must be perfectly logical and contain no contradictions whatsoever.
- 6) It must have zero and infinity as its logical limits.
- 7) It must be an information system, an energy system, a dynamic system.
- 8) Space, time and matter must be derivable from it.
- 9) It must explain the findings of quantum mechanics.
- 10) It must account for life, free will and the Cartesian mind-body problem.
- 11) It must form a perfect plenum, without a single gap.
- 12) It must form a perfect continuum, without a single gap.

- 13) It must make no “leaps”.
- 14) It must have cause and effect built into it.
- 15) It must account for causeless causes (first causes) and final causes.
- 16) It must account for the principle of sufficient reason.
- 17) It must account for eternal truths and constitute an eternal system.
- 18) It must be a perfect, perpetual motion system that can never “run down”.
- 19) It must account for calculus.
- 20) It must be completely stable under all mathematical operations, transformations and symmetries.
- 21) It must be mathematically perfect in every way.

Any version of mathematics that does not satisfy these stringent criteria cannot exist. Even if it's logically possible (which is highly doubtful), it will not be *ontologically* possible and certainly not *compossible*.

Any version of mathematics that does not satisfy the above conditions is unstable and annihilates itself. None of Tegmark's mathematical structures accords with these conditions. His mathematical Multiverse cannot exist. There is no sufficient reason for it and it defies the logic of mathematics. Mathematics must be complete, consistent and perfect. Ontological mathematics is necessarily unitary, not a plurality.

Tegmark's four-level Multiverse is not a rational hypothesis but an irrational one, based on pure randomness and mere possibility without any further scrutiny. Nothing in this system has any reason for its existence beyond possibility alone, regardless of whether it's likely or not, or consistent and compossible with anything else.

To use mere possibility as a justification for something to exist is to make existence a predicate, contrary to Kant's correct teaching. If anything is not forbidden (in Tegmark's view), it will definitely exist. Thus if God is not forbidden then, no matter how improbable, God must exist!

The whole point of Multiverse thinking is, when all is said and done, to avoid the need for a specific explanation of reality and, above all, to avoid conclusion that the universe is designed. Yet this way of thinking leads inexorably to the conclusion that a Designer, unless it's explicitly and formally impossible, *must exist*! So, Multiverse scientists would then have to demonstrate that God is technically impossible in scientific terms. Good luck with that!

Science can't even explain what time and space are, never mind prove that God can't and doesn't exist. Science is about instrumentalism and pragmatism, not about analytic, eternal truths of reason.

Illuminism is the assertion that only one mathematical system can satisfy all of the criteria listed above and thus constitute the fundamental ground of existence. That system is the one deriving from the generalised Euler Formula, which is why it's called the God Equation.

The generalised Euler Formula is the basis of life, mind, free will, consciousness, information, energy, motion, space, time, matter, causality, reason, perpetual motion ... absolutely everything. It's perfect, complete, consistent, and establishes a total plenum and continuum. The whole of mathematics can be derived from it. There is simply no other candidate, no other “mathematical structures”.

Tegmark is doing his best to defend the cause of mathematics within the straitjacket of the science establishment. However, his thinking is fatally constrained by the needs of scientific dogmatism. Rather than thinking properly about the real nature of mathematics, he has simply crowbarred a mathematical multiverse onto existing scientific Multiverse thinking, all of which is metaphysical, speculative and not actually part of science at all since no Multiverse theory can ever be experimentally tested!

Appealing to infinite universes where everything that can happen does happen is no explanation at all. It's a non-explanation, an anti-explanation, a claim that makes free will formally impossible (since all choices are made automatically in countless parallel universes), and renders consciousness an absurdity (what's the point of consciousness if not to freely ponder choices and choose one and not the others?).

Science despises life, mind, free will, consciousness, reason and design, and the Multiverse is the extremest product of this hatred. Tegmark has committed a catastrophic mistake by dragging mathematics into this scientific freakshow. But at least Tegmark is helping to show that the only real game in town is mathematics. If mathematics isn't the answer then there is no answer. If the universe is rational and ordered – which it is – then it has a rational answer, and that answer is mathematics.

The ontological significance of mathematics is the primary question of reality, and it ought to be the main question addressed by science. We don't need a scientific Grand Unified Theory of Everything since we already have one, a proper one – mathematics! All that's required is to map mathematics to the reality we experience.

The Mathematical War in Limbo

We can imagine all possible mathematical “structures” fighting it out in a pre-ontological Limbo. All those that have any defects (i.e. are inconsistent, incomplete, illogical, unstable, above the ground state, and so on) simply dissolve, cancel or are annihilated. One way or another, they all vanish, leaving only the “real deal”, the single complete and consistent system of mathematics that’s eternally stable and perfect under all circumstances. That unique system is the one that derives from the generalised Euler Formula.

If you throw countless unstable mathematical systems against one stable mathematical system, only the stable one will survive. A mathematical system of integers alone might seem possible but in fact is absolutely impossible. Mathematics all works together or doesn’t work at all. You can’t isolate parts of it and confer ontology on each, separate part.

Mathematics is not only the basis of holography, it is itself holographic, i.e. the whole of mathematics is in every part of mathematics. Integers make sense only in regard to all other numbers. If those numbers didn’t exist, nor would integers. They have no ontological independence and autonomy. They are not a separate “structure”, from which a universe can be built.

Ontological Mathematics

Pythagoras was the first ontological mathematician and performed the first – and dazzlingly brilliant – analysis of reality in terms of mathematics. Leibniz took Pythagoras’s ideas and made them more stunning still. Modern Illuminism has simply applied the discoveries of mathematics since Leibniz’s time and used them to refine Leibniz’s ontological Monadology.

Science and Instruments

Science tells you how to measure things experimentally. It fails to tell you what things are. So, you know that a clock measures time and a ruler measures space. Well, can you say from that what time and space actually are? Knowing how to measure something and knowing what that something is are two entirely different things. Science deals with instruments and instrumentalism, it does not deal with ontology, and that’s exactly what’s wrong with it. That’s why it can’t furnish the ultimate truths.

Newton provided a means for allowing scientists to measure things and was revered as a God. Leibniz explained what reality actually was and was mocked and rejected. Yet nothing is more certain than that Leibniz will be vindicated and seen as the greatest of all geniuses.

The Best Possible World

“[The best possible world is the one in which] the greatest effect is produced by the simplest means.” – Leibniz

“[The best possible world] is the one that is simultaneously simplest in hypothesis and richest in phenomena.” – Leibniz

The best possible world is in fact the one that originates in a single mathematical equation – the God Equation. This is both the simplest possible universe and the most productive possible universe. Simplicity does not count against richness in phenomena. On the contrary, it produces optimal richness! The key is to have a single ontological equation from which the whole of mathematics, and thus everything else, is derived.

“God makes the maximum of things he can, and what obliges him to seek simple laws is precisely the necessity to find place for as many things as he can put together.” – Leibniz

Illuminism is the assertion that the whole of mathematics can be generated from the generalised Euler Formula (the God Equation). A corollary is that any additional mathematical equation or formula would not only make the world more complex (and unnecessarily so), but also reduce the number of phenomena that could be produced. It would also be incompatible with the God Equation (as in Cartesian dualism).

“...this principle of general order ... is effective in physics ... because the sovereign wisdom, the source of all things, acts ... observing a necessary harmony.” – Leibniz

The maximum harmony is that produced from a single controlling law of ontology. Any additional law would produce only conflict and disharmony and raise a Cartesian interaction problem.

Complexity does not arise from complexity but in fact from absolute simplicity, the greatest simplicity possible.

“...more intricate processes take up too much ground, too much space, too much place, too much time that might have been better employed.” – Leibniz

“...the *means* are also in a sense *ends* ... they are desirable not only on account of what they do, but on account of what they are.” – Leibniz

The fundamental law must be universal and free from exceptions. It can't contain any possibility of error or contradiction. It must be eternally true. It must express the “law of continuity”, by which no change in nature can ever take place “by a leap”. It must enshrine the principle of sufficient reason, and thus generate the whole system of cause and effect. Yet it must also generate uncaused causes (first causes), which then cause everything else. The generalised Euler Formula accomplishes all of this.

“[God/Nature always acts] according to principles; always according to rules, [and never] according to exceptions.” – Leibniz

Leibniz emphasizes that nature should be efficient and “the greatest effect should be produced with ... the least expenditure.” Phenomena should be produced “by the simplest and most determinate ways.” The path chosen for any process should be the one “that is easiest or shortest”.

The best possible world is the simplest possible world.

The best possible world is the richest possible world.

The simplest possible world is the richest possible world.

The richest possible world is the simplest possible world.

These assertions are the absolute opposite of Multiverse thinking where the most complex

possible world is advocated, where there is no harmony, no reason, no efficiency, no shortest or easiest path, no elegance, no selection, no continuity and everything logically possible is deemed to exist somewhere at some time.

In Multiverse thinking, there are in fact no fundamental laws other than that everything not forbidden is compulsory. This isn't a mathematical or scientific law but rather an untestable metaphysical speculation. It has the effect of destroying meaning since nothing happens for any reason beyond its being possible, e.g. you never in fact make any choices or do anything for any reason because there are, so we are to believe, countless clones of you making all possible choices available to you. Thus choices are never made at all because every possible "choice" happens inevitably and automatically. Nothing is ever rejected or discarded. No selection takes place if *all* selections take place, i.e. if every possibility is actualized.

Imagine you are faced with ten choices. If you clone yourself nine times so that there are now ten of you, you can ensure that every choice is performed by one of you. But, of course, this means that you have not chosen anything at all. All you have done is allocate the choices to as many versions of you as are required to ensure that no choice is ever omitted. Since the quintessence of choice is to select one possibility and reject the alternatives, a Multiverse where all possibilities occur is one where no choices are ever made. Freedom is therefore an illusion. Scientists have thereby restored an absolutely deterministic machine universe!

There's no point at all in being conscious in such a world since we are all just robots. Given that we never choose anything, there should be no consistency in our behaviour (since choice is that which establishes a consistent pattern of behaviour). If Multiverse thinking were correct, all of us would be unconscious androids acting randomly as part of a vast cosmic process to ensure that no possibility was ever left unactualized.

Plainly, Multiverse thinking is about as absurd as it gets – yet is now favoured by almost the whole scientific community.

Nothing is more important than that this insane doctrine – as grotesque and mindless as Abrahamism (where "God" knows everything we are going to do in advance) – should be smashed to smithereens. Only someone on the autistic spectrum could conclude that he is not free and not conscious, and find this perfectly rational. It's sheer lunacy such as only a materialist could contrive to propose.

Causality

"A reason can be given for every truth, or as is commonly said, that nothing happens without a cause." – Leibniz

"One of the oldest doubts of mankind concerns the question of how freedom and contingency are compatible with the chain of causes." – Leibniz

How authentic freedom is explained within a system of sufficient reason and relentless causality is one of the greatest challenges of all. Any "grand unified theory of everything" must be able to account for freedom or is fundamentally false. It's science's signature attitude that it never once attempts to build freedom into any of its explanations of anything. It has waged a relentless war against freedom. The simple fact is that freedom is possible only in terms of idealism, and is inconceivable in relation to scientific materialism.

How does any system of causality get started? What is the "first cause"? To Aristotle and to the Abrahamists, the first cause is God. (The problem here is that God himself is thus not explained; what caused *him*? Why should we accept "God" as a stark fact of existence? What's the sufficient reason?)

Scientific materialists believe that causal chains are initiated not by God but by *randomness*. Random events get causal chains started: they are the first causes. However, they

themselves have no causes and no sufficient reason. They are formally inexplicable and more or less reveal that science starts with magic – with things appearing out of thin air for no reason. If this were true, we would expect things to be randomly jumping out of the magician's top hat at all times. No such randomness is ever observed.

A universe grounded in randomness as the *arche* – the fundamental “stuff” of existence – would itself be utterly random and chaotic, reflecting its core characteristic. We live in a universe of order, organisation and pattern. There is no suggestion that anything about the universe is random in any way whatsoever. To be frank, science's appeal to randomness as the first cause is more outrageous and incredible than Abrahamism's claim that “God” is the first cause. At least “God” has the qualities required to produce an ordered cosmos. Randomness has no such qualities. It's mathematically *impossible* for any perfectly random system to ever generate an ordered world.

In essence, atheistic scientists believed they had two choices to explain reality (and to act as first cause): God or randomness. As atheists, they eagerly chose the latter, without bothering to reflect on the sheer impossibility of this explanation. So absurd did this line of thinking become that, with Max Tegmark, it was turned into a four-level infinite *multiverse* (involving infinite versions of ourselves!), where every random event that was in any way possible (not forbidden) would occur somewhere at some time. As the familiar phrase goes, “If it's not forbidden, it's compulsory”.

So, science replaces God with the absurd combination of randomness and infinity, with infinite universes, infinite versions of us, infinite explorations of infinite random first causes. No random first causes are excluded. Every possible random first cause is given a whole universe to itself. This type of thinking is more or less insane. It contains no freedom, no mind, no consciousness, no choices, no decisions. It doesn't even contain life – unless we are to infer that life is itself randomness! It's a total non-explanation of existence. It's a deranged expression of infinite hatred of any notion of “God”.

Yet there's a simple way to escape from the devil and the deep blue sea, from Scylla and Charybdis – God on the one hand, and the random Multiverse on the other. The way out is of course *mathematics*.

What Illuminism asserts is that there are infinite uncreated, uncaused causes. These are mathematical information energy systems known as monads. They are alive and they have minds, but these minds are unconscious and act, in their default mode, like computers. Reality is literally made of mathematical monads – systems of perfect mathematical order. They are in no way random, and nor are they Gods, although they have the capacity to evolve into Gods.

Illuminism replaces one God (Abrahamism) and an infinite random Multiverse (scientific materialism) with infinite unconscious minds grounded in perfect mathematics (in the generalised Euler Formula, the God Equation). There's nothing random about monads. As mathematical systems, causality and order are built into them.

Abrahamists believe in one perfect, super conscious, intelligent being (God) as the basis of existence and the cause of everything else. It's inconceivable how such a being could ever be explained. Moreover, there's no sufficient reason why if one such God is possible then there should not be infinite such Gods since what was good enough to allow the existence of one must be good enough to allow the existence of infinite others.

In Illuminism, the perfectly actualised single God of Abrahamism is replaced by infinite unactualised, uncreated, unconscious mathematical souls. Through their mathematical relations and interactions with each other, all things are derived, including consciousness itself.

Illuminism isn't based on God and nor is it atheistic. It's based on reason, logic and ontological mathematics, and its central assertion is that every monad can be converted from bare

potential into perfect actualisation via an immense, cosmic dialectical journey. To put it another way, the end of a cosmic cyclical Age in Illuminism corresponds to the generation of infinite equivalents of the Abrahamic God if this God is defined as an all-knowing, all-seeing, perfect, rational, fully conscious being with no physical body. The universe is nothing but a mathematical God Factory. It's made of living, mental mathematical units that, in their unconscious default mode, act more like computers than organisms but which gradually convert themselves from machine-like entities into conscious, free entities deciding their own fate.

There are no other possibilities. You can have "God", randomness or living mathematics. What's it to be? Only living mathematics has a sufficient reason – the properties of mathematics itself, which allow "something" to be constructed from "nothing", for "something" to be mathematically structured "nothingness". The only conceivable answer to Leibniz's fundamental question of why there is something rather than nothing is that something is a mathematical expression of nothing. Just consider an equation as simple as $5 + 6 - 3 - 8 = 10 - 1 + 2 - 11$. Each side of the equation is different, and yet each side reduces to zero. So, the two sides of the equation are the same but different: something and nothing. It's precisely this quality of mathematics that makes it the basis of existence, able to generate an infinite universe of something by endlessly restructuring "nothing".

The Abrahamic God is manifestly silly. But the random Multiverse of science is no less silly. Mathematics, based on living monadic information systems (what we call "souls"), is the only rational foundation of existence.

Abrahamism and scientific materialism are irrational in the precise opposite ways. Abrahamism invokes an impossible perfect being who magically exists eternally. Scientific materialism invokes an infinite system of randomness, of every possible event being actualised, and with no random event having any explanation at all other than that it's possible. Science explains nothing at all because it makes no attempt to say why X happens rather than Y. It asserts instead that X and Y both happen ... in parallel universes. The fact that "laws" appear to operate in our universe is simply and wholly random. There is no explanation for such laws beyond that they are possible, hence must exist in Multiverse thinking. Science is a system of non-explanation. By explaining everything – via randomness – it explains nothing at all. What science does is make existence a predicate, i.e. it says that anything not forbidden must exist! In that case, it might as well revert to the famous ontological argument of philosophy and conclude that God exists! And isn't that the ultimate absurdity? We have actually come full circle.

The only escape route is mathematics.

All mathematical souls are deterministic because they always have a sufficient reason for their actions. They never do anything for no reason (randomly). The key to their behaviour, however, is that they are uncaused causes, hence exist outside any compulsory chain of scientific causality. They are causal agents, hence can freely inject their own actions into causal chains. Every time we choose to do anything, we are introducing our own causal agency into the world. According to science, this is impossible since, in that view, we are made of material atoms which are lifeless, mindless, purposeless and have no causal agency, hence are entirely controlled by any causal chain in which they find themselves, or by random quantum indeterminacy.

Souls

When all's said and done, what are souls. What are *we*? We are information receivers, information processors, information generators and information *experiencers*. We are both information subjects and information objects. And what is information? It's nothing but mathematical patterns, codes, waves, signals.

Go on, try to imagine anything at all that does not constitute some kind of mathematical information. Even your feelings are information. Music is information. Irrationality is information – *bad* information. Faith is information – absurd Mythos information based on feelings. The Torah, Bible and Koran are nonsense from beginning to end and have zero truth content, but what no one can deny is that they contain information ... bullshit information, but information all the same.

Human minds are mostly full of bullshit, but the more rational these minds become the more they filter out the bullshit and start coming into contact with real truth – the eternal Platonic truths of reason.

Leibniz said that everything has a reason. Everything is determined. How is that consistent with freedom? The answer lies in self-determination versus “other”-determination. When things outside ourselves – “other things” – determine our behaviour, we're not free. When we determine our own actions, we are free.

Science denies that anything is self-determining. It makes everything subject to other-determination, with randomness initiating everything.

Things that are self-determining have minds. Science denies that minds have any independence; it asserts that they are other-determined – by matter.

No question is more important than that of the existence of the soul – a dimensionless, immaterial, immortal, unconscious mind. In Illuminism, mind precedes matter. In science, matter precedes mind. It's all the difference in the world. It's the difference between free will, and a universe with no freedom and no meaning, where everything that can happen does happen.

When entities are self-determining, they are active; when they are other-determined they are passive. Everything is either passive or active. The more active you are the freer you are, and the more conscious you are. The most intelligent people are the freest and most conscious of all human beings. The biggest restraint on their freedom is the stupidity of the people with whom they are surrounded.

Science believes in a wholly passive, unfree world of deterministic machine processes, bafflingly underpinned by quantum mechanical indeterminacy. The greatest genius of all time wouldn't be able to explain how indeterminacy gives rise to total determinism and yet this is the core of modern scientific thinking. Science's attraction to “explanations” based on indeterminacy and randomness is pathological. It's driven by a compulsive hatred of any notion of mind, sufficient reason and design being fundamental to the operations of the world. Behind all references to mind and design, scientists see a coded reference to God and souls – utter anathema to atheistic scientists. In Illuminism, mathematical souls (monadic minds), but not God, underlie existence and it's their own inbuilt mathematical reason and causality that lead to the design of the universe.

So, what's it to be? Does our wondrously ordered cosmos reflect fundamental randomness, indeterminacy, chaos and probability, or fundamental “hidden variables” of cast-iron reason and causality?

Einstein famously said, “God does not play dice.” This was the truest thing he ever said

and is superior to his entire theory of relativity, which is one of history's greatest blunders, more or less suggesting that God *does* play dice since the principle of relativity stands in direct contradiction of the principle of objective reality.

"God" is mathematics, and mathematics is simply the principle of sufficient reason. Absolutely everything happens for a specific reason. There is nothing at all that is random and indeterministic. If any such things were possible, the universe would instantly unravel into absolute chaos where order, organisation and pattern were all impossible. If scientists were right, there wouldn't be a universe at all, hence they are 100% wrong! That's a fact. The scientists are as wrong as Abrahamists. Their only saving grace is that they are more usefully wrong.

"...when there is no judgment, there is no freedom." – Leibniz

"To will is to be brought to act through a reason perceived by the intellect." – Leibniz

"There is always a prevailing reason which prompts the will to its choice, and for the maintenance of freedom for the will it suffices that this reason should incline without necessitating." – Leibniz

We judge what is best for us. Unfortunately, none of us has perfect judgement and some of us (many of us!) have the judgment of beasts and lunatics.

The Final Battle of Armageddon: Science versus Math

Scientists are those who emphasise the “sensible” world – the domain available to the senses. This is the world in which experiments can be performed. Mathematicians are those who emphasise the “intelligible” world – the domain available to reason.

The sensible world is about contingent, temporal “truths of fact” and the intelligible world about necessary, eternal “truths of reason”. Scientists are technically extraverts since they look outside themselves, to the outer world, for truth. Mathematicians are introverts who look inside themselves, to the inner world, for truth. These are two radically different approaches to the nature of reality and they have nothing in common. Scientists emphasise “evidence” and mathematicians emphasise proof. The two are often regarded as more or less the same, but this is false. “Evidence” is fallible and interpretive; proof is eternal. A mathematical proof reveals something that has been true for all time. Scientific “evidence” provides nothing but a provisional view. For example, evidence might accumulate that all swans are white, and yet at any time a swan of a different colour might appear. So “truths of fact” are always interpretations. They are not truths at all.

What is more fundamental and certain? That $1 + 1 = 2$ or “the sky is blue”? The latter “fact” is highly questionable. The sky’s colour changes all the time depending on environmental factors, and colour is well-known to be a secondary, subjective property of an object rather than a primary, objective property.

Science is staggeringly subjective and yet manages to convince itself that it’s objective. Newton is perhaps the most famous scientist of all time, and yet nothing he said is technically true. What he did was create a highly successful and yet completely false ontology. His system – his model – is a simulacrum that provides a useful approximation in most circumstances, but that doesn’t change the fact that none of it is true. Approximation isn’t proof and isn’t truth. Scientists have never grasped this.

Science is all about the senses and mathematics about reason. It’s telling that science forms no part of mathematics, but mathematics is at the heart of science, without which it would be as useless as astrology. It never occurs to scientists, who revere experimentation, to explain why mathematics, which has no need at all for any experiments, powers their discipline. Scientists seem staggeringly uninterested in what mathematics means. How can you have the impudence to claim you are studying the truths of reality if you can’t even explain what mathematics is and what it’s doing at the core of science?

Those who look to the intelligible domain for answers are those who rely on an immaterial, mental entity – their own reason. Those who look to the sensible domain rely on their physical sensory organs, not reason, to guide their view of reality.

Your choice could not be clearer. Is ultimate reality rational and intelligible or is it irrational and sensory? Only the intelligible domain provides definitive answers. The sensible domain provides nothing but interpretations. Nietzsche’s analysis in this regard is definitive: “There are no facts, only interpretations.”

Rationalists look *in* to a mental world of reason, a mathematical domain from which we are ultimately made. Sensing types look *out* to a physical world, supposedly consisting of material objects from which we are allegedly made.

Do you see how different rationalists are from empiricists? Rationalists put mind first and explain matter as its derivative. They are idealists. Empiricists put matter first and explain mind as its derivative. They are materialists. They are the direct opposite of each other. They cannot

both be right. One side or the other has got things staggeringly wrong, completely upside down.

Rationalists are guided by intellect and empiricists by their senses. You have to ask yourself a simple question: is your reason a better organ for truth than your senses? If you prefer your senses over your reason you are ipso facto an irrationalist, hence you are a person of faith, not of reason. You have allied yourself to Abrahamists who also have contempt for reason and worship their feelings.

Nothing is more important than how you relate to ultimate reality. Do you do so via your feelings (faith), your senses (science) or your reason (mathematicians)?

Illuminism is all about rationalism; indeed *hyperrationalism*. We won't take any lectures from scientists about rationalism, who irrationally support sensory evidence over rational proof.

Illuminism places metaphysics over physics. Science rejects metaphysics. Mathematics and philosophy can work in perfect harmony, but science openly scorns philosophy.

Leibniz was a genius mathematician, scientist, philosopher and metaphysician. Newton was a genius scientist, competent mathematician and no kind of philosopher or metaphysician.

Illuminism follows the Leibnizian template, and science the Newtonian template. The Leibniz-Newton conflict in turn reflects the earlier conflict between Plato the rationalist and Aristotle the empiricist (though Aristotle was much less empiricist than modern scientists). Leibniz was firmly in the Platonic camp, though was happy to use Aristotle's logic and his most rationalist positions.

In Illuminism, any old metaphysics isn't accepted. It has to be metaphysics grounded in mathematics. Where science is sensory mathematical materialism, Illuminism is rationalist mathematical idealism. Science rejects a metaphysical domain. Mathematics *is* that metaphysical domain. Mathematics is rationalist, immaterial, mental, and beyond the senses and experiments. It's transcendent, immanent, immutable, eternal and Platonic. Plato's perfect domain of Forms is simply the domain of mathematics!

Illuminism is the religion, philosophy and metaphysics of mathematics. It has nothing to do with experiments, the senses, and with truths of facts. It's about mathematical proofs, rationalism and truths of reason. Mathematicians look to eternal, necessary truths and scientists to temporal, contingent observations.

It's staggering that so few scientists, mathematicians and philosophers actually understand the opposite nature of science and mathematics. These are two radically incompatible ways to approach the ultimate answers to existence and it's scandalous that science and mathematics are so often bundled together as though they were on the same team. They're not. They're as bitter enemies as religion and science. Mathematics is extremely close to religion, and science is extremely close to atheism.

To anyone who wants to consider religious truth, you will get no comfort from scientific materialism, which leads inevitably to atheism. Yet mathematics will give you everything you want regarding religion. It will even make a God of you!

Compulsory Existence

"Nothing can interfere with the possibility of that which involves no limits, no negation and consequently no contradiction." – Leibniz

These considerations are true of all monads ... all *souls*!

All properties of monads are simple and positive. None are formed by the negation of any properties. A monad has no negative (contradictory) properties or qualities. Hence all properties are mutually compatible and harmonious. None of its properties can be broken down. Monads have no hidden parts. It's impossible for simple, positive properties to be inconsistent or contradictory. Nothing can prevent such properties from existing.

In order for eternal rational truths to exist, there must be eternal rational minds to accommodate them. The only eternal truths are those of mathematics, hence all eternal minds are mathematical.

The Immaterial Creator

In the seventeenth century, considerable thought was given to how an immaterial being could create a material world, and why would he even want to. Why not create an immaterial world?

Some thinkers adopted a Platonic scheme whereby matter existed independently of God and he simply applied form to it, like Plato's Demiurge (thus God did not create the world out of "nothing").

In fact, the real task is to explain how dimensionless existence can give rise to dimensional existence – and that's a solely mathematical process.

The Principle of Sufficient Reason

Leibniz's famous principle is sometimes described as the "principle of the best". Leibniz also said that the principle held that "there is no effect without a cause." (He thus denied the possibility of random events such as science relies on to explain how things get started.) Causes will always be the best in the sense that they will always give rise to the simplest, most direct and most efficient processes. No cause will ever generate anything more complex, less direct and less efficient than it need otherwise be. Causality always obeys Occam's Razor (the principle of ontological economy); it never makes anything more complex than is required.

Given that Occam's Razor states, "Entities are not to be multiplied beyond necessity", the scientific Multiverse constitutes the worst possible violation of this fundamental principle. The Multiverse multiplies entities to the extremest possible degree, solely so that scientists can ignore any notion of design, choice and teleology in the universe.

Multiverse thinking is not compatible with any notion of causality. Multiverse thinking proclaims that there are infinite universes, all of which began randomly on the basis of mere possibility. But if these universes are imbued with randomness and quantum indeterminacy, why shouldn't the universes they generate be wholly random and indeterminate too? In other words, how does Multiverse thinking explain causality? Randomness is the opposite of causality. If you say reality is grounded in randomness, you have to confront the insurmountable task of how to explain how scientific laws and causality arise from quintessentially random activity. Leibniz put the principle of sufficient reason at the core of existence, hence was not faced with any such difficulty.

The true logic of Multiverse thinking is that you should have infinite random universes, and they should remain random forever. At no stage should causality appear since causality is logically incompatible with randomness. It's a category error to use randomness to explain causality. Equally, causality cannot be used to explain randomness. Causal processes are not random processes.

Quantum mechanics, when depicted as revolving around indeterminacy, makes no sense at all and it's impossible to see how a world of classical determinism can have its origins in acausal indeterminacy. Scientists invoke probability considerations, but even that is an invalid step. Why, in a random system, should anything be more probable than anything else? Why shouldn't all outcomes be equally probable? To refer to weighted probabilities is already to suggest that non-random elements are at work to make some outcomes more likely than other, but these are forbidden within a solely random system.

Why haven't philosophers torn scientific quantum gobbledegook to shreds? Of course, quantum mechanics isn't indeterministic at all when properly understood – in the manner described throughout the *God Series*.

Space

If, as Newton maintained, space would exist whether or not any objects existed then the implication is that space is an independent ontological substance. So what is it? What's it made of? Although the scientific description of space has moved on since Newton, it still hasn't come anywhere near explaining what space actually is and indeed how it exists at all.

The same argument applies to time. If time exists independently of everything else then it's an autonomous ontological substance. So, what is it and how does it interact with spatial substance? How come we can't probe and test "time substance" and build some huge

“accelerator” to explore it?

Devising warpable “spacetime”, as Einstein did, only compounds the problems. Science, of course, always reverts to instrumental definitions to avoid awkward questions, so it says that time is what you measure with a clock and space what you measure with a ruler. But even this is problematic if only spacetime exists. Shouldn’t we have a single “clock-ruler” to measure spacetime? To use clocks and rulers to perform measurements is to imply that time and space are different things, but if they are unified in spacetime then they are not different things, so why are we measuring them separately?

As ever, the science community simply ignores all philosophical problems and questions and blunders on towards the greatest folly of all – it’s grand unified theory of everything, which will of course explain nothing at all (and in fact they will never devise such a theory within their current, fatally flawed Meta Paradigm).

Mechanical Humans

If there is no immaterial mind to affect the human body then bodies are wholly material and thus they are machines subject to inescapable material causality, hence free will is impossible. This is what almost all scientists accept as reality. There is nothing more certain than our own free will, and yet free will is absolutely impossible if we are nothing but atoms subject to material causality.

“Thus the materialists have extended to man as well what the Cartesians maintain with regard to all other animals ... Those who point out to the Cartesians that the way they prove that animals are only automata could be taken as justifying someone who said that, metaphysically speaking, it is possible that all other men, except himself, are simple automata...” – Leibniz

Here, Leibniz anticipates modern philosophical “zombie theory”. How do we know that all the people we meet aren’t zombie automata who simply act as though they are free and human? Moreover, if *they’re* zombies, why shouldn’t we be zombies too? There’s no sufficient reason for us to belong to a different category from them.

Leibniz insisted that our own “internal experience” refutes the idea that we are materialist zombies. It’s impossible to explain our inner nature in terms of mechanics and material causality. Science has made no progress whatsoever in explaining our inner mental states ... and it never will since our inner mental states have nothing to do with the materialist paradigm.

It’s vital to understand that the “intelligible” world is an immaterial, mental world of reason. None of the eternal truths of reason have anything at all to do with matter and materialism. The “sensible” world is, on the other hand, all about matter, or, rather, the perception and illusion of matter. If we belong to the sensible world, we are mechanical automata with deterministic minds somehow derived from matter. If we belong to the intelligible world, we are free minds, and matter is a mental construct.

Nothing is more important in understanding reality than grasping whether we are free mental life forms or determined material machines. Reason is *mental*. Our senses, on the other hand, assume, in a “common sense” way, that there is a solid, persistent reality external to us, which has somehow produced us.

Either internal minds produce the illusion of external matter, or external matter produces the illusion of internal minds. Which is it?

Science takes the latter view, mathematical metaphysics the former view. These views are the exact opposite of each other. They have nothing in common. That’s why, in the end, any rationalist must seek the complete abandonment and abolition of scientific materialism, and its replacement with ontological mathematical idealism.

One of the greatest confusions in the present world is that mathematical idealism has been invalidly fused with scientific materialism (it’s a category error to bring the two together). Clarity of thought will not be possible until mathematics is fully extracted and separated from science and treated as the proper means of establishing our knowledge of ontology.

Science, dealing with “truths of fact”, explores the sensible world. Science denies the existence of an immaterial, intelligible Platonic domain. Mathematics, dealing with “truths of reason”, explores the intelligible world. The sensible world, in this view, is a derivation of the objective intelligible world and is deceptive, interpretative and subjective. Which view is right?

Science and mathematics are not allies – they are deadly enemies. If mathematics is ontological, science is wholly false. If science is ontological, mathematics is a bizarre abstraction with no authentic bearing on reality. Science is about the senses and evidence while mathematics

is about reason and proof. They are *opposites*.

The Choice

“Galileo and Wigner and lots of other scientists would argue that abstract mathematics ‘describes’ reality. Plato would say that mathematics exists somewhere out there as an ideal reality. I am working in between. I have this sort of crazy-sounding idea that the reason why mathematics is so effective at describing reality is that it *is* reality. That is the mathematical universe hypothesis: Mathematical things actually exist, and they are actually physical reality.” – Max Tegmark

Tegmark erroneously believes that Plato’s mathematical universe is an impossible ideal (i.e. unreal). He believes that Galileo refers to mathematics as an unreal abstraction. So, he believes he is being novel when he suggests that mathematics is “real”, i.e. neither an impossible ideal nor an unreal abstraction. Of course, the idea that mathematics is real (ontological) is as old as Pythagoras, and Tegmark is way behind the curve. Moreover, Tegmark, under the spell of scientific materialism, has completely failed to understand Plato’s position. For Plato, mathematics is immaterial and outside space and time. That does not make it an “impossible ideal”. It makes it a dimensionless frequency domain – a Singularity – exactly as in Fourier mathematics. It’s the physical world that is “unreal”. It’s a mathematical construct, projection or transform of mind (of the frequency domain). Tegmark has made the catastrophic error of trying to physicalise mathematics when he should have been trying instead to mentalise it. Tegmark simply couldn’t escape his scientific materialist empiricist belief system, and he tried to bring mathematics into it. Big mistake. Science is about the physical world and mathematics about the mental world. Mathematics, in its default ontological state, defines a dimensionless, immaterial Singularity of pure reason, of pure analytic sine and cosine waves, outside space and time.

The Difference Between Ontological Mathematics and Physics

Ontological Mathematics

Physics

A priori

A posteriori

Analytic

Synthetic

Eternal

Temporal

Necessary

Contingent

Deductive

Inductive

Eternal Truths of Reason

Contingent Truths of Fact

Rationalist and Idealist

Empiricist and Materialist

Mental

Material

Intelligible World

Sensible World

Precise

Ad hoc, Arbitrary, Provisional, Approximate

Complete and Consistent
 Incomplete and Inconsistent
 Based on Proof
 Based on Evidence
 Non-experimental
 Experimental
 Noumenal
 Phenomenal
 Based on Fourier Frequency Singularities
 Based on matter that comes from “nothing”
 Causal
 Acausal
 Deterministic
 Indeterministic
 Analytic
 Statistical
 Reflects the Principle of Sufficient Reason
 Irrational
 For thinking-intuitive types (such as Gödel)
 For thinking-sensing types (such as Einstein)
 Dimensionless and Dimensional
 Dimensional only
 Frequency and spacetime domains
 Spacetime domain only
 Reconciles the Cartesian extended (material) and unextended (mental) domains
 Reflects only the Cartesian extended domain
 Reflects *all* numbers
 Reflects only positive real numbers between zero and infinity (and negative numbers in
 relation to charge)
 Based on analytic Fourier mathematics and the generalised Euler Formula
 No known ontological root
 Accommodates zero and infinity
 Rejects zero and infinity
 Concerned with the “soul”
 Rejects the soul; concerned only with the body

Evidence versus Proof

Never get evidence and proof confused. Evidence is fallible and proof infallible. Evidence is temporal and proof is eternal. Evidence relates to truths of fact while proof refers to truths of reason.

Activity, Motion, Force

“...matter by itself is devoid of motion ... the origin of all motion is mind...” – Leibniz
 The original source of all motion in the physical universe was the Big Bang – a purely mental event.

When minds link to bodies, they can make the bodies move. Inanimate things – not directly linked to minds (souls) – cannot move under their own steam.

All the motion of atoms comes from the Collective Cosmic Mind.

“...motion is not given in bodies as a real entity in them ... whatever moves is continually recreated...” – Leibniz

The universe is recreated (recalculated) on an instant-by-instant basis.

What’s the true meaning of quantum mechanics? It’s that the universe is actually a cosmic mathematical wavefunction that recalculates or recreates itself every instant. The progress of the present instant (“now”) is simply the progress of this all-embracing wavefunction. The wavefunction reflects the tensed theory of time (or, rather, of motion). The past does not exist: it corresponds to previous states of the wavefunction that will never be repeated and to which nothing can ever return (i.e. time travel is absolutely impossible since it would require the reconstruction of a wavefunction that is in fact irreversibly *advancing*; it’s a uni-directional “arrow of time”). The future does not exist either. It is that which the wavefunction will become in due course, but not now.

It’s impossible for us to know exactly what the wavefunction will become, not the least because it contains freedom – our own! Freedom is part of the cosmic wavefunction and as consciousness erupts throughout the universe the wavefunction becomes more and more free.

The present (“now”) is the current state of the wavefunction. This is always changing. (“There is nothing permanent except change.” – Heraclitus) We live in a continuously changing present. Previous states are consigned to the past and lost forever, and states yet to be lie in the future, but we can have no idea what these states will be although the closer the future is to “now”, the closer it will resemble “now”.

“Secondary” Mathematics

“Given that the empiricist mind made judgements about mind by saying that it is derived from matter, the rationalist with a hint of skepticism about this assertion will say that mind makes only judgements upon itself and how, then, does it follow that it comes from something outside of itself if it can give no tangible evidence of that thing outside of itself (perceived from the senses as matter, but ultimately a noumenon). Therefore, the only things that mind can know about are things within the mind itself.” – P

Given that everything we know, or think we know, about matter originates in our minds, we can never have any evidence that matter is not a mental construct, so it becomes perverse to assert that something is primary of which we can have no direct evidence (evidence for the existence of matter is inferred from our senses, which are themselves all about mental states and perceptions), and something of which we have direct evidence – our own mental experiences, thoughts and perceptions – are secondary.

In essence, empiricists are those who work from their senses to what reality must be like (they inhabit the “sensible” world and reject the “intelligible” world), while rationalists are those who use their reason to work out what reality must be like (they inhabit the “intelligible” world and see the “sensible” world as a construct deriving from the intelligible world). The more you are a thinking intuitive, the more you will agree with rationalism, and the more you are a sensing thinker, the more empiricist you will be. The central question of course is whether reason or the senses provide the most sane, logical, reliable, intelligent way of apprehending reality. The answer is self-evident.

“Since the mind can only know of things within its own sphere of activity, it follows that it cannot make any certain judgments about anything outside of itself and that any judgments made upon its own thoughts are also subject to an infinite degree of leeway as to how mind will judge its own thoughts. The only thing not subject to this rule are the rules of tautology, where $1 = 1$, since that is true in itself and it follows that this constitutes the only certainty within mind-upon-mind judgements. The only things that are true within the mind are self-evident, and

from here we proceed to the truths of mathematics.” – P

Minds can know only mental things. There is nothing outside mind. Even if we granted that there were, such things would be wholly invisible and undetectable to mind, and mind could know nothing at all of them. It all comes down to Cartesian dualism. If mind and matter are two completely different substances, it's impossible for them to interact. Interaction includes knowledge, i.e. two wholly different substances that cannot interact cannot know anything of the existence of the other since nothing of the other would ever be detected.

If such detection were possible, it would mean that they *could* interact, hence were not in fact different substances after all. The corollary is that either the doctrine of independent mind or that of independent matter is wholly false. Either everything is mind (with matter being a mental output) or everything is matter (with mind being a material output).

Since our interaction with reality is 100% mental, it would be the most perverse of all conclusions to believe in a non-mental reality. There is literally not one scintilla of evidence that matter is not a mental construct. Dreams show how easy it is for minds to create “material” worlds that do not have any objective existence. If minds create the appearance of matter in dreams, why would we imagine that they cannot do the same in the waking state, but in conjunction with all other minds? In other words, individual minds create individual, subjective material dreamworlds, but all minds collectively create a collective, objective material world – the one we all inhabit!

Matter is nothing but a collective rather than individual mental output. Matter is what we get when we all dream collectively rather than separately. We all have two dream states: private dreaming (when we are asleep) and public dreaming (when we are “awake”). All that sleep does is break our link with the collective dream, while waking restores it. An out-of-body experience is what occurs when a person's body goes to sleep yet their mind remains linked to the collective dream, but now without any bodily restrictions. Death is a permanent break from the collective dream, and only reincarnation can then restore the link.

Objective Truth

The only things that are definitively and objectively true are those things that are eternally true and can never be contradicted. These are the analytic, tautological truths of mathematics. Everything else is interpretation, feeling, opinion, conjecture, belief and subjectivity. The more rational you are and the more you align yourself with mathematical knowledge and analysis, the closer you come to eternal truth and objectivity.

If you grasp “reality” through irrational Mythos (religious belief, for example), your mind will play tricks on you since you are minimizing its objectivity and maximising its subjectivity. The mind's highest ability is that which allows it to apprehend eternal truths. However, it usually operates in a much lower state, a subjective rather than objective state, where it's prone to delusion and fantasy, and where it constructs dreamlike fake realities.

Tautology

As Wittgenstein rightly noted, the whole of mathematics is just an analytic system of tautology. Mathematics is *holographic*. Any part of mathematics contains all the rest of mathematics. From absolutely any mathematical starting point, such as $1 + 1 = 2$, you can derive the whole of mathematics, since each tautology leads inexorably to the next.

Illuminism is the assertion that these tautologies are the flawless foundations of reality. Anything that was not tautologically true would be destroyed instantly by its own internal contradictions. Only perfect completeness and consistency can survive the test of reality itself. The slightest degree of incompleteness or inconsistency in the foundations of the universe would annihilate the whole universe instantly. It would be like a fatal fracture in the wing of a plane, but infinitely magnified and intensified. In fact, you couldn't build the wing in the first place.

The fact that the universe exists at all – that we see incredible order, organisation and pattern everywhere – is the proof that we live in a perfect, complete, consistent mathematical system. Any imperfection, incompleteness or inconsistency would instantly ripple throughout the whole universe and obliterate it, or turn it into meaningless chaos. Imagine trying to create a material world from atoms that contained fatal contradictions and mathematical errors, and were thus wholly unstable and unpredictable. You couldn't manage it, and in fact atoms wouldn't exist in the first place.

“This means that pure reason (in the form of tautological relations) is the mind reflecting upon its own very nature.” – P

That's correct. The mind is pure math, hence grounded in tautology. The tricky part to grasp is that minds *experience* information as well as *analysing* it. The mind inhabits the “sensible” world when it experiences it, and the “intelligible” world when it analyses it. Science is a hybrid, a flawed subject that seeks to analyse the sensible world, hence only produces unreliable knowledge of phenomena, of appearances. Science has nothing to say about the intelligible world, the world of noumena.

A mind is objective only when it reflects on mathematical tautologies; it's subjective when it says, for example, “Jesus Christ is God”. This is absolutely non-tautological. It's pure belief.

A mind is both subjective and objective. All objective truths are tautological and all subjective “truths” aren't. Objective mathematics provides the framework of existence, but subjective mathematics – the internal experiences and feelings we have – is how we actually live in this objective universe of math, and we don't experience ourselves as mathematical beings at all.

“Unconscious reason = will, and conscious will = reason.” – P

That's correct. We might also say: conscious reason = objective reason and unconscious reason = subjective reason (will). The only people who truly use conscious reason are mathematicians and metaphysicians. “Consciousness”, for most people, doesn't involve true reason but is just sophisticated sentience ... the rationalization of sentience, and uses Mythos rather than Logos. Most people are little more than apes walking on two legs.

Subjective reason may well seem irrational as far as objective reason is concerned because it's operating according to different rational criteria. As Nietzsche observed, will is all about power. Lies and irrationality which lead to greater power are, on that basis, justified rationally even though they are contrary to objective reason.

How does unconscious reason know itself? How can it reflect on itself and know what it

truly is? It has to become conscious and fully rational, and only then does it see that it is both subjective and objective, and that all objectivity is eternal and tautological while everything else is interpretation.

“Formal logic, then, just takes the structural relations of number theory and codifies them in another notation, but such logic is not pure given that it derives its structure from number theory.” – P

That’s correct. Logic is the inevitable product of tautological mathematics. Logic, when it exposes contradiction, shows where tautology has broken down (hence where something is not mathematically valid). Logic is really just a *test* of tautology and of combinations of tautological statements. Logic is derived from number theory, and number theory from the God Equation, which is none other than the principle of sufficient reason in its true mathematical form.

The principle of contradiction was one of Leibniz’s two great principles (the other being the principle of sufficient reason, and, in fact, the former can be derived from the latter, leaving just a single ontological principle of sufficient reason). Anything that involves a contradiction is mathematically false; it’s not tautological.

“Leibniz’s *Characteristica Universalis* and *Calculus Ratiocinator* were there all along as number theory – the language of pure thought. The only problem with Leibniz’s two concepts is that they are useless unless one can directly ‘read’ the analytic structure of a very particular thought and turn it into a precise expression, but since phenomenological-epistemological limitations say otherwise, all formulations of thoughts from the basic language of number theory will be arbitrary until the sciences can get deeper correlations via neuroscience and physics (by reading brain signals and their frequencies, comparing them to get a general idea of which frequency usually equates with what thought, and then stating said frequencies in terms of pure math). Even then, such a correlation still faces insurmountable epistemic problems since you can only get strong correlations at best. There will only ever be one way to know another in-itself as yourself (but still view the in-itself through your own perceptive lens), and that will always be through the universal identity expression $0 = 0$, which allows for intuition due to the rational structure of the universe. However, many people must have already taken the lead and formulated countless identities through number theory, which will one day find correspondences with future scientific discoveries. At the end of the day math is just intuition discursively carried out on paper.” – P

Leibniz’s ideas work for truths of reason, not truths of fact (or, rather, interpretations of fact as they should properly be called). The former are objective and the latter subjective. The statement “Jesus Christ is God” is a belief, not a rational assertion, hence is logically meaningless and simply false.

You cannot link objective and subjective knowledge. That said, a completely false idea, such as “Jesus Christ is God”, is an objective information signal that is entirely mathematically valid and tautological. The signal is valid, not the *content*! A dream consists of perfectly formed mathematical signals but the content is subjective fantasy, not objective reality.

It’s logically valid to say that Hamlet is a man, all men are mortal and hence Hamlet is mortal. The form is correct but not the content since Hamlet doesn’t exist: he’s a fictional character in a play.

All information is formed correctly – it’s pure mathematics – but most informational *content* is absolutely false, inaccurate or misleading. A false idea is information just as much as a true idea is. The difference is that true information has valid form *and* tautological content while false or dubious information has valid form but non-tautological content.

Mathematical signals can produce either sublime music (Beethoven) or sheer discord. They can also have true or false content. Content is true and certain when it belongs to eternal

truths of reason, and false or uncertain when it belongs to “truths” of fact. Christians say it’s a fact that Jesus Christ is God. Non-Christians deny it. Who’s right? How would we go about proving it once and for all? This is not a fact at all but a partisan assertion.

Form and content are two radically different things. Truths of reason are all about form while interpretations of fact are all about content. When truths of reason *are* the content then they coincide with the form, e.g. $1 + 1 = 2$ is both form and content, and the same is true of all mathematical statements. These are all objective and eternal.

Form is always objective and content usually subjective. All music can be analyzed in terms of form, and it will all seem much of a muchness (if you view each piece of music on something such as an oscilloscope). However, each piece of music has radically different subjective content and affects us differently. Subjective taste and feelings enter the picture. Similarly, “Jesus Christ is God” is subjective content that affects people differently.

Analyzing a thought mathematically verifies the form but it in no way validates the content. Phenomenology concerns content while ontology concerns form. To understand the mathematical form of Beethoven’s music tells you nothing at all about the phenomenology of Beethoven’s music.

Phenomenology is the study of phenomena, of things as they are perceived. It’s the philosophical investigation and description of conscious experience without reference to the question of whether what is experienced is objectively real. It is opposed to ontology, which is the study of being, the nature of things as they actually are. What Beethoven’s music *is* is an entirely different question from how Beethoven’s music is perceived. You can’t go from one to the other. You can’t learn anything about the one from the other. To be able to define the existence of Beethoven’s music with 100% mathematical precision wouldn’t provide a single clue as to how the music will be subjectively experienced. The line of thinking which asserts that a connection can in fact be made should be known as the “science fallacy” wherein scientific materialists believe that a precise definition of brain states (ontology) will tell us about mental states (phenomenology).

This goes to the absolute core of science’s inability to grasp what the mind is and what it does. Even if you could invent an external device that showed exactly what a person was thinking, you could never know what their internal experience of their thoughts was, what, so to speak, the “redness” of the thoughts was (the *qualia*). This internal experience is a causal factor in the next thought, so science could never produce a precise mental “causality”. It simply can’t get at the inner nature of thought.

As Leibniz said so tellingly, “Moreover, it must be confessed that perception and that which depends upon it are inexplicable on mechanical grounds, that is to say, by means of figures and motions. And supposing there were a machine, so constructed as to think, feel, and have perception, it might be conceived as increased in size, while keeping the same proportions, so that one might go into it as into a mill. That being so, we should, on examining its interior, find only parts which work one upon another, and never anything by which to explain a perception. Thus it is in a simple substance, and not in a compound or in a machine, that perception must be sought for.”

Here, Leibniz asserts that only a soul can think. The Leibnizian mill is like mathematics viewed from the outside. We can see exactly how the mathematical mill is constructed (ontology) but we can get no view of the experience from inside (phenomenology). Even if the mill could show us what thoughts were being thought, it could not reveal to us how they were being experienced (*qualia*). We can never experience what another person experiences. We can try to simulate it via empathy or sympathy, but often we will be hopelessly wrong. It’s extremely difficult, for example, to empathise with someone of a radically different personality type.

Souls are subjects, and to each soul everything else appears as an object. We can only ever be our own subject. We can never become a different subject, or experience anyone else's subjectivity. We are fundamentally locked inside our experiences and thoughts. We can transcend them only through tautology. Tautology has no subjective components whatsoever. 100% of subjects can agree that $1 + 1 = 2$, meaning that this is 100% objective. All truths of reason are of this type. No truths of fact are – they can all be challenged, hence none can be deemed unarguably true.

The statement “Jesus Christ is God” is ontologically identical to both a believer and non-believer. However, even if we could see its precise mathematical form, we would be no closer to knowing whether it was true or not. Ontologically, we might find that an additional logic flag is present in a mind whenever this statement is encountered. In a believer, the flag is set to “true” and in the non-believer it is set to “false”. So, a Christian experiences “Jesus Christ is God – true”, while a non-Christian experiences “Jesus Christ is God – false” (and there would also be intermediate fuzzy logic possibilities for agnostics and doubters). While this would allow us to ontologically separate Christians from non-Christians, it would do nothing to tell us about the statement's truth content. So, knowing precise mathematical forms of statements provides us with interesting information, but does not in any way help with epistemology, i.e. with gaining absolute knowledge of what is true and what is false.

Only tautology can ever be unarguably true. Everything else is phenomenology, not epistemology.

The bottom line is that tautology = epistemology = ontology = noumenology = mathematical form = objectivity. Phenomenology = mathematical content = subjectivity. No process can equate phenomenology and epistemology/ontology. No process can make non-tautology the same as tautology, or the synthetic the same as the analytic, or “truths” of fact the same as truths of reason. They belong to completely different categories.

Only tautology is true and has absolute meaning. The closer that a non-tautological statement is to tautology, the truer it will be. Thus, “Jesus Christ is God” is extremely far from tautology, hence can automatically be assumed to be false. The statements of science are also non-tautological, but since they are based on mathematics, they are far closer to tautology, hence are relatively truer, or more useful.

Mathematics – objectively – is nothing but a perfect, complete, absolutely consistent system of tautology. Subjectively, it is not tautological at all. Gödel's Incompleteness Theorem is remarkable because, by invoking self-reference, it brings about a collision between the objective and subjective aspects of mathematics, and these, like phenomenology versus ontology cannot be reconciled. They belong to different categories.

What Gödel did so ingeniously was to find a way to encode and insert a self-referential, subjective statement into an objective system, a phenomenological statement into an ontological system, a non-tautology into a system of tautology.

A self-referential statement such as, “All Cretans are liars, said the Cretan” is non-tautological, subjective and phenomenological. What Gödel proved is that such statements can be generated mathematically (or within some formalist system) – they have the correct form – but their truth content is unverifiable.

Far from meaning that mathematics itself is incomplete, inconsistent or both, Gödel's Incompleteness Theorem simply shows that we cannot prove all statements that have a valid mathematical form. “Jesus Christ is God” can be put into precise and valid mathematical form, but its content cannot in any way be mathematically verified since it's outside the system of mathematical tautology. All such statements are subjective, not objective.

It's precisely because of Gödel's Incompleteness Theorem that we live in a free,

subjective world and not a programmed, algorithmic machine where every statement can be absolutely proved and nothing can be free (since in a fully computerised world there would never be any doubt about which course of action to choose, meaning that everything would be strictly deterministic in a machinelike sense).

The Incompleteness Theorem is exactly the proof that ontology (objective form) and phenomenology (subjective experience) cannot be equated. The fact that “Jesus Christ is God” is a statement that can validly exist mathematically within a mathematical system of ontology does not make its content ontologically true. The form of the statement is ontologically real, not the content of the statement.

The Incompleteness Theorem reveals that absurd, irrational, subjective, ridiculous, meaningless, non-tautological mathematical statements can appear within a cast-iron, objective mathematical system.

The key to understanding this paradox is that *only mathematical form is objective*, not the content that can be generated by mathematics. Form is objective and content subjective. The only place where form and content are reconciled is within mathematical tautology itself. So, the statement, “ $1 + 1 = 2$ ” is both the form and the content.

Correspondence Theory of Truth

“The correspondence theory of truth states that the truth or falsity of a statement is determined only by how it relates to the world and whether it accurately describes (i.e., corresponds with) that world.” – Wikipedia

This is the basis of the scientific outlook and the notion that we inhabit a sensible, phenomenal world. If we in fact inhabit an intelligible, noumenal world then what appears to be true about the world isn't true at all, i.e. there is no correspondence between apparent truth and actual truth: “facts” are mere interpretations.

“Historically, most advocates of correspondence theories have been ontological realists; that is, they believe that there is a world external to the minds of all humans. This is in contrast to metaphysical idealists who hold that everything that exists is, in the end, just an idea in some mind. However, it is not strictly necessary that a correspondence theory be married to ontological realism. It is possible to hold, for example, that the facts of the world determine which statements are true and to also hold that the world (and its facts) is but a collection of ideas in the mind of some supreme being. ... One attack on the theory claims that the correspondence theory succeeds in its appeal to the real world only in so far as the real world is reachable by us. ... The rigorous idealist believes that there are no real objects. The correspondence theory appeals to imaginary undefined entities, so it is incoherent. ... The skeptic believes that we have no knowledge. The correspondence theory is simply false. ... Other positions hold that we have some type of awareness, perception, etc. of real-world objects which in some way falls short of direct knowledge of them. But such an indirect awareness or perception is itself an idea in one's mind, so that the correspondence theory of truth reduces to a correspondence between ideas about truth and ideas of the world, whereupon it becomes a coherence theory of truth. ... Either the defender of the correspondence theory of truth offers some accompanying theory of the world, or he or she does not. If no theory of the world is offered, the argument is so vague as to be useless or even unintelligible: truth would then be supposed to be correspondence to some undefined, unknown or ineffable world. It is difficult to see how a candidate truth could be more certain than the world we are to judge its degree of correspondence against. On the other hand, immediately the defender of the correspondence theory of truth offers a theory of the world, he or she is operating in some specific ontological or scientific theory, which stands in need of justification. But the only way to support the truth of this theory of the world that is allowed by the correspondence

theory of truth is correspondence to the real world. Hence the argument is circular.” – Wikipedia
Identical Things?

Leibniz’s Law asserts that if two objects are identical then they must have identical properties. So, if mind has a property that body does not have, then mind and body must be different. Scientific materialism claims that mental states are simply brain states, i.e. you can’t get mental states that are independent of brain states, or vice versa.

Descartes argued that dualism is correct because mind necessarily exists when thinking is taking place, but there is no necessity that the body exists during thinking (the very existence of the body can be validly doubted; something that can never be true of the thinking mind).

While the body can be divided into parts (it’s a plurality), the mind cannot (it’s a unity). Mind has the property of thinking, body does not. Thoughts exist outside space (real and imaginary), while the body exists in space (real and imaginary).

Modern arguments for dualism include: 1) the “privacy” argument (no amount of scientific materialist manipulation could ever breach the privacy of another person’s mind and allow someone to be inside another person’s mind), 2) the Leibniz’s Mill argument (if you could walk around inside an enormous working version of the human brain you would never encounter a single thought or even any sign of a thought: you would simply see material machinery in action), 3) the *qualia* argument (it’s impossible to explain in materialist terms what raw qualities are like – how would you explain “redness” to a blind person; how could you ever know that your red is the same as another person’s red; given that photons do not possess sensory colour (photons aren’t little coloured packets of blue, or red or green), how do we see colour at all?), 4) the knowledge argument (if you had complete knowledge of the physical brain and then added a single *quale*, this would constitute extra knowledge, which is non-physical), 5) the *inverted qualia argument* (we could just as easily experience sounds instead of colours, or colours instead of sounds – how is that possible if they are grounded in physicalism?), 6) the zombie argument (human zombies without consciousness could act identically to humans with consciousness, so what function does consciousness serve, and how can it be physical in origin if we can do without it physically?), 7) the *blindsight argument* (patients with the blindsight condition can see things without the experience and consciousness of seeing things: this demonstrates that consciousness is more than just physical processing), 8) the intentionality argument (thoughts have a non-physical quality of being about other things; of being directed at things and ends; they have content and meaning; they are teleological), 9) the reason and rules argument (conscious minds follow rules, reason and logic; they pursue truth – physical atoms don’t pursue anything; they rigidly and unthinkingly obey physical laws), 10) the unity and grainless argument (consciousness appears to be smooth and unified: there’s no sign of granularity, of its being made of cell like a body; what does DNA say about the mind?), 11) the *holism argument* (we understand the brain functioning as a collection of countless parts; we do not understand a mind functioning as a collection of countless parts – it operates holistically; consciousness exhibits a unitary, holistic identity), 12) the *self argument* (no atoms or physical objects have an inner identity – a sense of self – making them concerned for their own future and interests, and comparing themselves with others), 13) the free will argument (we self-evidently have free will, but free will cannot exist if the mind is derived from atoms acting according to inexorable, inescapable physical causality), 14) the *other people argument* (we do not treat other people as mere physical objects like cars or bricks; we assume they are more than just physical), 15) the *unpredictability argument* (there is absolutely nothing in scientific theory that requires consciousness or goes any way towards explaining consciousness; it seems a wholly redundant and inexplicable entity in relation to science; nothing in science would predict the existence of

consciousness, so it is therefore entirely unexpected and unpredicted; classical science gets by without any reference to consciousness whatsoever), 16) the argument concerning out-of-body experiences, paranormal experiences and ghosts (some individuals claim to have had experiences that would conclusively refute scientific materialism if regularly repeated), 17) the spirit argument (if a spiritual substance, such as God, existed outside of physical objects, why couldn't a spiritual substance, such as the soul, exist inside bodies? – religious beliefs always imply some sort of dualism).

Arguments against dualism include: 1) the interaction problem (if mind and body are substances with nothing in common, how do they interact), 2) the “no unity” problem (split-brain studies seem to reveal two minds, one in the left hemisphere and one in the right), 3) the double causation problem (if we are made of two separate substances, we must have two different causes of our physical and mental actions – why would they not conflict?), 4) the correlation problem (effects of alcohol, drugs and brain injuries suggest that mind and brain are intimately linked), 5) the evolution problem (Darwinism makes no reference to any unobservable mental substance and its evolution), 6) the spacetime problem (if mind exists in time it must exist in space since space and time are united in spacetime).

Two objects are identical if they have identical external *and* internal properties. However, no two monads could ever have identical *internal* properties. They are all unique souls.

The Truth

There are two ways in which the “truth” can be gauged. One is certain and involves analytic mathematics. The other is dubious and involves consensus (whether sensory consensus in the case of science, or faith consensus in the case of religion). If most people believe a thing, you are likely to believe it too. Christianity might be absurd, but, if you are brought up in a Christian world and continually told it's true, it seems a lot more plausible.

Illuminism is about grounding reality in the fundamental, analytic definitions of mathematics. This is reality. This is ontology. This is noumenology. Einstein said, “The most incomprehensible thing about the universe is that it is comprehensible.” There is only one answer to the comprehensibility of the universe – it's made of math. Einstein certainly didn't grasp this, saying, “As far as the laws of mathematics refer to reality, they are not certain, and as far as they are certain, they do not refer to reality.” This encapsulates how baffled scientists are by mathematics. Would any rational person say, “The uncertain laws of mathematics = reality; the certain laws of mathematics = unreality”? Why not, “The certain laws of mathematics = intelligible reality; the certain laws and encodings of mathematics when interpreted by subjective minds = the sensible reality of science and phenomenology”?

Einstein also said, “The human mind is not capable of grasping the Universe. We are like a little child entering a huge library. The walls are covered to the ceilings with books in many different tongues. The child knows that someone must have written these books. It does not know who or how. It does not understand the languages in which they are written. But the child notes a definite plan in the arrangement of the books – a mysterious order which it does not comprehend, but only dimly suspects.”

Well, the answer was staring Einstein in the face all along. All the “books” to which he referred are written in mathematics, which is why they reflect a “definite plan” and “mysterious order”. But mathematics is not at all incomprehensible. In fact, it's the only comprehensible thing that exists.

The next step is to analyse what phenomena a mathematical system can generate, thus

giving rise to phenomenology (of which science is an expression). Phenomena are in fact based on “translated” mathematics – mathematics turned into sensory data, or pain and pleasure signals, or languages. We then experience the world phenomenally rather than noumenally, and phenomena can be extremely deceptive and produce false ideas about the noumena underlying them.

Mainstream religion provides a ridiculous account of true reality. Mathematics doesn’t feature at all in religious Mythos. As for science, it gives a radically false account too since it tries to make reality accord with phenomena (sensory data) instead of noumena (intelligible mathematics). Seeing past phenomena to noumena – by using reason and intuition – is the most complex and challenging undertaking. This is the true basis of “enlightenment” – seeing what’s really there rather than what *appears* to be there.

How can a world of objective, rational mathematics generate so much apparent irrationality? The answer lies in mathematical coding. Consider the Latin alphabet with its twenty-six letters. We could imagine each letter being mapped to a specific number, e.g. a = 1, b = 2, c = 3, d = 4, e = 5, and so on. Thus, any combination of letters to form words is just a numerical string. If we give the “space” character a number too, then all sentences, paragraphs and books consist, underneath, of a fairly random sequence of numbers.

Many Jews claim that the Torah is mathematical via “gematria”, which Wikipedia describes as, “... a traditional Jewish system of assigning numerical value to a word or phrase, in the belief that words or phrases with identical numerical values bear some relation to each other, or bear some relation to the number itself as it may apply to a person’s age, the calendar year, or the like.”

Of course, the Torah has as much to do with mathematics as Shakespeare or Spiderman, i.e. nothing at all.

$1 + 1 = 2$ is a meaningful mathematical statement. However, “a” + “n” + “d” (to form the word “and”) = $1 + 14 + 4 = 19$ is mathematically well-formed, but what is its meaning?

What is the meaning of the statement “Jesus Christ is God”? This statement could be turned into a valid mathematical form, but there’s no connection between the meaning of the underlying numbers and the meaning we assign to the statement about Jesus Christ.

This is the crucial difference between mathematics itself and things derived from mathematics. Any piece of music, for example, is just a combination of mathematical notes (sound waves), but the meaning and quality we attach to any particular piece of music wildly varies. We have a *subjective* response to combinations of objective mathematical signals and functions when they are packaged in certain phenomenal ways.

Reality consists solely of *objective* mathematical signals and functions being used to encode other things (such as music, languages, sensory data), which are then interpreted wholly *subjectively*.

In other words, through coding, mathematics can be transformed from a dry, objective reality into an emotional, subjective reality.

The statement “Jesus Christ is God” has a legitimate underlying mathematical form and structure (as all things do or they wouldn’t exist at all), but the content of the statement is subjective. The statement is grounded in mathematics but its meaning has nothing at all to do with mathematics but with interpretation, conjecture, opinion, taste and belief.

What we have is a 100% mathematical world of perfect, objective mathematical signals and functions, but these are used to create languages, codes, musical notes, feelings and sensory data, which all have subjective meaning. Thus objective mathematics invariably gives rise to subjective interpretations that have no connection with objective mathematics – which is exactly why most people imagine they inhabit a non-mathematical universe.

There's an objective mathematical reality that we can analyse mathematically and which offers 100% truth and certainty ... and there is a subjective world derived from it which is subjective and interpretive and which doesn't – and never can – offer any truth and certainty.

The mathematical world is the intelligible, noumenal world studied by mathematics and metaphysics and reflects eternal truths of reason. The sensible, phenomenal world studied by science is subjective and reflects interpretations and “truths of fact”.

Truths of fact are never in fact true, unlike truths of reason. Imagine we asked everyone on earth what $1 + 1$ is equal to. Every person familiar with arithmetic would agree that $1 + 1 = 2$. There's nothing controversial or subjective about it. This is a 100% objective truth upon which 100% of people can agree. It's an absolute, incontestable truth, an eternal truth of reason. If, however, we asked everyone what colour the sky is, most people would say “blue”, but those who are blind wouldn't know, those who have total colour blindness would say it's some shade of grey, and those who have any colour-vision abnormalities might suggest other colours. This means that the colour of the sky is subjective, not factual at all. If everyone were colour blind, no one would know what “blue” even meant.

Science, sadly, takes things like the colour of the sky as objective truths when they are not objective at all, but highly interpretative. An objective truth is one with which 100% agree. Anything that does not command 100% agreement is subjective and interpretive. “Blue” is not at all a straightforward concept, as anyone who knows the philosophical difference between “primary” and “secondary” properties will tell you, and anyone who has studied the vision system and how we see what we see.

“Truths of fact” are never truths and always interpretations. The correspondence theory of truth is invalid. Truth is strictly analytic.

While “sensory” truths of fact are not objective truths, they command more consensus than “feelings” truths of fact.

Statements such as “Jesus Christ is God” or “the Koran is the literal Word of God”, or “the Jews are God's Chosen People” are all regarded as objective truths of fact by those emotionally attached to them and who believe in them, yet they are not regarded as truths or facts by anyone else. Indeed, they are usually considered stupid, irrational and 100% false.

The vast majority of human “knowledge” consists of dubious “sensory truths of fact” (science) and absurd “feelings truths of fact” (religion). Many people idiotically say, “Where's your evidence, where are your facts?” as if evidence and facts somehow reveal truth. They don't. Evidence and facts belong to the sensible world of deception, not to the true, intelligible world. “Proof” applies only to mathematical tautology. Actual truth and certain knowledge belong exclusively to mathematics.

We inhabit a false, illusory world of Maya, derived from a true mathematical world. We inhabit a phenomenal world of appearances, feelings and interpretations derived from a noumenal world of truth and certainty. We inhabit a mathematically *translated* world, a mathematically encoded world. We see objective mathematics (true reality) through the prism of subjective mathematics (false reality; interpretative reality).

Our task – if we want the ultimate answers to existence – is to transcend our senses, feelings and interpretations (all the stuff of the phenomenal, sensible world) and reach the noumenal, intelligible world accessible to reason and intuition. We must escape non-tautology and reach the eternal Platonic domain of tautology, of the eternal truths of reason. True reality is tautological reality, which is mathematical reality. Wittgenstein rejected math as “mere” tautology, empty of content about the world. In fact, objective mathematics provides the *form* of the world and mathematical encoding provides the subjective *content*. There isn't anything else. Only when we think in terms of objective mathematics do form and content coincide.

We must escape subjectivity and arrive at objectivity. Mathematics alone is objective. All codes derived from it (all things encoded mathematically but to which non-mathematical meanings are attached) are subjective.

Noumenology = Ontology = Mathematics = the ultimate answer to life, the universe and everything.

Although we inhabit a noumenal universe, we respond to it phenomenologically, i.e. subjectively and interpretationally. We don't see and interact with reality itself but with the *appearance* of reality, with a secondary encoding of reality, with a "Matrix".

We don't deal with the primary domain – mathematics itself – but with the secondary domain that subjective minds derive from it. Some mathematical signals are converted into smells, others into tastes, or sights, or sounds, or touch-sensations, or pleasure, or pain, i.e. all the stuff of our senses and feelings. The reason for this is simple: the secondary, encoded signals allow for a far more efficient processing of information. If you were confronted by true mathematical and ontological reality – an infinite cosmic mathematical wavefunction involving an infinite wave interference pattern based on complex numbers, you would have no idea how to interpret it meaningfully and manage it effectively. So, you convert it into secondary entities that you *can* understand and make sense of. That's what evolution has ordained.

True reality is noumenal, but we respond to it phenomenologically, and as soon we introduce phenomena we make reality subjective and interpretive. Phenomenology can never deliver absolute truth. Phenomenology is always opposed to direct ontology.

Phenomenology locks us into Maya, the secondary world of illusion and appearance. We can use our reason and intuition to escape from it, but only temporarily. We can't help but be subjects, interpreting reality subjectively.

Evolution is driven by survival and reproductive success – not by truth. Many stunningly stupid people are highly successful in terms of survival and reproduction. Intelligence and reason are rare, not standard.

Scientists slavishly worship sensory "evidence" but our sense organs could have evolved entirely different. For example, we might have evolved the ability to see the entire electromagnetic spectrum, or hear a vast spectrum of sounds, or feel the touch of individual atoms. If this had happened, we would have a wholly different response to the world and understanding of it. We are deluded by our senses. We can't help but believe we are seeing "reality", and yet if we could see microwaves, radio waves, infra-red waves, ultra-violet waves, x-rays, gamma-rays, the new reality would look absolutely nothing like our present reality, and our "science" and scientific hypotheses would be completely different. However, no matter what, tautological mathematics would never change. It's the one sure objective anchor amongst all the subjectivity.

We *can* know the objective, ultimate answers to reality, but we will grasp these strictly intellectually. We are not, however, pure intellectual beings. We are sensing, feeling, interpreting, subjective beings, searching for meaning. That will always be the case. And a good thing too – because otherwise we would be computers.

No one can ever have total objective knowledge of everything. We can certainly work out how reality objectively works (mathematically), but we will always experience it subjectively. The meaning we give to life is ultimately a subjective meaning.

Languages

All the letters used in human languages are just mathematical shapes made of straight lines and curves. When vocalised, human languages are just a kind of music, based on certain agreed sounds (notes), which vary from language to language and from accent to accent.

All languages derive from the proto-language of mathematics. Mathematics is strictly neutral and objective. All languages derived from it are partisan, fallible, subjective and interpretive.

Words in languages are based on consensus. We all agree what a table is, or a dog, or a mountain. There's no such thing as a private language. All languages are *necessarily* public. The problem is that while it's easy to agree what a table is, it's impossible to agree what or who "God" is. In the case of "God", everyone can have a fairly clear idea of what people mean by that word, but whether it accords to anything that actually exists or could ever exist is a different matter. This is the great problem of human language. We can start referring to fantasy things and concepts. We all know what a unicorn is, but no unicorn has ever existed. We all know who Hamlet is, but he's not a real person. Human beings can and do invent things. They engage in fantasy and Mythos and then assign reality and ontology to the fictions they construct. Subjective mental ideas begin to be treated as if they correspond to objective reality, and the more popular a false idea is, the more likely it is to be considered true.

People do not approach truth analytically. They approach it depending on what the people around them believe to be true, and what they would personally like to be true. People subscribe to "truth paradigms" – whether scientific, religious, economic, political, or social – and these are invariably not true at all, which is why we can get radical "paradigm shifts". As Schopenhauer said, "All truth passes through three stages. First, it is ridiculed. Second, it is violently opposed. Third, it is accepted as being self-evident."

People start to think and believe that "God" is a real, objective being rather than a character invented in a story. Bar a few heretics, the whole of Europe once believed in the God of Catholicism.

"Jesus Christ" is really just ye olde equivalent of Superman, Batman and Spiderman. The ancients didn't have comics, but they had stories of Gods and heroes that served much the same function. Unfortunately, they didn't grasp that they were just stories and not reality.

Most human beings use Mythos – fantasy and fiction – to define the most important things in their life, except, of course, they deny that they are just making it all up.

Logos is all about seeing beyond the fiction to the true facts. The more that humanity brings its beliefs, interpretations, opinions and conjectures into line with ontological mathematics, the more rational and objective it will be. To understand whether "God" exists, you have to perform a Logos analysis on this concept, not a Mythos one.

Jews say that Jehovah exists because it says so in the Torah ... the revealed Word of Jehovah. This is a 100% Mythos assertion. There's not one particle of Logos evidence that Jehovah exists at all as the alleged perfect Creator of the world. Not one shred!

All Mythos assertions should be consigned to the garbage. They are inventions of the mind that do not correspond to any reality.

Mythos certainly has a role in life, but not one that is ever associated with the truth. Mythos is for fun, entertainment and playtime.

"There seem to be only two kinds of people: Those who think that metaphors are facts, and those who know that they are not facts." – Joseph Campbell

Mythos is pure metaphor and Logos (as objective ontological mathematics) pure fact.

Set Theory

Here is wisdom. Set theory is just another manmade language that has nothing to do with ontological mathematics. Anything that does not define numbers according to the generalised

Euler Formula is false. The paradoxes that bedevil set theory all stem from a false understanding of the ontology of mathematics.

The logical positivists of the Vienna Circle sought to purge philosophy of anything that was neither directly observable through scientific experiment nor derivable through the laws of logic. This was itself immediately illogical since the laws of logic are not directly observable or amenable to scientific experiment. They are laws of the mind, of reason, of mathematics. They belong to the intelligible world, not the sensible world of scientific experiments. You can't logically make the intelligible and sensible realms equally valid. i.e. you can't emphasize the need for experimentation and then appeal to systems that don't need any experiments or observations at all. Either the intelligible world (mental world of reason) is primary, or the sensible world (physical world of the senses) is primary. It's nonsensical to say, as the Vienna Circle did, that both are primary.

Quite simply, mathematics is not based on logic or on set theory. The whole of mathematics, and indeed the whole of true physics, can be derived consistently and completely from the generalised Euler Formula alone. It contains absolutely no paradoxes or flaws. If it did, the universe would be inherently unstable and disintegrate in an instant.

The whole project of attempting to construct mathematics via logic and set theory – still going strong even now – is misconceived. Mathematics gives rise to valid logic. Logic does not give rise to valid mathematics. It produces inconsistency and/or incompleteness.

The Philosophy of Mathematics

“With the use of an ingenious proof [Gödel] was able to demonstrate that in any sufficiently complex system – in short, any system a mathematician would want to use – there are true statements that cannot be proven. Some thinkers despaired of this result. Others, like the formidable Wittgenstein, could never accept it. And still others misunderstood it as a torpedo to the hull of rationality. For Gödel, however, it was evidence of an eternal, objective truth, independent of human thought, which can only be apprehended imperfectly by the human mind.” – Rebecca Goldstein, *Incompleteness*

One of the most difficult subjects of all (because it's so ill-defined) is the philosophy of mathematics: the study of the concepts of mathematics, the justification of the principles of mathematics, the question of what mathematics actually is, how we come to have knowledge of it and what this knowledge signifies.

In Illuminism, the whole of ontological mathematics flows from a single equation – the God Equation. Everything is defined with respect to this equation. The God Equation defines mathematics as a complete and consistent set of energy waves (analytic sines and cosines). “Numbers” are qualifiers of sine and cosine waves. In particular, they define the amplitude, frequency (or wavelength) and phase of a wave. Waves are real objects, hence mathematics is about real things, not ideals or abstractions. Ontological mathematics is not set theory, it's not logic, it's not a manmade language and it's not a meaningless set of marks, as various philosophers of mathematics have claimed.

We and the world are made of mathematical waves. They do not exist independently of our thoughts. They are our thoughts! They exist both dimensionlessly and dimensionally and are the causal basis of the world. In terms of epistemology, we can attain knowledge of mathematics because we *are* mathematics. We are simply discovering our own inner workings and what defines us.

One of the most interesting questions in the philosophy of mathematics is the status of Gödel's Incompleteness Theorems. What, if anything, did Gödel actually prove? Are his theorems relevant to ontological mathematics itself, or do they simply prove the defects in the

various flawed, manmade theories and speculations concerning what mathematics is?

Gödel's entire approach was based on there being a system of arithmetic (as the simplest expression of mathematics) and a syntax for that system of arithmetic. He ingeniously translated the syntax into arithmetic, thus allowing the system of arithmetic to represent its own syntax. A sentence could then be validly constructed in terms of the system of arithmetic, the meaning of which was equivalent to (roughly speaking), "I am not provable."

Gödel's Incompleteness Theorems are in fact not about mathematics per se but about various views of what mathematics is, in particular "logicism". The *Pan Dictionary of Philosophy* defines logicism as: "The view, pioneered by Frege and Russell, that received mathematics, in particular arithmetic, is part of logic. The aim was to provide a system of primitives and axioms (which on interpretation yielded logical truths) such that all arithmetical notions were definable in the system and all theorems of arithmetic were theorems of the system. If successful the programme would ensure that our knowledge of mathematical truths was of the same status as our knowledge of logical truths. Arithmetic was eventually reduced to set theory, but this cannot be genuinely regarded as part of logic."

Gödel showed that in the *theory of arithmetic* – defined in logicist or similar terms – there would be true statements that could not be proved in the theory. This in fact is a statement about the *theory* of arithmetic, not about arithmetic itself. It indicates that the truth of an arithmetic or mathematical statement cannot consist merely in its proof from a set of axioms.

It's the *axiomatic* approach to the subject that Gödel showed was always incomplete or inconsistent. The whole of ontological mathematics is defined by a single analytic formula, not by a set of language rules, not by a contingent set of logicist or formalist axioms. Gödel's approach is thus completely irrelevant in relation to the God Equation. None of Gödel's conclusions is valid in this context. The God Equation is consistent and complete for strict ontological mathematical reasons, not for meta-mathematical reasons (with meta-mathematics being "statements about mathematics").

What is truly alarming about Gödel's position is that it has been taken to be something it definitely isn't. It's not about either arithmetic or mathematics in the ontological sense. It's about popular but dubious and ultimately fallacious approaches to addressing what arithmetic and mathematics are.

Gödel shows the inadequacy of *these approaches*, not the inadequacy of arithmetic and mathematics themselves, yet Gödel's conclusions are invariably taken to be statements about arithmetic and mathematics themselves.

Gödelian Incompleteness has absolutely nothing to do with arithmetic or mathematics in themselves. As Wikipedia says, "... in 1931, Gödel's incompleteness theorem proved definitively that PM [Russell and Whitehead's *Principia Mathematica*], and in fact any other attempt, could never achieve this lofty goal; that is, for any set of axioms and inference rules proposed to encapsulate mathematics, there would in fact be some truths of mathematics which could not be deduced from them."

What Gödel showed was that the various academic approaches to define and master arithmetic and mathematics in terms of other things (logic, axioms, syntax, language and so on) had no hope of conclusive success. Mathematics is the ur language. It can only be understood in terms of itself, not in terms of other things.

What makes ontological mathematics so powerful is that it's defined in terms of a fundamental mathematical formula, not in terms of anything non-mathematical.

The whole of the Gödelian debate has to be transformed. In essence, almost no one other than Gödel himself understood his result. People took dubious ways of defining arithmetic and mathematics to be arithmetic and mathematics themselves. To this day, most professional

mathematicians believe mathematics to be something to do with set theory. They couldn't be more wrong. They have fallen for the propaganda. They have to forget all of the fallacies and get back to mathematics *as mathematics*. Only an analytic formula – *one* fundamental analytic formula – can define mathematics ontologically.

It's time for a whole new philosophy of mathematics to underpin mathematics. It's time for radical new thinkers. Gödel showed what was wrong with the prevailing approaches to mathematics. He did not show what was wrong with mathematics itself. It's about time that people – especially mathematicians themselves! – understood the difference.

All wrong approaches to defining the ontology of mathematics will result in incompleteness or inconsistency. Only ontological mathematics itself can be objectively consistent and complete.

“Thus mathematics is either inconsistent or incomplete. The smart money is on incomplete.” – Stephen Hawking

Here Hawking commits the standard error of believing that Gödel's work relates to mathematics itself rather than to the axiomatic and logicist ways of trying to understand and define mathematics. How can mathematics possibly be inconsistent and incomplete? The universe cannot exist without a consistent and complete foundation. Any alternative would immediately destroy the universe through its contradictions and instabilities.

“But every error is due to extraneous factors (such as emotion and education); reason itself does not err.” – Gödel

All fallacious approaches to defining mathematics involve extraneous factors. Gödel is absolutely right. Reason – ontological mathematical reason – never errs and never *can* err.

Hilbert's Second Problem

“When we are engaged in investigating the foundations of a science, we must set up a system of axioms which contains an exact and complete description of the relations subsisting between the elementary ideas of that science. ... But above all I wish to designate the following as the most important among the numerous questions which can be asked with regard to the axioms: To prove that they are not contradictory, that is, that a definite number of logical steps based upon them can never lead to contradictory results. In geometry, the proof of the compatibility of the axioms can be effected by constructing a suitable field of numbers, such that analogous relations between the numbers of this field correspond to the geometrical axioms. ... On the other hand a direct method is needed for the proof of the compatibility of the arithmetical axioms.” – David Hilbert

“It is now common to interpret Hilbert's second question as asking in particular for a proof that Peano arithmetic is consistent.

“There are many known proofs that Peano arithmetic is consistent that can be carried out in strong systems such as Zermelo-Fraenkel set theory. These do not provide a resolution to Hilbert's second question, however, because someone who doubts the consistency of Peano arithmetic is unlikely to accept the axioms of set theory (which is much stronger) to prove its consistency. Thus a satisfactory answer to Hilbert's problem must be carried out using principles that would be acceptable to someone who does not already believe PA is consistent. Such principles are often called finitistic because they are completely constructive and do not presuppose a completed infinity of natural numbers. Gödel's incompleteness theorem places a

severe limit on how weak a finitistic system can be while still proving the consistency of Peano arithmetic. ...

“Gödel’s second incompleteness theorem shows that it is not possible for any proof that Peano Arithmetic is consistent to be carried out within Peano arithmetic itself. This theorem shows that if the only acceptable proof procedures are those that can be formalized within arithmetic then Hilbert’s call for a consistency proof cannot be answered. However, as Nagel and Newman explain, there is still room for a proof that cannot be formalized in arithmetic:

““This imposing result of Gödel’s analysis should not be misunderstood: it does not exclude a meta-mathematical proof of the consistency of arithmetic. What it excludes is a proof of consistency that can be mirrored by the formal deductions of arithmetic. Meta-mathematical proofs of the consistency of arithmetic have, in fact, been constructed, notably by Gerhard Gentzen, a member of the Hilbert school, in 1936, and by others since then. ... But these meta-mathematical proofs cannot be represented within the arithmetical calculus; and, since they are not finitistic, they do not achieve the proclaimed objectives of Hilbert’s original program. ... The possibility of constructing a finitistic absolute proof of consistency for arithmetic is not excluded by Gödel’s results. Gödel showed that no such proof is possible that can be represented within arithmetic. His argument does not eliminate the possibility of strictly finitistic proofs that cannot be represented within arithmetic. But no one today appears to have a clear idea of what a finitistic proof would be like that is not capable of formulation within arithmetic.”” – Wikipedia

Gödel’s work addresses the general Hilbertian approach to the foundations of mathematics. It’s about time that people stopped thinking that this Hilbertian approach, or something of its general flavour, is the only valid approach to determining the foundations of mathematics. Mathematics must be approached *as* mathematics. Mathematics is a real thing that defines our world. It’s not a system of axioms or logic or sets or marks on paper. To address mathematics correctly, it’s essential to address its ontology, i.e. in what way does mathematics actually exist in the world? And the answer is that it’s ontologically conveyed by Pythagorean-Leibnizian monads defined by the God Equation. This system must become the basis of mathematics, science and the philosophy of mathematics. All other approaches ultimately reflect the sort of deluded thinking that sustains the empiricist, materialist, finitist, positivist Meta Paradigm of science. Mathematics has found itself moving far too close to science. It should be exactly the other way around: science must come to mathematics.

Always question assumptions. Always question axioms. The more fundamental assumptions or axioms are, the more likely they are to be false!

Don’t mistake the map for the territory. Mathematics is the territory. Senses, feelings, axioms and languages are the map, and it’s not a very accurate map! Mathematics is the ontological form – what’s actually, objectively there. Everything else is the subjective content – what we imagine is there.

In the philosophy of mathematics, all of the following must be considered:

Form versus Content.

Objectivity versus Subjectivity.

The Territory versus the Map.

Truth and Facts versus the *Interpretation* of Truth and Facts.

Form versus Matter.
Mind versus Matter.
Form versus Chaos.
Order versus Disorder.
Frequency versus Spacetime.
The Singularity versus Spacetime.

Humans versus Machines

Humans are superior to machines because machines always have to be programmed with the axioms and rules of a formal system and these, as Gödel showed, will always be inconsistent or incomplete.

The Problem With Language

Is the statement “This statement is false” true or false? What’s the problem with this statement? Well, all truths of reason are strictly objective while the statement above is subjective. It’s making a self-referential statement as absurd as “I am beautiful”. How do you know you are? You’re simply making an assertion, not stating a truth of reason. The system of truths of reason never involves subjective, self-referential statements. Truths of reason must be provable objectively. No subjective statement is provable at all.

The statement “This statement is false” fails to make a statement that can be objectively tested. It’s a subjective assertion, an opinion, hence it’s neither true nor false and simply doesn’t belong to the system of truths of reason.

The only way you can address the truth or falsehood of the statement is through language analysis, but no language other than mathematics is infallible and rigorously logical and objective. When such a statement is converted into a pseudo-mathematical statement – in the manner advanced by Gödel in his Incompleteness Theorems – there’s no coherent outcome. That’s not surprising, given that mathematics alone is true, complete, consistent and coherent. Any approximation to mathematics fails. It’s the whole thing or nothing at all. Quite simply, in any system of objective truths of reason (pure mathematical statements), there’s no subjectivity. All self-referential statements are forbidden since they are inherently subjective.

Subjects can create statements that do not conform to any system of objective truths of reason. It’s precisely thanks to this that we do not inhabit a world of computers but a world of feelings, mistakes, errors, flaws, delusions, opinions, conjectures, hypotheses, beliefs, and interpretations.

Objective mathematics is all about objective truths of reason. Subjective mathematics is about “truths of fact”, “truths of belief”, “truths of opinion”, and so on. These are all outside the domain of eternal truths of reason. Self-reference and subjectivity play no part in the truths of reason.

It cannot be stressed enough that objective logic is completely separate from subjective “logic”. The former necessarily involves eternal truths of reason. The latter makes statements that reflect subjective statements that are never either true or false. The word “this” is self-referential in a statement such as “This statement is false”, hence is automatically outside the system of objective truth.

All users of language are subjects and language is primarily about the subject. Every sentence has a subject (what it’s about).

Objective mathematics does not involve subjective statements at all. There is no self-reference in objectivity. There is no “I”.

To itself, a monad is a subject. To any other monad, it’s an object. This dichotomy is vital to logic. Objective mathematics treats all monads as objects. Any subjective, self-referential statement is inadmissible in an objective system.

It's only thanks to monads being viewable as both objects and subjects that we have a world where free will is possible. There are four situations:

- 1) Pure Objectivity: objective monads interact with objective monads.
- 2) Subjectivity/Objectivity: subjective monads interact with objective monads.
- 3) Objectivity/Subjectivity: objective monads interact with subjective monads.
- 4) Pure Subjectivity: subjective monads interact with subjective monads.

These four situations all have their different systems of applicable logic. You commit an error whenever you extend the logic for one scenario to a different scenario.

According to science, there is no subjectivity (no autonomous, immaterial minds), only objectivity. This would produce a machine universe without any possibility of free will. With subjects, they can act on objective systems with a different kind of logic – the type associated with free will, belief, error, opinion, feelings and interpretation.

When subjects interact with each other, you get pure Mythos nonsense – religion, mythology, fiction, fantasy and propaganda.

When subjects rise above subjectivity and rationally enter the domain of objectivity, they become creatures of Logos (mathematics).

Gödel's Incompleteness Theorems are not far-reaching enough. They fail to make clear that the problem they are highlighting is actually that of objectivity versus subjectivity. What Gödel did, in effect, was to ingeniously insert a subjective statement into an objective system, resulting in logical mayhem. It's precisely because of this that we have a subjectively irrational universe full of feeling, sensation, belief and error. This would be impossible in a system of pure objective logic and mathematics. It's thanks to the dual character of the monad – subject and object – that the universe is so interesting.

The living (subjective) universe pursues messy dialectical logic rather than neat Aristotelian logic. The objective universe, on the other hand, is simply a flawless mathematical machine obeying perfect Aristotelian logic.

The subjectivity/objectivity dichotomy is fundamental to the workings of the universe, and existence cannot be understood without this mind-matter dual aspect. Science fails to address ultimate reality because it denies inherent subjectivity (mind) within the universe, hence treats reality as a dead machine (in which free will is impossible).

The world needs a new Gödel to precisely work out all of the logical differences between subjectivity and objectivity, and precisely delineate all of their logical limits of applicability. Are you such a person?

Gödel

Any system cannot be complete and consistent unless it logically accommodates zero and infinity. Gödel proved that all finitary systems (those that do not deal consistently and completely with zero and infinity) always generate insoluble paradoxes. Self-reference and infinite loops undo all finitary systems. Ironically, it's precisely subjectivity – self-reference – that allows the infinity logjam to be broken. Humans might well be puzzled by various paradoxical situations but they do not go into infinite loops, as computers do. Humans take a decision. They do not take a decision based on Aristotelian logic but rather Hegelian dialectical logic. This is something that no computer can do.

If we lived in a non-Gödelian – a finitary – universe, everything would long ago have degenerated into infinite loops or simply halted. The universe would be a “blue screen”, incapable of rebooting himself, hence stuck forever in a crash state. Luckily, we live in Gödelian mathematical universe that fully accommodates zero and infinity, and does not become frozen when confronted by zero and infinity.

Math

Mathematics is the only truth. All the rest is propaganda.

Gibberish

“A meaningful life is not being rich, being popular, being highly educated, or being perfect... It’s about being real, being humble, being able to share ourselves and touch the lives of others... It’s only then that we could have a full, happy and contented life...” – Anonymous

This is the gospel of the “Love and Light Brigade”, the sentimental, irrationalist feeling types, desperately seeking emotional accord with others.

In fact, a meaningful life is about doing everything in your power to become perfect (God), in alliance with all the rationalists around you.

The Conversion

People such as Max Tegmark and Roger Penrose *are* capable of producing a theory of quantum gravity, but this will happen within a mathematical Meta Paradigm, not a scientific one. It will be a Fourier theory, originating in the God Equation.

The world awaits the conversion of its smartest scientists – those who intuit the mathematical truth of existence – to ontological mathematics.

The Church of Science must fall and be replaced with the Temple of Mathematics: the Temple of Reason itself.

Above all, scientists will have to get used to dimensionless – *mental* – existence, ontologically separate from the spacetime material world. That is the *true* meaning of Fourier mathematics.

Drudgery

“Scientists are drudges.” – Steve Jones (scientist)

You’re not wrong, Steve! Science doesn’t have to be boring. Scientists made it so. They’re conservative bores who oppose all radical thinking that challenges the establishment creed of empiricism and materialism.

Common Sense

A great many people swear by “common sense”. Sadly, common sense is usually fantastical nonsense. It was once common sense for the whole of medieval Europe to be Catholic. It was common sense that the Earth was flat, that the Earth was stationary and the sun orbited it, and so on. Common sense is “common herd” sense, i.e. pure nonsense!

A vast number of absurd things are believed by humans simply because they are popular, or culturally acceptable, traditional, or have been brainwashed into people through childhood indoctrination.

It’s time to eradicate “common sense” and the brainwashing of children.

“The sky is blue” is another example of common sense, or common sensory experience, but this is an immensely problematic statement that can be challenged on multiple levels. Most of science is based on sensory common sense, but quantum mechanics, Multiverses, string theory in 11 dimensions, and so on, demonstrate that science can no longer sustain any connection with common sense. It’s becoming more and more abstractly mathematical.

Einstein famously said that common sense was the set of prejudices we acquire by age 18, and he was right about that. Most human “knowledge” is nothing but prejudice.

Logos and Mythos

Logos = ontology = noumenology = mathematics.

Mythos = phenomenology = secondary encodings of mathematics.

Science and religion both provide Mythos explanations of reality. Science has proved useful in terms of changing the world technologically, while religion has had a profound effect emotionally and politically on humanity.

Mathematics is the objective, neutral background and foundation for everything else – it's ultimate noumenal reality (which is why science can't escape from mathematics) – but it is not how we *experience* reality.

We don't process mathematics directly (as mathematics), we experience it second-hand, as phenomenal, interpretational and subjective information. We convert mathematics into sensory sensations, languages, fictions, fantasies, emotions, pleasure and pain signals, and so on.

Just as the world of houses of all different shapes, sizes and styles is based on the humble brick, so is our diverse world based on mathematical "bricks". And just as we don't live directly inside individual bricks but in houses made from bricks (the secondary expression of bricks), so we don't directly live inside mathematics itself (i.e., we don't encounter the mathematical bricks in their noumenal form), but the things built from mathematics (i.e., mathematical phenomena).

If we want to know how to build a house, we have to know about bricks and, equally, if we want to know how anything in the world works, we have to know about mathematical bricks. However, the problem is that we relate to reality via the houses, via the things made from the bricks, not via the bricks themselves. We invest meaning in the houses not in the bricks. An Englishman's home is his castle, not his bricks.

Mathematics, by itself, is devoid of meaning to ordinary minds. It's what mathematics generates, what we make from it, in which we invest meaning. The trouble is that we can use mathematical encoding to create fiction just as easily as fact, and we can then treat fictions as facts and believe in things that bear no resemblance to reality. Abrahamism is a classic example of a fantasy believed in by billions of people.

Mathematics generates meaning but minds do not experience it as meaning itself. It's what mathematics does, facilitates and accomplishes that's meaningful rather than what mathematics is. Mathematics as an abstraction is wholly meaningless to people. However, mathematics, as the quintessence of life, is nothing but *living meaning*. We are all souls – mathematical information systems; mathematical subjects – and it's up to us to define meaning. Of course, we all reach the same definition of meaning, the one identified by Nietzsche: we all wish to maximise our power, to take our Will to Power to its logical conclusion. That means we all dream of perfection, and perfection is what we call "God". We want to be with God (as in Abrahamism), to be absorbed by God (as in Eastern religion) or to become God (as in Illuminism). As for scientists, they want to devise a grand unified theory of everything, which will make them feel like Gods and have the knowledge of Gods.

The pursuit of perfection is what defines all things, and, when all things have accomplished this objective, we have ipso facto reached the best of all possible worlds. At which point, there's nowhere left to go, and after one instant of cosmic ecstasy (ontological orgasm), we commit "divine suicide" and begin again. We must always be in the process of becoming perfect. We cannot collectively be perfect for more than an instant because then the universe would come to a halt, and that can't happen in a permanently dynamical universe.

Mathematics as an abstraction sets the framework in which reality unfolds. Mathematics

as the essence of living souls is what obsessively seeks life, meaning and power. The arena it finds most suitable to that task is the secondary domain of the senses and feelings, with objective reason hobbling along at the back. Life chooses Mythos when it starts to become conscious. Why? Because it's much easier to understand and it appeals to the emotions. Only higher minds turn their consciousness to Logos and start to really understand the nature of existence. The highest minds are those that know they are pure mathematics. In them, mathematics has at last become self-conscious.

Equality

The mathematical equals sign is the basis of tautology. Anything written on the left of the equation is the same as anything on the right, just expressed in a different way. Since $1 + 1 = 2$, we can remove 2 from each side and get $0 = 0$. All tautology can be reduced to $0 = 0$. If $x = y$ then y (or x) can be removed from each side, leaving $0 = 0$. This is the great secret of how "something" is always a disguised version of nothing.

Blade Runner

The film Blade Runner features “Replicants” who are almost perfect android simulations of human beings. Are such Simulants possible? Can machines ever emulate living beings? The answer is no. You must have a soul to be alive. A soul is an uncreated, uncaused, eternal, indestructible mathematical monad. It’s Gödelian complete, i.e. a complete and consistent information system (of objective ontological mathematics). Monads alone are Gödelian complete. All finitary simulations and emulations of monads of Gödelian complete systems are incomplete and/or inconsistent, as Gödel proved with his Incompleteness Theorems. No non-Gödelian system can ever have the properties of a Gödelian system. No non-Gödelian system can ever be alive. All life is Gödelian. This in fact is a key definition of life.

The Arche

The whole of existence must be explained by just one thing – the *arche*, the fundamental substance of existence from which everything else derives. The brilliant ancient Greek philosophers knew this was the most important question of all and provided many ingenious suggestions as to what the arche is.

In M-theory, modern science has proposed one-dimensional energy strings vibrating within an eleven-dimensional mathematical space (comprising three “big” spatial dimensions, seven “small”, rolled-up, spatial dimensions and one temporal dimension) as the arche.

Illuminism states that the generalised Euler Formula is the arche. Everything derives from this extraordinary formula – which is none other than the God Equation. It’s the uncreated Creator of everything. It’s the First Cause, the Prime Mover.

The generalised Euler Formula is the *tertium quid* – the “third thing” – that’s the prerequisite for mind and matter, the underlying basis of both. The generalised Euler Formula is the ground of ontology. It’s nothing other than the mathematical expression of Hegel’s rational dialectic.

In his explanation of the dialectic in its most primordial form, Hegel said that there’s a thesis of “being”, an antithesis of “nothing”, and a synthesis of “becoming”. We thus have a universe of “becoming”. This dynamic quality of becoming is what propels the whole universe forward. It’s the engine of existence, and it drives existence all the way from “alpha” (being and nothing) to “omega” (the Absolute, perfection, the end of a Cyclic Age).

The Dialectic is the True God. The Dialectic is Unconscious Reason converting itself into Conscious Reason. Through the dialectic, reason becomes self-aware. And reason is nothing but mathematics.

Euler’s Formula is *exactly* “being”, “nothing” and “becoming” rolled into one. Euler’s Formula is a “something” which, because of its remarkable mathematical properties, always balances to “nothing” – and which is always in motion, hence always “becoming”. It’s this mathematical becoming defined by the generalised Euler Formula that’s the source of the principle of sufficient reason (for every fact there’s a reason why it’s thus and not otherwise), of cause and effect (why one state automatically succeeds the previous state), of the First Law of Motion, of the Conservation of Energy, of the tensed theory of time (motion) and the arrow of time (motion). It accounts for absolutely everything.

The generalised Euler Formula is none other than the perfect glue that unites Leibniz’s Monadology with Hegel’s dialectic and gives their union precise mathematical expression. This is true mathematical metaphysics and it stands in direct opposition to scientific materialism

(mathematical physics).

Well, what's it to be? – the dimensional 1-D energy strings of M-theory (matter), or the dimensionless 0-D energy monads of Illuminism (mind), defined by the generalised Euler Formula? These are the only two possibilities. M-theory is materialist and atheistic. Illuminism is idealist and religious.

Both theories are about the *arche*, and both are about mathematics. However, M-theory is about contingent, synthetic, incomplete and inconsistent mathematics (because it places physics first and mathematics second) whereas Illuminism is about necessary, analytic, complete and consistent mathematics (because it places mathematics first and physics second).

You have a very simple question to ask yourself. Do we live in a scientific or mathematical universe? Either science is primary or mathematics is. Science is the crown jewel of empiricism and mathematics the crown jewel of rationalism.

In other words, science versus mathematics is the true and final expression of the long philosophical dispute between the school of empiricism (defined in the modern age by the three British philosophers John Locke, Bishop Berkeley and David Hume) and the school of rationalism (defined by the three European philosophers Descartes, Spinoza and Leibniz).

The war for the soul of science was waged between Leibniz (on behalf of European rationalism and metaphysics) and Newton (on behalf of British empiricism). Newton won decisively, and that's why we now have an atheistic materialist science based on experiments rather than a religious idealist science based on mathematical proofs. That's why science rejects independent mind and free will.

As it turned out, pure mathematics itself carried forward the Leibnizian project and now science can do nothing without mathematics. Mathematics – a subject that doesn't involve a single experiment – is now the core of science, a subject supposedly defined by experiments. No scientist has ever explained this fatal contradiction at the heart of their subject and method.

Science is rationally untenable. It has become a quasi-religious faith. Its God is randomness, which science uses to explain everything. Science says that everything that can happen will happen, that anything not forbidden is compulsory. By explaining *everything* in this way, science explains *nothing*.

Science is no more philosophically credible than Islam, which states that the will of Allah is responsible for everything. Randomness is no improvement over "God". Only the principle of sufficient reason provides an explanatory system. Unlike scientific randomness, this principle says that everything that can happen *won't* necessarily happen, that everything that isn't forbidden *isn't* necessarily compulsory, and it explains exactly why some possible, unforbidden things do not happen. No absurd Multiverse is ever invoked.

Religion

The great rationalists were Descartes, Spinoza and Leibniz. Descartes was nominally a Catholic, Spinoza a Jew (before being declared a heretic and excommunicated) and Leibniz a Protestant (a Lutheran, but believed by most to be an atheist). In fact, all three were deists (although Spinoza could perhaps better be described as a pantheist).

How to Explain Reality

There are five ways to explain reality:

1) Abrahamism – religious Mythos; faith in unprovable, unobservable, irrational things is required. Faith is a feelings-based approach to explanation. People believe in things in which the rest of their community believe, in things which make them feel good and which provide a simple, story-based explanation of reality – which is why the Torah, Bible and Koran are all about stories and 100% devoid of mathematics, science, philosophy, technology and engineering. Faith is about “gut feeling”.

2) Eastern religion – religious intuition; mystical intuition is used to contemplate reality and is worked into a simple system of Mythos, meditation and ritualistic behaviour.

3) Physics (science) – a sensory approach to reality; seeing is believing; anything that isn't observable doesn't exist. The feelings, intuitions and Mythos of mainstream religion are replaced with the scientific method of matching hypotheses to experimental data.

4) Metaphysics – a thinking approach to reality; rational unobservables are accepted; no reliance is placed on the senses. Instead, reason is everything.

5) Mathematics – this is hyperrationalism. All reasoning is carried out strictly with regard to the tautological, analytic system of ontological mathematics. Any non-mathematical reasoning, any reason that cannot be linked to mathematics and grounded in mathematics, is rejected.

The approach you adopt will be based on whether you are a feeling type, sensory type, intuitive type or thinking type. In other words, you don't really have a choice.

Hyperrationalism is the approach best suited to intuitive thinkers. It's also the system with the incontestably greatest amount of intellectual horsepower and firepower. It can rationally rubbish all of the rival approaches. They cannot rationally rubbish it.

The Neoplatonic Psyche

In Neoplatonism, the “One” is the source of everything: the Singularity, as we would now say. Its first emanation is the Nous, which we might call the universal *superconsciousness* of the universe, where all Platonic Ideas are thought at once, with perfect clarity, integration, coherence, consistency, completeness and wholeness. This is “God consciousness” – the sort a mind attains when it achieves gnosis.

The Psyche is the first emanation of the Nous (hence the second emanation of the One). It’s the universal *consciousness* and is based on linear, sequential thinking rather than the holistic thinking of the Nous. The Nous is linked to intuition (networked thinking) and the Psyche to rational thinking (linear thinking). The Psyche is the Neoplatonic animating principle of the world, and all individuals souls come from it.

Aristotle equated mind with reason and considered it the highest property of the soul. However, in another sense, he didn’t think it belonged to the individual’s soul at all. Rather, to use Neoplatonic terms, it belonged to the unified Higher Psyche, while the vegetative and sensitive souls of plants and animals, respectively, belonged to the fragmented Lower Psyche. In other words, our basic bodily functions *separate* us, but our reason *unites* us and transcends us.

Plato, like Descartes later, held that body and soul are two different entities, with the body acting as the prison of the soul. Aristotle, conversely, argued that mind and body are intertwined in all living beings. They are inseparable (so bodily death means soul death, except for the rational part of the soul, which belongs to the rational World Soul or God, with the world as its body, and this can never die).

For Aristotle, growth, direction and purpose (teleology) are built into nature and operate everywhere. This is wholly rejected by scientific materialism, which is based on purposeless, mindless atoms.

The vegetative soul possessed by plants governs their growth (and eventual decay) and allows them to enjoy and utilize nutriment (nutrients). This soul does not have motion or sensation; it’s strictly passive.

The animal soul is what bestows motion and sensation on animals and allows them to be active.

The rational soul is that which elevates humans above animals and confers our consciousness and intellect on us.

Given that each higher form of soul possesses in full the attributes of the lower souls, only human beings possess all three types.

In another sense, our reason is the “divine spark” and actually belongs to God rather than to us.

One might say that the Aristotelian cosmos comprises matterless form (God, the World Soul) and formless matter (the chaotic “body” of the World Soul). Matterless form then unites with formless matter to produce *formed matter* – all the stuff of the material world, including living creatures. God (mind) and Chaos (matter) are the two cosmic bounds and everything else lies in between, part form (mind; soul) and part matter (body).

The Supreme Irony

What could be more ironic than that the explanation people desperately seek – the answer to life, the universe and everything – is the one they most fear and dread, and have least hope of understanding: *mathematics*!

Feeling types despise mathematics. Sensing types despise mathematical unobservables. Mystical intuitives base none of their intuitions on analytic mathematics. Non-mathematical thinking types regard mathematics as empty tautology completely divorced from the real world. Even the vast majority of professional mathematicians have rubbished the notion of mathematics being the final explanation of everything. Kurt Gödel was the last great mathematical thinker to glimpse the truth, and he loathed the mathematical establishment and thought they were engaging in an ancient conspiracy to keep humanity stupid, especially by suppressing the work of Leibniz.

Mathematics is how you hide the ultimate answer in plain sight, and yet it might as well be invisible. Only those with eyes to see have the Truth revealed to them. The true Book of Revelation is the cosmic, ontological Book of Mathematics.

You see, truth is not *given* to you, you have to take it. You have to be smart enough for it. Abrahamists claim that your faith is what counts. In fact, it's your reason, and most people are cursed with irrationality. Humanity will reach the Truth only through its smartest representatives. The common herd can never reach the Promised Land by themselves. Ontological mathematicians are the elite and sacred Knight of the Holy Grail.

Randomness

"Randomness means different things in various fields. Commonly, it means lack of pattern or predictability in events.

"The Oxford English Dictionary defines 'random' as 'Having no definite aim or purpose; not sent or guided in a particular direction; made, done, occurring, etc., without method or conscious choice; haphazard.' This concept of randomness suggests a non-order or non-coherence in a sequence of symbols or steps, such that there is no intelligible pattern or combination." – Wikipedia

Non-Randomness

Ultimately, things happen either for a reason or for no reason. But what of first causes, uncaused causes? What's the reason for them? The reason for them is that they are *eternal* (they belong to the unique class of eternal things). They have never not existed, so they can never have been caused by anything. That means they are reason itself, hence they are their own reasons, eternal reasons.

Science does not formally acknowledge eternity. It does not recognise eternal truths of reason. It rejects any transcendent Platonic domain outside space and time. It rejects mind as an independent entity. It rejects religion. Therefore, nothing is left to it except randomness: things happening haphazardly, unpredictably, patternlessly, disorderedly, aimlessly, purposelessly, inconsistently, incoherently, with no design, by chance and arbitrariness.

If science dares to suggest that any kind of rational and mental order underlies existence, it's immediately plunged into the "horror" of metaphysics, and a whole domain outside the scientific paradigm. Without randomness to turn to, science is finished, and yet, ironically, it's finished exactly by turning *to* randomness because ontological randomness is no different from magic. Things literally pop out of nothing for no reason. Does that sound like science or bullshit?

Here's the extraordinary fact. If randomness is ruled out as the basis of causality – and surely it must be since randomness is the absolute enemy of order, organisation, pattern and guaranteed effects following specific causes (rather than any old random effects) – then where

else is there left to go? You are in the arena of *design*. If not *conscious* intelligent design (that of “God”) then it must be unconscious intelligent design, and that means *mathematics*.

The world did *not* arise through chance. It did not randomly leap out of non-existence. It arose out of reasons that are *eternal*. The universe is not arbitrary. There is no random Multiverse. Randomness is an even worse and less rational explanation than “God” for the existence of our universe.

Numbers

In the dimensionless domain, all numbers are tied together by a mathematical formula (the God Equation). They have no units. In the dimensional domain, all numbers have units.

The derivative of a dimensional number (with units) is the same number, but now dimensionless (without units). Calculus is simply a dimensionality ladder. Integration adds dimensions and differentiation removes them.

When a black hole singularity is formed, dimensional numbers are “crushed” down to dimensionless numbers (which exist outside space and time; outside dimensionality).

The Big Bang involved dimensionality (space and time) being added to dimensionless frequencies in the Singularity of Mind. Mind is dimensionless – defined by dimensionless frequencies – while matter is dimensional – defined by spacetime functions derived from frequencies via inverse Fourier Transforms.

What are “event horizons”? They are the interfaces between dimensionless and dimensional numbers; they are where each type of number undergoes a phase transition to the other type. All the “undefined” and paradoxical problems of mathematics occur at the dimensionless domain (Singularity) controlled by zero and infinity. Zeno’s famous paradox concerns the category error of treating dimensional numbers in dimensionless terms.

The ultimate numbers are zeros – monads – which are inherently dimensionless and can never become dimensional. That’s why the Singularity is eternal and the material world is not.

The soul cannot be directly observed in the dimensional world because it’s intrinsically dimensionless.

Compossibility

A *possible* thing is something considered on its own.

A *compossible* thing is something considered with regard to other things.

Compossibility takes the “other” into account, the context. Possibility does not. So, a possible thing may not be compossible (if its existence is incompatible with other things). The more compossible a thing is – the more its existence is consistent with the largest number of other things – the more likely its actual existence becomes.

The Infinite Monkeys

It's said that there are infinite monkeys in the infinite Multiverse, each typing the works of Shakespeare, and one of them (actually, an infinite number of them) will correctly type Shakespeare's plays. (So how do we know that Shakespeare himself wasn't a typing monkey? ... that would certainly explain the huge amount of junk he produced!)

But Leibniz's doctrine of compossibility destroys this way of thinking. There are in fact *zero* universes in which any monkey would be engaged in typing for its whole life. It might be possible, but it certainly isn't compossible, hence it can *never* happen. By the same token, there are all sorts of things we could do, but we will never actually do them because they are not compossibilities: they are not consistent with the nature of our lives and our interactions with others. They belong to our fantasies. Isolated possibilities are not real possibilities. What is possible has to be stabilised by countless other possible things. Isolated possibilities that are not stabilised by other possibilities consistent with them are in fact *impossible*. They are squeezed out of actualisation by options that are more compossible.

Infinitely Divisible

If the continuum is "infinitely divisible" does that mean that we can keep dividing it forever ... or does it mean that if we *did* divide it infinitely, we would reach an indivisible layer of existence? If the continuum isn't grounded in the ultimately indivisible then the claim is being made that it has no "bottom" – it's a bottomless abyss. This means that reality isn't grounded in anything!

If reality is grounded in dimensionless entities then, even though we can keep dividing a dimensional continuum indefinitely, we know there's something at the bottom of it all.

The question, as ever, is how do we link the dimensionless and dimensional? They seem to suffer from Cartesian "substance dualism". How can they interact at all when they seem to have nothing in common? In fact, the relationship between the dimensionless and the dimensional is just another way of describing the mind-body problem. The mind is dimensionless and the body is dimensional.

The whole question of "divisibility" is highly problematic. This type of thinking reflects the idea of physical things through which we can theoretically pass a knife. Imagine cutting a block and keeping cutting it – can this process ever stop? However, now imagine cutting a mathematical function. What does it mean to cut Euler's Formula in two? It doesn't mean anything at all. It's not a possible action. If there are no material things but only continuous energy functions comprising "flowing points", dividing things in two becomes a kind of phenomenon or illusion.

Imagine a continuous energy wavefunction covering the whole universe. We can never divide this in *any way*. Within that wavefunction, there may be an object resembling a block of wood, and we may chop that block in two, but we haven't thereby altered the continuous wavefunction. All we have done is rejigged the wavefunction slightly.

We *cannot* divide the cosmic wavefunction. We cannot divide energy, which provides a continuous energy field (into which no gaps can ever appear).

By the same token, there are no such things as indivisible atoms. Atoms – such as they are – belong to a continuous cosmic energy wavefunction. There are no atoms surrounded by void. Atoms are just "excitations" of an energy field.

The real issue concerns dimensionless energy and dimensional energy. Both types of

energy are continuous and indivisible, but they belong to radically different categories. Dimensionless energy belongs to the frequency domain of Fourier mathematics, and dimensional energy belongs to the spacetime domain of Fourier mathematics.

Energy can never be divided at all, so the whole question of infinite divisibility never actually arises.

Descartes and Leibniz both emphasized the issue of non-extension versus extension. In fact, the real issue is more subtle. All energy is based on the “flowing point”, and this is always technically extended.

Leibniz, in his published work, was insistent that reality must be based on the unextended, the indivisible. The indivisible must have “no parts”. Only at the end of his life, did Leibniz see that the indivisible *could* have parts, but parts that are so necessarily interlocked that they can never be separated, and whose net effect is zero. Zero is dimensionless on its own, and is also dimensionless as the *net effect* of dimensional or extended processes that balance each other.

It’s not so much about dimensional versus dimensionless existence since all of existence is based on the flowing point – which is dimensional. Rather, it’s based on whether dimensional existence is perfectly, immediately and automatically balanced, so that it cancels to zero, or whether this balance takes places much more sluggishly and complicatedly, so that it might not be apparent that any balance is actually taking place.

Mind and matter are in fact *both* grounded in the dimensionless *and* the dimensional. Everything comes from the dimensional flowing point, so to that extent everything is dimensional, including mind (which could be interpreted as meaning that everything is ultimately material). However, all flowing points are perfectly balanced between positive and negative elements, so all balance out, leaving zero, the number of dimensionless existence (meaning that everything is ultimately *mental*).

The difference between mind and matter is subtle and originates in the *immediacy* with which a net dimensionless effect is achieved. Mind depicts the situation where a net dimensionless effect is *immediately* achieved. This corresponds to the *frequency* domain – which is crucially *outside* space and time. Matter, on the other hand, depicts the situation where a net dimensionless effect is not immediately achieved. This corresponds to the *spacetime* domain, where the space and time elements conspire to obstruct the *immediate* balancing of positive and negative elements, or make this balancing much more complex.

Another way of thinking about mind and matter is via the individual minds versus many minds. When energy is restricted to an individual mind, it’s “dimensionless”. When energy is shared *between* minds, it’s “dimensional”. Private energy is dimensionless and public energy is dimensional.

When a private mind generates its own “dimensional” content (as in our dreams), there are no strict dimensional rules. Space and time do not operate correctly, and “causality” is emotionally driven rather than physically driven.

When the cosmic collective of minds generates dimensional content (as in the material world we inhabit), there *are* strict rules of dimensionality. Rigid spacetime rules become inescapable. “Physical” causality can’t be overcome by our emotional states, except in special circumstances such as the placebo effect.

Leibniz’s monads are indeed indivisible, but they are not “without parts” as Leibniz initially said. Rather, they have parts that cancel or balance out, leaving a dimensionless monad that’s a pure frequency domain. No part can be extracted from a monad. It’s the whole, or nothing.

The Flowing Point

The fact that reality is based on the flowing point rather than the static point is crucial. The static point is dimensionless in every conceivable regard. We might call it “true” nothing. The flowing point, on the other hand, is dimensional; it has extension. However, the generalised Euler Formula defines all flowing points, and this formula has the inherent effect of balancing everything to zero. So, even though flowing points are always dimensional, they have a *dynamic dimensionality* that cancels to zero. (Just imagine a flowing point moving around the circumference of an Euler circle.) This is a “false” zero since it is in fact a concealed something rather than nothing at all.

Reality is grounded in “false” zero – “something” that cancels to “nothing”. Reality is grounded in dimensionality – both positive and negative – that balances out to zero (the dimensionless).

It’s not the case of everything originating from the dimensionless but of everything originating from the dimensional, which is perfectly balanced between positive and negative dimensionality, producing a *net* dimensionless effect. We are dealing with a classic dialectical situation:

Thesis = dimensional existence = “something” = “being”.

Antithesis = dimensionless existence = “nothing”.

Synthesis = dimensional and dimensionless existence = something and nothing = being and non-being = *becoming*.

This is the core engine of the famous Hegelian dialectic.

It is the effect of *balance* between positive and negative, of positive and negative being able to perfectly cancel each other, that underpins *dimensionless* existence. Dimensionless existence doesn’t exist in its own right. It relies on dimensional components balancing each other.

Consider Descartes’ famous definition of extended existence versus unextended existence. The crucial error he committed was not to contemplate “negative” extension. If you put negative extension right on top of a corresponding amount of positive extension, what do you get? Mathematically, you get zero: non-extension. But now “non-extension” is no longer something ontologically separate from extension but something that results from the *balancing of positive and negative extension*. It is grounded in extension rather than being something distinct from extension.

Leibniz too, until his final years, simply never considered the implications of positive and negative extension and how this radically changed everything. Zero now becomes the perfect balance point within a dimensional environment rather than something distinct from dimensionality.

Consider the origin in a Cartesian coordinate system. It’s where the whole system perfectly balances. It’s the Singularity!

We are dialecticians. We make maximum use of opposites. Thus far in the *God Series*, we have portrayed a dialectical struggle between mathematical idealism (Illuminism) and mathematical materialism (science). Mathematical idealism revolves around zero and infinity and the dimensionless point, and fully accepts negative and imaginary numbers as ontologically real.

Mathematical materialism, on the other hand, rejects zero and infinity and, loosely speaking, is based on the flowing point (dimensional) rather than the static point (dimensionless). It uses imaginary numbers in calculations, as long as they “vanish” by the end of the calculation, and negative numbers tend to be shunned except where they relate to “charge”. Negative energy, negative dimensionality and negative extension are typically regarded as unreal.

So, it’s time to reach the synthesis of these two views. Mathematical materialism is right that the smallest possible object is based on dimensionality, but it’s wrong to reject negative dimensionality. When negative and positive dimensionality are added together in perfect balance,

dimensionless existence is the outcome. They do not “annihilate” each other. Rather, they co-exist, but with no dimensional net effect. Moreover, they can exhibit dimensionless functionality – which is what we call thinking (mental activity), which occurs outside spacetime dimensionality.

Mathematical materialism is right that everything is grounded in dimensionality, but wrong that dimensionless existence is impossible.

Mathematical idealism is right that dimensionless existence is the key to the operations of the universe, but wrong that zero (and hence infinity) are ontologically real in their own right (rather, they are resultants, or net effects).

Mathematical materialism and mathematical idealism reach their synthesis in ontological mathematics – the mathematics of what really exists. There’s no room for abstraction in ontological mathematics.

Above all, the mathematical point – the perfect zero – does not exist ontologically (as a thing in itself). It’s the hole – the Singularity – at the origin of the cosmological Cartesian coordinate grid: the perfect balance point of positive and negative dimensionality.

The true ontological point is the flowing point, the most basic expression of energy/movement/information. Flowing points, depending on context, can relate to the mental or physical. They are the “tertium quid” – the third thing, the common ground – that underlies the mental and the physical.

Leibniz defined the monad as the most basic unit of existence. In fact, it’s the flowing point – the perfect mathematical expression of energy – that truly fulfils this role. The monad comprises an indestructible, indivisible set of flowing points, perfectly balanced to create the dimensionless monad. The material world is a spacetime set of flowing points.

What is the Biggest Number?

There are two basic ways to generate infinity:

- 1) Divide any number by zero.
- 2) Consider a process in which you take any number, no matter how big, and add one to it. Can this process ever end? If not, infinity must exist. No end can ever be established.

Infinity never comes to an end. It's never bounded or constrained. It cannot be contained. It's a divergent process. Or is it?

If both 1) and 2) are ontologically forbidden then we in fact live in a finite rather than infinite universe.

To forbid 1), we simply have to establish that the smallest conceivable point is dynamic rather than static (hence dimensional rather than dimensionless). In a causal system of mathematical energy, where energy is always flowing according to a precise mathematical formula, energy is necessarily dimensional (it has extension because it is never at any precise location but has already moved on to the next location in the mathematical sequence, hence is made of flowing points rather than stationary points).

We thus have two mathematical systems:

1) Abstract mathematics where the dimensionless static point can be defined, hence infinity is possible.

2) Ontological mathematics where only the dimensional flowing point truly exists, hence infinity arising from division by zero is impossible. There is no static point per se in ontological mathematics, although Euler flowing points always have a net effect of zero, hence are net static, dimensionless points. We thus have the paradox of a system of points full of motion (indeed perpetual motion) that has a net result of no motion. This paradoxical system of points corresponds to Leibnizian monads – *souls*. Only if you could divide by whole monads (zeros) could you generate infinity. In reality, all division takes place in terms of finite numbers of flowing points, hence always generates a *finite* result.

As for handling the $(n + 1)$ process with no end, this leads to true infinity only if it is *linear*. If, however, it's circular then there comes a stage when $n + 1$ does not in fact generate a brand new number, but actually brings us back to zero.

Mathematicians such as Professor Doron Zeilberger describe infinite mathematics as meaningless, as abstract nonsense, a fiction of the mind, an illusion. Eventually, they say, you reach the biggest possible number and when you add one you go back to zero and start again! (That is, everything is circular rather than linear.) The biggest number is absolutely, unimaginably enormous and will never be directly encountered. Nevertheless, it exists.

Mathematicians such as Zeilberger believe that this circular system is much less ridiculous than linear infinity, which is weighed down with disastrous paradoxes. Mathematicians advocating the same sort of position as Zeilberger are known as finitists, classical finitists, strict finitists, and ultrafinitists. They all accept the existence only of finite or countable mathematical objects.

So, is the universe infinitely big or just unimaginably big? How could we ever tell the difference?

How can we ever know the practical difference between an unimaginably large number (the “Zenith” number; the Myriad) and infinity? Again, we have two systems of mathematics:

1) An abstract, linear mathematics where we can keep adding 1 to generate ever-larger numbers, with no end.

2) An ontological, cyclical mathematics where we reach a highest possible number, and the addition of 1 takes us back to the start.

So, what of the “point at infinity”? In the Riemann Sphere, the point at infinity is opposite the origin (at zero). We can regard these two points as abstract limits between which ontological reality actually unfolds. Actual reality is based on a smallest possible number (the size of *any* flowing point of energy – all flowing points have exactly the same size) – and the highest possible number (the Zenith number), for which the next number in the sequence is zero.

Even though, ontologically, we can’t get below a certain number (the Nadir number), or above a certain number (the Zenith number), this doesn’t mean that we can escape “abstract” zero or infinity. We can still generate infinite loops from which we can never escape. Even the Zenith number itself can be infinitely repeated! This is why the universe can never end.

As things stand, we’re in a world where abstract zero and infinity insist on intruding in an otherwise finite world. What’s for sure is that we need a new mathematical genius – a new Gödel or Leibniz – who will be able to give a much clearer definition of the domain of zero and infinity, and to provide clear-cut “rules of engagement” when it comes to specifying the applicability of zero and infinity. There can be little doubt that many paradoxes that are currently encountered regarding zero and infinity aren’t “real”, but are artefacts of the domain of zero and infinity not being properly defined.

For example, is division by zero an operation that can actually happen, or is it formally prohibited ontologically? In a system of flowing points, division by zero won’t normally occur. However, if two flowing points of equal but opposite dimensionality are superimposed on top of one another, it’s hard to see how division by zero can be prevented, leading to the paradoxes of infinity. Ontologically, what seems to happen is that a phase change from mental to material, or vice versa, occurs whenever zero and infinity are encountered.

A new mathematics will need to be introduced to properly and rigorously examine this mind-matter phase change. This is exactly where geniuses are required. Geniuses are those who create new vistas. The masses can never peer into the true nature of things.

One of the primary tasks of meritocracy is to identify geniuses and extract the last drop of their genius for the greater benefit of the human race. In today’s world, many geniuses fail to prosper because the prevailing system is so hostile to them. If they succeed, it’s through pure luck, or because they just happened to have produced an idea that could be commercially exploited by rich capitalists.

Nadir, Zenith, Zero and Infinity

If the Nadir Number is the smallest possible “dimensional” number, and the Zenith Number the highest possible dimensional number, how do these two numbers relate to the two dimensionless numbers zero and infinity?

We seem to have a bizarre gap – a No Man’s Land – between the Nadir Number and zero, and between the Zenith Number and infinity.

The answer to this enigma lies in the science of black holes, with their singularities and event horizons. When a dying star has a density so enormous that its radius is less than a specific value known as the Schwarzschild radius then it will inevitably and irreversibly collapse down to a black hole singularity where curvature is infinite. In other words, once a certain, well-defined point of no return is reached, anything dimensional will collapse all the way down to a point (to zero).

We might think of the Nadir Number in the same way as the Schwarzschild radius. Even though there are numbers hypothetically smaller than the Nadir Number yet larger than zero, these, in effect, are simply “gobbled up” instantly, and the Nadir Number collapses straight to zero, via all intermediate points, which present no barrier at all. They are rolled up in one step.

“Within the Schwarzschild radius, nothing – not just light – can escape. Imagine a sphere that has a radius equal to the Schwarzschild radius and which surrounds the collapsed star. Such an imaginary spherical surface is known as the event horizon and is an artificial boundary in space which marks the point of no return. Outside the horizon gravity is strong but finite and it is possible for objects to escape its pull. But once within the horizon, an object would need to travel faster than light to escape, and this is not allowed. Thus the event horizon is a rather unpleasant concept that allows one-way traffic only.” – Jim Al-Khalili, *Black Holes, Wormholes and Time Machines*

Event horizons are what allow the finite (the dimensional) and the “zero-infinite” (the dimensionless) to co-exist.

Collapse

The range from the Nadir Number (smallest possible dimensional number) to zero is an event horizon. We can’t look inside this zone and see exactly what goes on. What we know is that any Nadir Number (and the Schwarzschild radius constitutes the Nadir Number for black hole formation) can have only one terminus once it is surpassed – the dimensionless point (zero).

By the same token, any process that reaches and surpasses the Zenith Number will converge automatically on infinity. Contrary to what Einstein’s theory of relativity claims, any physical object whose dynamic mass surpasses the Zenith Number will automatically be converted into a photon and have infinite energy.

The opposite of black hole formation is white hole formation. Just as black hole formation governs collapse from the Nadir Number to zero (the dimensionless point), so white hole formation involves the expansion of a dimensionless point to the Nadir Number. We could call this Big Bang “inflation”.

Also, any dimensionless infinite-energy process must be made dimensional to enter physical reality and this involves the “collapse” of infinity to the Zenith Number.

All of these dimensional-dimensionless phase transitions are concealed by “event horizons”. We will never be able to observe the precise details of dimensionality transitions. We can apply reason to these transitions, but we will never have any empirical scientific evidence,

contrary to the basic tenets of the scientific method.

We might also apply “event horizon” thinking to mind-matter transitions, to intuition and even to death and reincarnation.

Quite simply, ontology involves a system of mathematical one-way valves to handle key dimensionality phase changes. Abstract mathematics is prone to error because it doesn’t recognise these processes. For example, does it make sense to talk of “division by zero”, as happens in abstract mathematics, or should we in fact be talking about irreversible collapse after a certain finite number is exceeded, followed by the formation of an ontological event horizon?

You never get a bare division of anything by zero. Instead, you get a process of ontological collapse leading automatically to the formation of a dimensionless point, which is always associated with zero and infinity.

Another critical point to bear in mind is the double-sided coin relationship of zero and infinity. Zero is always a portal to infinity, and infinity always a portal to zero. If we think of zero as the beginning (origin; alpha point) of a process, and infinity as the end (terminus; omega point) then, when we reach infinity, the next step will be zero, i.e. we finish the present process and begin the next. This is the basis of the eternally oscillating, Cyclical Universe. The instant before the Big Bang of a new universe was the Big Death (or Crunch) of the previous universe.

So, ontologically, the Nadir number and zero go together. They are linked by event horizons and irreversible collapse. Exactly the same is true of the Zenith number and infinity.

The Total Number of Monads

A fascinating question that arises is whether the total number of monads is infinite, or is equal to the Zenith number. Well, how would *you* answer that question?

Points

“Static” points are frequency domains (monads; singularities). These have a dimensionless resultant effect. All the dimensional things of the material world come from these dimensionless monads, i.e. all spacetime entities come from frequencies, as in Fourier mathematics.

All monads (souls) are uncreated, autonomous, eternal frequency domains. They are outside space and time and have no *net* dimensionality. The material world is a Fourier creation of the frequency domain, i.e. dimensional matter comes from dimensionless energy (pure frequencies).

Logical Zero versus Ontological Zero

Strictly speaking, zero is logically possible but not ontologically possible. What *can* exist is a resultant zero, a net zero, a *logical* zero. Zero is a logical effect of the balance between non-zeros of opposite sign. It doesn’t exist in its own right as a thing in itself, but as the resultant of other things.

If zero can’t exist ontologically then neither can infinity. As before, we can get a resultant infinity, a net infinity, a logical infinity, but not an ontological infinity. Neither zero nor infinity are ontologically “countable”. They are therefore not numbers at all, but logical resultants. That’s why they’re subjective and not objective. That’s why they can’t appear in the physical world but only in the mental, dimensionless world.

The Brain and Strange Loops

“Hartmann is forced to adopt the conclusion, which had been previously announced by Schelling, that the brain is the condition, or necessary prerequisite, for the origin, not of mind as such, but of Consciousness. Mind acts independently in the Unconscious; but it cannot become cognizant of itself, and therefore cannot be emancipated from its servitude to the Will, till it has deluded the Will into building up a brain.” – Francis Bowen

It's certainly true that Mind and Consciousness are different. Standard consciousness is mind filtered through the prism of space and time. Space and time can have the effect of slowing down and localising thought, eventually to the extent that it can reflect upon itself.

Animals think much faster than humans. They think instinctively (unconsciously), which is instantaneous thinking, as it is in the dimensionless domain of pure Mind. Consciousness arises when thought is slowed down, allowing choices to be pondered and made. Unconscious animals can never ponder choices. They always act instantly – by programmed instinct.

Hartmann is wrong that the Will is deluded into building up a brain. That in fact is its *objective*. The whole purpose of self-optimizing existence is to create choice, planning and freedom through consciousness. Consciousness needs a brain until such time as a mind can achieve enlightenment (gnosis) and be conscious without a brain or body. This is “God consciousness.”

“Consciousness, says Hartmann, is not a continuous and fixed state, but a process; it is an action frequently repeated, a constant becoming conscious.” – Francis Bowen

This is somewhat an anticipation of Douglas Hofstadter's hypothesis in *Gödel, Escher, Bach: An Eternal Golden Braid* where he explains consciousness in terms of a “strange loop”. Of this, Wikipedia says, “A strange loop, technically called tangled hierarchy consciousness, arises when, by moving only upwards or downwards through a hierarchical system, one finds oneself back where one started. ... Strange loops take form in human consciousness as the complexity of active symbols in the brain inevitably lead to the same kind of self-reference which Gödel proved was inherent in any complex logical or arithmetical system in his Incompleteness Theorem. Gödel showed that mathematics and logic contain Strange Loops: propositions that not only refer to mathematical and logical truths, but also to the symbol systems expressing those truths. This leads to the sort of paradoxes seen in statements such as ‘This statement is false,’ wherein the sentence's basis of truth is found in referring to itself and its assertion, causing a logical paradox. Hofstadter argues that the psychological self arises out of a similar kind of paradox. We are not born with an ‘I’ – the ego emerges only gradually as experience shapes our dense web of active symbols into a tapestry rich and complex enough to begin twisting back upon itself. According to this view the psychological ‘I’ is a narrative fiction, something created only from intake of symbolic data and its own ability to create stories about itself from that data.”

In *I Am a Strange Loop*, Hofstadter defines a strange loop as, “...not a physical circuit but an abstract loop in which, in the series of stages that constitute the cycling-around, there is a shift from one level of abstraction (or structure) to another, which feels like an upwards movement in a hierarchy, and yet somehow the successive ‘upward’ shifts turn out to give rise to a closed cycle. That is, despite one's sense of departing ever further from one's origin, one winds up, to one's shock, exactly where one had started out. In short, a strange loop is a paradoxical level-crossing feedback loop.”

Infinity itself may constitute a strange loop, i.e. as you get further and further from zero, you unexpectedly end up right back at zero. When people talk of infinity, they invariably say that

you can always add “1” to a number to get a larger number. But imagine that you add “1” and unexpectedly find yourself back at zero. In that case, infinity is actually a “bounded infinity.”

There are an infinite number of points between the south and north poles but if you circumnavigate the world from the south pole via the north pole, you will end up back at the south pole. Now, if you’re wearing a pedometer, the number shown on it will keep going up if you decide to keep going, but you are in fact simply retracing your steps. So, you have actually run out of “infinity” – you have exhausted it – in terms of new ontological locations. What you are doing now is generating an artificial infinity that involves endlessly repeating an ontological bounded infinity. We thus have to be very clear about the distinction between ontological infinity and “repetitive infinity” whereby, instead of resetting a number to zero even though we have in fact gone back to the start, we keep “adding 1” to it (we are treating it linearly rather than circularly).

The Absolute and The Manifest

“If the Absolute is to become manifest to itself, then, in respect to its objective, it must appear as dependent upon something else, upon something foreign to itself. This dependence, however, is not of the essence of the Absolute, but belongs merely to its manifestation. ... not the mental states themselves, but the Consciousness of them, is conditioned by an affection of the organism; and if the empiricists had restricted their assertion to the latter point, there would be nothing to object to in their doctrine.” – Schelling

Objective versus Subjective Reason

When people encounter the concept that the universe is made of reason, they think that ought to imply that the universe is machinelike or computerlike. The universe – without animal life – is exactly that. Look at the stars, planets, moons and how they seem to belong to a giant clockwork machine, or computer programmed system. The behaviour of the cosmos in this regard reflects *objective* reason and is what scientists study.

However, conscious human beings certainly wouldn’t be regarded as models of reasonable conduct and rationality. How can humans be built of reason in the same way as the orderly cosmos is? The difference is that humans – all humans – *do* behave rationally, except there are difference grades and sophistication of rationality. Take feelings, for example. People who operate according to their feelings are those who obey a very rational principle: what causes them pleasure and happiness (good feelings) is good and what causes them pain and distress (bad feelings) is bad. Good feelings should be pursued and bad feelings avoided. That is a perfectly rational – albeit crude and simplistic – formula for trying to optimize one’s life.

People who are locked into their sensory experiences (scientists and the like) follow the naive rational principle that it’s more reasonable to believe in the existence of things you can see and experience than things you can’t see and experience.

Intuitives obey the rational principle that it’s reasonable to accept the intuitions that come to them because if these intuitions aren’t real and true yet are overwhelmingly convincing then why should non-intuitions that are equally overwhelmingly convincing be deemed any more real and true? What is your criterion for reality and truth if you can’t trust your own mind and what it presents to you?

So, feelings, sensing and intuition are all perfectly rational – but they are not *optimally* rational. For feeling types, what makes you feel good might prove bad for you and what makes you feel bad might be immensely beneficial. For sensing types, your senses might be presenting you with powerful illusions – a Matrix world – rather than reality itself, and thus you will form completely the wrong idea of what reality is (as scientists do). Intuitives might not be able to

accurately interpret what that their intuitions are really telling them and instead convert them into mystical Mythos (as in the case of Eastern religion).

The point here is that reason can be subjective as well as objective. Subjective reason uses some subjective criterion or another – feelings, sensations or intuitions – to gauge the optimal way forward. However, these subjective criteria are sub-optimally rational. They are intended to provide a basis for deciding the best way forward, but they are all deeply flawed, hence do not accomplish the desired outcome, and might in fact have the opposite effect. In other words, subjective reason – based on *primitive* rational criteria – can seem exactly the same as downright irrational behaviour. Yet nothing is truly irrational: everyone has a reason for what they do. The issue is whether they have a good reason rather than bad, whether they have effective and productive reasons for their actions and plans.

How reason becomes more effective and productive is exactly described by the Hegelian dialectic. The dialectic does not take us from an irrational to a rational world. Rather, it takes us from a sub-optimal rational world to an optimal rational world. In some sense, the point of the dialectic is to slowly and steadily reveal what is wrong with using feelings, sensations and intuitions as rational criteria. They are simply not rational enough.

What the dialectic ultimately achieves is the evolution of rational *thinking*, especially self-reflective thinking. True consciousness is the same as reason becoming self-aware, and rationally and deliberately knowing how to optimize itself. Rational thinking allows reason to work out and plan how to optimize itself – something not available to the feeling, sensing and intuitive types who operate essentially instinctively, without any reflection and planning.

The ability to think is always last to appear in evolutionary terms. On Earth, only humans can think (by which we mean logically rather than by feelings, sensations or intuitions), and only a small fraction of them can think well. Animals are stuck on the level of instinctively reacting to pleasure and pain, to what their senses tell them, to instinctive intuitions. Animals have instinctive logic but no self-reflective logic that can lead to long-term planning and control of the environment, such as humanity has achieved.

When logical thinking first appears, it too is sub-optimal and is usually pressed into the service of feelings, sensations or intuitions – where it can reach entirely wrong conclusions. People of faith, for example, use their logic, such as it is, to justify what they would like to believe about the world rather than to reflect what the world is actually like (which they don't like because it causes them distress).

So, rational thinking and logic have to evolve and the intellectual history of humanity is the dialectical record of that evolution.

The best human thinkers are the most rational and logical, those least influenced by feelings, sensations and wild intuitions. In them, reason is optimizing itself, understanding itself, becoming conscious of itself. Conscious reason provides the platform from which we can strive to make ourselves into Gods. Without it, we don't have a prayer (!). The purpose of Higher Humanity is therefore to make the world as rational as possible, and to reduce the influence over our world of feelings, sensations and mystical intuitions that are not harnessed to logic.

In summary, non-living things are governed by machine-like reason – which is nothing but the laws of objective mathematics. Living things are affected by an internal, subjective expression of reason, which is sub-optimal and can only be made optimal through the Hegelian dialectic – a process that is as simple as possible, and infallible, but only over an immense period of time. The dialectic is horrifically messy (which is why the history of our world has been so brutal and ugly), but it gets the job done in the end.

The endpoint – the omega point of the dialectic – is when subjective reason evolves into objective reason and every subjective mind can grasp the most objectively rational thing to do in

every circumstance, uninfluenced by feelings or sensations. Intuitions might be welcomed, if they can be harnessed rationally. In fact, the very highest form of thinking – “God thinking” – is intuitive rational thinking. That’s why INTJ and INTPs have the most highly evolved minds in the world.

Our world currently reflects the values of extraverted sensing and feelings types – the absolute opposite of what is needed. Human destiny will be 100% different when it reflects the values of introverted intuitive and thinking types.

The New World Order is about the revaluation of all values. Specifically, it’s about ending the dominance of extraverted values – which are superficial, narcissistic, egoistic and individualistic – and replacing them with introverted values which are profound and communitarian, fair to all. It’s about bringing an end to the Mythos world of crass, childish feelings and sensations, of faith and primitive sensuality, and replacing it with a Logos world of reason and intuition.

The evolutionary trajectory of humanity could not be any clearer. The dialectic demands that INTJ and INTPs become those who plan and shape the world’s future. To leave it to “markets” is insane. A market reflects the dialectic in its most primitive form. Rational, logical planning reflects the dialectic in its optimal form. Well over 2,000 years ago, Plato designed the kind of perfect State (his famous *Republic*) that would deliver paradise on Earth. It’s not run by markets, or rich elites, or the privileged, or capitalists, or libertarians, or religious fanatics, or people of faith, or New Age people blabbering on about “unconditional love” and “love and light”. No, it’s run by the Guardians, the Philosopher Kings, the supreme meritocrats, the best thinkers. They serve the public interest, not their own. They serve the General Will, not their particular wills. They own no property. They are not rich. They are the ultimate selfless heroes, the greatest human beings of all.

Humanity’s tragedy is that it has had the answer to all of its ills for thousands of years – the Meritocratic Republic ruled by the smartest, hardest working and least selfish people – and yet it has turned its back on it. Instead, it has preferred a world of privilege and inequality, of rule by rich elites and psychopaths (who are usually one and the same).

Those who know that Plato’s answer is the right one no longer have any choice. We must decisively break free from the rest of the human race. We will advance to divinity – our divine heritage – and the rest will sink back into the swamps whence they came. They simply failed to evolve. In them, reason did not become sufficiently conscious.

Physics versus Abstract Mathematics versus Ontological Mathematics

Physics believes itself to reflect the mathematics of what actually exists. Being materialists and empiricists, physicists believe that the sensory, material world of space and time is all that exists. Any mathematics that deals with the immaterial and non-sensory is regarded as unreal. For physicists, physics is the mathematics of reality while mathematics itself is solely about hypothetical things, many of which can never be realised because they are not compatible with reality.

The quantum mechanical wavefunction is said to be unreal because it involves imaginary numbers and various operations have to be applied to it to make it “real”. So, the wavefunction per se is an abstract mathematical entity and the probabilistic wavefunction derived from it is treated as physical (ish). However, here we have a serious philosophical problem akin to mind-matter Cartesian dualism. How can unreal, abstract mathematics be the underpinning for real, concrete physical mathematics, and is a probability “cloud” real anyway? How is reality determined? Who decides? How can the real emerge from unreality? What’s the rational cause of this miracle, this magic? And, of course, physics actually abandons cause and effect at this level and refers to randomness, probability, indeterminacy and statistics. Determinism is completely rejected.

Physical mathematics is dynamic while abstract mathematics is static. Physical mathematics is constrained by physical constants while abstract mathematics is not.

Ontological mathematics is the dialectical synthesis of physical mathematics and abstract mathematics:

- 1) Thesis – Physical mathematics: dynamic, always observable and constrained by ontological constants such as the speed of light, gravitational constant and Planck’s constant.
- 2) Antithesis – Abstract mathematics: static, can be wholly unobservable and is unconstrained by ontological constants.
- 3) Synthesis – Ontological mathematics: dynamic, both observable and unobservable, constrained by ontological constants.

With 3), the speed of light, gravitational constant and Planck’s constant are no longer constants of physics but mathematical ontological constants, i.e. they are built into ontological mathematics (which they are not in relation to abstract mathematics).

Physical and abstract mathematics are both wrong: they are approximations to the truth of mathematics: *ontological* mathematics, the mathematics of actual existence. They get many things right because they partake of several features of ontological mathematics but they also get many things wrong because they do not do so completely and consistently.

“Paper mathematics is, then, a language like any other since it interprets ontological mathematics.” – P

Not true. It partakes of true mathematics (unlike any human language) but makes several false assumptions. For example, in ontological mathematics there’s a cosmic speed limit and anything that does not reflect this is ontologically absurd. Abstract mathematics can easily cater for tachyons (imaginary mass particles that exceed the cosmic speed limit), but no such things could ever actually exist. So, abstract mathematics allows ontologically impossible (forbidden)

situations to be discussed because it does not obey the restrictions that define ontology.

Physics dismisses permissible ontological things that cannot be observed, and abstract mathematics accepts impermissible ontological things. Ontological mathematics sits in the middle. It caters for the unobservable but not for the impossible. Physics is mathematics that is too constrained (to accommodate the sensory world) and abstract mathematics is not constrained enough, hence can deal with ontologically impossible situations.

Physical mathematics: dimensional mathematics only; constrained by physical constant; dynamic.

Abstract mathematics: dimensional mathematics (uncertain towards dimensionless mathematics); not constrained by physical or ontological constants; static.

Ontological mathematics: dimensional (spacetime) mathematics and dimensionless (frequency) mathematics: constrained by physical and ontological constants; permanently dynamic (everything is always in motion: there can be no static state for anything that exists).

It might be said that the equations that control motion are static, but, on the other hand, they are directing permanent motion, so they are always put into effect through motion, hence are not static.

Thesis: Frequency mathematics: dimensionless.

Antithesis: Spacetime mathematics: dimensional.

Synthesis: Ontological mathematics: dimensional and dimensionless, with the interaction between dimensional and dimensionless mathematics being enacted by Fourier mathematics, the core of ontological mathematics.

The Pythagoreans

For the Pythagoreans, the tetrad (the number 4) was the cosmic number, the root of all things, the fountain of Nature and the most perfect number. God himself, the Number of Numbers, was, in his cosmic aspect, the tetrad. (In his inner aspect, he is the Monad, the One that is the source of all.) The decad (the number 10) is composed of 1, 2, 3 and 4. Since the decad is made of the first four numbers (a tetrad of numbers) then the decad is mystically linked to the tetrad and partakes of its sacred nature and perfection.

The tetrad is symbolic of God. The decad is God's other number and is in some sense just the tetrad in a different aspect. The tetrad, being halfway between 1 and 7, is the centre of the week. There are four seasons of the year. There are four classical elements. The four positive integers (1, 2, 3 and 4) constitute the basic ratios of musical harmony. The tetrahedron is the first perfect geometric solid.

In one scheme, the Pythagoreans described the human soul as a tetrad since the soul has four powers: mind, science (knowledge), opinion, and sense (or we might say intuition, thinking, feelings and the senses in Jungian terms).

The tetrad was considered the nexus number that connects all beings, elements, numbers, and seasons.

Music and Mathematics

“Music theorists sometimes use mathematics to understand music, but music has no axiomatic foundation in modern mathematics. Mathematics is ‘the basis of sound’ and sound itself ‘in its musical aspects... exhibits a remarkable array of number properties’, simply because nature itself ‘is amazingly mathematical’. Though ancient Chinese, Egyptians and Mesopotamians are known to have studied the mathematical principles of sound, the Pythagoreans of ancient Greece are the first researchers known to have investigated the expression of musical scales in terms of numerical ratios, particularly the ratios of small integers. Their central doctrine was that ‘all nature consists of harmony arising out of numbers’.

“From the time of Plato, harmony was considered a fundamental branch of physics, now known as musical acoustics. Early Indian and Chinese theorists show similar approaches: all sought to show that the mathematical laws of harmonics and rhythms were fundamental not only to our understanding of the world but to human well-being. Confucius, like Pythagoras, regarded the small numbers 1,2,3,4 as the source of all perfection.

“To this day mathematics has more to do with acoustics than with composition, and the use of mathematics in composition is historically limited to the simplest operations of counting and measuring. The attempt to structure and communicate new ways of composing and hearing music has led to musical applications of set theory, abstract algebra and number theory. Some composers have incorporated the golden ratio and Fibonacci numbers into their work.” –

Wikipedia

Musical Intervals

“The term musical interval refers to a step up or down in pitch which is specified by the ratio of the frequencies involved. For example, an octave is a music interval defined by the ratio 2:1 regardless of the starting frequency. From 100 Hz to 200 Hz is an octave, as is the interval from 2000 Hz to 4000 Hz. The intervals which are generally the most consonant to the human ear are intervals represented by small integer ratios. The intervals which are perceived to be most consonant are composed of small integer ratios of frequency. The octave (2:1), fifth (3:2), and fourth (4:3) are the intervals which have been considered to be consonant throughout history by essentially all cultures, so they form a logical base for the building up of musical scales.” –

<http://hyperphysics.phy-astr.gsu.edu/hbase/music/mussca.html#c2>

The Tetraktys

The four levels of the Tetraktys ($1 + 2 + 3 + 4 = 10$) incorporate the basic elements of all of arithmetic. The Tetraktys comprises the Monad, the Dyad, the Triad and the Tetrad and sums to the Decad.

“The tetraktys... is a triangular figure consisting of ten points arranged in four rows: one, two, three, and four points in each row, which is the geometrical representation of the fourth triangular number. As a mystical symbol, it was very important to the secret worship of the Pythagoreans. ...

1. The first four numbers symbolize the harmony of the spheres and the Cosmos as: (1) Unity, (2) Dyad – Power – Limit/Unlimit (peras/apeiron), (3) Harmony (4) Kosmos.

2. The four rows add up to ten, which was unity of a higher order (The Dekad).

3. The Tetractys symbolizes the four elements — fire, air, water, and earth.

4. The Tetractys represented the organization of space:

1. The first row represented zero-dimensions (a point).

2. The second row represented one-dimension (a line of two points).

3. The third row represented two-dimensions (a plane defined by a triangle of three points).

4. The fourth row represented three-dimensions (a tetrahedron defined by four points)” – Wikipedia

The tetraktys was the source of all mortal and material things. It contained the nature of the universe, hence was also called *kosmos*. Other names for it were *pan* (all; everything), *ouranos* (the sky and heavens of ancient Greece, imagined as a solid dome, decorated with stars) and *decad*, the sacred ten. The tetraktys encapsulates and symbolizes all harmonies because 1, 2, 3 and 4 are used in the harmonic ratios of fourth; fifth, and octave. Apollo’s Oracle at Delphi, being the source of divine truth and reason, was also called the tetraktys.

The tetraktys stood for the ordered world (the cosmos), its harmonious nature and divine origin. Order in the universe is everywhere manifest, at every level. The heavens exhibit wondrous order, as does the soul and music, and the whole of nature, especially in the perfect solids. Beautiful numerical proportions and relations appear everywhere. The task of the wise is to observe these proportions, unlock their numerical secrets and reflect their divine harmonies.

Pythagoras was the supreme mathematical sage and master of the divine tetraktys. The Pythagoreans offered a prayer to the Tetraktys (also known as the Mystic Tetrad): “Bless us, divine number, thou who generated gods and men! O holy, holy Tetractys, thou that containest the root and source of the eternally flowing creation! For the divine number begins with the profound, pure unity until it comes to the holy four; then it begets the mother of all, the all-comprising, all-bounding, the first-born, the never-swerving, the never-tiring holy ten, the keyholder of all.”

Pythagorean initiates were required to swear a secret oath by the Tetraktys: “By that pure, holy, four lettered name on high, nature’s eternal fountain and supply, the parent of all souls that living be, by him, with faith find oath, I swear to thee.” And they also took an oath to Pythagoras himself, discoverer of the Tetraktys: “I swear by him [Pythagoras] whom to our mortal race did bring the Tetraktys, of ever-flowing nature, the spring.”

The infinitely seductive song of the Sirens was also referred to as the tetraktys, and was said to reflect, for mortal ears, the divine, cosmic Music of the Spheres.

Kabbalah with its Tree of Life depicting ten spheres of emanation (the sephirot) is derived

from the divine Tetraktys.

The Tetragrammaton (the Hebrew four-lettered name of God) reflects the Mystic Tetrad and Tetraktys.

The Tetraktys is the cause of all things and makes all things, hence is the intelligible God fully expressed. The Monad is God in himself. The Tetraktys is God and his world.

Plutarch gave an alternative number for the cosmic Tetraktys: 36 (rather than 10). This consisted of the first four odd numbers ($1 + 3 + 5 + 7 = 16$) added to the first four even numbers ($2 + 4 + 6 + 8 = 20$).

“Pythagoras seems to have honoured most of all the study of numbers... likening all things to numbers. For number holds up all other things and there is a proportion of all numbers to each other. ... Monad is the beginning of number and number is the plurality composed of monads.” – Aristoxenus

“[The Pythagoreans] construct the [whole of] nature out of numbers.” – Aristotle

“All things that are known have number; for it is not possible for anything to be understood or known without it.” – Philolaus

“Nothing could be known if everything were unlimited.” – Philolaus

Numbers

Numbers are the basis of epistemology and ontology. They unite knowledge and existence in a single entity. The studies of knowledge and existence are unified in mathematics and in mathematics alone. Only mathematics allows absolute knowledge of existence to be attained. Number is the extension or repetition of unity.

“The beginning of all things is the monad. From the monad there arises the indeterminate dyad which then serves as the passive material to the monad, while the monad serves as active cause. From the monad and indeterminate dyad there arise numbers; from numbers, points; from points, lines; from lines, plane figures, solid figures; from solid figures, perceivable bodies compounded of the four elements, fire, water, earth, air.” – the Pythagorean system

“What is the wisest? Number ... What is most beautiful? Harmony. ... the whole universe is number and harmony.” – the Pythagorean view

“Whatever is anxiously sought after by the lovers of learning, was brought to light by Pythagoras.” – Iamblichus

“Of rational animals one kind is a God, another man, and another such as Pythagoras.” – Iamblichus

Many Pythagoreans came to wonder if Pythagoras was not in fact Apollo (God of Reason), incarnate on Earth. When asked whether he was wise [*sophos*] like a God, Pythagoras replied, “No, I am a lover of wisdom [*philo-sophos*].” And thus the very word and ideal of “philosophy” was born. And philosophy, for Pythagoras, was mathematics. That’s exactly what philosophy should be now: the epistemology, ontology and phenomenology of mathematics.

Given that the ancient Greeks didn’t embrace a complete mathematical system and tried to work out reality based on an invalid, incomplete, inconsistent subset of mathematics – exactly as modern scientists do with their obsession with positive real numbers – there’s not really much point in trying to find hidden truths in how they analysed the world. The Pythagoreans got the

“big picture” right about reality, but many details were handled wrongly. Illuminism is dialectical. Illuminism is much more accurate now than it was thousands of years ago when Pythagoras first began to formalize it. The idea that the ancients had access to superior knowledge than is available to everyone today is absurd and itself a Mythos.

Leibniz was much taken with the binary system as the explanation of everything (1 = “being” and 0 = “nothing” or “non-being”). The Pythagorean “Dyadic system” is just a precursor of this type of thinking. It might be fascinating, but it’s imbued with mysticism. Moreover, the Dyadic system sees 1 and 2 as somehow extremely ontologically different, but given that $2 = 1 + 1$ (thus the Dyad is just two Monads!), it’s hard to see how this line of argument is productive.

The Quadrivium

“The quadrivium (plural: quadrivia) comprised the four subjects, or arts, taught in the Renaissance Period, after teaching the trivium. The word is Latin, meaning “the four ways” (or a ‘place where four roads meet’), and its use for the four subjects has been attributed to Boethius or Cassiodorus in the 6th century. Together, the trivium and the quadrivium comprised the seven liberal arts (based on thinking skills), as opposed to the practical arts (such as medicine and architecture).

“The quadrivium consisted of arithmetic, geometry, music, and astronomy. These followed the preparatory work of the trivium made up of grammar, logic, and rhetoric. In turn, the quadrivium was considered preparatory work for the serious study of philosophy (sometimes called the ‘liberal art par excellence’) and theology. These four studies compose the secondary part of the curriculum outlined by Plato in *The Republic*, and are described in the seventh book of that work. The quadrivium is implicit in early Pythagorean writings... As Proclus wrote:

““The Pythagoreans considered all mathematical science to be divided into four parts: one half they marked off as concerned with quantity, the other half with magnitude; and each of these they posited as twofold. A quantity can be considered in regard to its character by itself or in its relation to another quantity, magnitudes as either stationary or in motion. Arithmetic, then, studies quantities as such, music the relations between quantities, geometry magnitude at rest, spherics [astronomy] magnitude inherently moving.” – Wikipedia

Schrödinger

There are two versions of the Schrödinger equation of quantum mechanics: 1) time-dependent (the main one), and 2) time-independent (which is seen as a special case of the first). For simplicity, the equations are usually specified with regard to one spatial dimension.

In ontological mathematics, the basic dimensionality is not one spatial dimension but, rather, the complex plane, with one real spatial dimension and one imaginary spatial dimension. Imaginary space takes the role of “time” and so time is *always* present in the wavefunction.

In Illuminism, motion is a primary feature of existence. A “time-independent” spacetime wavefunction is relevant only where the same pattern is repeated over and over again so that it looks as if the solution is static (hence unchanging through time). Of course, if the system were to be examined at the microscopic level, it would be seen to be changing constantly and not to be static at all. It’s static *over* time (an averaged period of time) but not *through* time (on an instant-by-instant basis).

Ontological mathematics makes complex numbers the primary descriptors of spacetime, while physical mathematics tries to use real numbers only.

No Infinity

There’s no such thing as infinity in a dynamical ontological mathematical system. The

flowing point has a tiny size, enough to make zero and thus infinity impossible. The universe has a smallest number that can be reached (zero is not a true number but rather a *resultant* number), and that means there's also a largest possible number (the inverse of the smallest number), after which mere repetition occurs.

Is 1 divided by 0 a different infinity from 2 divided by 0, or simply the first infinity doubled (extended; repeated)? Once we have one infinity, do we simply keep repeating it? Is there a hidden ontological means to link "countable" and "uncountable" infinities?

Subjective and Objective

The miracle of math – and the basis of life – is that it can be both subjective and objective. In terms of the dimensionless frequency domain, mathematics is devoted to pure thinking and subjective experience, i.e. a mathematical monad is something that has feelings towards the mathematical information it receives, processes and generates. If a computer could have feelings about the information it was processing, it would be alive too. Of course, a computer simply executes programs created by beings external to it, and over which it has no say, interest, motivation, intentionality or feelings, so a computer remains nothing but a machine.

Mathematics generates the objective, non-feeling, material world of science through the mechanism of Fourier mathematics, which generates a spacetime domain from a frequency domain. Mathematics produces an objective world and then experiences it subjectively. This is the basis of life.

Plato and Descartes had the right idea about this mind-matter dualism (mind = dimensionless, immaterial existence, and matter = dimensional, material existence), but they did not have the mathematics at their disposal to account for it. Leibniz intuited the answer, but he too lacked the full mathematical tools necessary. Only with the advent of Fourier mathematics at the start of the nineteenth century was it possible to fully explain the interaction of mind and matter. Fourier mathematics (as recognised by the Illuminati alone and not by the general mathematical community, and not even by Fourier himself), turned the Cartesian unextended, thinking domain into an eternal monadic frequency domain (the Soul Domain), and the Cartesian extended, material domain into a spacetime domain (the World).

Reality is nothing but the Fourier interaction of the Soul Domain and the World, proceeding by way of Fourier reincarnation (i.e. we keep acquiring, via reincarnation, new mathematical spacetime bodies to which our mathematical frequency minds can link). That's the entire basis of mind-matter dualism, except it's not a dualism at all but a mathematical monism. Fourier mathematics is a single mathematical system with a dual, interlinked outcome. That dual outcome is what is interpreted as dualism. Dualism is philosophical nonsense since two genuinely dualistic substances cannot interact. Quasi-dualism grounded in mathematical monism overcomes this objection.

The miracle of existence is the miracle that Fourier mathematics can present the same information in two radically different ways. Is it not extraordinary that dimensionless frequency functions can be converted into dimensional spacetime functions, and back again? If this were not possible, life as we know it would be impossible. We are all the expressions of Fourier mathematics, and the sooner everyone understands that, the better. Everything else is pure nonsense.

The Arrogance

Stephen Hawking was disappointed with the discovery of the Higgs boson because, he said, it eradicated a scientific mystery. Life would be far more interesting, he thought, if the particle was shown not to exist because that would mean the standard model of physics would have to be rethought, a great challenge for science.

Hey, Stephen, haven't you been keeping up: the standard model fails to account for 95% of the universe and has completely failed to support a quantum theory of gravity. Moreover, it literally does zero to explain mind, life, consciousness and free will. The idea that the standard model of physics will survive is moonshine. The whole of physics will be replaced in due course

– by ontological mathematics.

The Higgs boson is not the explanation of why matter has mass. The answer to that is ontological Fourier mathematics. The Higgs boson is simply a particle mandated by Fourier mathematics. As ever, scientists wholly misinterpret the facts and evidence. As always they always use contingent arguments, and never necessary arguments.

Is

To refer to “is” (e.g. “What exists is....”) is traditionally interpreted as referring to “being”, to enduring objects.

Yet this isn’t what “is” means at all. “Is” can never be dissociated from “now”, the present instant. It has no relevance to the past or future (which don’t exist). The only things that exist are those that exist right now. Whatever exists right now will be in a slightly different context in the next instant: everything in the universe will have changed very slightly. The right word for this process is “becoming”, not “being”.

“Is” refers to the current state of the process of becoming, not to any enduring thing. When Heraclitus conceived of ultimate reality as fire, he captured the right sort of notion. Fire is never static; it’s always in motion. Any specific element of the fire is not fixed and static. To say that what exists is fire is to say that the world is always changing and becoming.

If Heraclitus was the philosopher of becoming, Parmenides was the philosopher of being. Parmenides denied the possibility of real change and of multiplicity. The true logic of being leads to a single object that fills everything and never changes. Bizarrely, the best candidate for this is “nothing”.

The difference between Parmenidean being and Parmenidean non-being is simply that the former is an object that does nothing and the latter is not an object, but also does nothing. There would be no way of differentiating the two states since to investigate either would require separate things that change, and no such things exist (according to Parmenides).

A limitless block of something that does nothing and a limitless block of nothing that does nothing are functionally identical: neither supports meaningful existence.

With fire, Heraclitus introduced a substance that was always changing and always becoming. An ever-changing fire is, however, no improvement over nothing if all it does is change, but not to any end (i.e. it has no teleology).

Imagine a limitless block of Chaos – an ever-changing substance. Well, if all it does is randomly change, it still hasn’t given any meaning to existence.

What is required is a substance that has a purpose and strives towards that purpose. Yet for anything to have purpose, it must possess the quality of mind since mindless objects, as in scientific materialism, are devoid of teleology. Modern science specifically ridiculed Aristotle for putting teleology at the heart of his version of science, and rejected this wholesale (thus, catastrophically, removing mind from science). Plato and Aristotle viewed the world as a living organism: modern science sees it as a dead machine.

It’s no good simply having a purpose. You must be able to achieve it, at least in principle. You must be able to take meaningful steps towards it. So, for existence to be meaningful we need a substance of becoming rather than being, a substance with teleology (hence mind), and an ordered substance that can quantitatively and qualitatively alter the state of its order to increase its order and organisation, and thus its power. It must be able to convert potential into actuality.

Any substance with teleological mind automatically has life since life is precisely that which strives towards goals. If a non-teleological mind existed that did not strain towards any object, it would be no different from the block of static something that does nothing, the block of nothing that does nothing, and the block of dynamic something that does nothing but change to

no end. Mind is pointless unless it's teleological. If a computer had a mind, it would be a pointless mind, since the computer has no goals, so would not set out to accomplish anything (it would not exhibit the quality of life).

Why does science utterly fail to account for life? Because life involves mind and teleology, both of which are fundamentally rejected by scientific materialism. Science simply has no conceivable tools for investigating mind, life, consciousness and free will since none of these feature, or can feature, in the Meta Paradigm of materialism and empiricism.

Science has never once attempted to rationally explain how lifeless, mindless, purposeless, non-conscious atoms, wholly lacking free will can give rise to living creatures with minds, purposes, consciousness and free will. Which means that, in ultimate terms, science is a 100% false account of reality. It absurdly talks of the process of "emergence", which it might as well redefine as "magic" since we are expected to believe that life simply emerges from non-mind just by mixing atoms around in more complex ways. A complex arrangement of atoms can never be any better than a simple arrangement of atoms at generating something it does not possess (life, mind, purpose, consciousness, free will). In either case, complex or simple, the task is rationally impossible.

Existence is therefore about a living substance, a mental substance, a changing substance of becoming, a teleological substance, a substance of order and organisation rather than chaos, a substance that can convert potential into actualisation, a substance that can become more ordered and organised, thus increasing its actuality and power.

There is of course only one candidate that fits the bill: living mathematics, conveyed by countless, autonomous living mathematical minds: monads – souls!

Being: Something, such as an object, an idea, or a symbol, that exists, is thought to exist, or is represented as existing; the totality of all things that exist.

The word "being" is essentially false. Everything is becoming. We should refer to "becomings" rather than "beings".

I am becoming

You are becoming

He, she or it is becoming

You are a living mathematical process or organism, not an enduring object. A monadic soul is a permanent becoming. It's always changing. It has an identity but that identity is carried through change. We are the result of our experiences. Our changing experiences create our identity. It's not our identity that creates our changing experiences.

"Is" never refers to anything enduring (since there are no such things). "Is" always refers to the present state of affairs, and that state will be different in the next instant. "Is" is being permanently subverted and altered.

Energy

A word that can be used for living mathematics is *energy*. Existence consists of nothing but mathematical energy seeking to optimise itself, which, since it comes in monadic collections of energy rather than one vast, unitary block of energy, means that each individual energy collection (monad) is striving to optimise itself. This leads to a competitive dialectic, and to all the conflict, binary oppositions and “evil” that we encounter in the world.

Schopenhauer conceived of ultimate reality as a single mental entity outside space and time (the Will). His view was derived from the living Oneness that underpins Buddhism.

If such a system were true, there would be no conflict in the universe. You would simply have one organism (“God”) trying to optimise itself, and it would have nothing else with which to compare how it was doing, and nothing else with which to compete and conflict.

If there’s a sufficient reason for one entity outside space and time, there’s a sufficient reason for any number of that entity outside space and time. If one monad is possible, countless monads are too, exactly as envisaged by Leibniz. A universe of countless autonomous centres of power all trying to maximise their power, and all either competing or cooperating with each other, leads to exactly the world we observe. Enlightened people see cooperation with others as the best way to increase their mutual power. Unenlightened people pursue the master-slave dialectic and want to be put others down to raise themselves up.

Confusion

“To be honest, nothing has been as confusing to me as quantum mechanics interpretations and the contemporary issues in the philosophy of science.” – P

That’s because the conventional interpretations of quantum mechanics are absurd.

Certainty?

“Not to be absolutely certain is, I think, one of the essential things in rationality.” – Bertrand Russell

In fact, the whole point of the principle of sufficient reason – the basis of rationality – is to be absolutely certain. If you don’t want to be certain, be an irrationalist, like the scientific materialists, the analytic (linguistic) philosophers or the religious believers! Everyone can be 100% certain that $1 + 1 = 2$.

Resolution

The ancient Greeks more or less solved every problem. The only thing they lacked was powerful enough mathematics to rationalise it all.

Everything that is infallibly true has been infallibly true *forever*. These are the eternal truths of reason. Their very existence proves conclusively that the universe is immortal. They are analytic, necessary, *a priori* and deductive, not synthetic, contingent, *a posteriori* and inductive.

Mathematical Idealism

Illuminism is what you get when you apply mathematics to idealism, rationalism and metaphysics rather than to materialism, empiricism and physics (as in science).

The Big Question

Peter Higgs was awarded a Nobel Prize when the CERN Large Hadron Collider confirmed the existence of the Higgs boson. This is utterly inconsequential in relation to the biggest question of all – is mathematical idealism, rationalism and metaphysics correct, or mathematical materialism, empiricism and physics? The ultimate nature of existence, and the possibility of a Grand Unified Theory of Everything hinges on exactly this question. Metaphysics (“what comes after physics”) is about an unobservable mathematical domain – a frequency Singularity defined by the God Equation. Physics demands that everything be observable.

So, here’s a challenge for a world-historic genius such as Kurt Gödel (with his Incompleteness Theorems) or John Bell (with his Inequality Theorem). Can you construct a definitive, incontestable, logical test to settle whether ultimate reality must be observable or unobservable? Conventional science can offer no way forward since it’s predicated on everything being observable. (It simply begs the question.)

The Truth

“Nobody wants the truth. Nobody wants honesty. They are embarrassed to admit they were tricked.” – K

That’s why it’s so difficult to change anything. People would prefer to believe the lies they were raised with rather than accept rational truths that contradict those lies. It’s fundamental human nature: fear of change, of anything different, anything unusual and challenging.

Eternal Structures

“In Illuminism, can we consider the subject-object divide to be rooted in the nature of a monad’s contents? This is the way I have come to see it so far: The objective universe is made up of analytic *a priori* structures and for all intents and purposes is Gödel-complete. However, the way in which a monad perceives the objective domain through its evolving contents is Gödel-incomplete since the relations (not the actual mathematical laws!) that exist within the monad’s contents are oftentimes arbitrary and unique.” – P

Yes, that’s right. The monadic mathematical framework is the fixed part (or what we might loosely call “being”), but, within each monad, the mathematical informational content is evolving uniquely (“becoming”). Being is objective and eternal; becoming is subjective and contingent. At the Omega Point of dialectical becoming, a monad realises what the total eternal laws of being are. This is the condition known as *gnosis* – when God-consciousness is attained. Once you know the eternal laws of existence, you have the power of God since you know exactly how everything works and you can effortlessly work out and resolve any problem.

Hidden Variables

Hidden variables are hidden to the senses, not to reason. They're completely *visible* to reason, as part of ontological mathematics. There's nothing "spooky" or "speculative" about them. The supreme irony is that quantum physics, which scoffs at "hidden variables", is based on an unreal, unobservable, abstract wavefunction, which has to have various things done to it to make it "real". Even then, it's not deterministically, causally real. Rather, it's indeterministic, random, probabilistic and statistical. What would you rather have? – an intelligible, rational, real mathematical world of hidden variables, or an unintelligible, non-causal, unreal wavefunction that renders reality probabilistic? Well, does God play dice or not?

Possibility and Compossibility

A "possible" thing is something we consider in isolation, in no context or environment, with no regard to other things.

A "compossible" thing is something we consider with regard to other things, in a context or environment.

It's self-evident that compossibility is a much more powerful, realistic and valid way of thinking about existence than mere possibility. Scientific Multiverse thinking is quite happy to construct whole universes for those things that are possible but not compossible, the worst conceivable breach of Occam's Razor.

The Selfish Gene

According to the Selfish Gene theory, living organisms exist only to propagate their genetic material. This attributes an intentionality, a purpose, a *teleology*, to the allegedly lifeless, mindless, purposeless atoms of genes.

Of course, it's not genes that determine our behaviour. It's our behaviour that determines the fate of our genes. If we choose not to reproduce, our genes are removed from the gene pool. It's as simple as that.

The Science Delusion

1) Newtonian Gravity – indistinguishable from voodoo. Effects are transmitted across any amount of space instantaneously without any medium of transmission or any possible mechanism of transmission.

2) Einsteinian Relativity – the wholesale destruction of objective reality. It's no longer possible to say consistently and absolutely what dimensions anything has because it's all relative, dependent on the viewpoint of the subject. Subjectivity becomes the cornerstone of science. There is no reality principle.

3) Quantum Mechanics – reality is said to derive from an unreal, unobservable wavefunction rooted in total indeterminacy. Cats are said to be simultaneously alive and dead. Wavefunction “collapse” results from observation, implying that observers determine reality. This is a system of absolute subjectivity and the total denial of the reality principle. The moon doesn't exist when no one is observing it.

4) Multiverse – there are infinite copies of you in infinite parallel universes. All possibilities are actualized in one universe or another. There is no meaning in any universe, no choice, no free will. Anything that can happen will happen, and does so mechanically.

5) The Machine Universe – mind is created by matter and is subject to inexorable, inescapable, deterministic atomic laws. Free will is impossible.

Are you swallowing the science Mythos? How easy it is to con the gullible, ignorant, credulous masses.

Science is a quasi-religion, demanding faith, not reason. In many ways, science is obscenely silly, making irrational claims that are as preposterous as any of those of religion.

Any system of thought that cannot account for the most obvious fact of our existence – our own free will – is pure junk.

What is the central problem of science? It's that it's getting more and more mathematical, and mathematics is the über-rationalist subject. The trouble lies not with the mathematics itself – in fact, the more mathematical the better! – but with the philosophical ideology used to interpret the meaning of mathematics. Science is based not on idealism and rationalism but on materialism and empiricism, with the latter being the direct opposite of rationalism. So, scientists attempt to apply a formally irrational set of dogmas to interpret the exclusively rationalist discipline of mathematics. This, of course, is absurd, and leads to precisely the deranged claims made by modern science.

Never forget, the mathematics of science is generally robust – hence why science is so successful. It's the *interpretation* of the mathematics that's so ludicrous and false.

To understand what mathematics means, you have no choice but to use rationalist thinking of exactly the type used by Leibniz. Leibniz was a genius in mathematics, philosophy, science and law. Newton was hopeless at philosophy, not an elegant mathematician (which is why no one uses his “fluxions” method of calculus), but certainly a talented physicist. He was also a religious maniac, believing himself the last prophet of God.

The world chose to follow Newton and not Leibniz. When it comes to its senses, it *will* follow Leibniz. The more people perceive the truth to be mathematical, the more they will join the camp of Leibnizian rationalism – *Illuminism*.

Divine Intoxication

Are those filled with the “holy spirit” drunk? No alcoholic spirit is stronger than the Holy

Spirit.

Paradox

Classical science depicted energy in mutually exclusive terms: as particles (concentrated) or waves (spread out). Quantum mechanics then produced the concept of wave-particle duality, despite its inherent logical impossibility within the framework of science.

It is in fact rationally easy to reconcile particles and waves within the framework of Aristotelian and Leibnizian metaphysics. Energy is delivered by flowing points (particles), which have an inner *form* – that of mathematical wave motion. In other words, the fundamental assumption of classical science that particles move, by default, in straight lines is false. All particles in fact move as waves. The only thing that is real is a cosmic wavefunction, and all the phenomena of physics derive from this all-embracing wavefunction.

It's the Form of the wave (built into all particles) that's the true secret of science. Mainstream science has only started to encounter this truth via quantum mechanics, which it has, however, wholly failed to understand, hence why there's no agreed interpretation of what quantum mechanics means.

Science says things are *both* waves and particles. In fact, all things are flowing points (particles) that follow an inbuilt wave trajectory.

Things aren't particles *and* waves. They are particles following wave trajectories – an entirely different concept.

Since wave mathematics is at the heart of reality, it means that interference patterns are fundamental to reality, and that's why the famous twin-slit experiment of quantum mechanics delivers the results it does, results that are incompatible with any possible model of classical physics.

The extraordinary thing is that, despite the advent of quantum mechanics, scientists still cling to classical thinking. They refuse to see the truth – that particles have an inbuilt wave program that directs their motion and behaviour. The results of the twin-slit experiment are self-evident as soon as it's realised that particles are guided by pilot waves built into their motion.

The pilot wave, it must be noted, is not something that physically extends from the particle (as some people have suggested). Rather, it is *implicit* in the particle, guiding it in a precise, analytic wave trajectory.

Max Born won the Nobel Prize for his outrageous probabilistic interpretation of quantum mechanics – one of the most disastrous ideas of all time. (It's amazing how often the Nobel Prize has been awarded to people for ludicrous ideas that will in future be regarded as mad or comical.)

Rather than having real particles being guided by a precise inner wave program (which in fact is the generalised Euler Formula), Born proposed that particles aren't real things at all but, rather, they are entities that pop in and out of existence according to a probabilistic wavefunction that itself has no physical reality. Incredibly, this remains the standard way of thinking about quantum "reality". Reality has been abolished and replaced by a bizarre probability cloud, from which particles miraculously appear in random locations. No links at all can be established between where a particle is at one instant and where it will be at the next instant. It could randomly show up on the other side of the universe! There is no sufficient reason in Born's system. Formal causality has been swept away and replaced by probability.

In fact, it's extremely difficult to account for how anything deterministic can happen at all in a universe grounded in absolute randomness (in the sense that even though probability applies, there is nothing to prevent incredibly improbable things from happening and indeed any such absurdity must actually happen if it's not explicitly forbidden).

Why isn't the universe just a big mush of chaos with things randomly popping up

anywhere at any time and causing crazy things to happen? The fact that our universe is so stable and predictable is the surest proof that Born's probabilistic model is utterly false. Crazy things simply aren't happening, even though they ought to be given Born's model, which lacks basic causality.

Born's line of thinking leads to the most absurd claim of all – that a cat in a box can be both dead and live until someone opens the box and takes a look. Schrödinger invented this paradox precisely to illustrate how idiotic the probability interpretation of the wavefunction was. Ironically, and tragically, his paradox is now taught to illustrate the *truth* of Born's interpretation! This really is the silliest idea of all time, outside of religion. It might even be worse than any religious claim. No one observed the Big Bang, so how did the universe ever escape from the same fuzzy superposition of states that allegedly applies to the unfortunate cat? The degree of illogic in this proposal is simply staggering. How did observers ever come into existence in the first place? Who observed the observers? You might as well go back to Bishop Berkeley's position that God is the ultimate observer who guarantees reality!

In Illuminism, wave motion is programmed into authentic particles. For Max Born, an unreal wavefunction generates particles according to probability, and these same particles could then reappear absolutely anywhere in the universe since the wavefunction caters for the whole observable universe and assigns a non-zero probability to every location. In truth, all of Born's particles should be referred to as "virtual" since none of them has any causal, trackable continuity.

Einstein, like Schrödinger, rejected this probabilistic interpretation and asserted that there must be hidden variables that explained things properly. The scientific establishment then turned on him – one of their own! – and dismissed him as an out-of-touch old man. That's gratitude for you.

David Bohm resurrected the hidden variables concept and he too was ridiculed. However, the great genius John Bell showed that the hidden variables theory was not at all ridiculous and could not be refuted by its opponents.

Illuminism, based on rational mathematical unobservables would be classed as a hidden variables theory that gets rid of Max Born's probability interpretation of quantum mechanics.

John Wheeler said, "No elementary particle is a phenomenon until it is registered." This either means that elementary particles are noumena until they are registered (which no scientific materialist would ever accept!) ... or that they simply don't exist as real things until they are observed, in which case there is no such thing as objective reality. Science has abolished the reality principle! In 2000, Wheeler said, "...we still do not know 'how come the quantum?'".

That's the least of his problems. How come "reality"?!

The Final Synthesis?

Thesis = Illuminism

Antithesis = M-Theory, String Theory and Supersymmetry

Synthesis = Omega Illuminism

Constants?

There are various universal constants, all deriving from the mathematical properties of the generalised Euler Formula. These global constant apply everywhere.

What about *local* constants? In conventional calculus, positional constants disappear during differentiation and can't be recovered during integration. Do such local, positional constants actually exist? In a *dynamic* universe, an expanding universe, there are in fact no such constants, hence this calculus problem never arises. Everything changes other than the global

constants and the laws of mathematics.

Dimensionality: The Subtlety

“In mathematics, the dimension of an object is an intrinsic property independent of the space in which the object is embedded. For example, a point on the unit circle in the plane can be specified by two Cartesian coordinates, but one can make do with a single coordinate (the polar coordinate angle), so the circle is 1-dimensional even though it exists in the 2-dimensional plane. This intrinsic notion of dimension is one of the chief ways the mathematical notion of dimension differs from its common usages.” – Wikipedia

M-theory and Illuminism

In an important sense, M-theory and Illuminism aren't radically different.

M-theory says that reality is based on one-dimensional energy string loops while Illuminism says that reality is based on one-dimensional energy Euler circles (but just as a 1-D string loop might be said to exist in a 2D real plane, an Euler circle equivalently exists in a 2D complex plane). To this extent, they are highly compatible.

However, M-theory sets its strings within an eleven-dimensional stage consisting of three "big" spatial dimension, seven "small" spatial dimensions and one time dimension. The strings, it must be emphasized, do not explain this 11-D arena, hence this is an ad hoc add-on.

In Illuminism, the Euler circles generate the space and time arena via their own properties, hence fully explain space and time without recourse to any ad hoc theoretical additions. In Illuminism, space is based on real numbers and time on imaginary numbers. Space and time are both types of space: conventional space is real space while time is imaginary space, orthogonal to real space. There are as many temporal dimensions as spatial dimensions and no dimensions are "rolled up" into near non-existence (as happens in M-theory). In M-theory, time is the source of change and movement. In Illuminism, time is static (it's just a type of space) and energy itself is the source of change and movement, and provides the arrow of time (motion).

In M-theory, there's no such thing as dimensionless existence. In Illuminism, dimensionless existence – meaning that positive and negative dimensional elements completely balance each other – is the ground of reality. Dimensionless existence is the basis of mind (soul). In M-theory, mind (soul) is absolutely rejected.

Illuminism is a rationalist, idealist, analytic theory of mathematics.

M-theory is an irrationalist, empiricist, materialist, synthetic theory of physics.

M-theory, if it became rationalist and analytically mathematical, would simply convert into Illuminism, and humanity would at last have its final theory of everything.

The Photoelectric Effect

Einstein won the Nobel Prize for Physics for explaining the photoelectric effect. The photoelectric effect involves shining light (of high enough energy) on to a metal, causing electrons to be emitted from the metal and thereby generating an electric current.

It was observed that light below a threshold frequency would lead to no emission, no matter how intense the light was (with intensity being the number of photons per second for a specified area). On the other hand, low intensity light above the threshold frequency would succeed in releasing electrons.

This means that the total amount of energy (the intensity of light) isn't what's important but rather the specific energy per light photon. You could use a massively intense light beam, and keep increasing it, and it would have no photoelectric effect whatsoever if the photons were of insufficient frequency. However, if the photons were of the right frequency then increased intensity would lead to an increased photoelectric effect.

Every metal has a characteristic threshold frequency of light with which to create the photoelectric effect.

It's often said that the photoelectric effect provides convincing evidence for the particle nature of light in certain circumstances. However, it could be said that what the photoelectric effect really demonstrates is a profound difference in the energy of a light beam judged in terms of amplitude on the one hand and frequency on the other. A high amplitude, low frequency light

beam is very different from a low amplitude, high frequency light beam. A high amplitude, low frequency light beam might have an enormous amount of energy, but be unable to budge a single electron from an atom on the surface of a metal. A low amplitude, high frequency beam can deliver much more energy at a specific point and thus release an electron.

The photoelectric effect was a conundrum because classical physics said that the emitted electrons must acquire their kinetic energy from the light beam shining on the metal surface. If the light were envisaged in terms of ocean waves, and the electrons in terms of pebbles on the beach then, obviously, the more intense the ocean activity, the more pebbles should be displaced on the beach. This is the standard, classical wave understanding of light. The problem with this view is that it likens light to water and says that the only way of changing the energy of the ocean is via changing the amplitude of the waves. But what if there were countless different types of ocean, each with a different inherent energy? An ocean with a high inherent energy but low amplitude might be able to shift far more pebbles than an ocean with a low inherent energy but high amplitude.

It's therefore not at all self-evident that Einstein was right to invoke wave-particle duality to explain the photoelectric effect. The effect could be explained just as well by saying that light waves must be considered in terms of two different properties – amplitude and frequency. A low frequency, high amplitude light wave has very different effects from a high frequency, low amplitude light wave. A light beam of extremely low intensity but extremely high frequency could be interpreted as an ocean with gentle waves but water molecules of high energy. A light beam of extremely high intensity but extremely low frequency could be interpreted as an ocean with dramatic waves but water molecules of low energy.

The Principia

“*Philosophiæ Naturalis Principia Mathematica*, Latin for ‘Mathematical Principles of Natural Philosophy’, often referred to as simply the *Principia*, is a work in three books by Sir Isaac Newton, in Latin, first published 5 July 1687.” – Wikipedia

Well, everything is certainly grounded in mathematical principles. It's a shame that Newton plainly didn't understand what this meant ontologically.

Wave-Particle Duality

What *is* a wave? What *is* a particle? Bizarrely, these have never been analytically defined by modern science. Historically, a wave was regarded as something “spread out”, and a particle as something “concentrated”. But what is it that’s spread out and what is it that’s concentrated? Science has never said.

What are light waves? What are light particles (photons)? How can they be one and the same thing? Modern science simply proclaims this “a mystery” (wave-particle duality). Well, that’s helpful, isn’t it? It might as well have called it “magic”.

So, what’s really going on?

A wave is actually a “flowing point” tracing out the trajectory dictated by the generalised Euler Formula. It’s this trajectory that constitutes the wave. The “active” part of the wave is the flowing point itself at its current location on the wave trajectory. It’s the flowing point that constitutes the “particle”.

The scientific claim that entities are sometimes waves and sometimes particles is irrational, ridiculous and almost insane. Science actually claims that something can be both spread out and concentrated at the same time, depending on how you’re observing it. So, your subjective observation becomes the crucial factor, thus contradicting the reality principle that demands that things be objectively real in all circumstances.

Of course, science has a long track record of making absurd, illogical claims. Einsteinian relativity claims that something can be simultaneously contracted and not contracted depending on who’s observing it. Quantum mechanics, in the standard interpretation, claims that observable reality is based on an unobservable, unreal wavefunction.

It’s astounding that science has been allowed to get away with these lunatic claims for so long. The trouble, as ever, was that science chose to follow the anti-rationalism of Newton rather than the rationalism of Leibniz. Newton was happy to propose successful equations as “reality”, even though no valid rational interpretation could be attached to these equations. Newton’s theory of gravity was itself the most egregious example, more or less making an appeal to “God” in order to make any sense at all. Newton was wrong about absolute space, absolute time, the speed of light, atoms and gravity – and yet he’s regarded by many as the greatest physicist of all time. He could scarcely have been more wrong. Except for one thing – his equations were highly successful.

It’s one of the incredible features of mathematics that equations based on total fallacies can nevertheless provide an extremely good approximation to the true equations of reality. So, for example, the assumption of Newtonian absolute space and time works extremely well at speeds low compared with the speed of light. It’s precisely the success of fallacious equations that makes them so hard to overturn.

Leibniz produced mind-blowingly good rational arguments about reality, but no equations on a par with Newton’s equations of motion, so Newton became a hero and Leibniz forgotten, even though only Leibnizian rationalism can make sense of existence.

There is no wave-particle duality as science understands it. Rather, there are flowing points that have the concentrated characteristics of particles and these move in a wave trajectory determined by the generalised Euler Formula. The generalised Euler Formula acts as a guiding wave or pilot wave for the flowing point (the “particle”).

Classic wave interference effects are produced via the interaction of pilot waves. The interference pattern then establishes a new trajectory for a flowing point to follow. However, the

flowing point always arrives at any location as a particle. We thus easily explain why quantum entities seem to travel as waves and arrive as particles. In fact, quantum entities always travel as particles and are always following a pilot wave.

Aristotle and Leibniz would call the flowing point “matter” and the pilot wave its “form”. True quantum mechanics is therefore all about wave *forms* determining the trajectories of “material” particles. These forms correspond to mathematical minds within the flowing points.

Science denies that particles have any minds or forms, so is left to invoke randomness and indeterminacy as the ground of reality. In Illuminism, there’s no randomness or indeterminacy, and reality is grounded in absolute mathematical rationalism and causality.

In science, particles’ behaviour results from collisions between particles. In Illuminism, particles’ behaviour results from the inner mathematical forms of particles. So, science is about factors (forces) external to particles whereas Illuminism is about factors (forces) internal to particles.

As ever, we come down to metaphysics versus physics. Forms are rational unobservables, hence metaphysical (mathematical), and science rejects all such unobservables.

Some scientists, such as Einstein and Bohm, referred to “hidden” variables. Well, there’s nothing hidden about mathematical metaphysics, about rational unobservables. These may be hidden to sensory scientific observation, but not to rational thinking, to which they are fully visible.

The difference between Illuminism and scientific materialism is that the former is based on everything having a metaphysical, mathematical inner form that is intelligible but not sensible, i.e. available to reason but not to the senses. Science, by rejecting rational unobservables, rejects mathematical metaphysics and rationalism. Science is a formally irrationalist system of thought, privileging the senses over reason, the sensible over the intelligible.

All the absurd claims of science flow from the fact that it’s constrained to explain everything in terms of its ideological, quasi-religious stance whereby everything is interpreted according to science’s Meta Paradigm of empiricism and materialism, and its two mantras that a) absence of evidence is evidence of absence and b) anything that cannot be observed does not exist.

Illuminism has no difficulty accommodating the unobservable Aristotelian-Leibnizian form (soul); indeed it’s predicated on such a form. Science dismisses form 100%. Particles, according to science, have no “interior”, no inner *mathematical* structure. What you see is what you get. Scientists can search for the Higgs boson, but they can’t search for the *form* of the Higgs boson that actually gives it its properties. Form, like mind and soul, is unobservable, beyond the reach of any experiment. Science can “see” only the outside of things, never the inside. Rationalism is about working out what this inside must be like in order for the universe to appear as it does. Science is about denying any inside and trying to work out what the universe is like based purely on external appearances.

Illuminism is about arriving at a rational understanding of an intelligible, noumenal universe. Science is about arriving at an instrumental, pragmatic understanding of a sensible, phenomenal universe. These are two radically different undertakings.

Illuminism is all about rational, definitive, analytic answers. Illuminism pursues the “rational method” based on the exercise of reason.

Science is all about sensory, contingent, synthetic “answers”. Science pursues the “scientific method” based on experiments.

Rational proofs are eternal and unarguable. Experimental data is provisional and interpretive.

So, which side are you on? – that of rationalist mathematics or irrationalist science.

The Most Complex Object in the Universe

It's often said that the human brain is the most complex object in the universe. In fact, the generalised Euler Formula is the most complex object there could ever be. It gives rise to space, time, the material world, Fourier Series, Fourier Transforms, the Quantum Universe and holography. Moreover, not only does it give rise to the brain itself but also to mind, the soul and even God. And that's why it's called the God Equation. In fact, it's the authentic Creator, the true first cause. It's the uncreated Prime Mover. It's ontological mathematics itself. God isn't a mathematician. God *is* mathematics.

That might seem horrific and incredible to most people, but it's the best possible news. It has two astounding consequences. Firstly, it means that existence has a definitive answer – the eternal truths of mathematics. Secondly, since ontological mathematics concerns universal mathematical optimization, it means that all mathematical “unities” (Leibniz's monadic souls) will eventually become perfect, i.e. they will be transformed into Gods.

We all have a divine future, but some of us will reach divinity much earlier than others, and the others will become Gods only because we made it possible for them. The “first Gods” are the “God makers”. In Illuminism, the first God of all – the God who makes it possible for everyone else to become God – is named Abraxas. He was not always God. He *became* God. He did so through his own efforts and merits. He wasn't born as God. He didn't inherit divinity. He made himself divine. *He created himself*. He was the ultimate meritocrat.

In terms of human history, Abraxas refers to Simon Magus, Grand Master of the Illuminati. He certainly wasn't the most intellectual Grand Master of the Illuminati, but he was the most powerful and was able to transcend the human condition and prove it from beyond the grave – in the object known to history as the Holy Grail, the Illuminati's most precious artefact and the most priceless object in human history.

By an astounding irony, the false Messiah (Jesus Christ) was a contemporary of Simon Magus and lived in the same part of the world.

The Five Numbers

When $x = \pi$, the Euler Formula gives rise to the famous Euler Identity:

$$e^{i\pi} + 1 = 0$$

This is the most stunning and unexpected relationship between, arguably, the universe's five most critical numbers. It's true to say that the power and significance of Euler's Identity hasn't been scratched yet. No genius has yet unpicked its lock and allowed its riches to flood forth. But, as always, it's only a question of time. Someone, somewhere will have an overpowering intuition and see some incredible ramification of the Euler Identity that will open up vast new vistas of mathematics.

The Euler Formula

Euler's Formula, ontologically, is all about energy, motion and information. Euler's Formula has two completely different aspects: dimensionless and dimensional, or, to put it another way, frequency versus space and time.

Euler's Formula always involves the motion of a one-dimensional “point” – the flowing point that carries causality with it. In the frequency domain, the flowing point that goes right round the circumference of the Euler circle passes through positive numbers and negative numbers, through real numbers and imaginary numbers, with all the numbers equally balanced

and none privileged over any of the others. The net outcome is a system of “somethings” that automatically balance to zero, leaving a resultant of nothing, i.e. dimensionless existence. So, the frequency domain is an unchanging, eternal, Platonic dimensionless domain. It’s also the platform for the immortal soul. The soul cannot perish precisely because it’s composed of indestructible, dimensionless frequencies that endure forever.

The generalised Euler Formula generates all ontologically possible frequencies. These frequencies are associated with *perfect* sine and cosine waves. Sine functions are “odd” while cosine functions are “even”. Sine and cosine functions are orthogonal to each other. Sine waves are associated with imaginary numbers (time = imaginary space) and cosine waves with real numbers (space). There are also negative sine and cosine frequencies, ensuring that all frequencies, both positive and negative, balance to zero.

These sines and cosines provide a perfect set of Fourier basis frequencies. Using these, Fourier mathematics allows us to construct extended functions in space and time, the functions associated with the material rather than mental world.

Perfect cosine waves are none other than photons (light or space particles for which no time, or alternatively imaginary space, passes). Perfect sine waves are *chronons* (time or imaginary space particles for which no real space passes). All functions that involve mixtures of sines and cosines are spacetime functions.

So, via the generalised Euler Formula and Fourier mathematics, we have absolutely everything we need to create a mental universe of frequencies and a material universe of spacetime functions. Moreover, using Fourier transforms and inverse Fourier transforms, we can effortlessly swap between the two domains (this is the basis of mind controlling matter, and matter affecting mind, and is thus the answer to Cartesian dualism).

Bear in mind that Fourier mathematics is the basis of the wave mechanics of quantum physics and also of holography.

Illuminism asserts that quantum mechanics, when understood properly, is actually holography, and we live in a cosmic hologram. Souls are themselves holographic.

So, there you have it – exactly how a universe of mind and matter, frequency and spacetime can be constructed from nothing but the generalised Euler Formula. No other ingredients are required. The God Equation supplies everything required for a living universe of self-optimizing, dialectical souls.

We’re not pretending for a moment that the concepts we have described here are not staggeringly difficult, especially for those with little mathematical experience. We have tried our best to use as few mathematical equations as possible, and to describe ontological mathematical concepts rather than present detailed mathematical arguments. Whether we have succeeded in conveying the power of the God Equation to mathematical novices is for our readers to decide.

What we hope we have done is provide a glimpse of how everything – all the baffling mysteries of existence – can rationally be traced back to just one all-powerful mathematical equation.

We have presented our case, and we have shown exactly where we think scientific materialism has gone catastrophically wrong. If you believe we are wrong then you must be able to show where we are in error. Of course, to do so, you will need to understand everything we have said and to see some disastrous flaw in our logic (of exactly the kind we have attributed to scientific materialism). We expect that exceptionally few people will have a firm grasp of what we have said, and even fewer will be able to use our material productively. Most people who offer criticisms of our work plainly haven’t understood any of it. It’s like being criticised by

geocentrists or flat earthers.

If the *God Series* gives just one solitary genius the platform he needs to reveal his brilliance to the world, we will have fully succeeded in our mission. We hope, above all, to inspire a generation of elite ontological mathematicians who will collectively destroy scientific materialism once and for all and replace it with living mathematics.

Never forget, scientific materialism logically leads to atheism. Ontological mathematics, however, is all about eternal minds – *souls*.

Mathematics is the only valid foundation for any rational Logos religion. All religions based on faith, revelation, gurus and Mythos are absurd and deranged. Abrahamism is 100% false and Eastern religion, while intuiting a few key truths, is about 50-75% false. It's mathematics or nothing if you want to understand existence and have tenable religious views.

The Euler Universe

Ontological Euler mathematics, leading to Fourier mathematics, quantum mechanics and holography, explains *everything*. Far from being completely alien to scientific materialism, ontological mathematics shares many of the same features. Indeed science could be regarded as nothing but an empiricist materialist misunderstanding and misinterpretation of a quintessentially rationalist, idealist universe.

Science, for all of its claims to be otherwise, is ultimately just a philosophy and even a quasi-religious faith – the irrational faith in a sensible rather than an intelligible universe.

Einstein famously said, “The most incomprehensible thing about the universe is that it is comprehensible.” What he should have said is, “It’s incomprehensible that a sensible universe – a universe based on fallible senses – is comprehensible, but it’s self-evident that an intelligible universe – a universe based on reason – is comprehensible.”

You simply cannot rely on your senses if you want definitive answers to existence. If you want answers, you must use your reason, not your senses. Mathematics is the ultimate subject of reason. It has no connection whatsoever with the senses. The whole of mathematics can be worked out without ever looking at the world.

Science is about the senses while mathematics is about intellect and reason. Which side would you rather be on? Mathematics is rational and science fundamentally irrational since it refuses to accept reason over the senses.

The Monad

What is a monad? It’s a basic unit of ontological mathematics. In a simplified universe consisting only of a 2D complex plane, it’s defined by the generalised Euler Formula ($\mathbf{A} \mathbf{e}^{i(\mathbf{f}\mathbf{x} + \phi)} = \mathbf{A} \cos(\mathbf{f}\mathbf{x} + \phi) + i \mathbf{A} \sin(\mathbf{f}\mathbf{x} + \phi)$), catering for all possible values of A and f. The whole system balances to zero. As a net zero, the monad is dimensionless, hence not physical. It cannot be directly experimentally observed under any circumstances. It’s a system of pure frequencies outside space and time – a mind, a *soul*. A mind is simply that which processes information associated with the countless linked frequencies defined by the generalised Euler Formula. A mind *experiences* this information subjectively, i.e. it feels it and derives pleasure or pain from it. In other words, it can discriminate between “good” and “bad” information. Good information is that which a mind feels is enhancing its power; bad information is that which a mind feels is reducing its power. All monads are in exactly the same game – self-optimization via maximising their feeling of power. How to assess power is a dialectical enterprise.

In a higher dimensional universe, we simply add additional complex planes in all conceivable orientations with respect to the first. This makes the mathematics more complicated, but it in no way changes the basic principles.

The most stable higher dimensional universe is six-dimensional, involving three orthogonal complex planes (equating to three spatial and three temporal dimensions). This contrasts with the asymmetric, four-dimensional universe of Einsteinian physics (involving three spatial dimensions and just one temporal dimension). Illuminism always balances space and time (since there’s no sufficient reason why one should be privileged over the other), while Einstein introduces a fundamental asymmetry between space and time, without providing any sufficient reason. Time, as we said throughout the *God Series*, has never been understood by physicists and this constitutes one of its greatest unsolved problems if not the greatest of all. If you can’t get time right, you can’t ultimately get anything right.

The Grand Unified Theory of Everything

The generalised Euler Formula is the simplest possible rational explanation of existence since it involves just a single equation and has no extraneous elements.

In accordance with Leibniz's requirements for the "best possible world", the Euler universe is the simplest conceivable hypothesis and "richest in phenomena". This is exactly how a rational God would choose to build the universe. However, it is of course the God Equation that gives rise to God, and not God that gives rise to the God Equation. The God Equation is eternal, immutable, all-knowing (insofar as it's the basis of all knowledge) all-powerful, and everywhere. It is the uncaused First Cause. It's uncreated. Hence it has precisely the properties of God!

God is not a person, "it" is a self-optimizing mathematical equation that acts as a God Factory and makes Gods of all of us. You don't have to kneel to it and worship it. You don't have to have faith in it. It doesn't need any prophets or holy texts. It doesn't sentence people to hell. It doesn't torture people or set "moral" Commandments. You can't have a personal relationship with it. It's pointless to pray to it. It won't hear you and will never answer. It doesn't bang on about "love and light" and "unconditional love".

Thanks to the fact that it uses a mathematical dialectical process to optimize the universe, the God Equation is amoral, Darwinian, unsentimental, and never intervenes to stop brutality, savagery, and evil. These are all eliminated in the end – thanks to the dialectic reaching its divine omega point.

The existence of evil in the world is entirely to be expected when the God Equation is God, but not when some morally perfect, all-powerful Creator is proposed to be God. There's no such being.

Since it has only one element, the Euler universe is in best accord with Occam's Razor. There's nothing outside it. It explains *everything*. It leaves no gaps, hence obeys Leibniz's principle of continuity. It's the basis of Leibniz's principle of sufficient reason, and of all causality. It enshrines all of Plato's immutable, eternal laws (considered mathematically).

The Euler universe is a monism but with a dual aspect (of mind and matter: a dimensionless frequency domain and dimensional spacetime domain, linked by Fourier mathematics).

The Euler universe brings together mind, matter, consciousness, the unconscious, free will, mathematics, science, philosophy, metaphysics, religion and psychology, hence is a true theory of everything (unlike anything science will ever have to offer). It's the Neoplatonic "One", the source of all, the Singularity from which all other things emanate.

The God Equation does everything that science signally fails to do ... it explains everything, including the *origins* of everything. It makes no reference at all to faith, magic, randomness or probability. It provides a sufficient reason for why everything is such and not otherwise.

The God Equation is the only possible answer to existence, to "life, the universe and everything". In *The Hitchhiker's Guide to the Galaxy*, Douglas Adams famously said that the answer was "42". In fact, it's not a number but an equation that's the answer. Whether or not people *understand* the answer is an entirely different issue. To all irrationalists, it's the worst possible answer, and they will unquestionably reject it. They will go on believing in their absurd prophets, gurus and sacred texts. They will continue to worship their Mythos Gods. They will continue to be "believers" and have faith in whatever nonsense floats their boat. Some will continue to believe in mystical intuitions and some will solemnly declare that "love" is the

answer. The scientific materialists will continue to reject anything their senses can't grasp. Such is life.

The truth, as we often say, simply isn't for everyone. The truth belongs to the rational, intelligible world. If you prefer to subscribe to faith, feelings, mystical intuitions and your senses, you will never enter the mansions of truth, objectivity and reality. You will never enter the Logos domain but remain forever in the Mythos realm, telling yourself unbelievable little stories to comfort you. The irrational scientists will keep saying, "Show us the evidence" and we will keep replying: "Use your reason, not your senses. There is nothing to see. Reason isn't a sensory organ."

Only the Gods can grasp the truth. Mathematics is the truth. The God Equation generates all numbers and controls all numbers. It's thus in complete accord with Pythagoras's incredible intuition when he said, "All things are numbers; number rules all."

Your task is not to worship the God Equation as God. Your task is to understand the God Equation and use its power to make yourself God! The God Equation, not meditation, is your route to gnosis and divinity.

How many are equipped to fathom the mysteries of the God Equation? Precious few. Only the authentic Elect.

Pindar said, "Neither by land nor by sea shalt thou find the road to the Hyperboreans." The road to Hyperborea is the road of the God Equation, and only the finest Grail Knights can discover it. It's hidden in plain sight, but only those with eyes to see and ears to hear can detect it.

"Read Euler: he is our master in everything." – Pierre-Simon Laplace

"He [Euler] calculated without any apparent effort, just as men breathe, as eagles sustain themselves in the air." – François Arago

"Like a Shakespearean sonnet that captures the very essence of love, or a painting that brings out the beauty of the human form that is far more than just skin deep, Euler's equation reaches down into the very depths of existence." – Keith J. Devlin

"Perhaps the most surprising thing about mathematics is that it is so surprising. The rules which we make up at the beginning seem ordinary and inevitable, but it is impossible to foresee their consequences. These have only been found out by long study, extending over many centuries. Much of our knowledge is due to a comparatively few great mathematicians such as Newton, Euler, Gauss, or Riemann; few careers can have been more satisfying than theirs. They have contributed something to human thought even more lasting than great literature, since it is independent of language." – Edward Charles Titchmarsh

"Our jewel ... one of the most remarkable, almost astounding, formulas in all of mathematics." – Richard Feynman on Euler's Formula, of which Euler's identity is a special case.

Why did Feynman – who acknowledged how astounding Euler's Formula was – fail to recognise that it was the very basis of existence? Because he was a fanatical believer in the religion of scientific materialism and refused to accept mathematics as a higher religion.

"There is a famous formula, perhaps the most compact and famous of all formulas – developed by Euler from a discovery of de Moivre: $e^{i\pi} + 1 = 0$. It appeals equally to the mystic, the scientist, the philosopher, the mathematician." – Edward Kasner and James R. Newman in *Mathematics and the Imagination*

"Gentlemen, that is surely true, it is absolutely paradoxical; we cannot understand it, and we don't know what it means. But we have proved it, and therefore we know it must be the truth." – Benjamin Peirce [on Euler's Identity]

“One of the most frequently mentioned equations was Euler’s equation, $e^{i\pi} + 1 = 0$. Respondents called it ‘the most profound mathematical statement ever written’; ‘uncanny and sublime’; ‘filled with cosmic beauty’; and ‘mind-blowing’. Another asked: ‘What could be more mystical than an imaginary number interacting with real numbers to produce nothing?’ The equation contains nine basic concepts of mathematics — once and only once — in a single expression. These are: e (the base of natural logarithms); the exponent operation; π ; plus (or minus, depending on how you write it); multiplication; imaginary numbers; equals; one; and zero.” – Robert P. Crease, in “The Greatest Equations Ever” at *PhysicsWeb* (October 2004)

“Although to penetrate into the intimate mysteries of nature and thence to learn the true causes of phenomena is not allowed to us, nevertheless it can happen that a certain fictive hypothesis may suffice for explaining many phenomena.” – Euler

Euler – like so many mathematicians – didn’t understand the ontology of his own work and referred to it as “fictive”, yet with the remarkable and inexplicable property (if it were indeed fictional) of explaining real phenomena.

“Madame... I have come from a country where one can be hanged for what one says.” – Euler

Perhaps Euler was too scared to state the truth. The Christian authorities have never reacted with anything but hostility to those who contradicted Scripture.

“Mathematicians have tried in vain to this day to discover some order in the sequence of prime numbers, and we have reason to believe that it is a mystery into which the human mind will never penetrate.” – Euler

Yes it will, and it will come about through the generalised Euler Formula. That’s cosmic irony for you! The generalised Euler Formula will solve the Riemann Hypothesis. It alone can do so.

The Eternal Universe

The universe – the *eternal* universe – must be grounded in eternal, not contingent, truths. Mathematics alone provides eternal truths, hence mathematics is the foundation of existence. It's absolute, infallible, flawless, perfect, and ubiquitous.

Mathematics is the ground of reality. Mathematics is the arche, the fundamental stuff of existence. Mathematics is just another word for reason in its purest and most objective form. The universe is intelligible because mathematics is intelligible. Mathematics is fundamentally *mental*, not physical. Mathematics is the basis of rationalism and idealism, and opposed to empiricism and materialism (i.e. opposed to conventional science). Mathematics requires no experiments. Mathematics is deductive, not inductive. Mathematics is about rational proof, not about sensory "facts" and "evidence". Facts and evidence are interpretations: proofs are eternal truths.

Science, to be plausible when it comes to the ultimate questions, *must* be based on rationalist mathematics and not on the empiricist scientific method that is incapable of addressing infallible, absolute truths.

Science, by preferring its "method" over mathematics, is simply wrong. Leibniz put mathematics above scientific experiments and it's to Leibniz's conception of science that we must return. The age of Newtonian science, in which "observations" (performed on the sensory world) are paramount, must be closed.

Ultimate reality is about intelligibility, not sensibility. Ultimate reality is about reason, not about the senses. Reason belongs to the intellect, not to the sense organs. Do you understand reality with your intellect or with your sense organs? Your sense organs are simply gatherers of data from the world external to you. It's your intellect that orders, organises, interprets and rationalises this data.

Science has placed the sense organs – the data gatherers – above the intelligence to which the sensory data is sent and processed. How crazy is that? How can the data gatherers be more important than the data processor? It's like worshipping the camera lens rather than the pictures the camera takes (chosen by an intelligent camera operator).

The fact that data is gathered at all already presupposes an underlying intelligence. The intellect is something to which the sense organs send data, but the intellect itself is not subject to the senses. It is empirically *transcendent*.

The best that science can do, with its empiricist method, is comment on sensory data and interpret it. What science can *never* do is address the non-sensory, i.e. the unobservable intellect itself. It can't talk about its structure, its modes of operations, its ontology. Yet it's precisely the intellect that reflects the fundamental mathematical basis of reality.

God

What is "God"? God is Mathematics. God, therefore, is *not* conscious and God is not "good". God is a rational process of optimisation and actualisation, expressed through the fundamental living units of mathematics – monads (immortal and indestructible mathematical minds/souls).

The Will

For Schopenhauer, the unitary Cosmic Will – outside space and time – underlies all things. In Illuminism, Will is replaced by Mathematics (also outside space and time), and is not unitary but is instead composed of a myriad of mathematical minds (monads).

Existence

The only thing that can have *necessary* existence is something that is also nothing. It's precisely this connection with nothing that places no existential barrier in the way of something, i.e. as long as *something* has a *net effect of nothing*, there is no sufficient reason to prevent its existence. You cannot prevent nothing. "Something" has a barrier to overcome; "nothing" does not. If something didn't exist at all, there would be nothing. Nothing is the default, hence in no need of explanation. It's "something" that must be explained, and it can be explained only as a special type of "nothing", a particular expression of "nothing".

There is only one thing that's exactly and perfectly both something and nothing, a something that is nevertheless mathematically null. That thing is the generalised Euler Formula, the most beautiful, efficient and effective thing of all, the quintessence of existence, of ontological mathematics, of information, of energy, of dimensionless (mental) and dimensional (material) existence.

Living Information

What's the difference between the atoms in the dirt and rock all around us and the atoms in our body? The latter are bound by a special, unique biological code for each of us – DNA. DNA is a mental code produced by Fourier mathematics. Dead matter cannot organise itself into life. Only living minds can organise dead matter into bodies that support life. They do so through the DNA information code: the information code of life on Earth.

What We All Are

We are uncaused, uncreated, self-solving, self-optimizing, immortal mathematical beings. That's the way it is. Too bad if you don't like it. Better to be on the side of the truth rather than against it.

To all those who deny that we are pure mathematics, we simply say, *so tell us what we are without invoking faith, revelation, mysticism, feelings, the senses or Mythos*. Good luck with that.

Mind and Matter

Scientists believe that, *macroscopically*, matter behaves in a wholly orderly way, according to rigid laws. They can't accept the existence of independent, immaterial mind since they associate this with disorder and non-lawful behaviour (i.e. with the possibility of free will). However, when it's understood that mind is mathematical (wholly defined by mathematics) then it becomes easy to see why the mental domain is one of complete order. Free will does of course exist but it arises purely from mathematical laws. Free will is derived from mathematics operating in its subjective rather than objective mode; from mathematics operating in its living, autonomous aspect rather its collective, objective form (which is what we associate with the material world). The only way to link mind and matter, and have a universe of law and order, is through mathematics.

Young Mathematicians

Why is most work of genius in mathematics and science performed by young men in their early twenties? The answer is that they have not yet been locked into the rigid paradigmatic thinking of the establishment. They are still capable of thinking differently, innovatively, rebelliously and subversively. Once they join the establishment, all of that is lost. They become part of the bureaucratic, conformist, groupthink machine.

It has been said, with humour, that the young are *too stupid* to know that something won't work. In fact, the reverse should be stated: the old have become too stupid to know that something *will* work.

Groupthink makes people stupid. Becoming part of the establishment makes people stupid. Accepting the prevailing paradigm makes people stupid. Becoming a careerist and worrying about others' opinions and criticisms makes people stupid. The most stupid people of all are those who dismiss new ideas out of hand because they are incompatible with old ideas.

Establishment scientists would completely reject our dismissal of Einsteinian relativity and the Copenhagen interpretation of quantum mechanics. They would no doubt say that we were "idiots" who had got it completely wrong, even though they will then admit that relativity is rather bewildering, and indeed incompatible with quantum mechanical non-locality, and that no one knows what quantum mechanics really means, and that relativity and quantum mechanics appear to be wholly irreconcilable and, therefore, cannot be combined into a single theory. What's for certain is this: two theories can be combined only if they have the same rational basis and talk the same language. Science's two greatest theories plainly do not.

Well, we have presented our case in the *God Series*, and countless scientists have presented theirs ad nauseam in schools and universities, books and TV documentaries. You will have to decide for yourself who is right, you will have to use your reason, not your prejudices and groupthink.

Ultimate Truth

Plato believed that ultimate truth resided in abstract, metaphysical Forms (Ideas) beyond space and time, in an intelligible but not a sensible domain, which we could reach via our reason. This was the only way in. No other paths were available.

The sensible world in space and time was an inferior copy of this perfect intelligible world. For Plato, the more we escaped our senses and made use of our reason, the closer we came to absolute truth, and to the ultimate source of everything – the Form of the Good (more or less the Form of God).

Illuminism accepts Plato's system, with one critical alteration. Mathematics is the Form of the Good, the Form of God that exists quintessentially in a domain beyond space and time (the Fourier frequency domain).

Things such as the Platonic Forms of Beauty and Justice are purely subjective and interpretive. They have nothing to do with the eternal truths of reason. Only eternal things can belong to the Platonic domain, and opinions, tastes and beliefs do not qualify. Only the laws of mathematics and the ontological, living units of mathematics (monads) are necessarily eternal. They alone are truly Platonic. They alone constitute absolute, infallible, immutable truth and knowledge. All else is speculation, conjecture, opinion, belief and interpretation. Mathematical reason alone reveals the eternal truths of mathematics. No faith and no experiments will help you to reveal absolute truth. The religious on the one hand and the scientific on the other are both wrong.

The mathematical metaphysical rationalists were and are humanity's Knights of the Holy Grail, finding their way to the source of incontestable truth. All other "knights" are false, mere pretenders and charlatans.

Mathematics is perfection. Nothing else is or can be.

A beautiful young woman has a perfect "Form". She is a *physical* form. In Plato's system, she is a physical expression of the abstract Form of Perfect Beauty: an eternal immutable Form outside space and time.

For Plato, this abstract Form was more real than the beautiful girl. She was its mortal product, while it was immortal. We can't see it with our eyes, but we can understand it with our reason.

Later, many thinkers rejected Platonic "realism" – Forms as real things, real existents – and Forms thus came to be regarded as unreal abstractions that could change reality only through their intellectual, conceptual influence (although it's hard to see how something "unreal" can achieve any effects at all).

Science regards Forms or Ideas as unreal abstractions. Yet the quantum mechanical wavefunction is also an unreal abstraction, from which a real world somehow (and inexplicably) arises. Science simply ignores this inconvenient contradiction.

In Illuminism, Forms or Ideas are real things – *mathematical* things. Eternal souls convey the eternal laws of mathematics. Souls are real! Souls are the foundations of ontological mathematics and of existence itself.

Where do ideas come from? The mind. They are not in space and time. They are not physical things. They are mathematical Fourier functions composed of different combinations of sine and cosine waves.

On 9/11, three thousand Americans died because of the ideas in the heads of nineteen Muslim fanatics. Six million Jews died because of the ideas in the minds of Nazi leaders. Nothing

is more powerful than ideas, and nothing is deadlier. Ideas are capable of making humanity extinct or of destroying the planet, so nothing is more important than understanding ideas, and promoting good ideas over poor ones, rational ideas over the irrational.

Ideas are information. Information is all about mathematical sine and cosine waves, the carriers of information. An “idea” is simply a mathematical Fourier function. When someone “learns” an idea, it means that they have performed a copy of that function and now have their own internal access to it. Of course, the copying process is never perfect and that’s why ideas are continually changing, evolving, mutating. It’s why ideas are misunderstood and misinterpreted.

If you could analyse the mathematical forms of people’s ideas and beliefs, you would see all sorts of subtle mathematical differences, leading to different subjective experiences and understandings of these ideas.

In Plato’s system, each Idea is perfect, eternal and immutable, so absolute knowledge of everything is possible. In Illuminism, Ideas are dynamic, mutable and evolving towards a dialectical endpoint. The only ideas that are fixed, static and eternal in the true Platonic sense are the eternal truths of reason: the laws of mathematics and the living units (monads) of mathematics.

The eternal laws of mathematics define the cosine and sine waves that are the carriers of information, but the information they carry is temporal, provisional, mutable, contingent, always evolving. Mathematically, this corresponds to the Fourier function that defines an idea continually altering through the addition or subtraction of cosine and sine waves of different frequencies, amplitudes and phases.

If there were a fixed Fourier function for every idea, Plato would be right, but there isn’t. Ideas, other than immutable eternal truths, are subject to continuous dialectical evolution. Only when they have all reached their omega point – at the end of time – does the universe correspond to Plato’s vision of absolute reality, at which point we are all Gods and we all have absolute knowledge of everything, and absolute power.

Our whole world is shaped by ideas, meaning that our world is shaped by the mind, not by matter. Plato argued that the sensible world is a poor copy (simulacrum) of the physical world. Ideas are stamped on recalcitrant matter. A rock (matter) does nothing. Only when matter has a controlling mind is it capable of action, at which point matter becomes “body” – organised matter designed to host a mind. We affect the material world through our immaterial minds.

The Contradiction

Einstein’s principle of relativity contradicts the true first law of motion (that everything moves forever at the same speed: the speed of light). Nothing can ever be stationary or define itself as a stationary frame of reference. All true frames of reference move at exactly the same speed.

The Necessary Universe

The universe did not come from absolute nothingness (non-existence). If such a thing were possible, the universe could just as easily return to nothing at any instant, or new universes could erupt at any instant at any place. Therefore, the universe has existed forever. It’s founded in eternal, necessary things.

Only one thing is truly eternal and necessary – mathematics. It’s truths have always been true and always will be true. Mathematics can never not *be*. Mathematics is the only thing that can make “something” and “nothing” the same, and this is the solution to the mystery of existence. There is *something* because, mathematically, this is an inevitable consequence of the laws and numbers of mathematics. But *something* must balance to *nothing* via the exact matching

of positive and negative. Symmetry demands it. The universe must always be in its ground state of nothing.

Mathematics can be neither created nor destroyed. Mathematics itself is the true basis of the First Law of Thermodynamics. It's also the basis of the First Law of Motion. Indeed, it's the basis of *all* laws. All authentic laws are mathematical and all of them are analytic, eternal, *a priori* and necessary.

Mathematics is the *only* way to explain everything. Nothing else can rival or challenge mathematics.

"How often have I said to you that when you have eliminated the impossible, whatever remains, however improbable, must be the truth?" – Sherlock Holmes (Arthur Conan Doyle)

Mathematics is the answer to existence. To many people that might seem absurd. Yet all rival solutions are logically impossible. Mathematics, no matter how improbable non-mathematicians might find it, is the truth. To ontological mathematicians, of course, no answer could be more obvious or welcome. Thanks to mathematics, we can all become Gods! Living mathematics is literally a self-perfecting system, a system that solves itself by finding the optimal answer – divine perfection, a community of Gods living in heaven itself.

The Final Theory of Everything

All meaningful questions of philosophy, science and religion reduce ultimately to questions of mathematics. Everything else is Mythos nonsense. Everything else is interpretation, belief and opinion.

The Necessary Basis of Existence

All great religions and philosophies posit a fundamental and necessary ground of existence.

Hinduism says it's "Brahman", a kind of divine noumenon (thing in itself), which is then subject to an illusory veil of phenomena (Maya), causing the individual soul (the atman) to forget that it's actually Brahman (atman = Brahman), or, to be more exact, capable of *becoming* Brahman.

Buddhism invokes a Cosmic Oneness as the root of all things. Schopenhauer's philosophy says it's the cosmic Will, outside space and time. Hegel says it's Mind/Spirit (*Geist*). Hartmann says it's the Unconscious. Nietzsche says it's the Will to Power (inside space and time). Jung says it's the Collective Unconscious. Pierre Teilhard de Chardin says it's the Omega Point (on which the Noosphere is converging). Neoplatonism says it's the "One". Pythagorean-Leibnizian Illuminism says it's numbers and monads (living, ontological mathematics).

Taoism says that the *Tao* gives birth to the Universe (the "One"), which in turn gives rise to Yin and Yang (the "Two") from whose interaction comes Chi/Energy (the "Three"), which gives rise to Everything (the "Ten Thousand Things"). In Hegelian terms, *Geist* would be equated to both Chi and Tao itself; yin and yang would be the thesis and antithesis phases of the dialectic, and would be supplemented by the synthesis phase, allowing a dynamic dialectical drive to a final convergence (Omega Point; Absolute State) where everything is rendered dialectically perfect.

In Abrahamism, "God" is the necessary root of all things. Yet we can immediately see that there's nothing necessary about this God: he's not necessary in a scientific, mathematical or dialectical universe and he's not necessary in the universe described by eastern religion. A universe can easily be understood and explained without the Abrahamic God. It cannot be explained without mathematics.

Even science – the subject of contingency, provisionalism, randomness and statistics – has to posit something that is necessary: the cosmic wavefunction. This wavefunction is not considered to be *real*, and yet modern science cannot dispense with it. The cosmic wavefunction, as described by the Copenhagen school, describes everything that can happen, every conceivable possibility until the end of time. So, for science, the existence of possibility is what is essential – what is *necessary* – rather than the existence of actual, ontological things.

Illuminism is based on real, ontological, eternal mathematical souls (monads) that express the complete laws of mathematics. Monads are actual, permanent things. Science, on the other hand, is based on *potential* things rather than *actual* things. This, manifestly, is absurd. A thing either exists or it doesn't. A "potential thing" isn't an existent. Potential applies to what actual things *might* do. You cannot have potential and possibility existing *independently* of actual things, which is essentially what science claims.

Science says that something can come from nothing (Big Bang theory), but what it means by "nothing" is extremely bizarre. Where Illuminism says that "nothing" is dimensionless existence composed of mathematical monads (net zeros), and that dimensionless existence (the mental frequency domain) is the necessary precursor of dimensional existence (the spacetime material world), science claims that "nothing" is an unreal mathematical wavefunction expressing all conceivable possibilities, and that this is the precursor of the spacetime material world.

Where Illuminism says that the spacetime universe is produced by a *real* universe that exists *outside* space and time (a mental, frequency domain), science says that the spacetime

universe is produced by an *unreal* wavefunction that doesn't exist anywhere, which makes science a theory of magic!

In Illuminism, dimensionless existence logically precedes a dimensional universe, a mental universe precedes a material universe, a noumenal universe precedes a phenomenal universe, an intelligible world precedes a sensible world, an *unobservable* world precedes an *observable* world. Illuminism reflects idealist and rationalist thinking where mind is primary and matter is secondary.

For science, with its Meta Paradigm of empiricism and materialism, there's no such thing as a dimensionless universe, an intelligible rather than sensible universe, a noumenal rather than phenomenal universe, an unobservable rather than observable universe: there is no *mental* domain. Matter is primary and mind secondary. So, science says that an unreal, unobservable mathematical wavefunction that exists nowhere randomly generates the material universe of space and time.

In Illuminism, the Big Bang is about mental reality creating material reality. In science, the Big Bang is about *unreality* creating reality. This is pure "magic thinking". It's a scientific act of faith. It's so ridiculous that scientists might as well believe in the Abrahamic God, an equally ludicrous and magical concept.

How can potential things exist? How can unreal things exist? How can an unreal mathematical wavefunction exist? How can it exist *nowhere*? How can space and time be preceded by unreality, potential, randomness, possibility and a mathematical wavefunction that exists nowhere, isn't real, and yet is somehow essential to the workings of the universe and absolutely necessary? That's the logic of the madhouse.

Such "logic" exists for one reason alone – to deny that mind is primary and matter its product. As soon as you get rid of this blind, mindless prejudice, you can have a fully rational, logical, intelligible, mathematical universe of mind that can be fully explored using the God Equation and Fourier mathematics. There's no mystery at all as soon as mind rather than matter is acknowledged as the basis of existence. When will scientists abandon their quasi-religious faith in materialism and empiricism and face the facts?

As ever, we come back to the ontological status of mathematics as the supreme issue confronting science. In Illuminism, the universe is made of mathematics (mathematical monadic minds). In science, mathematics is some unfathomable abstraction that somehow has some form of existence even though, according to science, it's unreal and has no location. Science makes a cosmic mathematical wavefunction necessary and essential to the workings of science and then accords it no reality and no place where it can reside, and makes it full of potential things rather than actual things. You might as well believe in ghosts. Science's understanding of the fundamental ground of reality is insane. It wholly contradicts the entire basis of materialism and empiricism, upon which science is predicated. The cosmic mathematical wavefunction of science is neither material nor observable. It can't be experienced in any way, and it has no location in space or time, or indeed anywhere.

How can you have a dogmatic materialist empiricist ideology that is grounded in non-materialism and non-empiricism? How can scientists expect anyone to buy into this nonsense?

All of these illogicalities and contradictions vanish as soon as rationalism and idealism are accepted as the true means to understand reality. As soon as mind rather than matter is accepted as the primary reality, the state preceding the Big Bang becomes one of pure mathematics, encoded within countless living, eternal mathematical minds ... singularities ... *souls*.

It's All There

Everything must be present in that which is necessary, in the *arche*, the fundamental stuff of existence. Life must be present, as must mind, subjectivity, free will and the evolutionary root of consciousness (consciousness is simply the highest expression of the unconscious mind, not something different from the unconscious).

According to science, life, mind, free will, the unconscious and consciousness are not in any way present in atoms, the building blocks of matter. These qualities and properties all miraculously “emerge” (by magic, presumably).

When it comes to explaining the fundamentals of existence, science is a staggeringly absurd system. It has no truth content and no explanatory value whatsoever. Science can *never* reach the inner nature of things, the true kernel of existence.

The Scientific Excess

There are more scientists working today than in the whole of human history, and they are better educated than at any time in history – so why haven't they produced a final scientific theory of everything? It's because it's *impossible* within the prevailing Meta Paradigm of science. Scientists, being fundamentally irrational and philosophically illiterate, simply can't grasp this. They keep retreading the same old materialist and empiricist ground, hoping, like Mr Micawber in Dickens' *David Copperfield*, that "something will turn up". It won't.

Einstein said, "Insanity is doing the same thing over and over again and expecting different results." Yet that's exactly what science does. It's *insane*!

Science is now the proof that more is less. Most scientists are on the autistic spectrum. They have no imagination, no intuition and are locked into the sensory world. They see reality in machinelike terms rather than living terms. Science suffers from catastrophic groupthink, conformism and careerism. Scientists are terrified of saying anything heretical for fear of being shunned and having their funding cut off. Science is controlled not by any quest for truth but by those holding the purse strings, who reflect bureaucratic concerns.

Russian sci-fi writer Yevgeny Zamyatin said, "True literature can exist only where it is created, not by diligent and trustworthy functionaries, but by madmen, hermits, heretics, dreamers, rebels, and skeptics."

True science has exactly the same requirement. Instead of that, it's *wholly run* by dull, unimaginative, diligent and trustworthy drudges and functionaries. Many of them are even proud of that and boast about it. Truth is more or less their lowest priority, and "success" their highest. ("Success has always been the greatest liar." – Nietzsche.)

Science now operates by a consensual principle of the "wisdom of the crowd". Geniuses who cause paradigm shifts are not required. In fact, they're regarded as nutcases. Science crawls along incrementally, like a slug, allowing dim-witted careerists to think they understand their subject and are masters of it. In fact, they are as clueless about the true nature of reality as Abrahamists.

Science has no central authority. Instead, it has a central Meta Paradigm. Science has placed the scientific experimental method on its highest pedestal, and this is worshipped as the God of Science. Yet it's a false God. The true God of science is mathematics, and mathematics has no need at all of experiments!

In order for science to undergo its necessary Reformation, it must relegate the scientific method and elevate mathematical rationalism. It must stop being materialist and empiricist and become idealist and rationalist. It must stop being obsessed with observation and start being obsessed with reason. Physics must bow to metaphysics.

Traditionally, metaphysics has been associated with unverifiable philosophical speculation while the scientific method has provided the link between hypotheses and the "real world". Yet, true metaphysics isn't philosophy. Rather, it's *mathematics*. And mathematics (metaphysics) is the foundation of physics. What lies beyond physics isn't speculation but mathematics, the essence of rationalism. As long as metaphysics sticks to mathematics alone, it can reveal every authentic secret of existence.

Physics and metaphysics must merge under one umbrella: ontological mathematics, thus linking mind and matter, noumenon and phenomenon, the intelligible and the sensible, the rational and the empirical, the unobservable and observable.

Kant claimed that mathematics and science were "synthetic *a priori*". In fact,

mathematics is analytic *a priori* and science is synthetic *a posteriori*. Mathematics tells us all about the intelligible world and science about the sensible world. Mathematics deals with eternal truths of reason, and science with contingent truths of fact.

The Universe

Schopenhauer thought that the universe was ultimately just one thing – the Cosmic Will, outside space and time; a single Mind, so to speak. Schopenhauer had a wholly negative view of this Mind, seeing it as irredeemably malevolent and evil. If anything, he regarded it as the Devil, the *unconscious* Devil. In Schopenhauer's system, there's no "God" to make things better. The universe proceeds on its evil way forever, producing nothing but evil, wickedness and suffering. In many ways, this is the most pessimistic Gnostic vision of reality. There's only a monstrous Demiurge and no True God.

Schopenhauer's philosophy is Buddhism treated negatively rather than positively, pessimistically rather than optimistically. Why *should* Buddhism result in a good outcome rather than bad? There's no sufficient reason. Why do Buddhists insist on having a benevolent religion?

Schopenhauer is remarkable for producing an entirely negative religion. No Buddhist can say Schopenhauer is wrong. After all, he was enormously smarter than the Buddha. There's simply no contest. Read both and judge for yourself.

In Illuminism, it's not a single mind that underlies reality but rather an infinite number. One philosophical issue then arises. What of the Mind of minds, the Monad of monads, the Singularity of singularities? Each individual mind is uncreated and uncaused but what about the whole collection, the set of monads? The whole set cannot exist without its individual members, so it's caused and created by its members. The whole set is therefore secondary, not primary.

Can you get a secondary mind, a secondary person? Can the set of all monads be a Super Mind comprising all of its members? Does it evolve as all of its constituents evolve? Can it actually feed back to its constituents and accelerate their evolution, as in Teilhard's Noosphere? In other words, do we all help each other's evolution? Are we all indissolubly bound together?

We either progress together, or we all drag each other down. We either proceed as a collective or we destroy each other as competing individuals.

"We must, indeed, all hang together, or assuredly we shall all hang separately." – Benjamin Franklin

This is why it's imperative to beat the anarchists, the libertarians, the individualists, the egoists, the narcissists, the selfish ones, the self-obsessed ones, the super rich, the free-market capitalists, all those who oppose the General Will, the Collective Good, the Commonwealth. They are actively holding us all back, and obstructing the evolution of humanity.

Abrahamism and Consciousness

Abrahamism states that consciousness (the consciousness of “God”), precedes everything else. Taoism, Hinduism, Buddhism and Illuminism all deny that consciousness is primary. Consciousness is the *culmination* of an evolutionary universe. “God” comes at the end, not at the beginning.

Dreams

Close your eyes. Picture a castle. Change the details. Make it as exotic and complicated as you like. This castle exists nowhere other than in your mind. It looks like lots of real castles, and yet you simply imagined it. Real castles are made of atoms. What are castles in your mind and dreams made of?

Well, ultimately, they must be based on the same things or they couldn’t possibly resemble each other. In your dreams, you can’t tell your dream reality from actual reality. It’s as vivid and lifelike to you as the real thing. How can this be unless your dreams somehow reflect the same processes as reality?

In Illuminism, this paradox is easily resolved. The “real world” involves *collective* Fourier mathematics and the dream world involves *individual* Fourier mathematics. They obviously resemble each other because they’re using the same mathematical procedure. If they didn’t, how could dreams ever be mistaken for reality?

Science makes no attempt to address this fundamental issue, which goes to the heart of the nature of reality. In any discussion of “reality” and quantum mechanics, dreams must be brought into the mix and accounted for.

Fourier mathematics is at the root of everything in our world. What are the sense organs? They are “spacetime” Fourier organs that feed back information to the Fourier frequency domain of mind. It’s purely because mind and matter are linked by Fourier mathematics that they can interact at all.

If We Could “See”

If we could see far and deep enough, all we would ever see is mathematics in operation. There’s nothing else. How could there be? Try to imagine any moving object in the universe not obeying mathematics. It’s actually impossible.

Hocus Pocus

Scientific Multiverse theory is motivated purely by the intention to prove that God does not exist and yet ends up proving that God *does* exist!

It can’t be stressed enough that science is so ignorant of philosophy that it keeps digging up philosophical positions that have long since been refuted. Scientific Multiverse theory is actually the most extreme version of the medieval Catholic ontological argument. Catholic theologians sought to prove that God must exist and the issue eventually came down to whether or not existence is a predicate (and Kant famously said that it’s not).

To understand what this means, consider a unicorn. What’s the difference between the concept of a non-existent unicorn and the concept of an existent unicorn? The two unicorns have exactly the same properties (predicates; things that can be asserted of them), so you in no way change the concept of a unicorn by saying it exists. The concept is the same whether or not the unicorn exists. Existence does not add anything to the concept. Existence simply instantiates the

concept in the physical world; it does not add to the concept or change the concept.

You can say that a unicorn exists or does not exist. You cannot say that an existent unicorn is conceptually any different from a non-existent unicorn. They are exactly the same by definition. If it exists, you can find an instance of it, and if it does not exist, you can't. That's the only difference.

Existence isn't a predicate. It's a test of whether the concept is instantiated or not, not a property of the concept.

In Multiverse theory with its statements that a) everything that can happen will happen and b) anything not forbidden is compulsory, it's asserted that every actualisable possibility is instantiated. This means that if God is possible and is not forbidden then he *must* exist – exactly what science was seeking to prove was impossible!

Science, in order to exclude God, must prove that he's "forbidden" and this of course is scientifically impossible. God is no more impossible than a Big Bang Singularity. In fact, maybe he *is* the Big Bang Singularity! How could science know given that science can't explain the Big Bang either?

Memory

What is a memory? Where is it? How can a memory be stored and retrieved? Yet again, it's all about Fourier mathematics and the interaction between the spacetime Fourier domain and the frequency Fourier domain outside space and time. A memory can be stored forever in the frequency domain. The memory system is all about converting spacetime data into frequency domain data for storage outside space and time, and then retrieving it and converting it all back into spacetime data – all handled via Fourier mathematics.

It's because memories aren't stored in space and time that even after severe brain damage and removal of huge chunks of the brain, a patient might still have access to a full set of memories. However, if any of the key brain structures involved in laying down Fourier memories and then retrieving them again are damaged then partial or complete memory loss can occur.

There are two main Fourier memory systems: one for short term and one for long-term memories. Depending on what brain systems suffer damage, different memory problems can occur. There are even cases of people being unable to lay down any new memories at all and being stuck forever in the present.

The Scientific Religion

“Knowledge, absolutely sure of its infallibility, is faith” – Yevgeny Zamyatin

This is true of science, but not mathematics. Mathematics, the subject of eternal truths that can never be contradicted, is infallible by definition, whether anyone likes it or not. That which is necessary, foundational and eternal must be infallible.

Zero Truth Content

Abrahamism, Karmism and scientific materialism all have zero truth content. However, at least science is useful. It produces a false model of reality, but one that we can use to lead us to the true language of reality – *mathematics*. Mathematics is the subject of the Gods, and science is for barbarians. Mathematicians can transcend their senses, scientists can't.

Saint Paul in Athens

Saint Paul preached in Athens – and was ignored. The pagan intelligentsia, accustomed to the great philosophies of Plato and Aristotle, knew a charlatan when they encountered one.

But Paul learned a powerful lesson. He had to make the tale of a Jewish Messiah more pagan. He had to add far more of Mithras into the mix. And that's what he did.

Christianity is actually Mithraic Judaic Messianism. When the Protestants came on the scene, they got rid of Mithras and returned to Jewish Messianism. If the Protestants had any sense of history and philosophy, they would classify themselves as a Jewish sect; in fact, a Jewish heresy.

There are some Jews who believe that Christ was the long foretold Jewish Messiah. This is essentially the position of Protestant Christians. By rejecting the paganism present throughout Catholic Christianity and accepting only the Scriptures, the Protestants unwittingly converted themselves into Jews. After all, the Old Testament is simply the Torah by another name. And the New Testament is about a Jew who believed in the Torah, and was written by Jews.

If you accept the Jewish Bible, supplemented by the preachings of a Jewish heretic (Yehoshua ben Yosef aka Jesus Christ) then you're a heretical Jew. It's that simple.

The Simulacrum

In postmodern hyperreality, the simulacrum is the point at which reality and fantasy merge. All authentic experiences dissolve. Reality vanishes. All that's left is propaganda. We become the consumers of our own delusions, and believe in them implicitly.

Science is the simulacrum of mathematics. Truths of reason (actual truth) are replaced by truths of fact (contingent interpretations).

Science is the “authentic fake”, the copy without an original. Except there is an original – the intelligible world of mathematics.

Sub Specie Aeternitatis

The universe can be viewed in two ways: *sub specie aeternitatis* (“under the aspect of eternity”) and *sub specie temporis* (“under the aspect of time”). The first is the eternal mathematical perspective and the second the contingent scientific perspective. Only the mathematical view is valid.

For anything to qualify as the answer to existence, it *must* be eternal. It must be valid at all times. Only one subject has necessarily eternal properties: *mathematics*. Mathematics alone

provides an eternal view of reality. It alone is true.

The Test

Science has never been able to account for human free will. Much worse, it either actively believes that humans do not have free will (a total absurdity) or it does not think that this issue is a big deal. Frankly, any theory that cannot provide an explanation for human free will is ipso facto false. Hence science is fundamentally in error.

The Absolute Truth

Anything that is true forever has *existed* forever. The only thing for which that is true is mathematics. Only the eternal, immutable, analytic tautologies of mathematics are Platonically true.

The Unknown God

The ancient Greeks worshipped the “Unknown God” to ensure they didn’t inadvertently leave any god out of their pantheon. Think of how tolerant that made them towards the ideas and beliefs of others. They always left a space in which others might be right. Contrast that with the Abrahamists. They fanatically toppled any other gods as “idols”. They left no space in which others might be right. They were, and are, intolerant maniacs with no place in a civilised world.

The Tipping Point

Science will never make any great progress from now on until it undergoes a radical paradigm shift, one that fully addresses the fatal contradiction between experimental empiricism and rationalist mathematics. A tipping point is required. The graveyards must become filled with the old, irrationalist empiricists and must be replaced by new gleaming ontological mathematicians – the glorious and heroic rebels and revolutionaries against the stultifying, careerist establishment.

The Newtonian world will vanish once and for all and we will enter the palace of the light of reason designed by Leibniz, the greatest genius in human history.

The Wisdom of Leibniz

“Music is the pleasure the human mind experiences from counting without being aware that it is counting. Music is nothing but unconscious arithmetic.” – Leibniz

“What *is* is what must be.” – Leibniz

“It is unworthy of excellent men to lose hours like slaves in the labour of calculation which could be relegated to anyone else if machines were used.” – Leibniz

“Theologian: But what is to love? Philosopher: To be delighted by the happiness of another.” – Leibniz

“Everything that is possible demands to exist.” – Leibniz

“I am convinced that the unwritten knowledge scattered among men of different callings surpasses in quantity and in importance anything we find in books, and that the greater part of our wealth has yet to be recorded.” – Leibniz

“Even in the games of children there are things to interest the greatest mathematician.” – Leibniz (the originator of Game Theory?!)

“Nature does not make leaps.” – Leibniz

“I have seen something of the project of M. de St. Pierre, for maintaining a perpetual peace in Europe. I am reminded of a device in a cemetery, with the words: *Pax perpetua*; for the dead do not fight any longer: but the living are of another humour; and the most powerful do not respect tribunals at all.” – Leibniz

“There are two famous labyrinths where our reason very often goes astray. One concerns the great question of the free and the necessary, above all in the production and the origin of Evil. The other consists in the discussion of continuity, and of the indivisibles which appear to be the elements thereof, and where the consideration of the infinite must enter in.” – Leibniz

“I have said more than once, that I hold space to be something purely relative, as time is; that I hold it to be an order of coexistences, as time is an order of successions.” – Leibniz

“In whatever manner God created the world, it would always have been regular and in a certain general order. God, however, has chosen the most perfect, that is to say, the one which is at the same time the simplest in hypotheses and the richest in phenomena.” – Leibniz

“If there were no best among all possible worlds, God would not have created one.” – Leibniz

“I do not believe that a world without evil, preferable in order to ours, is possible; otherwise it would have been preferred. It is necessary to believe that the mixture of evil has produced the greatest possible good: otherwise the evil would not have been permitted.” – Leibniz

“The combination of all the tendencies to the good has produced the best; but as there are goods that are incompatible together, this combination and this result can introduce the destruction of some good, and as a result some evil.” – Leibniz

“This miracle of analysis, this marvel of the world of ideas, an almost amphibian object between Being and Non-being that we call the imaginary number.” – Leibniz

“We never have a full demonstration, although there is always an underlying reason for the truth, even if it is only perfectly understood by God, who alone penetrated the infinite series in one stroke of the mind.” – Leibniz

“Music is a hidden metaphysical exercise of the soul, which does not know that it is philosophizing.” – Schopenhauer

The Solution

For existence to have an answer, existence *must* be mathematical, hence rational. The game is to bring about a transition from a physics-based understanding of reality to a mathematical one. This is the ultimate scientific revolution. It’s about leaving behind the Newtonian approach to physics and embracing Leibniz’s approach.

The Machine

“A house is a machine for living in.” – Le Corbusier

A body is a machine for a soul to live in.

A universe is a machine for a God to live in.

The Romans

The Romans killed Jesus Christ, not the Jews. Roman capital punishment was by crucifixion, Jewish capital punishment by stoning. The Romans executed Jesus Christ because he called himself the King of the Jews and raised an armed insurrection against the Romans in Jerusalem on the Passover, even riding into the city on a donkey to let the Jews know that he was their scripturally prophesied Messiah.

The idea that the Romans would have killed a Jew because other Jews requested it is comical. The Romans despised the Jews and would *never* have done their dirty work for them. The Romans killed Yehoshua ben Yosef because he rose in armed revolt against *them*, and not for any other reason. Why do Christians relentlessly lie about this?

Jesus Christ, like his namesake Joshua (Yehoshua) and (later) Mohammed, was a warrior prophet, not a peacemaker. Because the Roman garrison in Jerusalem militarily crushed him, thus making a mockery of his military credentials, his obsessed, deranged followers simply switched the narrative away from military defeat in this world to inevitable success in the next world, to a paradise of peace and harmony. They rewrote his whole story to present him as a sacrifice to “God” rather than as an insolent, crazy Jew who, like so many others of the era, thought he could topple the power of Rome.

In the end, the Romans, after enduring several such rebellions, destroyed the Jewish

Temple, razed the city, and banished its Jewish citizens to the rest of Judea.

Jesus Christ was no different from the various self-proclaimed Jewish Messiahs who were lunatic enough to challenge Rome. He was one of a long line of Jewish nutters suffering from “Jerusalem Syndrome”: the idea that they were on a special mission from Jehovah. The lunatic asylums are full of such people.

The Big Tell

Parents eventually have to admit to their children that Santa doesn’t exist and is just a childish fantasy figure. Why don’t they go on to admit that “Jesus Christ” is just a childish fantasy figure too?

Parents think it’s “cruel” to stop their children believing in Santa. Do they also think it’s “cruel” when others tell *them* to stop believing in Jesus?

Isn’t it time for all Abrahamists to grow up and leave behind their imaginary friends and childish comfort blankets?

We live in an infantilized world of children in adult bodies.

Science: An Intellectual Disgrace

In the nineteenth century, physicists asserted that ultimate reality was 100% deterministic. God definitely did not play dice. In the twentieth century, physicists asserted that ultimate reality was 100% indeterministic. Probability replaced causality. Dice was all that God played. What position will science hold by the end of the twenty-first century? Given that it has a 100% range, i.e. whatever is white today might be regarded as black tomorrow, or vice versa, who can say?

How can science claim to be a rational endeavour when it has no sure principles or foundations, when it has no analytic, *a priori*, necessary truths, when it does not operate according to a principle of sufficient reason and rejects ultimate causality, replacing it instead with probability?

Science is fundamentally *irrational*. It was driven to reverse its position on determinism by 180 degrees – the biggest possible change of position – because it dogmatically chooses to be empiricist rather than rationalist.

In the twentieth century, it became evident that science could hold onto determinism (a non dice-playing God!) only if it accepted the reality of “hidden variables”. Hidden variables are inherently beyond experimental physics, hence are metaphysical (or mathematical, as we would say).

With the advent of quantum mechanics, science had a very simple choice. It could remain a deterministic subject, but now acknowledging the limitations of the experimental method, and turning to mathematical metaphysics as the deterministic underpinning of physics. It would thus become science in the Leibnizian sense.

Or it could fanatically cling to the experimental method as the be-all and end-all, the sine qua non of science, and reject mathematics, metaphysics, rational unobservables, hidden variables and rationalism. It would thus remain science in the Newtonian sense (Newton being he who did not “feign hypotheses”).

No one ordered science to pursue the latter path. It chose to. It elected to reject causality, sufficient reason and determinism and become random, indeterministic, acausal, irrational, probabilistic and statistical. It did not do so for scientific reasons. It did so for philosophical and even quasi-religious reasons. Scientists chose to have absolute faith in empiricism and materialism and to reject the fundamental truth that quantum mechanics was revealing: that existence is rationalist and idealist.

Science gave up being science in the twentieth century. It was faced with a terrible dilemma, namely is the core of science (i.e. the experimental method: empiricism) the key to understanding reality or is the mathematical method (i.e. rationalism) the true path for science to pursue? Mathematics does not require any experiments at all. To turn to mathematics allows the admission of hidden variables to science, and the end of the primacy of the experimental method.

Faced with the choice between experimental verification and mathematical proof, science opted for the former and thus became irrationalist, formally rejecting determinism, causality and sufficient reason. This is the greatest intellectual blunder in human history. The antidote is ontological mathematics. Ontological mathematics posits that we live in a mathematical universe, not a scientific universe, an intelligible universe rather than a sensible universe.

The experimental method is the means for probing the sensible world, and mathematics the means for probing the intelligible world. Science once believed that the sensible world was also an intelligible world, hence the experimental method and determinism could happily coexist without contradiction. Quantum mechanics destroyed this hope.

If you believe that the world is sensible rather than intelligible (what you see is what you get; there are no hidden variables) then you must adopt an indeterministic, probabilistic approach to reality and say that the universe is ultimately grounded in randomness, i.e. you must accept the Copenhagen interpretation of quantum mechanics, which asserts, in effect, that the moon does not exist when no one's looking at it, and God is a dice player.

If you accept that the world is intelligible rather than sensible (there are truths of reason that no sensory experiment can ever reveal) then we live in a rational, determined, causal system of ontological mathematics, and quantum mechanics is not probabilistic at all: the wavefunction is ontologically real, not unreal (hence is in no need of being turned into a probability function).

The most controversial Nobel Prize ever awarded was to Max Born, the inventor of the probabilistic interpretation of quantum mechanics. This interpretation was the only one that could save the empiricist, materialist Meta Paradigm of science, but it destroyed once and for all the principle of sufficient reason, determinism and causality in science.

The issue could not be clearer: is the quantum mechanical wavefunction ontologically real or unreal? If unreal, Born's approach is correct. If real, Born was guilty of inverting the truth and turning science into an irrational joke. But how can reality be grounded in an unreal wavefunction? That's pure madness. And that's where science stands today. Only ontological mathematics can restore sanity to the madhouse.

The Heretics

"If we have no heretics we must invent them, for heresy is essential to health and growth." – Yevgeny Zamyatin

"The world is kept alive only by heretics: the heretic Christ, the heretic Copernicus, the heretic Tolstoy. Our symbol of faith is heresy." – Yevgeny Zamyatin

There are no heretics in modern science because science is now the Church. This secular Church funds scientists, and none are prepared to stand up against the Church (the hand that feeds them). You need no Inquisition to enforce orthodoxy when you have funding committees!

The New World

"Now I no longer live in our clear, rational world; I live in the ancient nightmare world, the world of square roots of minus one." – Yevgeny Zamyatin, *We*

On the contrary, the mathematical world, where the square root of minus one resides, is the quintessence of reason.

The Nobel Prize

Alfred Nobel wanted his annual prizes to be awarded "to those who, during the preceding year, shall have conferred the greatest benefit on mankind." In relation to the Nobel Prize for Physics, he said it should be given "to the person who shall have made the most important 'discovery' or 'invention' within the field of physics." It's up to the Nobel Committee to define what a "discovery" is and they have resolutely pursued a materialist and empiricist understanding, thus defining the nature of the scientific establishment. They have thereby inflicted huge damage on the progress of science and held back its inevitable transition to mathematics. Any prize decided by a committee is sure to be counterproductive and an obstruction to the pursuit of truth. Geniuses are infinitely far ahead of bureaucratic committees of careerist functionaries. Nobel wanted to advance humanity. Instead, he has held it back. His ridiculous prize should be abolished before it does any more harm. It entrenches all the false doctrines of science's faith-based Meta Paradigm. Science is simply the latest false prophet.

The Living Dead

“It is an error to divide people into the living and the dead: there are people who are dead-alive, and people who are alive-alive. The dead-alive also write, walk, speak, act. But they make no mistakes; only machines make no mistakes, and they produce only dead things. The alive-alive are constantly in error, in search, in questions, in torment.” – Zamyatin

Scientists are the “dead-alive.”

What is Science?

Science is sensory mania. Science is the sensory revulsion for the unseen and intelligible. Science is the worship of the sensible world over the intelligible world. Science is the doctrine that the senses are more reliable than reason. Science is anti-rationalism. Science is as bad and deluded as the other great enemy of rationalism: religious faith.

Hidden in Plain Sight

Euler’s Formula has been staring all mathematicians and scientists in the face for centuries. It’s hidden in plain sight. Of course, you have to be able to “see”. You have to be able to ask the right questions. Without that, all the knowledge in the world won’t help you.

It took the genius of Euler to establish Euler’s Formula. It took the greater geniuses of the Illuminati to see its true significance.

Dark Star

One of the greatest sci-fi movies is John Carpenter’s *Dark Star* (his first film). Wikipedia provides the following summary:

“*Dark Star* is a 1974 American comedic science fiction motion picture directed, co-written, produced and scored by John Carpenter, and co-written by, edited by and starring Dan O’Bannon. ... In the middle of the 22nd century, mankind has reached a point in its technological advancement that enables colonization of the far reaches of the universe. Armed with artificially intelligent ‘Thermostellar Triggering Devices’, the scout ship *Dark Star* and its crew have been in space alone for twenty years on a mission to destroy ‘unstable planets’ which might threaten future colonization.

“The ship’s crew consists of Lt. Doolittle, Sgt. Pinback, Boiler, and Talby. Commander Powell, their superior officer, was killed by a faulty rear seat panel, but remains on board the ship in a state of cryogenic suspension. The crew perform their jobs in a state of abject boredom as the tedium of their task has driven them around the bend, with only each other, an increasing number of (sometimes comical) systems malfunctions (for example, an explosion in a storage bay has destroyed the ship’s entire supply of toilet paper) and the soft-spoken ship’s computer for company. They have attempted to create distractions for themselves – Doolittle, formerly an enthusiastic surfer, has constructed a musical bottle organ, Talby spends all his time in the ship’s observation dome watching the universe go by, Boiler enjoys smoking cigars and target practice with the ship’s laser rifle, while Pinback enjoys playing practical jokes on the other crew members, maintains a video diary, and has also adopted a ship’s mascot in the form of a mischievous alien ‘beachball’ that refuses to stay put in the food locker and forces Pinback to chase it all over the ship.

“Pinback’s video diary includes an earlier entry in which he states that he is actually liquid fuel specialist Bill Frugge, who accidentally took Pinback’s place on the mission after

failing to rescue Pinback from committing suicide by wading into a fuel tank before the mission.

“While navigating an asteroid storm, en route to their next target (the Veil Nebula), the *Dark Star* is hit by a bolt of electromagnetic energy, resulting in an onboard malfunction and bomb #20 receiving the order to deploy. With some difficulty, the ship’s computer convinces bomb #20 that the order was in error and persuades the bomb to disarm and return to the bomb bay. To the complete disinterest of his crewmates, Talby decides to investigate the fault, and discovering a damaged communications laser in the airlock, Talby dons a spacesuit to investigate further. While attempting to repair the laser, Talby is blinded and inadvertently triggers a more serious problem, causing extensive damage to the ship’s main computer and damaging the bomb release mechanism on bomb #20.

“On arrival at their target planet, the bomb is armed in the usual way, but then the crew discovers they cannot activate the release mechanism and attempt to abort the drop. Bomb #20 becomes belligerent and refuses to disarm. Its detonation countdown is in progress and it refuses to abort the countdown sequence. The other crewmembers attempt to talk the bomb out of blowing up. Doolittle revives Commander Powell, who advises them to teach the bomb the rudiments of phenomenalism, resulting in a memorable philosophical conversation between Doolittle and the bomb. Bomb #20 aborts its countdown and retreats to the bomb bay for contemplation, and disaster appears to have been averted. Pinback addresses the bomb over the intercom in an attempt to finally disarm it.

“Doolittle has mistakenly taught the bomb Cartesian doubt, and as a result, later in the film, the bomb determines that it can only trust itself and not external inputs, states ‘Let there be light,’ and promptly detonates. Pinback and Boiler are killed instantly. Commander Powell is jettisoned into space encased in a large block of ice, and Talby is taken away by the Phoenix Asteroids (a cluster of glowing asteroids he had a fascination with) to circumnavigate the universe. Doolittle, who previously expressed his love of surfing and how much he misses it, finds an appropriately shaped piece of debris and ‘surfs’ down into the atmosphere of the planet, burning into an incandescent speck.”

Dark Star: Phenomenology

The most brilliant part of the film occurs when Doolittle uses phenomenology in his attempt to persuade Bomb 20 not to explode.

Scene: [Aboard the *Dark Star*, a spaceship on a mission to blow up unstable stars and planets. The bombs used to explode the unstable stars and planets are intelligent. Bomb 20 has falsely received the startup sequence to detonate.]

Pinback: You can't explode in the bomb-bay. That's foolish. You'll kill us all. There's no reason for it.

Bomb: I am programmed to detonate in nine minutes. Detonation will occur at the programmed time.

Pinback: Wouldn't you consider another course of action? For example, just waiting around a while, so we can disarm you?

Bomb: No.

[A few minutes later, Doolittle, in a spacesuit, arrives to talk directly to the bomb.]

Doolittle: Hello Bomb, are you with me?

Bomb: Of course.

Doolittle: Are you willing to entertain a few concepts?

Bomb: I am always receptive to suggestions.

Doolittle: Fine, think about this then. How do you know you exist?

Bomb: Well, of course I exist.

Doolittle: But how do you know you exist?

Bomb: It is intuitively obvious.

Doolittle: Intuition is no proof. What concrete evidence do you have that you exist?

Bomb: Hmmm... well... I think, therefore I am.

Doolittle: That's good... that's very good. But how do you know that anything else exists?

Bomb: My sensory apparatus reveals it to me.

Doolittle: Ah, right!

Bomb: This is fun!

Doolittle: Now listen, listen... here's the big question. How do you know that the evidence your sensory apparatus reveals to you is correct? What I'm getting at is this. The only experience that is directly available to you, is your sensory data, and this sensory data is merely a stream of electrical impulses that stimulates your computing centre.

Bomb: In other words, all that I really know about the outside world is relayed to me through my electrical connections.

Doolittle: Exactly!

Bomb: Why, that would mean that... I really don't know what the outside universe is like at all, for certain.

Doolittle: That's it! That's it!

Bomb: Intriguing. I wish I had more time to discuss the matter.

Doolittle: Why don't you have more time?

Bomb: Because I must detonate in 75 seconds.

Doolittle: Now, Bomb, consider this next question very carefully. What is your one purpose in life?

Bomb: To explode, of course.

Doolittle: And you can only do it once, right?

Bomb: That is correct.

Doolittle: And you wouldn't want to explode on the basis of false data, would you?

Bomb: Of course not.

Doolittle: Well then, you've already admitted that you have no real proof of the existence of the outside universe...

Bomb: Yes, well...

Doolittle: So you have no absolute proof that Sergeant Pinback ordered you to detonate?

Bomb: I recall distinctly the detonation order. My memory is good on matters like these.

Doolittle: Of course you remember it, but, But... all you're remembering is merely a series of sensory impulses which you now realise have no real definite connection with... with outside reality!

Bomb: True, but since this is so, I have no proof that you're really telling me all this.

Doolittle: That's all beside the point. I mean, the concept is valid no matter where it originates.

Bomb: Hmmm...

Doolittle: So if you detonate in ...

Bomb: ... 9 seconds ...

Doolittle: ... you will be doing so on the basis of false data!

Bomb: I have no proof it was false data.

Doolittle: You have no proof it was correct data!

Bomb: I must think on this further.

[Bomb returns to bomb-bay]

Computer [feminine voice]: Attention, attention. The Bomb has returned to the bomb-bay. The destruction sequence is aborted.

Pinback: Alright, Bomb, prepare to receive new orders.

Bomb: You are false data.

Pinback: Huh??

Bomb: Therefore, I shall ignore you.

Pinback: Hello, Bomb?

Bomb: False data can only act as a distraction, therefore, I shall refuse to perceive you.

Pinback: Hey, Bomb!

Bomb: The only thing which exists is myself.

Pinback: Snap out of it, Bomb!

Bomb: In the beginning there was darkness, and the darkness was without form, and void.

Pinback: Huh, hello? Bomb?

Bomb: And in addition to the darkness there was also me. And I moved upon the face of the darkness, and saw that I was alone...

Pinback: Hey, Bomb?

Bomb: Let there be light...

[Explodes, in a tremendous burst of light, destroying the space-ship...]

Scientific Materialism and Dark Star

Scientists are faced with exactly the same problem as Bomb 20. How do they know what external reality is? As Doolittle says: “Now listen, listen... here’s the big question. How do you know that the evidence your sensory apparatus reveals to you is correct? What I’m getting at is this. The only experience that is directly available to you, is your sensory data, and this sensory data is merely a stream of electrical impulses that stimulates your computing centre.” Scientists have placed absolute trust in their senses as the means to find out about external reality, but they have no sufficient reason to conclude that we live in a sensible (material) world rather than an intelligible (mental) world. Why would any rational person place more confidence in their non-rational senses rather than in their own reason to reveal ultimate reality?

Scientists are afraid of relying on something that cannot be verified, and it’s true that religion and philosophical metaphysics have both been highly speculative. However, mathematics is not speculative at all. Mathematics is the perfect rational system. Ontological mathematics is consistent and complete. It establishes an absolute plenum (“fullness”), outside which nothing can exist (non-mathematical existence is thus impossible). It’s stable under all conceivable transformations, hence can never be destroyed, an essential quality for anything that underpins eternal existence.

We can never know, from a sensory perspective, how our universe began. However, we can certainly work it out using our mathematical reason. And if we can work that out, we can work out *everything*.

“I am the way and the truth and the life. No one comes to the Answer except through me.”
– Ontological Mathematics

Well, what answer should Bomb 20 have given? What *is* the answer to existence? Eventually, if it were rational enough, the bomb would realise that the God Equation is the answer: the final, incontestable, absolute and infallible answer. The answer to why the universe is ordered, organised, intelligible and rational is that it’s literally made out of living reason!

Anyone who claims to have provided the answer to life, the universe and everything must show exactly why the answer is that and not anything else. Only one thing can accomplish this – the principle of sufficient reason itself. This principle is ontologically expressed through ontological mathematics defined by the God Equation, via which everything has a precise causal reason why it is thus and not otherwise. We live in a universe of pure reason – of pure *living* reason, which has two aspects: objective Aristotelian logic (machinelike), and subjective Hegelian dialectical logic (the messy logic of life striving to understand itself and realise what it truly is).

Objective reason is infallible. Subjective reason is fallible, and its task is to go on a great cosmic journey to arrive at the truths of objective reason: the eternal, infallible truths of reason.

The Two Pillars

Science rests on two incompatible and contradictory pillars:

- 1) Experimentation (empiricism)
- 2) Mathematics (rationalism)

In order to effect some sort of accord between 1) and 2), scientists have historically chosen to accept a subset of mathematics based on positive real numbers only (and negative numbers in the case of “charge”). Imaginary numbers, complex numbers, negative numbers (except those relating to charge), zero and infinity are all rejected as “unreal”.

The prevailing probabilistic interpretation of quantum mechanics flows solely from this peculiar and illogical rejection of complete and consistent mathematics, and the selection of only a positive real numbers subset. This is not a scientific but rather a philosophical choice. *It has nothing to do with science.*

So, science is essentially defined by one pillar – the experimental method – while the second pillar (of mathematics) is regarded as a mere aid or tool, very much a secondary, second-class pillar.

Science makes no attempt to explain why some of mathematics is “real” and the rest (the vast majority) unreal. Mathematics has no clear-cut status at all within science. Science has no theory regarding what mathematics actually is, where it originates and how it’s so essential to science. The status of mathematics is in point of fact the greatest challenge facing science. The whole of science is ultimately bogus unless it can explain math.

What if instead of accepting only a dubious, incomplete and inconsistent subset of mathematics, science included *all* of mathematics (consistent and complete)? This would usher in a scientific revolution.

What if the first pillar (the experimental method) were relegated to a mere aid or tool, and the second pillar (mathematics) elevated to the primary expression of science? This would bring about the desperately needed paradigm shift in science that would deliver, at long last, a grand unified theory of everything. With mathematics as primary, hidden variables, rational unobservables and mathematical metaphysics are all admitted to the scientific table, something unthinkable within the current paradigm.

Pillar 1 – the experimental method – is Newtonian.

Pillar 2 – mathematics – is Leibnizian.

Science must choose between 1 and 2). Hitherto, science has chosen 1), but the scientific Meta Paradigm has now conclusively failed. Science’s two most successful, experimentally verified theories – quantum mechanics and relativity theory – are formally irreconcilable.

Science, in its current incarnation, has reached the end of the road. It cannot get any closer to ultimate truth and to a final, conclusive theory. Science must now reject Newton and pillar 1) and embrace Leibniz and pillar 2). Science must be redefined as ontological mathematics. Reality is mathematical, not scientific. Reality is intelligible (mathematical), not sensible (scientific). Pythagoras and Plato understood this almost 2,500 years ago. It’s about time science caught up!

Science believed it could escape philosophy and metaphysics. This naive belief has been confounded. The experimental method – the highest expression of the empiricist and materialist

philosophy – has failed. All thinkers must now turn to the opposite school – rationalism and idealism. However, there's one vast difference between the old and new versions of rationalism and idealism. The former was the province of philosophers, while the latter requires ontological mathematicians. Mathematics has become the language of metaphysics, not ingenious philosophical speculation. Mathematical proof is now what counts. It replaces philosophy and experimental science. We can be certain of the new school of rationalism and idealism because we can be certain of mathematics. We have no need of experiments, except as a secondary check. Mathematics now becomes all-powerful, and mathematics is of course not reliant on experiments at all. The obsession with experimentation must end. Mathematical proof is the road to certainty.

Max Tegmark is a classic example of a theoretical physicist who knows that mathematics is the way forward, yet he lacks the nerve and courage to abandon the scientific establishment. Rather than get rid of the failed empiricist materialist Meta Paradigm of science, he simply chooses to bolt a mathematical Multiverse beneath the traditional scientific Multiverse. He still talks of trying to experimentally “verify” his system even though the entire point of mathematics is that it has nothing to do with experimentation. You can't claim that the solution to existence is mathematics and then prattle on about the need for non-mathematical experiments. It's one or the other. Science must choose. It's Newton or Leibniz. It can't be both.

To all scientists, we say this – you are not rational unless you acknowledge the primacy of ontological mathematics over physics. You are a prisoner of irrational belief in your senses. As Nietzsche observed, you are a drudge and plebeian thinker who doesn't have the patrician imagination to conceive of something beyond the senses. What lies beyond the senses? – reason. What lies beyond physics? – mathematical metaphysics. It's mathematics that's the ground of physics, not the other way around. Mathematics can explain physics. Physics cannot explain mathematics.

The world needs a new generation of thinkers, those who have the guts to go beyond the likes of Tegmark, stuck between two conflicting systems of physics and mathematics and unwilling to choose between them. The world needs audacious thinkers who refuse to bow to the establishment drones and drudges.

“Successful research impedes further successful research.” – Keith J. Pendred

The success of science is precisely what has stopped the greatest thinkers from admitting that science has failed when it comes to addressing the biggest questions.

“A science is any discipline in which the fool of this generation can go beyond the point reached by the genius of the last generation.” – Max Gluckmann

The tragedy is that science is now run by fools. And, actually, the great geniuses such as Pythagoras, Plato, Leibniz and Gödel are still far beyond all current scientists.

“The progress of science is strewn, like an ancient desert trail, with the bleached skeletons of discarded theories which once seemed to possess eternal life.” – Arthur Koestler

Scientists think this is a compliment. It's not. Science is utterly arbitrary and ad hoc. It has no necessary truths. Any theory can be refuted at any time. You can't rely on any scientific “truth”. It might not be true tomorrow. Science once advocated 100% determinism and now advocates 100% indeterminism at the quantum level. Science is therefore not a rational subject. Only a lunatic would imagine that science is the Way of Truth.

“Modern science is generally practised by those who lack the flair for conversation.” – Fran Lebowitz

And the flair for rational thinking.

“Science is the religion of the suburbs.” – Oscar Wilde

Science is for drudges and drones.

“The real accomplishment of modern science and technology consists in taking ordinary men, informing them narrowly and deeply and then, through appropriate organization, arranging to have their knowledge combined with that of other specialized but equally ordinary men. This dispenses with the need for genius. The resulting performance, though less inspiring, is far more predictable.” – John Kenneth Galbraith

This is the whole problem with science. Geniuses have been replaced by pedestrian functionaries, with no imagination, no intuition, and wholly ignorant of philosophy and the history of thought.

“Science is a cemetery of dead ideas.” – Miguel de Unamuno

And the final grave is reserved for the experimental method as the keystone of science.

The Final Damnation

“A scientific truth does not triumph by convincing its opponents and making them see the light, but rather because its opponents eventually die and a new generation grows up that is familiar with it.” – Max Planck

No statement could be more scathing about the scientific “method”. In truth, there is no *method* in science. There are only believers in various theories. Science is a popularity contest, a fashion parade. It has nothing to do with Truth.

The “Turn”

All of thinking – the whole of Logos – must take a “mathematical turn”. Henceforth, mathematics should be the baseline for everything. All languages that are non-mathematical should automatically be understood to be approximate, imprecise, contingent, interpretive, and no source of definitive, absolute truth. Mathematics is the language of Truth. It’s the only one. Other languages are capable of asserting truth only to the extent that they can now and again reflect mathematics.

Nietzsche was the great destroyer of all claims to truth and showed how we are caught in deceptive webs of language. Nietzsche would be the greatest philosopher and thinker of all time – all of his conclusions absolutely correct – were it not for one thing: *mathematics*.

It’s mathematics that refutes Nietzsche, that refutes nihilism, skepticism, cynicism, relativism, existentialism postmodernism, absurdism, deconstructionism, materialism, empiricism and faith.

Mathematics, and mathematics alone, is the language of absolute, eternal, immutable, indestructible Truth. Any statement that is not mathematical or describing mathematics philosophically cannot be wholly and unarguably true. It’s immediately swallowed up by interpretation, belief, conjecture, hypothesis and opinion.

The Non-Saviour

Modern science says that the only thing saving the universe from absolute chaos is that quantum mechanics says some outcomes are more probable than others, and, from this, a kind of quasi determinism can be established, i.e. we don't know exactly what is going to happen (there's no absolute causality or principle of sufficient reason), but we can know what is probably going to happen, and if improbable things happen we simply say, "That's because of the indeterminacy."

Modern science is just about the most bogus, irrational subject there is. It completely denies that the universe is grounded in reason and causality, which means, of course, that it must be grounded in unreason and non-causality – sheer chaos and randomness, from which order and intelligibility could never arise under any circumstances.

The *deus ex machina* that science pulls out of the hat to save its irrationalist system is *probability*. From probability, a pseudo rationalist system is constructed, and 99% of scientists have declared that they are perfectly happy with randomness, acausality and probability over causality. Of course, what they never explain is why mathematics – the ur rationalist subject – is at the heart of modern science and how, in a system of pure randomness and chaos, anything is more probable than anything else. Why aren't all outcomes equally probable? If this were the case, science would degenerate into total randomness and there would be no cosmos at all since order would be impossible.

Science refuses to account for its mathematical engine and the greater probability of some outcomes over others, hence is just a ridiculous Mythos that no intellectual could take seriously.

Modern science is the only formally irrationalist system of thought. All other systems of thought have placed rational consciousness (God) or rational unconsciousness ("Nature" or mathematics) at the root of existence. Modern science, in a 100% contradiction of classical science, places irrational chaos, randomness and acausality at the root of existence, and then dishonestly invokes "probability" to allow it to escape from the prison of perpetual disorder it has constructed for itself.

Science rejects mind, conscious or unconscious, and says everything in our world is grounded in lifeless, mindless, non-conscious material atoms that miraculously sprang out of nothing at all (or a magician's top hat) ... out of non-existence. Science is intellectually offensive and an insult to reason. It's based on faith, magic, miracles and pure unreason. It's the gospel of irrational disorder (chaos) inexplicably ordering itself.

Science is the debunked, discredited faith of empiricism and materialism, of the slavish worship of the human senses over eternal reason. What a disgrace. Science must be wholly replaced with rationalist, intelligible, causal ontological mathematics.

The whole purpose of scientific irrationalism is to defend scientific ideology, dogmatism, and faith – the Meta Paradigm of materialism and empiricism. Above all, to preserve the importance of the experimental and observational method. As soon as rational "hidden variables" (rational unobservables), such as those Einstein advocated to prevent God "playing dice", are admitted, experiment and observation are deposed as the centrepiece of science and replaced by mathematics. Mathematics, of course, requires no experiments or observations, hence is the "Devil" to the experimental "God" of science.

Science is an irrational faith system that is based on the worship of things that can be brought into human sensory experience, and the rejection of everything that can't (which are called "non-existent" even though science cannot provide one shred of evidence that

mathematical hidden variables are non-ontological, and it would be wholly irrational to conclude, as science does, that mathematics contains sensory ontological elements and non-sensory non-ontological elements).

Life

What is the defining basis of mind and life? It's something incredibly simple – the regular, constant, uniform flow of self-organizing, self-solving, self-optimising mathematical information within an autonomous, indestructible, immortal container.

If there were only a single cosmic container, there would be only one mind, a Cosmic Mind, an evolutionary God all on its own. However, there are myriad autonomous containers – monads (souls).

We live in a universe of minds, all either competing or cooperating to enhance their power. This is what the dialectic is all about.

Consciousness is simply information that has become self-aware: an inevitable outcome of any living system of self-optimising information. Such a system creates language, concepts, the notion of “I”, and hey presto, you have consciousness.

Creation or Evolution?

Existence predicated on conscious reason (a being called “God”) is Creationist.

Existence predicated on unconscious reason (mathematical monads) is Evolutionary and Dialectical.

The first is about reflecting the perfection of the Creator and slavishly worshipping and obeying him (a slave morality for the followers, and master morality for God and his representatives). The second is about things self-optimising and becoming perfect themselves (an autonomous morality).

The first is all about the being of God. The second is all about becoming God.

The Flow

In order for the cosmos to exhibit life and mind, life and mind must be present in elementary form in the basic stuff of existence. So, what is this most elementary form of life and mind? The basis of life and mind is simply the *flow of information*.

Life is ontological mathematics. Mind and life are the same thing. Any mental thing is alive. Anything alive has mental activity. Even apparently dead things are just living things at one step removed. They are under collective rather than individual mental control.

Science says life and mind can emerge from non-life and non-mind. This is absurd and impossible. Of course, science has never given any definition of life and mind. Illuminism does: life and mind are the ordered flow of information. This information is precise, analytic mathematical information obeying just one formula: the generalised Euler Formula (God Equation). Random, chaotic, disordered, indeterminate, probabilistic information – the basis of scientific materialism – can never be the basis of life and mind. Absolute, infallible order and precision are required, and only math can deliver them.

It might seem odd to people, but math and life are exactly the same thing. We live in a 100% mathematical universe, and that means a 100% living and mental universe. Matter is an illusion collectively generated by myriad monads via Fourier mathematics.

Consider DNA, the basis of life on Earth. What is it? It’s just a double helix mathematical information code!

Information is structure. Information is pattern. Information is order. Information is organisation. Information is causal flow. Information is reason. Information is mathematics. Flowing information is mind Flowing information is life.

The Ignoramus

“Mathematics may be defined as the subject in which we never know what we are talking about, nor whether what we are saying is true.” – Bertrand Russell

Everything about the statement is false. It’s staggering that so many mathematicians follow Russell’s lead.

“Mathematics takes us into the region of absolute necessity, to which not only the actual world, but every possible world, must conform.” – Bertrand Russell

How is this consistent with Russell’s previous comment saying that humans don’t know what they’re talking about mathematically? What a fool!

“I like mathematics because it is not human and has nothing particular to do with this planet or with the whole accidental universe – because, like Spinoza’s God, it won’t love us in return.” – Bertrand Russell

There’s nothing “accidental” about the universe. And mathematics has everything to do with our world.

Tesla

“Today’s scientists have substituted mathematics for experiments, and they wander off through equation after equation, and eventually build a structure which has no relation to reality.” – Nikola Tesla

Here we see the classic scientific error that “reality” is exclusively sensory. In fact, it’s the sensory world that fails to be “real”. The real world is the intelligible world of mathematics.

More’s the Pity

“One of the most amazing things about mathematics is the people who do math aren’t usually interested in application, because mathematics itself is truly a beautiful art form. It’s structures and patterns, and that’s what we love, and that’s what we get off on.” – Danica McKellar

It’s about time you got interested in the ontology of mathematics, which is the ontology of the soul itself.

“Secondary” Mathematics

“Language is remarkable, except under the extreme constraints of mathematics and logic, it never can talk only about what it’s supposed to talk about but is always spreading around.” – Howard Nemerov

Mathematics is the ur language, and the sole language of existence and truth.

?????

“In mathematics the art of proposing a question must be held of higher value than solving it.” – Georg Cantor

Only to a lunatic. The astute question is the *first part of the solution*. You would not ask the question if you didn’t crave its answer. Unanswered questions are valueless.

Math

“All science requires mathematics. The knowledge of mathematical things is almost innate in us. This is the easiest of sciences, a fact which is obvious in that no one’s brain rejects

it; for laymen and people who are utterly illiterate know how to count and reckon.” – Roger Bacon

But mathematics has no requirement for science. Math is the boss.

The Falsehoods

“Logic and mathematics are nothing but specialised linguistic structures.” – Jean Piaget

On the contrary, mathematics is the ur language, to which all other languages are clunky and fallible approximations.

“Mathematics is a game played according to certain simple rules with meaningless marks on paper.” – David Hilbert

Mathematics is no game. Mathematics is existence itself.

The Many Names

“Mathematics is the art of giving the same name to different things.” – Henri Poincare

Mathematics is how to write $0 = 0$ in countless different ways.

The Hate

Leonardo da Vinci said, “Study without desire spoils the memory, and it retains nothing that it takes in.”

People hate mathematics because they have no desire to study it and can’t see any point to it in their day-to-day lives. If they realised it was the basis of their soul, they would regard it in a whole new light.

The Core of Life and Mind

Life = information flow.

Mental activity = life = information flow.

The mathematical universe is all about the flow of information.

The soul is all about the flow of information. *As above, so below.*

The more efficiently and powerfully information flows, the greater the life and mind involved.

The greater the organisation and structure of information, the greater the life and mind involved.

It's impossible to provide any definition of mind or life that is not based on structured information flow, hence which does not involve mathematics.

It's the great secret of the universe that mathematics itself is structured information flow, hence is the ground of life and mind. Nothing else involves structured information flow. Only mathematics is alive and mental.

The mystery of existence is the mystery of mathematics. But mathematics isn't mysterious. It's the most rational, knowable and intelligible thing there can possibly be.

The Question

What did God do in the *eternity* before Creation – when he was *completely on his own*? Why did he suddenly decide to create? What was his sufficient reason? Surely he wasn't *bored*.

Panlogism

“In philosophy, panlogism is a Hegelian doctrine that holds that the universe is the act or realization of Logos. According to the doctrine of panlogism, logic and ontology are the same study.” – Wikipedia

“The real is rational; the rational is real.” – Hegel

Hegel depicted a universe of dialectically evolving reason. That's exactly what living mathematics is. Hegel thought that the rational dialectic would culminate with the *Absolute* – the Absolute Mind/Spirit and Absolute Knowledge. This is equivalent to Absolute Reason, the Omega Point of Reason, its optimisation, its maximum expression.

Mathematics is none other than the Principle of Sufficient Reason as ontology. Reason exists. Reason is alive. Reason is Life. Reason is Mind. However, it's evolving Reason, not set Reason. Reason must optimize itself in order to know itself, to be self-aware – conscious.

God Consciousness is the ultimate and highest expression of reason. Once you have attained God Consciousness, you understand all of existence, hence all of mathematics.

Pythagorean philosophy, Platonism, Neoplatonism, Leibnizianism, and Hegelianism are all major stageposts on the road to God consciousness. The *God Series* itself is the culmination of all of these. It's the closest thing yet to expressing the Mind of God. In a hundred years, it will be updated, and it will keep being updated every hundred years as humanity converges on divinity.

The less rational you are, the further you are from divinity. The stupid are the damned, the beasts.

People of faith have zero chance of becoming God. They are the endarkened. There's not one true aspect to any belief system. They contain zero truth content.

Mathematics is 100% true. If it's not math, it's false!

The Dialectic

Absolutely everything is about information, and information is conveyed by only one thing – sinusoidal waves (sines and cosines). All information in the universe – whether it concerns atoms, quarks, Higgs bosons, DNA, sound waves, dark energy, dark matter, black holes, the Big Bang, light, and even thoughts – is about combinations of sinusoids. A single theory of sinusoidal information (based on the God Equation) is the explanation of everything.

What is the dialectic? It's the cosmic march of information, of information optimising itself, through the brute force method (in the absence of conscious planning) of bringing opposites together (thesis and antithesis), making them fight it out and then using the outcome of the battle (the synthesis) to act as a new thesis, and then going through the whole cycle again and again. This process, across countless iterations, and often taking backward steps, eventually converges on an Omega Point – perfection itself where all theses and antitheses are rationalised in one cosmic synthesis. This is the ideal answer to everything, with which no rational person can disagree. This is the ultimate expression of the General Will, the fairest and most just solution for all. It's the opposite of Satanic anarchy, libertarianism and free-market capitalism.

The Illuminati are the guardians, the knights, of the Omega Point. We are the ones driving the Dialectic towards its inevitable rational conclusion. We are opposed by many irrational antitheses, but we shall overcome them all. Such is the nature of the rational dialectic. Are you on the side of the final victors, or are you futilely opposing them, like dinosaurs?

Objectively, information is perfect. Subjectively, it's imperfect. It starts out with limitless potential and no actualisation, and its dialectical task is to turn all potential into actualisation, to turn bare monads (souls) into full Gods.

A critical part of the dialectic is for information and reason to become self-aware – so that they can then rationally plan their own optimisation. The Illuminati are those who now understand the cosmic dialectic so well that we can see exactly how it must be directed to make humanity divine. This is what the *God Series* is all about. This is the blueprint for human perfection, for a *Star Trek* future, without Abrahamists, Karmists, anarchists, libertarians, free-market capitalists, scientific materialists and all right wingers, traditionalists, conservatives and fundamentalists.

We are the future. We are the Plan. We are the Dialectic. We are the Illuminati.

Life is about the better and better, the more and more optimal, structuring and organisation of information. It's about the quality of networks. A soul is a massive, complex network of sine and cosine waves. Every network has an optimal expression – an omega point, its “God point”. Subjectively, this corresponds to the experience of being God.

The Evolution of Illuminism

Pythagoreanism – all things are numbers. Mathematics is the arche, the first principle, the root of all. This view has never altered in Illuminism.

Platonism – the division of reality into the intelligible (mental) and sensible (material) domains. The material world is a flawed, inferior, perishable copy of the perfect, immutable domain of Forms.

Neoplatonism – at the bottom of everything is the ineffable “One”. Its emanations – Nous (Logos) and Psyche – structure and order the world.

Leibnizianism – the universe is made of monadic souls – immaterial, dimensionless points (mathematical singularities). The material world is a projection of mathematical mind.

Hegelianism – the universe is dialectically evolving reason (logos), converging on its Absolute condition (Absolute Knowledge; Absolute Mind/Spirit)

Modern Illuminism is all of the above put in a precise, analytic mathematical framework of the generalised Euler Formula and Fourier mathematics.

Non-Privilege

“Something” cannot be privileged over “nothing”, and vice versa. What system can ensure that something and nothing are perfectly balanced? The only system is mathematics. The question of why anything exists is, oddly enough, the same question as why nothing exists. It’s hard to grasp, yet absolutely true, that the universe, which seems so resolutely “something”, is just as resolutely “nothing”.

True reality comprises an immaterial, dimensionless Singularity outside space and time, the objective properties of which all balance to exactly zero.

When Leibniz asked why there is something rather than nothing, he could just as easily have asked why is there “nothing” rather than “something”.

The Singularity is not quite absolute nothingness because although all of its properties ultimately balance to zero, they do so over a succession of states and not over a single state. This is the meaning of the dynamic Euler Formula. The generalised Euler Formula delivers a perfect zero but only across all of its different states within its cycle. Each individual state is “something” but the overall state for a full cycle of states is “nothing”, so something and nothing are perfectly reconciled in this system.

We inhabit a living information system. All of its objective “material” properties are zero overall, but its informational properties are always something. Monads – mathematical subjects – are dimensionless points overall (“nothing”) and yet they are always “thinking” (experiencing and processing information) because whenever you examine a monad, you will always find it in a definite state (“something”).

The universe is always nothing averaged across all of its states and yet is always something at each particular state. This is the mathematical, informational universe of becoming. This is the universe of “now”, of the present instant, of the present state. Current states immediately cease to exist (they become the “past” and cannot be recovered under any circumstances), and future states do not exist because they have not yet been reached (and cannot be jumped to under any circumstances). All that exists is the current state at this instant (“now”, the “present”, the razor’s edge between the non-existent past and non-existent future). Everything is mathematically calculated at this instant for this state, and there is nothing else.

The only thing that can deliver this is the generalised Euler Formula, the God Equation

that controls existence, that controls “something” and “nothing”.

“Nothing” it must be stressed, is the overall state, the average state, the state averaged across all states, the general state. “Something” is the current state, the particular state.

People have traditionally thought in binary terms: of either something or nothing. In fact, they should have been thinking in terms of averaged states (*nothing*) versus unaveraged states (*something*). “Nothing” is a perfectly symmetrical state and “something” is the asymmetric state that exists at any particular instant. The average of all asymmetric states is a perfectly symmetric state.

Both a monad and the universe itself (*as above, so below*) are a succession of asymmetric states (“something” states) that average overall to a perfect symmetric state (“nothing”). Everything is cyclical. The end of the cycle is what closes the symmetry. The Big Bang event is the first asymmetric state of a cyclical cosmic Age, and the Big Crunch is the final asymmetric state of a cyclical cosmic Age, which seals the symmetry for the entire cosmic age, the full cycle. The universe is based on nothing but circles, but these are complex Euler circles, involving imaginary numbers as well as real numbers.

The Adventure

The human adventure is only just beginning, and it’s purely thanks to the Euler Formula. This is what separates us from the beasts and provides us with the platform to be Gods.

The Light

In the Book of Genesis, “God” says, “Let there be light.” But what *is* light? Light is the basis of real space and comprises analytic cosine waves. Time is the complement of light and comprises analytic imaginary sine waves. (The counterpart of light particles – photons – are time particles: *chronons*.)

From light and time we get spacetime (or light-time, to be more accurate). The material world is all about the interaction of photons and chronons, light and time, space and imaginary space. The whole shebang is mathematical.

If “God” were real, we all know what he would really have said:

“Let There Be Mathematics!”

